BATTLETECH™

BATTLEMECH MANUAL

CATALYST game labs™

YOUR AVATAR AWAITS!

- *Standing from seven to sixteen meters tall, and weighing from twenty to one hundred tons*
- *Powered by an armored and shielded fusion reactor*
- *Skeleton of honeycombed, foamed aluminum core wrapped with stressed silicon carbide monofilament and sheathed by a rigid, titanium-steel shell*
- *Locomotion generated via bundles of polyacetylene-fiber myomer muscles*
- *Protected by aligned-crystal steel over a layer of boron nitride impregnated with diamond monofilaments*
- *Mounting a swath of powerful weapons from charged particle beams to lasers, missiles to rapid-fire autocannons*
- *All at the command of the noble elite, the MechWarriors*

The modern BattleMech is the end result of more than three thousand years of battlefield technology development. Combining awesome destructive power and unparalleled maneuverability, the BattleMech is perhaps the most complex machine ever produced. The undisputed master of thirty-first century warfare, the BattleMech seems destined to reign supreme for centuries to come.

—Excerpts from a promotional pamphlet originally distributed by Defiance Industries of Hesperus, Lyran Commonwealth, 3007

Using the most up-to-date rules text, the *BattleMech Manual* has been designed from the ground up to cater to the *BattleTech* player wanting to engage in an all-'Mech battle. Includes a variety of optional rules, terrain and more.

Welcome to the most table-usage-friendly *BattleTech* rulebook ever published!

Used under license from

Topps

CATALYST game labs

©2023 The Topps Company, Inc. All Rights Reserved. BattleMech Manual, BattleTech, Alpha Strike, BattleMech, 'Mech and MechWarrior are registered trademarks and/or trademarks of The Topps Company, Inc., in the United States and/or other countries. Catalyst Game Labs and the Catalyst Game Labs logo are trademarks of InMediaRes Production, LLC. Printed in China.

BATTLEMECH MANUAL

CATALYST game labs™

TABLE OF CONTENTS

INTRODUCTION 5
Vs. *Total Warfare* 5
Optional Rules 5
GAME TERMS 5
'Mech 5
Four-Legged 'Mech 5
Movement Points 6
Target 6
Target Number 6
Modified Target Number 6
Attacker Movement Modifier 6
Target Movement Modifier 6
Margin of Success/Failure (MoS/MoF) 6
Armor Value 6
Damage Value 6
Location 6
Slot 6
Level 6
Mapsheet 6
Playing Area 6
Scenario 6
COMPONENTS 7
Counters 7
Dice 7
Mapsheets 7
'Mech Record Sheet 7
SCALE 8
A Note on Realism and Scale 8
Level 8
THE MECHWARRIOR 9
Skills 9
Damaging A Warrior 9
PLAYING THE GAME 10
DEPLOYING 'MECHS 10
SEQUENCE OF PLAY 10
Initiative Phase 10
Movement Phase 10
Weapon Attack Phase 10
Alternate Initiative 10
Moving Multiple 'Mechs 10
Segmented Fire 11
Physical Attack Phase 12
Heat Phase 12
End Phase 12
VICTORY CONDITIONS 12
UNEQUAL NUMBERS OF 'MECHS . . . 12
MOVEMENT 13
MOVEMENT MODES 13
Standing Still 13
Walking 13
Running 13
Piloting Skill Rolls 13
Movement Dice 13
Jumping 14
Sprinting (Optional) 14
MOVEMENT BASICS 14
Facing 14
Ground Movement 15
Terrain 16
Minimum Movement 16
Immobile 18
Stacking 18
JUMPING MOVEMENT 18
PRONE BATTLEMECHS 19
Dropping to the Ground 19
Moving While Prone 19
Standing Up 19
MOVEMENT IN WATER 19
MOVEMENT ON PAVEMENT 20
Road Movement 20
Bridge Movement 20
Skidding 20
RECKLESS MOVEMENT 20
OTHER ACTIONS 21
Hidden 'Mechs 21
Lifting Items (Simplified) 21
TAG Designation 21
COMBAT 22
LINE OF SIGHT 22
Levels and Height 22
Intervening Terrain 22
Other 'Mechs and LOS 23
Water Hexes 23
FIRING ARCS 24
Forward Arc 24
Left Side Arc 24
Right Side Arc 24
Rear Arc 24
Arcs and Torso Twists 24
Arcs and Reversing (Flipping) Arms 24
ATTACK DECLARATION 24
Reversing (Flipping) Arms 24
Expanded Arm Flipping (Optional) 25
Torso Twists 25
FIRING WEAPONS 25
Making the Attack 25
G.A.T.O.R. 25
Base Target Number 25
Target Number Modifiers 25
Combat in Water 27
Prone 'Mechs 30
Specialized Attacks 30
Rolling to Hit 31
Ammunition Expenditure 31
HIT LOCATION 32
Attack Direction 32
Determining Hit Location 32
Damage Resolution 32
PHYSICAL ATTACKS 34
Making A Physical Attack 34
Different Levels 34
Physical Attacks And Prone 'Mechs 35
Physical Attacks And Water 35
Charge Attacks 35
Club Attacks 36
Death From Above Attacks 36
Kick Attacks 38
Physical Weapon Attacks 38
Punch Attacks 38
Push Attacks 40
Damage Resolution 40
DAMAGE 41
DAMAGE RESOLUTION 41
Step-By-Step Damage 41
DAMAGE TIMING 43
HULL INTEGRITY AND BREACHES . . 43
LOCATION DESTRUCTION 43
Transferring Damage 44
Head Destruction 44
Leg Destruction 44
Torso Destruction 45
CRITICAL HITS 45
Floating Criticals (Optional) 45
Applying Critical Hits 46
CRITICAL HIT EFFECTS 47
Ammunition 47
Arm Blown Off (Arm) 47
Cockpit (Head) 47
Engine (Torso) 47
Foot Actuator (Leg) 47
Gyro (Torso) 48
Hand Actuator (Arm) 48
Head Blown Off (Head) 48
Heat Sinks 48
Hip (Leg) 48
Jump Jet/UMU (Leg/Torso) 48
Leg Blown Off (Leg) 48
Life Support (Head) 48
Lower Arm Actuator (Arm) 48
Lower Leg Actuator (Leg) 49
Sensors (Head) 49
Shoulder (Arm) 49
Upper Arm Actuator (Arm) 49
Upper Leg Actuator (Leg) 49
Weapons And Equipment 49
DESTROYING A 'MECH 49
HEAT 50
HEAT POINTS 50
Building Up Heat 50
Dissipating Heat 51
Recording Heat Build-Up 51
EFFECTS OF HEAT 51
Movement 51
Weapon Attacks 52
Shutdown 52
Ammunition 52
Damage To MechWarriors 52
OTHER ACTIONS 53
PILOTING SKILL ROLLS (PSR) 53
Making Piloting Skill Rolls 53
Piloting Skill Roll Timing 55
DISPLACEMENT 55
Domino Effect 55
FALLING 56
Location After a Fall 56
Facing After a Fall 56
Falling Damage to a 'Mech 57
Falling Damage to a MechWarrior 57
Accidental Falls From Above 57
DAMAGING A MECHWARRIOR 58
DUMPING AMMUNITION 58
TAG DESIGNATION 58

TABLE OF CONTENTS

THE BATTLEFIELD 59
PLANETARY CONDITIONS. 59
TERRAIN TYPES 59
Clear. 60
Jungle, Light 60
Jungle, Heavy 60
Pavement. 60
Rough 60
Rubble. 60
Sand. 60
Water (Depth 0) 60
Water (Depth 1) 60
Water (Depth 2) 60
Water (Depth 3+). 60
Woods, Light. 60
Woods, Heavy. 60
TERRAIN MODIFICATIONS 61
Ice (Simplified) 61
Mud (Simplified) 61
Smoke, Light. 61
Smoke, Heavy. 61
Snow, Deep. 61
Swamp (Simplified) 61
WEATHER CONDITIONS. 62
Extreme Temperatures 62
Fog 62
Light. 62
Rain 62
Snow 63
Wind 63
FIRE AND SMOKE (SIMPLIFIED) . . 63
Wind Direction. 63
Starting a Fire 63
Effects of Fire 63
Spreading Fire 64
Extinguishing Fire. 64
UNDERWATER COMBAT 64
Attacking Underwater. 64
Hull Integrity And Breaches. 65
URBAN COMBAT. 66
BUILDING TYPES 66
Construction Factor 66
Building Levels. 66
Basements. 66
Simplified Buildings 66
MOVEMENT EFFECTS 67
Moving Into Buildings. 67
Skidding. 68
COMBAT EFFECTS. 71
Attacking Buildings 71
Attacking 'Mechs Inside Buildings 71
Combat Within Buildings. 72
Collapse 73
SPECIAL CASE RULES . . 75
BATTLEFIELD SUPPORT. 75
Before Play Begins. 75
During Game Play. 75
DROPPING 'MECHS (SIMPLIFIED) . . 79
Drop Times 79
Arrival 79
EJECTION (SIMPLIFIED) 81
FORCED WITHDRAWAL 81
Crippling Damage. 81
Broken Morale (Simplified—Optional) 81
HIDDEN 'MECHS 82
Point-Blank Shots 82
DESIGN QUIRKS 82
Positive Design Quirks. 82
Negative Design Quirks 86
BattleMech Quirk Listing 89
Master Unit List. 89
WEAPONS AND EQUIPMENT 96
BASIC INFORMATION 96
WEAPON TYPES 96
ARTILLERY. 97
Arrow IV Homing Missiles (Simplified). 97
AUTOCANNONS 98
Autocannon 98
LB-X Autocannon (LB-X) 98
Light Autocannon (LAC) 98
ProtoMech Autocannon (PAC) 98
Rotary Autocannon (RAC) 98
Ultra Autocannon (UAC). 99
FLAMERS 99
Flamer. 99
ER Flamer 99
Heavy Flamer 99
GAUSS RIFLES 99
Gauss Rifle. 99
Anti-Personnel Gauss Rifle. 99
Heavy Gauss Rifle 100
Hyper-Assault Gauss Rifle. 100
Improved Heavy Gauss Rifle 100
Light Gauss Rifle 100
Magshot Gauss Rifle. 100
Silver Bullet Gauss Rifle. 100
LASERS 100
Laser 100
Binary Laser (Blazer) Cannon. 100
Chemical Laser. 100
Extended-Range (ER) Laser 101
Extended-Range (ER) Pulse Laser 101
Heavy Laser. 101
Improved Heavy Laser. 101
Pulse Laser. 101
Re-engineered Laser 101
Variable-Speed Pulse (VSP) Laser 101
X-Pulse Laser. 101
M-POD 101
MACHINE GUNS. 101
Machine Gun (MG) 101
Heavy Machine Gun (HMG). 101
Light Machine Gun (LMG) 102
Machine Gun Array (MGA). 102
MISSILE LAUNCHERS 102
Advanced Tactical Missile (ATM) Launcher 102
Enhanced LRM Launcher (NLRM) 102
Extended LRM Launcher (ELRM). 103
Improved One Shot (I-OS) Missile Launcher . 103
Long Range Missile Launcher (LRM) 103
Medium Range Missile (MRM) Launcher . 103
Multi-Missile Launcher (MML) 103
One-Shot (OS) Missile Launcher 103
Rocket Launcher (RL). 103
Short Range Missile (SRM) Launcher 103
Streak LRM (SLRM) Launcher 104
Streak SRM (SSRM) Launcher 104
Thunderbolt Missile Launcher 104
Torpedo (SRT, LRT, NLRT) Launcher. 104
NARC MISSILE BEACONS. 104
Narc Missile Beacon 104
iNarc Missile Beacon 104
PARTICLE PROJECTOR CANNONS . . 105
PPC. 105
ER PPC. 105
Heavy PPC 105
Light PPC 105
Snub-Nose PPC 105
PPC Capacitor. 105
PLASMA WEAPONS. 106
Plasma Cannon 106
Plasma Rifle. 106
RIFLES (CANNONS) 106
SPECIAL MUNITIONS. 106
Autocannon Munitions. 106
Missile Munitions 107
Narc / iNarc Pods. 107
PHYSICAL WEAPONS 108
Claws. 108
Flail. 109
Hatchet. 109
Lance. 109
Mace 109
Retractable Blade 109
Sword 109
Talons 109
Vibroblade. 109
ELECTRONICS. 109
Active Probes 109
Artemis IV Fire-Control System 110
Artemis V Fire-Control System 110
C^3 Computer System. 110
C^3 Boosted System. 112
C^3I Computer System 112
ECM Suites. 112
MRM "Apollo" Fire Control System. 113
Target Acquisition Gear (TAG) 113
Targeting Computer. 114
Watchdog Composite Electronic Warfare System (CEWS) 114
STRUCTURAL COMPONENTS 114
Armor 114
Cockpit Systems 115
Engines. 115
Gyros 115
Maneuverability Aids. 116
Structure 117
MISCELLANEOUS EQUIPMENT 118
Actuator Enhancement System (AES) 118
Anti-Missile System (AMS). 118
Laser Anti-Missile System (L-AMS) 118
Armored Component 118
Cellular Ammunition Storage Equipment (CASE) 118
CASE II. 119
Coolant Pod 119
WEAPONS & EQUIPMENT TABLES 120
WEAPON TYPES 120
COMMON MISCONCEPTIONS 128
PLAYING THE GAME 128
MOVEMENT 128
COMBAT. 129
DAMAGE. 130
HEAT 131
OTHER ACTIONS 132
SPECIAL-CASE RULES 132
WEAPONS AND EQUIPMENT 132

CREDITS

BattleTech Original Design
Jordan K. Weisman
L. Ross Babcock III
Sam Lewis

BattleTech Line Developer
Ray Arrastia

BattleTech Line Editor
Aaron Cahall

Project Development
Keith Hann
Assistant Development
Ray Arrastia
Randall N. Bills
Mary Kaempen
Additional Writing
Randall N. Bills
Keith Hann
Chris Hart

Production Staff
Art Director
Brent Evans
Assistant Art Director
Ray Arrastia
Graphic Design & Layout
Ray Arrastia
David Kerber

Cover Art
Alex Iglesias

Interior Artwork
Ray Arrastia
Doug Chaffee
Matthew Cross
Ken Fairclough
Mark Hayden
Alex Iglesias
Alex Immerzeel
Michael Kormarck
Marco Mazzoni
Florian Mellies
Neil Roberts
Anthony Scroggins
Florian Stitz
Franz Vohwinkel

Additional Review and Proofing
Doug Buchanan, Rich Cencarik, Joshua M. DeBoe, Johannes Heidler, Andy Holtkamp, Alex Kaempen, Jan Prowell, Craig Reed, Scott Roberts, Geoff Swift, James Topa, Chris Wheeler, Matthew Wilsbacher, Peter Wort, Piotr Wrzosek.

Online Feedback
Thanks to everyone online that provided initial *BattleMech Manual* suggestions:

ActionButler, Asmo, Bartholomew bartholomew, Bosefius, c_gee, cavingjan, cmoreland, CrazyGrasshopper, Daemion, Daryk, Descronan, Empyrus, fuzbuckle, Hammer, Hellraiser, Hobbes, I am Belch II, Ice_Trey, jackpot4, JadedFalcon, jimdigris, Karasu, Kharim, Lionheart, MadCapellan, Maingunnery, Martius, nckestrel, NeonKnight, Nicolai_Malthus, Pa Weasely, Probable Koz, sadlerbw, Sartris, SCC, Scotty, SeeM, Sigil, skiltao, Stinger, Tai Dai Cultist, Vampire_Seraphin, Weirdo, Welshman, Wolf Reaper, worktroll, Wrangler.

Special Thanks From Keith Hann
To Ray, for a phone call out of the blue that got this ball rolling, and for making this baby look beautiful; to Randall, for trusting some random freelancer to run terrible experiments on his baby; to CA, Muninn, and Liao for invaluable help and fun IRC times; to c_gee, for a great chapter idea and putting some random stranger up; and lastly, to the helpful General and his staff, who lived up to their legendary reputation for polite, civil discourse and provided excellent feedback while piecing together this book.

Special Thanks From Randall Bills
This is the fourth *BattleTech* rulebook I've had the honor of working on across almost twenty years. It's because of you, the amazing community of *BattleTech* players, that this wonderful universe is still growing and experiencing a renaissance in so many ways. Thank you for the privilege of being a part of this brilliant experience.

FIND US ONLINE:
precentor_martial@catalystgamelabs.com
(e-mail address for any BattleTech questions)
http://bg.battletech.com/
(official BattleTech web pages)
http://www.CatalystGameLabs.com
(Catalyst web pages)
http://store.catalystgamelabs.com
(online ordering)

Printed in China.

Corrected seventh printing.
Published by Catalyst Game Labs,
an imprint of InMediaRes Productions, LLC.
5003 Main St. #110
Tacoma, Washington 98407

TOTAL WARFARE CREDITS & ACKNOWLEDGEMENTS

Project Development: Randall N. Bills • **Development Assistance:** Herbert A. Beas III

Additional design and development
The following people have been involved in the creation and development of BattleTech *rules, either by writing material that was assimilated into the main body of the rules, serving as the* BattleTech *line developer in the past, or otherwise contributing to the game in a major way.*

Samuel B. Baker, Herb Beas, Randall N. Bills, Forest G. Brown, Chuck Crain, Chris Hartford, Clare Hess, Scott Jenkins, J. Andrew Keith, James R. Kellar, Dale Kemper, L.R. "Butch" Leeper, Bryan LiBrandi, Jim Long, David McCulloch, Jim Musser, Bryan Nystul, Mike Nystul, Blaine Pardoe, Boy F. Peterson Jr., Rick Raisley, Jerry Stenson, Christoffer Trossen, Wm. John Wheeler.

Acknowledgements
To the core group of people who, to one extent or another, have significantly contributed to keeping *BattleTech* not simply alive these last five years, but thriving once more: Ray Arrastia, Herb Beas, Loren Coleman, Warner Doles, David M. Stansel-Garner, Diane Piron-Gelman, Jason Hardy, Chris Hartford, Jason Knight, Chris Lewis, David McCulloch, Ben Rome, Matt Plog, Paul Sjardijn, Peter Smith, Scott Taylor, Christoffer Trossen, Øystein Tvedten, and Jason Vargas.

Playtesters/Proofers/Fact Checkers
The playtesters for any game line are a necessary component in the creation of a solid product. However, when tackling something on the scope of *Total Warfare*, which involves the melding of twenty years worth of rules into a unified whole that far exceeds it constitute parts (particular when set against the millions-of-words background of the vast scale of *BattleTech* history and continuity) their roles are not only greatly expanded, but become vital. A testament to the strength of the *BattleTech* community, their dedication knew no bounds: the product is far superior due to their significant contributions.

Joel Agee, Brian Alter, Ray Arrastia, Ron Barter, Terren "Tel Hazen" Bruce, Rick Casarez, Doug Davidhizar, Brent Dill, Benjamin Disher, John "Quentil" Dzieniszewski, Dave Fanjoy, David Ferguson, Thomas Ferrell, Bruce Ford, Eugen Fournes, John "AngelRipper" Frideres, Aaron "Bear" Gregory, Eric Hawkins, Glenn Hopkins, Ken Horner, John "Bleusman" Hudson, Darryl Hunt, Stephen "Blayde" King, Rod Klatt, Nadine Klatt, Michael "Konan" Koning, Peter LaCasse, Edward "Tenaka Furey" Lafferty, Mark Maestas, Eberhard "Rote Baron" von Massenbach, Brian McAuliffe, Tim McAuliffe, Mike Miller, Jeffrey "Fej" Morgan, Darrell "FlailingDeath" Myers, Lou "Nukeloader" Myers, Andrew Norris, Mike Pfister, Max "Medron Pryde" Prohaska, Ben Rome, Richard Sardiñas, Christopher K. Searls, Chris Sheldon, Alyson Sjardijn, Paul Sjardijn, Scott Stansbury, Joel "Septicemia" Steverson, Rob Strathmann, Terry Sweetapple, Dr. Geoff "97jedi" Swift, Lee Thoms, Roland Thigpen, Todd "BlueWeasel" Thrash, Andrew Timson, John Unchelenko, Chris Wheeler, Paul Wolf, Patrick Wynne; Gravedigger Arms Limited: James Humpula, Darlene Morgan, Aaron "Gravedigger" Pollyea, Jeff Skidmore, Charles "IronSphinx" Wilson; The Singapore Longshoremen: Jason Chodara, Leonard Conklin, Dan Eastwood, Tom Evans, Rick "Rick Steele" Remmer, Chris "Chungalo" Smith, Sam "Wasp" Snell, Jeremy Zuvich; Seacoast BattleTech Society: Debra Atkinson, David Baughman, James Bernard, Josh Carter, Alex Ellis, Korac MacArthur, Clayton Poland; Steel City MechWarriors: Rich Cencarik, Rich Darr, Brian Golightly, Dave Lang, Drew Martin.

And to all the players that fought through the endless delays to get on-line questions answered over the years; your questions generated numerous FAQs off of which much clarification came about in this book.

©2017, 2023 The Topps Company, Inc. All Rights Reserved. BattleMech Manual, Classic BattleTech, BattleTech, 'Mech, BattleMech, MechWarrior and Topps logo are registered trademarks and/or trademarks of The Topps Company, Inc. in the United States and/or other countries. No part of this work may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, without the prior permission in writing of the Copyright Owner, nor be otherwise circulated in any form other than that in which it is published. Catalyst Game Labs and the Catalyst Game Labs logo are trademarks of InMediaRes Productions, LLC.

INTRODUCTION

—Excerpts from a promotional pamphlet originally distributed by Defiance Industries of Hesperus, Lyran Commonwealth, 3007

- *Standing from seven to sixteen meters tall, and weighing from twenty to one hundred tons*
- *Powered by an armored and shielded fusion reactor*
- *Skeleton of honeycombed, foamed aluminum core wrapped with stressed silicon carbide monofilament and sheathed by a rigid, titanium-steel shell*
- *Locomotion generated via bundles of polyacetylene-fiber myomer muscles*
- *Protected by aligned-crystal steel over a layer of boron nitride impregnated with diamond monofilaments*
- *Mounting a swath of powerful weapons from charged particle beams to lasers, missiles to rapid-fire autocannons*
- *All at the command of the noble elite, the MechWarriors*

The modern BattleMech is the end result of more than three thousand years of battlefield technology development. Combining awesome destructive power and unparalleled maneuverability, the BattleMech is perhaps the most complex machine ever produced. The undisputed master of thirty-first century warfare, the BattleMech seems destined to reign supreme for centuries to come.

The *BattleMech Manual* has been designed from the ground up to cater to the *BattleTech* player wishing to participate in an all-'Mech engagement. Using the most up-to-date rules text, the *BattleMech Manual* focuses on presenting the complete standard rules for BattleMechs in as clean and concise a manner as possible.

VS. TOTAL WARFARE

The *BattleMech Manual* is an alternate presentation of the *BattleTech* rules as seen in *Total Warfare*, not a new edition. Most of the rules as published in both books are the same, though the *BattleMech Manual* rewords many of them for clarity or reduced word count. It also incorporates all relevant errata as of its publishing date.

Additionally, the *Manual* adds rules and items of equipment from sources other than *Total Warfare*, such as combat-dropping BattleMechs and expanded terrain options. As many players regularly use such rules, the *Manual* includes them to provide a one-stop BattleMech play experience. However, some of these rules appear in a simplified or stripped-down form compared to their original form, in the interest of presenting a taste of the wider *BattleTech* ruleset but not overwhelming the reader with complexity.

OPTIONAL RULES

Throughout the text, you will find rules labeled "**(Optional)**". These are variants that expand upon or slightly twist the existing rules to allow for a potentially more interesting game. As their label indicates, these rules are entirely optional: the use of each must be clearly detailed and agreed on by all players prior to setting up the game.

BEYOND THE BATTLEMECH MANUAL

While the *BattleMech Manual* is self-contained and there's literally hundreds of hours of enjoyment within its pages, there is an epic body of additional weapons, equipment and rules that cover a huge swath of the *BattleTech* experience: from vehicles to infantry to aerospace elements, much less a plethora of additional optional rules for expanding your BattleMech experience.

For many more optional rules to expand your *BattleTech* games, see *Tactical Operations* (*TO*), *Strategic Operations* (*SO*), *Interstellar Operations* (*IO*), *Campaign Operations* (*CO*), and *Alpha Strike* (*AS*).

GAME TERMS

The following terms describe important concepts used in the *BattleTech* rules.

'MECH

This is the common abbreviation for "BattleMech", and will be used throughout the text. Unless specified otherwise, this term refers to both biped and four-legged BattleMechs.

FOUR-LEGGED 'MECH

Often simply called "quads," rather than having two arms and two legs, these 'Mechs instead have four legs. Notable examples are the *Scorpion*, *Goliath*, and *Tarantula* 'Mechs.

STRATEGIC RULES

The players will find numerous "Strategic" sidebars in the pages of the *BattleMech Manual*. These are all optional rules. However, these are set-off in a sidebar style because there's a distinct difference. Unlike the rest of the optional rules that are concerned with the modification of specific mechanics within the game—a little more movement, or a different way for a BattleMech to stand up—these concentrate on different ways of running the game as a whole. Whether they're provided to help speed up game play or because it's an option that a large part of the community regularly uses: they're ultimately about ways to make the *BattleTech* gaming experience better.

MOVEMENT POINTS

Commonly abbreviated as "MP," the number of Movement Points a 'Mech generates in a turn is based on the movement mode the 'Mech chooses that turn. A 'Mech spends Movement Points in a Movement Phase based on how often it changes its facing, the terrain in each hex traveled through, and other factors (see *Movement*, p. 13).

TARGET

A target is anything a 'Mech may attack, typically an enemy 'Mech. A friendly 'Mech can never be the target of a direct attack, but may be damaged through friendly area-effect attacks. See *What Is A Target?* on page 22.

TARGET NUMBER

Any action requiring a dice roll has a Target Number, representing the action's difficulty. If the roll result equals or exceeds this Target Number, the action succeeds. If the result is less than the Target Number, the action fails. Base Target Numbers for each action are specified in the rules sections for that action. If the Target Number is 2 or less, the action automatically succeeds; if 13 or higher, it automatically fails.

MODIFIED TARGET NUMBER

A Target Number may be modified up (making the task more difficult) or down (less difficult), based on various circumstances such as terrain, range, and so on. The final number against which a player makes a dice roll is called the modified Target Number. For example, weapon attacks use the MechWarrior's Gunnery Skill as the base Target Number, which is then modified by factors such as how far the attacker and defender moved.

ATTACKER MOVEMENT MODIFIER

This modifier is applied when a 'Mech makes an attack during the same turn in which it moved, increasing the Target Number and thus making a successful attack more difficult. The exact modifier depends on the movement mode the attacker used that turn (see *Attacker Movement Modifier*, p. 25).

TARGET MOVEMENT MODIFIER

If a target moved during the turn in which an attack is made against it, the number of hexes it entered and the type of movement it used may increase the Target Number. The more hexes moved, the more difficult it is to hit the target (see *Target Movement Modifier*, p. 25).

MARGIN OF SUCCESS/ FAILURE (MOS/MOF)

Most actions in *BattleTech* have simple pass/fail results. The outcomes of some actions, however, depend on the amount by which the roll succeeds or fails. To determine the Margin of Success or Margin of Failure, sometimes abbreviated MoS or MoF, compare the roll result to the Target Number. If the result is greater, the difference is the Margin of Success. If the result is less, the difference is the Margin of Failure. If the dice roll matches the Target Number, the result is a success with a margin of 0.

ARMOR VALUE

Armor Value is defined as the total number of armor circles in a given location as shown in the Armor Diagram on a 'Mech Record Sheet. Some rules also use the terms "points of armor" or "armor points" to reference the number of armor circles in a location, or the number of remaining armor circles in a location.

DAMAGE VALUE

Damage Value is the amount of damage each weapon inflicts on a target. The Weapon and Equipment Tables list the Damage Values of each weapon.

Some weapons do not deal their full Damage Value to a single location. Instead, the points of damage from the weapon are divided into Damage Value groupings and then each grouping is assigned to a separate location (see *Cluster Weapons*, p. 96).

LOCATION

A BattleMech is comprised of eight locations. These are the left, right and center torsos, the head, and the four limbs (two arms and two legs for biped 'Mechs, four legs for quad 'Mechs). Damage is always applied to specific locations, and all 'Mech components are stored in these locations.

SLOT

On the Critical Hit Table of a 'Mech Record Sheet, each line within a location (six in the head and legs, twelve in other locations) is referred to as a "slot". Slots are where the weapons, equipment, and structural components that make up a 'Mech are placed. When critical hits are assigned, they are assigned to specific slots, disabling or even potentially exploding whatever is in that slot (see *Critical Hits*, p. 45). Many items occupy multiple slots.

LEVEL

Level refers to the height of ground terrain; i.e. a Level 2 hill hex, a Level 5 building hex, woods rising two levels above the underlying level of the hex and so on (see *Level Change*, p. 15). The term will also be used when referring to the height of a 'Mech; i.e. a 'Mech rises two levels above the underlying level of the hex.

SUBLEVELS AND DEPTHS

A subset of levels, *Sublevel* refers to any level below 0, while *Depth* refers to water hexes.

MAPSHEET

This general term refers to a single 18 by 22-inch (15 x 17 hex) playing map as printed by Catalyst (or FanPro/FASA) in the various map sets, HexPacks, and Map Packs, or available electronically at bg.battletech.com.

PLAYING AREA

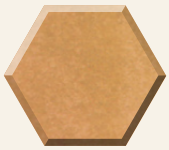
This general term refers to the total playing area of a given game, which consists of one or more mapsheets.

SCENARIO

This term usually describes a game with specific forces and set-up conditions. It can also be used to describe a single game—for example, a one-off game between the players.

TERRAIN HEX ICONS

The forests, rivers, hills, buildings and rough areas on a *BattleTech* mapsheet represent a typical mixture of the terrain found on the habitable worlds of the Inner Sphere. The following symbols represent each type of terrain as described. Specific rules regarding the effect of terrain on movement and combat appear in *The Battlefield* chapter (p. 59).



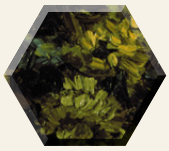
CLEAR

Clear terrain represents fields, meadows and other grasslands. The ground is firm and may be gently rolling, but its Level does not change significantly from one side of the hex to the other.

If a hex is not clearly marked as containing another terrain type, assume it is clear.

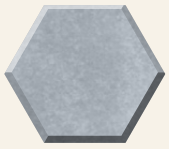
JUNGLE, LIGHT

Jungle terrain contains thick trees and shrubs, hanging vines and dense undergrowth that makes movement extremely slow and difficult.



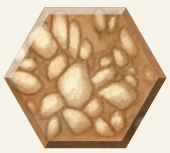
JUNGLE, HEAVY

Heavy jungle terrain is an even denser, taller growth of vegetation, impeding movement further.



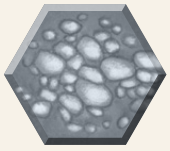
PAVEMENT

A Paved hex offers a fairly smooth and very hard surface. Paved hexes typically include roads, sidewalks, and landing fields made of asphalt, cement, or even cobblestone.



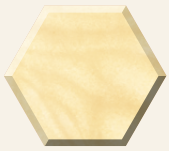
ROUGH

Rough terrain represents broken, rocky and jumbled ground. Though firm, the unevenness of this type of terrain makes it more difficult to cross than clear terrain.



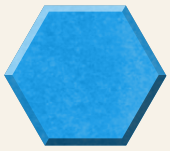
RUBBLE

Weapons fire, fire damage and physical damage inflicted by BattleMechs can reduce a building to a rubble terrain hex. A Rubble hex is difficult to move through.



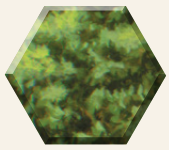
SAND

Sand terrain simulates the deep, shifting sands common to deserts and beaches throughout known space.



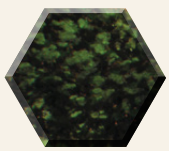
WATER

Streams, rivers, swamps, ponds and lakes are covered by Water terrain hexes. A Water hex is defined by depth levels (see *Level Change*, p. 15).



WOODS, LIGHT

Light woods terrain is covered with sparse trees of up to 12 meters in height. BattleMechs cannot cross this terrain as easily as clear terrain.



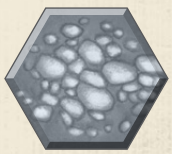
WOODS, HEAVY

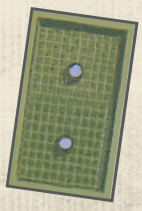
Heavily wooded terrain is covered thickly with 12-meter-tall trees, making movement through these areas very difficult. Light woods often border heavy woods.

COMPONENTS

BattleTech often employs a number of physical components, such as the following.

COUNTERS





Certain terrain features, such as rubble—or buildings, if players wish to add buildings to a mapsheet that does not have building hexes already printed on it—can be represented on the map by cardboard or paper counters. You can make your own, or purchase generic counters for use. Rather than printing such features on the maps, the use of counters to represent them allows the players to alter their locations before the game begins, based on the requirements of the scenario being played or by mutual player agreement for added variety.

DICE

BattleTech requires players to use two six-sided dice. If the situation requires the player to roll one die, the rules indicate this in shorthand as 1D6. Unless otherwise noted, the abbreviation 2D6 means the player rolls both dice and adds the results together.

MAPSHEETS

The 18- by 22-inch mapsheets used in *BattleTech* are divided into six-sided areas called hexes (short for hexagons); whether paper or game board-style, both are called "mapsheets" and treated identically in the rules. The players use these hexes to regulate movement and combat by moving 'Mechs from hex to hex during a turn.

'MECH RECORD SHEET

Players use a 'Mech Record Sheet to track damage done to a 'Mech during combat.

Open Information: Unless a scenario says otherwise, record sheets are always open information that your opponent can review at any time.

ARMOR DIAGRAM

1 The Armor Diagram on the top right-hand side of the record sheet shows the arrangement of a BattleMech's armor plating, while the Internal Structure Diagram directly below it shows the arrangement of a 'Mech's internal structure. Each circle on the Armor Diagram represents a point of armor. As armor is destroyed, the player marks off the appropriate circles. When all the circles in one location are filled in, damage transfers to the appropriate internal structure location, as shown on the Internal Structure Diagram. When all the circles in an internal structure location are filled in, that location is destroyed. This is covered in detail in the *Damage* chapter (see p. 41).

The Armor Diagram also shows the front and rear armor of the 'Mech's three torso locations (the arms, legs and head do not have separate rear armor locations). The Damage Transfer Diagram, which appears at the bottom of the Critical Hit Table, shows where the player must transfer damage when an already destroyed location takes additional damage.

For ease of reference, next to the name of each location on both the Armor Diagram and the Internal Structure Diagram, a line is provided to fill in that section's starting value. On pre-generated record sheets, the number of armor and internal structure points is already listed.

'MECH DATA

2 Located in the upper left-hand corner, this section of the record sheet lists the BattleMech's most important statistics, including type, tonnage, movement, and weapons inventory.

WARRIOR DATA

3 This section lists the name, skills, and condition of the MechWarrior piloting the BattleMech.

CRITICAL HIT TABLE

4 The Critical Hit Table shows the physical location of the BattleMech's critical equipment, weapons and ammunition when applying critical hits. A BattleMech is usually destroyed through critical damage.

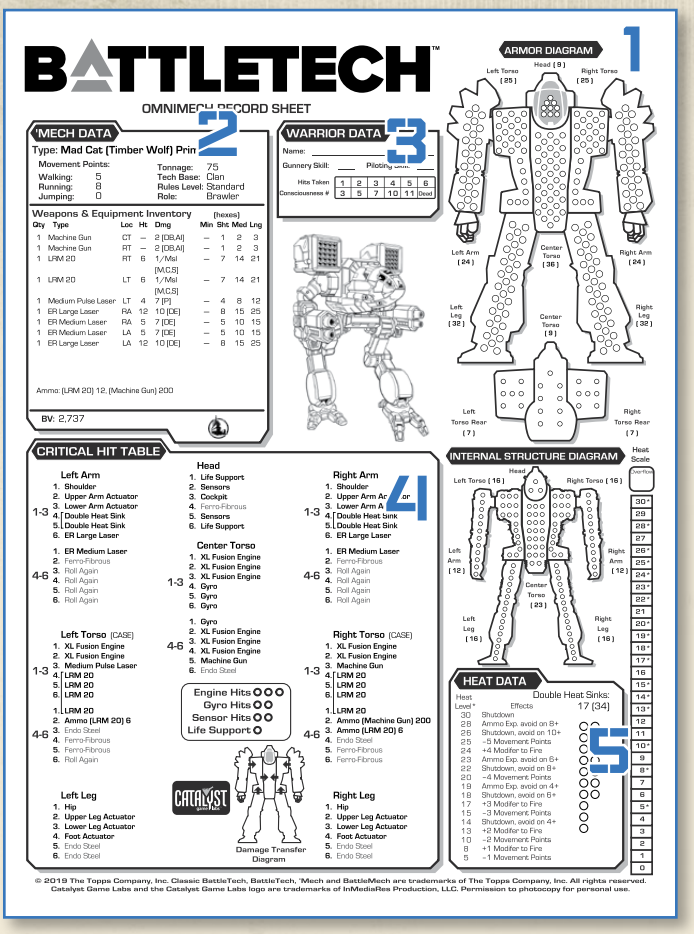
HEAT DATA AND HEAT SCALE

5 Located in the bottom right-hand corner of the Record Sheet, the Heat Data and Heat Scale help the player track the 'Mech's internal heat buildup, as well as indicate how many heat sinks a 'Mech mounts (circles underneath the numerical value in the Heat Data section allow players to mark off heat sinks as they are damaged). As heat builds up, the player checks off the boxes in the Heat Scale from low to high (usually with a pencil, as heat will fluctuate up and down the Heat Scale throughout a game).

Heat is a limiting factor. At certain levels of heat buildup (noted on the Heat Scale with asterisks), corresponding information in the Heat Data section describes the heat's effect on the 'Mech's operation. The blank space marked "Overflow" at the top of the Heat Scale is used to record heat generated in excess of 30 points.

SCALE

One *BattleTech* turn is ten seconds of real time. A standard *BattleTech* mapsheet measures 18 by 22 inches. Each hex on the mapsheet represents an area 30 meters across (roughly 100 feet).



© 2019 The Topps Company, Inc. Classic BattleTech, BattleTech, 'Mech and BattleMech are trademarks of The Topps Company, Inc. All rights reserved. Catalyst Game Labs and the Catalyst Game Labs logo are trademarks of InMediaRes Production, LLC. Permission to photocopy for personal use.

A NOTE ON REALISM AND SCALE

Given the 30-meter area of each hex stated above, players may note various oddities in the weapon ranges presented in this book, such as the fact that the standard 'Mech-scale machine gun of the distant thirty-first century only reaches out to 3 hexes (90 meters). As today's machine guns have effective ranges of some 2,000 meters, this may seem somewhat absurd.

The reason for this is simple: *BattleTech* is a game. Because *BattleTech* mapsheets are only seventeen hexes long, recreating real-world ranges on a table would require more than seven mapsheets laid end to end, for a playing space greater than twelve feet in length. Few people have that much table space. Nor would it provide players with any tactical maneuvering room: anywhere a player might move a 'Mech on the map, an attacker could hit it. As such, while we may safely assume "real" *BattleTech* weapons have exceptional ranges, range abstractions are an absolute necessity unless one is regularly able to rent a tennis court for game time.

LEVEL

The level of a hex is the height to which it rises above the prevailing terrain. All terrain has a level, but the level of each hex is independent of the type of terrain it contains, such as woods or water. If it is not marked on the map, assume a hex's level is 0.

This height is expressed in terms of levels. Level 1 is approximately six meters high, waist-high to a 'Mech. Level 2 terrain is approximately twelve meters high. This is considered as being

the height of all 'Mechs for gameplay purposes. Level 3 terrain is approximately eighteen meters high, and so on. The level of a hex is considered equal to the highest level present in it.

Sublevels: Hexes with levels lower than 0 are referred to as sublevels. Such hexes are marked in sublevels that correspond to levels in reverse. For example, a Sublevel 1 hex is six meters deep, while a Sublevel 2 hex is twelve meters deep, and so on.

If any part of a hex contains a sublevel, the entire hex is considered to be the deepest sublevel marked in it. The exception to this rule occurs if the hex also contains a level greater than 0, in which case the level of the hill takes precedence as described above.

THE MECHWARRIOR

The soldiers who pilot BattleMechs are called MechWarriors.

SKILLS

MechWarriors use two skills in combat: Piloting and Gunnery. Each skill has a rating; the lower a MechWarrior's skill rating, the better the MechWarrior is at the skill—and the lower their base Target Number will be when using that skill during gameplay.

Piloting Skill represents a MechWarrior's ability to control their machine. This skill includes keeping a 'Mech from falling down and striking targets in physical combat.

Gunnery Skill helps determine how easy or difficult it is for the MechWarrior to hit with their 'Mech's ranged weapons (see *Firing Weapons*, p. 25).

DEFAULT SKILL RATINGS

The average Inner Sphere MechWarrior has a Gunnery Skill of 4 and a Piloting Skill of 5. The average Clan MechWarrior is a superior pilot and marksman, with a Gunnery Skill of 3 and a Piloting Skill of 4. Unless otherwise stated in the scenario being played, all MechWarriors of these respective factions have these skill ratings.

Battle Value: In scenarios balanced using the Battle Value system, the improved Gunnery and Piloting Skills of Clan MechWarriors given above must still be paid for in BV. See the Master Unit List at *www.masterunitlist.info* for a comprehensive list of units and accompanying Battle Values.

PILOTING SKILL ROLLS (PSR)

When a 'Mech attempts a dangerous maneuver, or whenever the MechWarrior might lose control of the 'Mech, that 'Mech's controlling player must make a Piloting Skill Roll, commonly abbreviated as PSR (see *Piloting Skill Rolls*, p. 53).

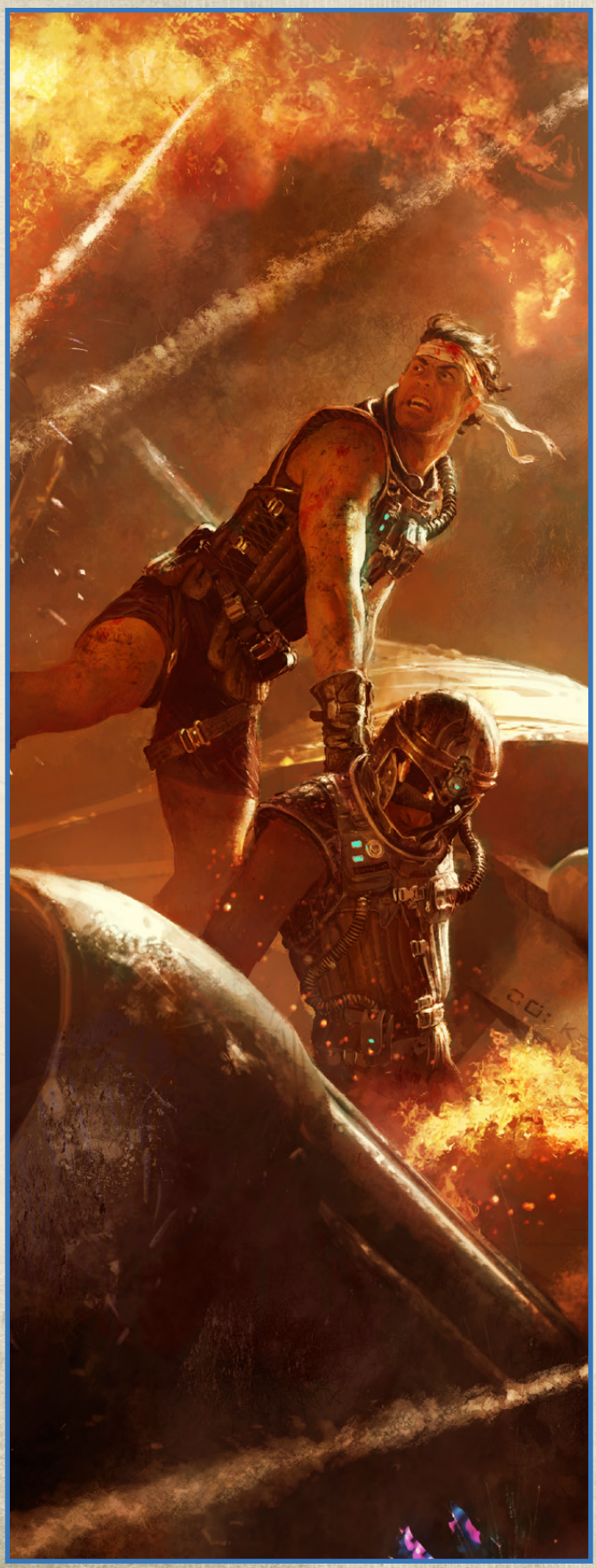
Additionally, a 'Mech's base Target Number for physical attacks is equal to the Piloting Skill rating of its MechWarrior. When adjusted for movement, damage, and other factors, this number becomes the modified Target Number for such attacks (see *Physical Attacks*, p. 34).

GUNNERY SKILL RATING

The base Target Number for ranged weapon attacks is equal to the Gunnery Skill rating of the MechWarrior making the attack (see *Firing Weapons*, p. 25).

DAMAGING A WARRIOR

Falls, ammunition explosions, heat, and other perils can wound and eventually kill a MechWarrior. See *Damaging a MechWarrior*, page 58.



PLAYING THE GAME

A *BattleTech* game consists of a series of turns, each being ten seconds of real time. This chapter breaks down how to play a game of *BattleTech* from start to finish.

DEPLOYING 'MECHS

There are no formal rules to how each side should deploy their 'Mech(s) on the mapsheet at the start of the game. Typically, instructions to this effect are laid out by a gamemaster (if any), or by the scenario being played. If there are no such instructions, players should come to an agreement regarding deployment before the game begins, or simply use the following informal guidelines:

- Both sides roll Initiative.
- The winning player chooses a side of the playing area they will enter; the losing player will enter the opposite side.
- Each 'Mech, when it is their turn to move, enters their side walking, running, or jumping, as announced by its controller, counting MP expenditure beginning with the first full hex of the playing area.

SEQUENCE OF PLAY

Each turn consists of several smaller segments, called phases. During each phase, players may take one type of action, such as movement or combat. The players execute the phases in a given order, with all players completing a phase before anyone moves onto the next phase. Details on each phase are fully explained in separate sections later in this book.

Each turn includes the following phases, performed in the following order:

1. Initiative Phase
2. Movement Phase
3. Weapon Attack Phase
4. Physical Attack Phase
5. Heat Phase
6. End Phase

STRATEGIC

ALTERNATE INITIATIVE

Occasionally other Initiative systems are used, due to preference for a different style or as a way to speed up play in larger games. For example, one common system deals playing cards to all players. These cards are shuffled by a gamemaster at the start of each turn, then flipped up in one at a time, activating each player in the order of the cards.

As long as the system is random and not weighted towards one side or another—and provided all players agree to its use—such alternates can work well.

INITIATIVE PHASE

One player from each side rolls 2D6 and adds the results together to determine their side's Initiative. The side with the higher result wins Initiative for that turn. Re-roll all ties.

MOVEMENT PHASE

The side that lost Initiative must act first. They choose one of their 'Mechs and assign it a movement action: this is called a movement selection. The side that won Initiative then does the same. Movement alternates between sides until all 'Mechs have been moved, with any Piloting Skill Rolls (see p. 53) required due to movement made the moment they are called for. If either side has more 'Mechs than the other, it may need to move more than one 'Mech at once (see *Unequal Numbers of 'Mechs*, p. 12).

STRATEGIC

MOVING MULTIPLE 'MECHS

Many scenarios have players move more than one 'Mech at a time (i.e. a lance or even all the 'Mechs on a side), since this greatly speeds up movement. Such Initiative modifications are optional, and so must be agreed on before play begins.

Remaining still is a valid movement action and still uses one movement selection. A player may assign a movement action to any 'Mech that has not been destroyed, even if it is completely immobile. For example, a 'Mech whose warrior is unconscious can still be given a movement action (in order to use up a movement selection), even though it must remain immobile.

WEAPON ATTACK PHASE

The side that lost Initiative must act first. They choose one of their 'Mechs to declare fire first: this is called a fire selection. The side that won Initiative then does the same. Fire selection alternates between players until all fire has been declared.

If either side has more 'Mechs than the other, the players of that side may need to declare fire for more than one 'Mech at once (see *Unequal Numbers of 'Mechs*, p. 12).

See the *Combat* chapter, starting on page 22, for how to resolve attacks, and the *Damage* chapter, starting on page 41, for how to apply the damage from those attacks.

DECLARING WEAPONS FIRE

A player may declare fire with any 'Mech that has not been destroyed. For example, a 'Mech with no weapons or whose warrior is unconscious can still declare fire (in order to use up a fire selection), even though it cannot actually attack. Declaring that a 'Mech will not fire is a valid action and still uses one fire selection.

When declaring attacks, the player controlling the 'Mech chosen declares whether it will twist its torso or flip its arms, and in which direction. They must declare any attacks they plan to make, specifying which weapons they will fire and at what target(s). If a

weapon uses special ammo loads, such as LB-X cluster munitions, or can make special types of attacks, such as double-firing an Ultra autocannon, or can produce any other unusual effects, the player must declare any use of these at this time.

Players may not change an attack declaration once made.

TORSO TWIST

Torso twists are made when declaring a weapon attack; this affects physical attack arcs as well. A 'Mech that has torso-twisted changes its own firing arc, but this has no effect on damage it receives. See *Torso Twists*, page 25.

RESOLVING WEAPONS FIRE

After all attacks have been declared, players resolve weapons fire one 'Mech at a time. Again, the side that lost Initiative must act first.

All weapon attacks occur at the same time. As such, all declared attacks must be made, even if the intended target is destroyed before all attacks against it have been resolved, and a 'Mech always gets to make its declared attacks for the phase, even if that 'Mech or its weaponry winds up being destroyed in that phase.

The order in which a 'Mech's weapons are rolled and resolved is up to that 'Mech's controller. The damage inflicted by each weapon hit is fully resolved, from start to finish, before moving on to the next hit. If a single weapon attack inflicts multiple hits (possible with missiles, rapid-fire weapons, and the like), then each hit from that attack is fully resolved before moving on to its next hit.

All weapon attacks by a 'Mech should be resolved before moving to the next 'Mech.

STRATEGIC

SEGMENTED FIRE

As seen in the process outlined at left, after all players have declared their attacks, normally each player completely resolves the Weapon Attack Phase actions of each 'Mech one at a time, from resolution to damage, before moving on to the next 'Mech. As an option to speed up play, after all players have made their attack declarations, all players may then make all to-hit rolls for all attacks declared that phase, before any damage by any 'Mech is resolved.

After all attacks have been resolved, all Consciousness Rolls required due to damage inflicted in this phase are made (see p. 58). After this comes all Piloting Skill Rolls required (see p. 53), each taking into account all modifiers resulting from damage inflicted that phase.

PHYSICAL ATTACK PHASE

Players repeat the steps given for the Weapon Attack Phase, except for torso twisting, with all damage from physical attacks taking effect before the Heat Phase. See page 34.

HEAT PHASE

Players adjust their 'Mech's Heat Scales to reflect any heat built up or lost during the turn and resolve any effects caused by excessive heat. In particular, 'Mechs shutdown due to heat make their restart attempts in this phase. See the *Heat* chapter, page 50.

Heat may also damage a warrior, forcing one or more Consciousness Rolls (see p. 52).

END PHASE

A variety of miscellaneous actions can occur during this phase, some optional and some mandatory, if applicable. An action's description will state whether it occurs during the End Phase, but the following summarizes all such actions:

MANDATORY

- Any MechWarrior who was unconscious during the Initiative Phase of this turn rolls 2D6 to see if they regain consciousness (see *Recovering Consciousness*, p. 58).
- Torsos that were twisted return to a forward-facing position (see *Torso Twists*, p. 25).
- Reversed arms return to a forward-facing position (see *Reversing (Flipping) Arms*, p. 24).
- A pilot of a submerged 'Mech with critical damage to its life support system takes 1 damage (see *Life Support (Head)*, p. 48).
- A hex that started on fire this turn fills the first two hexes downwind from it with smoke (see *Effects of Fire*, p. 63).
- Smoke produced by extinguished fire hexes is removed from play (see *Extinguishing Fire*, p. 64).
- Cargo being unloaded always completes its unloading in this phase (see *Lifting Items (Simplified)*, p. 21).
- Ammunition announced the previous End Phase as being dumped is dumped at this time (see *Dumping Ammunition*, p. 58).

OPTIONAL

- Any player with a 'Mech which will attempt to clear a jammed rotary autocannon next turn must declare that attempt at this time (see *Rotary Autocannon (RAC)*, p. 98).
- Heat sinks can be switched on or off (see *Building Up Heat*, p. 50).
- Machine gun arrays, stealth armor, the auto-eject system, and any equipment listed in the *Electronics* section (see p. 109) can be switched on or off.
- 'Mechs can be voluntarily shut down, or restarted if voluntarily shut down in a previous End Phase (see *Shutdown*, p. 52).
- Ammunition dumping can be announced at this time (see *Dumping Ammunition*, p. 58).
- Searchlights can be switched on or off (see *Light*, p. 62).

VICTORY CONDITIONS

Players repeat the sequence of play given above until one side achieves victory. Under normal circumstances, the side with the last surviving 'Mech(s) on the map wins. If the last 'Mechs from each side are destroyed simultaneously in the same turn, or if the last 'Mechs from each side cannot move and have no ability to damage one another, the game is a draw. Players may set other victory conditions by mutual agreement before play begins. In addition, some scenarios may have special victory conditions of their own.

UNEQUAL NUMBERS OF 'MECHS

The Movement Phase, Weapon Attack Phase, and Physical Attack Phase require each player to alternate moving or declaring attacks with their 'Mechs. When both sides have an equal number of 'Mechs, each player simply takes a turn moving or declaring a single 'Mech's action, then the other player declares movement or an action for one 'Mech, and so on. If the number of 'Mechs on each side is not equal, however, this procedure must be altered.

If, prior to any pair of movement or attack declarations, one team has at least twice as many 'Mechs left to declare for as the other team, the team with twice as many 'Mechs declares for two 'Mechs each time, rather than one. If one team has at least three times as many 'Mechs, it declares for three each time, and so on.

The following two tables provide examples of how this works across different force sizes.

UNEQUAL NUMBERS OF 'MECHS EXAMPLES

Move No.	Side B Units Left to Move	Side A Units Left to Move	Side B Moves	Side A Moves
1	10	18	1	1
2	9	17	1	1
3	8	16	1	2
4	7	14	1	2
5	6	12	1	2
6	5	10	1	2
7	4	8	1	2
8	3	6	1	2
9	2	4	1	2
10	1	2	1	2

Move No.	Side B Units Left to Move	Side A Units Left to Move	Side B Moves	Side A Moves
1	5	14	1	2
2	4	12	1	3
3	3	9	1	3
4	2	6	1	3
5	1	3	1	3

MOVEMENT

In the Movement Phase, BattleMechs have the opportunity to change their position on the mapsheet.

Buildings: Movement dealing specifically with buildings is largely covered in the *Urban Combat* chapter (see *Movement Effects*, p. 67).

Hidden 'Mechs: If using the *Hidden 'Mechs* rules (see p. 82), a hidden 'Mech is not counted when making movement selections unless the player plans to move that hidden 'Mech that turn. A player that plans to move it that turn must reveal it and place it on the map at the start of that turn's Movement Phase. Hidden 'Mechs revealed by enemy 'Mechs during the Movement Phase cannot move during that phase.

STRATEGIC

PILOTING SKILL ROLLS

As you move your 'Mech about, you'll almost certainly be called on to make a Piloting Skill Roll (a "PSR") to avoid falling. Typically this is due to entering difficult terrain, such as water and rubble, but damaged 'Mechs may fall when navigating even clear hexes. As PSRs can be called for as a result of combat as well, we've placed the rules for them in the *Other Actions* chapter (see p. 53). Players should familiarize themselves with the PSR rules before beginning play.

STRATEGIC

MOVEMENT DICE

To help remember which 'Mechs moved in what way, movement dice are often used. Commonly, white dice are used to mark 'Mechs that walked, black dice for those that ran, and red dice for those that jumped (if using the optional Sprinting rules, yellow dice are typically used when sprinting). The number displayed on the die is the target movement modifier generated as a result of the move, with "6" traditionally indicating a TMM of 0 (or immobility). Also common is the rule that—like taking your hand off the piece you are moving in chess—once a movement die is placed, that move cannot be changed.

It's best to use dice of a different size and/or color than those used for rolling attacks, to prevent confusion. During the End Phase, remove all movement dice from the board.

MOVEMENT MODES

'Mechs are assigned movement actions in initiative order (see *Sequence of Play*, p. 10). Every 'Mech must be assigned one (and only one) of the movement modes available to it. In general, the more hexes a 'Mech moves, the harder it is to be hit. Modes that provide more Movement Points (MP) thus help keep a 'Mech safe. However, such modes generate more heat, and make it so that a 'Mech has a harder time hitting with its own attacks.

Minimum Movement: A 'Mech that intends to use the *Minimum Movement* rule (see p. 16) must run.

Prone 'Mechs: As a 'Mech is assigned a movement mode first, and then moves, a 'Mech that begins the Movement Phase prone (see p. 19) must declare whether it will walk or run before it attempts to stand.

Underwater Maneuvering Units (UMUs): A 'Mech must use the Jumping movement mode to use UMUs. For details on their use, see *Movement in Water*, page 19.

XXL Engines: A 'Mech with an XXL engine generates heat even if standing still, and more heat than normal for each other movement mode. See *XXL Engine* on page 50 for details.

STANDING STILL

A 'Mech that stands still remains in the hex in which it started the turn. It may not expend MP that turn, even to change its facing.

Choosing this mode does not make a mobile 'Mech immobile (see *Immobile*, p. 18). However, this mode can be assigned to any 'Mech, even immobile ones. As "choosing" this mode for an immobile 'Mech still uses a movement selection, this is a good way to put off assigning actions to your more useful mobile 'Mechs until later in the initiative order.

- **Target Number Modifier as Attacker:** 0
- **Target Number Modifier as Target:** 0
- **Heat Generated:** 0

WALKING

A walking 'Mech may expend a number of MP up to its Walking MP rating (even spending 1 MP to turn a hexside facing counts as walking). 'Mechs may walk backward.

- **Target Number Modifier as Attacker:** +1
- **Target Number Modifier as Target:** Based on hexes traveled (see the Attack Modifiers Table, p. 28)
- **Heat Generated:** 1 (total, not per hex traveled)

One-Legged 'Mechs: Walking is the only movement mode beyond standing still that a one-legged 'Mech can use.

Triple-Strength Myomer: A 'Mech with active TSM (see p. 117) increases its Walking MP by 2. Recalculate its Running MP based on this new total, using the procedure described below.

RUNNING

A running 'Mech may expend a number of MP up to its Running MP rating. A 'Mech cannot move backward while running.

A 'Mech's Running MP rating is equal to its Walking MP times 1.5, rounding up, unless the 'Mech is using special equipment such as MASC or a supercharger (see below).

- **Target Number Modifier as Attacker:** +2
- **Target Number Modifier as Target:** Based on hexes traveled (see the Attack Modifiers Table, p. 28)
- **Heat Generated:** 2 (total, not per hex traveled)

Critical Damage: If a 'Mech uses Running MP, each of the following forces a Piloting Skill Roll (see p. 53) after its movement in order to avoid falling: critical hit to the gyro, critical hit to a hip.

If damage reduces a 'Mech's Walking MP rating, the player must recalculate the 'Mech's Running MP rating.

MASC: A 'Mech that has successfully activated MASC that turn (see p. 116) may run at up to twice its Walking MP.

Pavement: A 'Mech using running movement on a paved surface risks skidding (see *Skidding*, p. 20).

Supercharger: A 'Mech that successfully activates a supercharger that turn (see p. 116) may run at up to twice its Walking MP.

If a 'Mech activates both a supercharger and MASC, it receives a Running MP equal to 2.5 times its Walking MP. If a 'Mech uses a supercharger with active triple-strength myomer in the same turn, the TSM modifier is added before applying the supercharger modifier.

Water: 'Mechs cannot enter a Depth 1 or deeper water hex while running (whether the 'Mech is already in water or not). However, a 'Mech can use running movement to change facing in water and/or move from a water hex to a land hex.

JUMPING

Only 'Mechs that possess Jumping MP and are standing at the start of the turn may select this mode; they may then expend a number of MP up to their Jumping MP rating. As jumping features numerous exceptions to the basic movement rules, see *Jumping Movement* on p. 18 for the full details.

- **Target Number Modifier as Attacker:** +3
- **Target Number Modifier as Target:** +1, plus a further modifier based on hexes travelled (see the Attack Modifiers Table, p. 28).
- **Heat Generated:** 1 heat point for every hex jumped; improved jump jets generate half this much heat (round the final amount up). However, using jump jets always generates a minimum of 3 heat points, no matter how far a 'Mech jumps or what jump jet type is used.

Critical Damage: If a 'Mech uses Jumping MP, each of the following forces a Piloting Skill Roll (see p. 53) to avoid falling upon landing: loss of a leg, critical hit to the gyro, critical hit to a hip or any actuator in a leg location.

'Mechs lose one Jumping MP for each jump jet destroyed.

SPRINTING (OPTIONAL)

Sprinting is an optional movement mode. To sprint, a 'Mech must have two intact hip actuators.

A 'Mech's Sprinting MP is twice its current Walking MP. Sprinting generates 50% (round down) more heat points per turn than the 'Mech generates when running.

A 'Mech that sprints may not make any deliberate attacks that turn. Accidental charges as a result of skidding are still possible, and a 'Mech may still fire weapons to generate heat, though such firing has no chance of doing damage. Additionally, the 'Mech may not spot for indirect fire or take any action that would require it to sacrifice its attack. All restrictions that apply to a running 'Mech apply to a sprinting 'Mech.

Attacks against a sprinting 'Mech receive an additional –1 Target Number modifier. Piloting Skill Rolls made for a sprinting 'Mech apply a +2 Target Number modifier.

A 'Mech equipped with MASC or a supercharger may engage one or both systems and sprint during the same turn. Engaging one system provides MP equal to the current Walking MP multiplied by 2.5. However, any 'Mech that tries to sprint and use MASC or a supercharger must make a Piloting Skill Roll (see p. 53; apply the +2 modifier for sprinting) to avoid falling; the roll is made at the end of the 'Mech's movement.

If a 'Mech engages both MASC and a supercharger in the same turn and sprints, the 'Mech's MP is equal to its current Walking MP multiplied by 3. However, the 'Mech must make a PSR as described above for both the MASC and the supercharger.

MOVEMENT BASICS

After being assigned a movement mode, a 'Mech then spends the Movement Points (MP) provided. A 'Mech need not spend all its MP each turn, but no matter how few are spent, the 'Mech's movement mode does not change. MP left at the end of the turn are lost (i.e. you can't "bank" MP).

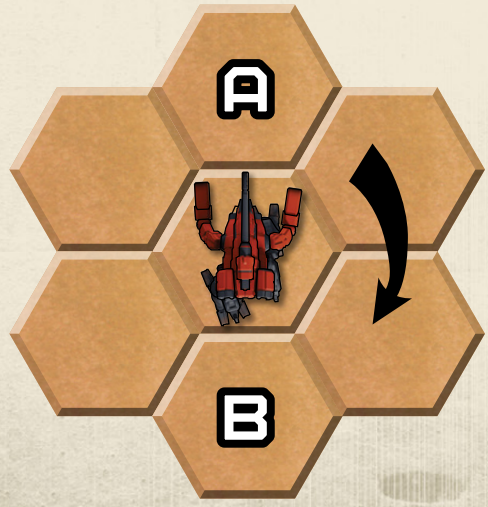
Key Reference: Movement Costs Table (see p. 17).

FACING

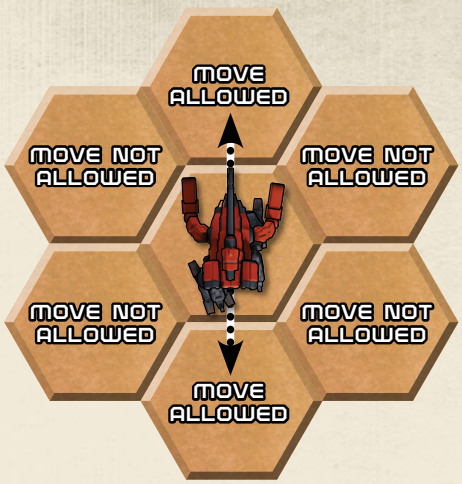
Every hex on the map has six edges, called hexsides. Every 'Mech must face one of those six hexsides: this is known as its facing. A 'Mech faces the way its feet are pointing.

Changing facing costs 1 MP per hexside. For example, in the diagram below, a 180-degree turn (three facings, from Hex A to Hex B) costs 3 MP. 'Mechs not clearly facing one hexside at the end of the Movement Phase must be realigned to one of the two closest hexsides by the gamemaster (or the opposing player, if there is no gamemaster).

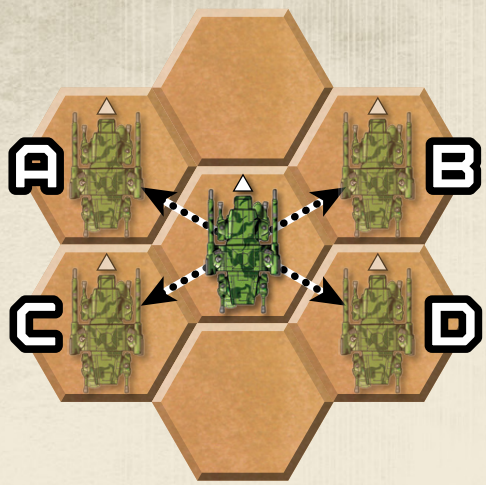
Torso Twisting: A 'Mech can twist its torso. This only changes its weapon arcs, not its facing, and occurs in the Weapon Attack Phase (see p. 25).



• FACING CHANGE DIAGRAM •



• GROUND MOVEMENT DIRECTION DIAGRAM •



• LATERAL SHIFT DIAGRAM (1 ADDITIONAL MP) •

GROUND MOVEMENT

A 'Mech can move forward into the hex it is facing, or backward into the hex directly to its rear. To move into any other hex, it must first change its facing (unless it's a quad 'Mech; see below).

To enter a clear hex costs 1 MP. Terrain in a hex often raises this cost (see *Terrain*, below).

Lateral Shift: A quad 'Mech that still has all four of its legs may move laterally (sideways) without changing its facing. A quad 'Mech making a lateral shift moves into an adjacent hex that is not directly to its front or rear, while retaining its facing, for the same MP cost as if it walked forward into it. A lateral shift costs 1 MP in addition to the cost of moving into the target hex.

A backward lateral shift—moving into hexes C or D in the diagram above—is considered backward movement; a 'Mech must walk to do this.

Partial Hexes: 'Mechs cannot voluntarily move into or through half- or quarter-hexes. 'Mechs forced into such hexes are considered destroyed for the remainder of the scenario.

BACKWARD MOVEMENT

A 'Mech can only move backward if it uses Walking MP that turn. Such a 'Mech can move forward, backward, or both in the same Movement Phase.

'Mechs moving backward may not change levels.

Backward Level Changes (Optional): If using this optional rule (see at right), a 'Mech may make a level change while moving backwards.

TERRAIN

While a 'Mech always spends at least 1 MP to enter a hex, terrain often raises this MP cost, as shown under the MP Cost Per Hex/Terrain Cost column of the Movement Costs Table (see p. 17). For example, entering a heavy woods hex costs 3 MP: the base 1 for entering any hex, and then another 2 for the heavy woods.

Difficult Terrain: Entering a hex containing certain terrain types requires a Piloting Skill Roll (see p. 53) to see if the 'Mech falls. Such information appears on the Movement Costs Table (see p. 17).

Water: If jumping into or out of water, see page 19. Otherwise, see *Movement in Water*, page 19.

LEVEL CHANGE

A 'Mech can move forward into a hex up to two levels (or depths) higher or lower than the hex the 'Mech just left. This costs an extra 1 MP per level (whether up or down).

Backward Movement: 'Mechs moving backward may not change levels.

Prohibited Level Change: You can't move a 'Mech into a hex three or more levels higher than the hex it just left. Not even a forced level change (such as from a charge or push) can force this to occur.

Similarly, you can't *willingly* move a 'Mech into a hex three or more levels lower than the hex it just left. However, a 'Mech can be *forced* downward any number of levels. Being forced two or more levels lower results in an automatic fall. No 'Mech may "voluntarily fall" from a greater level in order to circumvent the maximum allowable level change downwards.

If a 'Mech is forced into a hex that already has a 'Mech, see *Displacement*, page 55.

BACKWARD LEVEL CHANGES (OPTIONAL)

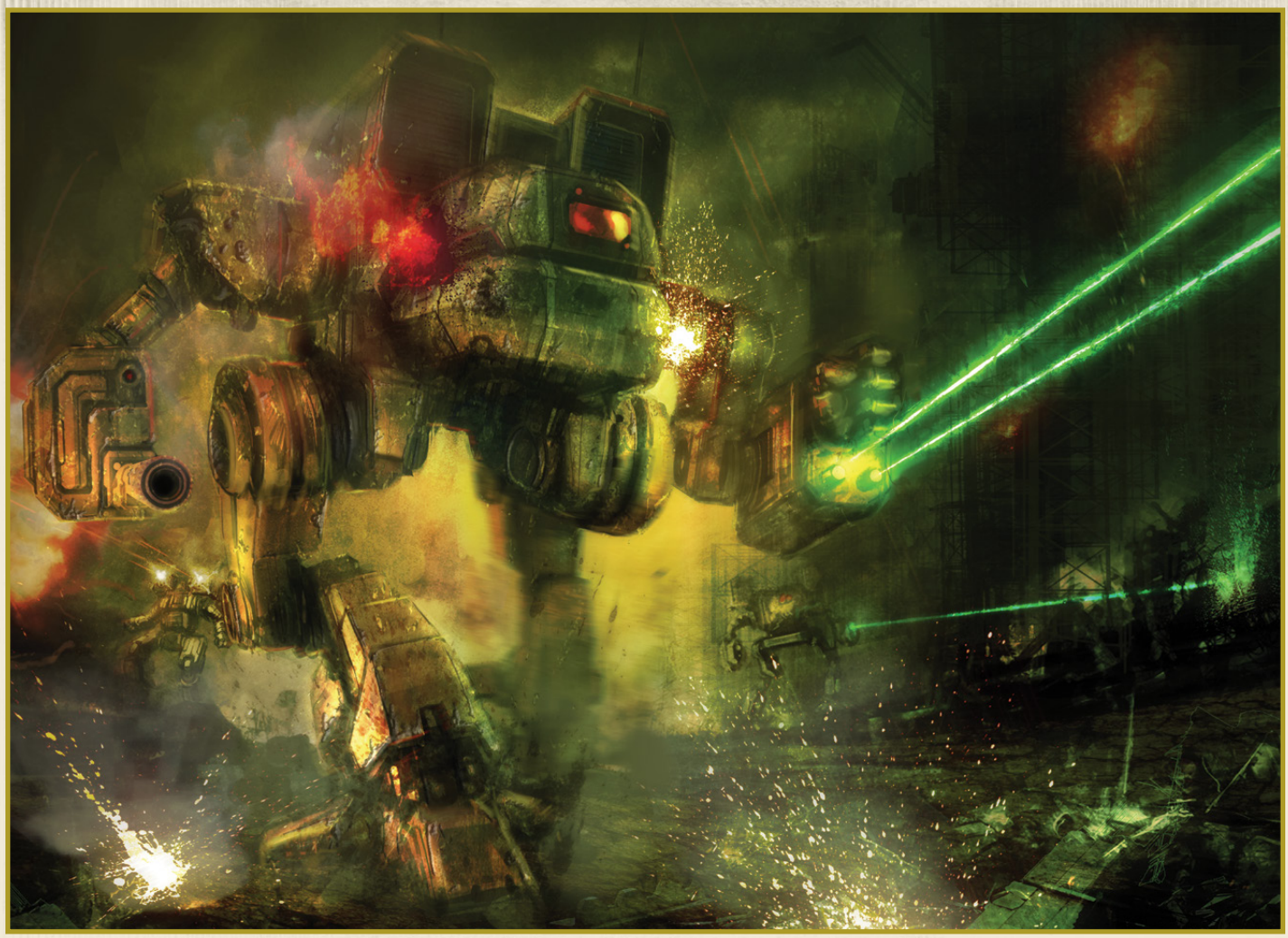
This optional rule allows a backward moving 'Mech to change 1 level per hex. A Piloting Skill Roll (see p. 53) is required each time the 'Mech changes a level in this way.

If the roll fails, the 'Mech falls either into the hex it was moving into or the hex it was moving out of, whichever is lower. The change in levels is not taken into consideration for determining the number of levels fallen.

STACKED TERRAIN

Multiple terrain types can occur in the same hex. For example, a Level 2 hex might be rough, but might also be a rubble hex due to a destroyed building. In such situations, all modifiers (including Piloting Skill Rolls required) are cumulative.

Conflicting Terrain: Maps may feature conflicting terrain. If any part of a hex contains a sublevel, the entire hex is considered to be the deepest sublevel marked in it. The exception to this rule occurs if the hex also contains a level greater than 0, in which case the level of the hill takes precedence.



OCCUPIED HEXES

A 'Mech can move through a hex occupied by a friendly 'Mech, but cannot enter a hex occupied by an enemy 'Mech. There are some unusual cases/exceptions:

- You can jump over hexes containing an enemy 'Mech (see p. 18). You can also try to jump onto an enemy 'Mech: this is known as a Death From Above attack (see p. 36).
- You can deliberately try to walk or run into an enemy 'Mech. This is a charge attack (see p. 35).
- A skidding 'Mech might skid into or through hexes containing enemy 'Mechs (see p. 68).
- If conducting a 'Mech drop (see *Dropping 'Mechs (Simplified)*, p. 79), you can deliberately place your 'Mech in the same hex as an enemy 'Mech.
- A 'Mech attempting to enter a hex containing an enemy hidden 'Mech (see p. 82) immediately stops before it can enter the hex.

Additionally, a 'Mech can never end its move in a hex with another 'Mech (friend or foe; see *Stacking*, p. 18).

PARTIALLY-OCCUPIED HEXES (OPTIONAL)

When using this option, a 'Mech can enter and then exit a hex occupied by an enemy 'Mech in the same move if the enemy 'Mech is abandoned (see p. 49), immobile (see p. 18), or prone (see p. 19).

MINIMUM MOVEMENT

A 'Mech can move into the hex directly in front of it, even if it does not have the MP normally required. A 'Mech can only do this if it meets all of the following conditions:

- It has at least 1 MP to spend
- It is not prohibited from entering the hex
- Such movement is the only MP it expends that turn (e.g. it cannot change its facing in the same turn)

To enter a hex in this fashion, a 'Mech must use running movement. 'Mechs using this rule can enter hexes that normally could not be entered by a running 'Mech (for example, Depth 1 or greater water hexes). Any Piloting Skill Rolls required for running still apply.

Prone 'Mechs: A prone 'Mech with only 1 MP available that turn can spend this MP to make a single attempt to stand using this rule, even if it only has one leg. See *Standing Up*, on page 19.

MOVEMENT COSTS TABLE

Movement Action/Terrain Type	MP Cost Per Hex/Terrain Cost	Piloting Skill Roll
Cost to Enter Any Hex	1	—
Terrain Cost When Entering Any New Hex		
Clear	+0	—
Paved/Bridge	+0	—
Road	+0*	—
Rough	+1	—
Sand	+0	+1
Light woods	+1	—
Heavy woods	+2	—
Light jungle	+2	+1
Heavy jungle	+3	+2
Water		
Depth 1	+1† (Level change MP cost not included)	–1
Depth 2+	+3† (Level change MP cost not included)	+0 (+1 if Depth 3+)
Level change (up or down)		
1 level	+1	—
2 levels	+2	—
Rubble	+1	+0
Light building	+1	‡
Medium building	+2	‡
Heavy building	+3	‡
Hardened building	+4	‡
Additional Movement Actions		
Facing change	1/hexside	—
Dropping to the ground	1	—
Standing up	2/attempt	+0
Lateral Shift (quad 'Mechs only)	+1	—
Terrain Modifications		
Ice	+1	+3§
Mud	+1	+1§
Deep Snow	+1	+1§
Swamp	+1	+1§
Weather Conditions		
Light Fog	+1	—
Heavy Fog	+2	—
Full Moon Night	+1**	—
Moonless Night	+2**	—

* If traveling along road; otherwise cost of underlying terrain.
† MP cost to move along the bottom of the water hex.
‡ Piloting Skill Roll required to prevent damage. The roll is modified for the 'Mech's movement per the Building Movement Modifiers Table (see p. 67).
§ This penalty applies to all PSRs made in this hex (including the PSR made when entering it).
**A 'Mech with an activated searchlight ignores this penalty.

IMMOBILE

An immobile 'Mech cannot make movements of any kind. Attacks against an immobile target apply a –4 Target Number modifier, and aimed shots (see p. 30) can be made at the target.

- **The following are immobile:** buildings; hexes; 'Mechs shutdown, abandoned, missing all four limbs, or with an unconscious MechWarrior.
- **The following are not immobile:** mobile 'Mechs that choose to stand still, prone 'Mechs, 'Mechs missing two legs (biped or quad), 'Mechs reduced to 0 MP by actuator damage and/or heat, a 'Mech with its gyro destroyed.

Unless a situation specifically states the target is immobile, it is not.

When making movement selections (see *Movement Phase*, p. 10), immobile 'Mechs may still be selected, even though they can do nothing but stand still.

Piloting Skill Rolls: An immobile 'Mech forced to make a Piloting Skill Roll (see p. 53) automatically fails, unless it is prone.

STACKING

At the end of the Movement Phase (after any 'Mechs destroyed that phase are removed from play), only one 'Mech can be in any single hex. This is called the stacking limit, and no 'Mech may deliberately violate it (unless conducting a 'Mech drop; see p. 79).

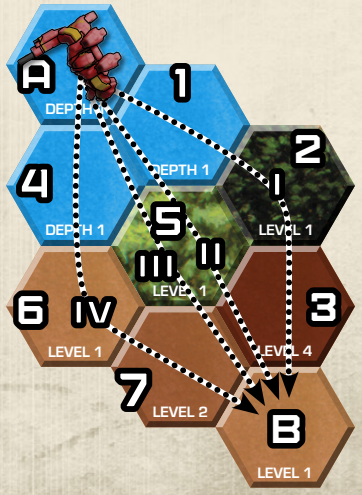
If the stacking rules are violated involuntarily, one of the 'Mechs in the hex will be forced out of that hex (see *Displacement*, p. 55).

Buildings: The stacking rules do not apply to 'Mechs in the same building hex but on different levels of it. Only apply the stacking limit to each level of the building hex (see *Building Levels*, p. 66).

JUMPING MOVEMENT

Jumping features numerous exceptions to the movement rules. To jump, a 'Mech must be standing at the start of the turn.

A jumping 'Mech can move 1 hex for each available Jumping MP. Terrain does not affect this MP cost: if a hex is legal to enter, it costs 1 MP to move over or into it, period. A 'Mech may jump into the same hex it began the turn in, for 1 MP.



• JUMPING DIAGRAM •

A jumping 'Mech ignores terrain (and other 'Mechs) in hexes it jumps over. As such, it need not make Piloting Skill Rolls if moving over hexes that normally force a PSR (though any PSR-related effects for the hex it lands in are applied). A jumping 'Mech also ignores facing: it can jump in any direction for the same MP cost, regardless of its initial facing, and when it lands it chooses any facing desired.

A 'Mech with at least 1 Jumping MP may jump down any number of levels. However, a 'Mech cannot enter a hex with a level higher than the sum of its current Jumping MP plus the level of the hex its jump began in.

A 'Mech can only jump to its destination hex via the shortest path (i.e. the fewest number of hexes required to get there). If more than one such path exists, the player chooses which of these paths their 'Mech takes. If all such paths are illegal (for example, there is only one path, and one of the hexes in the path has a level too high for the 'Mech to enter via jumping), the 'Mech cannot jump to the destination hex at all.

In the Jumping Diagram at left, the 'Mech in Hex A has a Jumping MP of 6. The 'Mech jumps to Hex B, four hexes away. Because the 'Mech is jumping, it spends only 1 MP for every hex that it moves, ignoring all terrain costs (including levels) for the hexes it passes over and for the hex in which it lands. (To reach Hex B with the facing shown by walking or running, the 'Mech would have had to spend at least 11 MP). As it lands, the player can face the 'Mech in any direction desired, at no extra cost.

The 'Mech is starting in Depth 1 water (Hex A), giving it a level of –1. Two of its jump jets are in its legs and so cannot be used, as they are submerged. Its other jump jets are located in the torsos, however, so it can still jump with a Jumping MP of 4.

At first, it appears the 'Mech could have jumped into Hex B by at least four different hex paths that are equally short (four hexes), as indicated on the diagram. There are actually two more paths (1-5-7-B and 4-5-3-B), but in this example only the primary four paths will be discussed.

Path I: 1, 2, 3, ending Hex B.
Path II: 1, 5, 3, ending Hex B.
Path III: 4, 5, 7, ending Hex B.
Path IV: 4, 6, 7, ending Hex B.

The player cannot choose Path I and II, however, because a hill in the intervening terrain (Hex 3) has a level of 4. The 'Mech has 4 Jumping MP, which would normally be enough, but as it is standing in Depth 1 water it has a starting level of –1, which means it can only jump over Level 3 terrain and lower this turn. As such, the 'Mech cannot jump over Hex 3. If Paths I or II were the only paths available, then the 'Mech could not jump to Hex B at all, because the shortest possible routes would both be illegal.

While the trees in Hex 5 rise two levels above the underlying terrain, they only rise to Level 3 (1 (underlying terrain) + 2 (trees) = 3). The 'Mech's Jumping MP equals or exceeds that level (4 Jumping MP – 1 = 3), but the 'Mech could cross that path regardless of the height of the trees, as trees do not affect jumping. As such, the player may still choose either Paths III or IV.

If the 'Mech had 5 or more Jumping MP available, it could have traveled to its destination along Paths I or II, because the 'Mech's Jumping MP (5) plus the level of the starting hex (–1) would equal or exceed the level of the intervening hilly terrain (4).

Death From Above: A 'Mech cannot jump into a hex that already contains a 'Mech unless it is making a Death From Above attack against an enemy 'Mech. See page 36 for details.

Into Water: A 'Mech jumping into a water hex Depth 1 or deeper makes a Piloting Skill Roll (see p. 53), applying no modifiers for the depth of the hex. If the PSR succeeds, the 'Mech drops out of the air safely to the bottom of the water hex. If it fails, the 'Mech tumbles to the bottom instead, falling a number of levels equal to the hex's depth (halve the resulting damage as normal due to falling in water; round down).

Out of Water: Jump jets cannot be fired while submerged. As such, a 'Mech standing in Depth 1 water may not fire jump jets located in its legs, and a 'Mech standing in Depth 2 or deeper water cannot jump at all. For example, a 'Mech with a Jumping MP of 5 that has one jump jet in each leg and one in each torso location may only use 3 Jumping MP when jumping out of Depth 1 water.

Mechanical Jump Boosters: This piece of equipment (see p. 116) provides Jumping MP, but operates quite differently from jump jets. A 'Mech using mechanical jump boosters to jump generates no heat, must begin and end its jump with the same facing, cannot jump down levels, can use its full Jumping MP even when submerged, and cannot execute death from above attacks. A 'Mech cannot use both jump jets and mechanical jump boosters in the same turn.

Partial Wing: This item modifies a 'Mech's jump capabilities. See page 116 for details.

PRONE BATTLEMECHS

A 'Mech might drop to the ground to take cover, or be forced to the ground due to falls. Whether it fell or chose to drop, such a Mech is prone. Prone 'Mechs rise 1 level above the underlying hex.

Torso Twist: A prone 'Mech cannot torso twist.

DROPPING TO THE GROUND

A 'Mech may choose to drop to the ground if it did not using jumping movement that turn. This action creates no additional heat, causes no falling damage, and costs 1 MP. The 'Mech drops face down with the same facing it had while standing and is thereafter prone. To regain its feet, it must attempt to stand (see *Standing Up*, below).

MOVING WHILE PRONE

While prone, a 'Mech can only change its facing or attempt to stand (see *Standing Up*, below): it cannot leave its hex. If a 'Mech begins the Movement Phase prone, it must declare whether it will walk or run before it makes any movement (such as attempts to stand); it may not jump.

STANDING UP

A 'Mech may attempt to regain its feet when on the ground. A 'Mech can attempt to stand even if missing one leg, or both arms A 'Mech missing all its legs, or one leg and both arms, cannot attempt to stand.

If a 'Mech begins the Movement Phase prone, it must declare whether it will walk or run before it attempts to stand. Each attempt to stand, successful or not, costs 2 MP. A 'Mech may attempt to stand in the same Movement Phase that it fell, as long as it has enough MP left and did not jump that turn.

To stand, the 'Mech's pilot must make a successful Piloting Skill Roll (see p. 53). If the PSR fails, the 'Mech suffers a 0-level fall (see p. 56); use the same facing it had on the ground as its initial facing when rolling on the Facing After Fall Table (see p. 57). The 'Mech may make repeated attempts to stand as long as it has the MP required for each attempt.

When a 'Mech successfully stands, it may select any facing at no cost (regardless of its facing while on the ground), and may continue to expend any remaining MP in that phase.

Arm Damage: This has no effect on a 'Mech's attempt to stand, with the caveats given above.

Four-legged 'Mechs: A prone quad 'Mech need not make a Piloting Skill Roll to stand unless one or more legs have been destroyed.

Heat: Each attempt to stand creates 1 heat point, which is cumulative with the heat created by the movement mode the 'Mech chose that turn.

Leg Destruction: A prone biped 'Mech with only one leg (or a prone quad 'Mech with only two legs) may still attempt to stand. However, it may only make one attempt per turn. This attempt to stand is always considered running, and can still be made even though a 'Mech with one leg can normally only walk.

As an exception to the normal rules, only one Piloting Skill Roll is required for this attempt, regardless of the number of PSRs that might normally be required. All modifiers to this PSR are still cumulative, however.

Minimum Movement: A prone 'Mech with only 1 MP available can use this rule (see p. 16) to make a single attempt to stand.

CAREFUL STAND (OPTIONAL)

This optional rule allows a 'Mech to take its time when trying to stand up and thus improve its chances. The 'Mech must have at least 3 Walking MP available, and the careful stand attempt takes its entire Movement Phase, whether successful or not. The 'Mech is considered to have walked.

When making a careful stand attempt, the Piloting Skill Roll to stand up applies a –2 Target Number modifier.

MOVEMENT IN WATER

Water hexes deeper than 0 have a depth that functions as a level change (see p. 15). 'Mechs entering such water hexes must pay:

- the base 1 MP for entering a hex
- plus the MP cost for entering water of that depth (1 MP if Depth 1, 3 MP if Depth 2 or deeper)
- plus the cost for any level change

Entering a water hex of Depth 1 or deeper forces a 'Mech to make a Piloting Skill Roll (see p. 53). This includes moving between water hexes, but not standing up or making facing changes within one. The modifier applied to this PSR depends on the hex's depth (–1 if Depth 1, +1 if Depth 3 or deeper).

A 'Mech standing in Depth 1 water has its legs submerged, and has partial cover (see p. 26). Underwater 'Mechs are completely submerged: this means a Depth 2 or deeper water hex if standing, or Depth 1 or deeper if prone.

'Mechs cannot enter a Depth 1 or deeper water hex while running (unless using *Minimum Movement*; see p. 16). However, a

A 'Mech that begins the turn standing in a clear hex wishes to move into a neighboring waterway, consisting of a series of Depth 2 water hexes. Because the 'Mech wants to enter a water hex with a Depth greater than 0, the 'Mech cannot run: its options are only walking or jumping.

The 'Mech chooses to walk. The first hex requires 6 MP to enter (1 for the hex, 3 MP for entering Depth 2+ water, and 2 more MP for moving from a clear hex to a Depth 2 hex: a 2-level change). A Piloting Skill Roll is also required (no PSR modifier is applied by the water, as the hex is not Depth 3+).

If the 'Mech has enough MP left (and assuming it doesn't fail its PSR), it can then immediately continue to the next hex in the pond. This would cost 4 MP (1 for the hex, and 3 MP for entering Depth 2+ water). As the 'Mech is moving into a water hex, another PSR is required, again with no modifier for the water.

The next turn, as the 'Mech is in water, it cannot run unless it only intends to change its facing or if the first hex it moves into is out of the pond and onto land. It also could not jump, as a Depth 2+ hex means the 'Mech is submerged, and jump jets cannot be used when submerged.

'Mech can use running movement to change facing in water and/or move from a water hex to a land hex.

Hull Breaches: Submerging one or more locations without armor will automatically breach those locations (see p. 65).

Jumping: To resolve jumping into and out of water, see *Jumping Movement*, page 18.

Underwater Maneuvering Units (UMUs): If a 'Mech is standing and submerged at the start of the turn, it may use its UMUs (see p. 116). When doing so, it counts as having jumped that turn, and ignores the additional MP cost for entering a water hex. This means that a water hex with no obstructions or level changes would be treated as a clear hex, costing 1 MP to enter. It always costs just 1 heat point to use UMUs (regardless of how far a 'Mech moves).

UMUs cannot be used to move from a land hex into a water hex, or vice versa, and have no effect on a 'Mech's movement when it is not completely submerged.

MOVEMENT ON PAVEMENT

Movement on pavement is largely handled in the same way as movement through clear terrain, but with a few important exceptions.

ROAD MOVEMENT

A road is simply a paved hex that passes through terrain of some other type. A 'Mech is traveling on a road if it moves from one hex to the next on that road.

'Mechs traveling on roads pay only 1 MP per hex; however, any MP costs for changing levels are still required. 'Mechs may move through prohibited terrain while traveling on a road, but must begin and end their movement on the road and remain on it while traversing such terrain.

BRIDGE MOVEMENT

Roads that cross a water hex, or span a distance in which the underlying terrain of the hex is lower than the level of the bridge, are considered bridges.

Bridges are classified as light, medium, heavy, or hardened in the same manner as buildings, and have the same range of Construction Factors (CF) (see *Urban Combat*, p. 66). If the bridge type is not defined, assume it is a medium bridge with CF 40. Bridges can be attacked in the same manner as buildings. When a bridge's CF is reduced to 0, it collapses.

Capacity: As with buildings, each hex of a bridge will only support 'Mechs with a total tonnage equal to or less than the bridge's current CF. If the current CF is exceeded, the bridge hex collapses, and any 'Mech on it falls as normal (see *Falling*, p. 56).

If a bridge covers more than a single hex, CF values are assigned to each hex. The destruction of a single hex of a multi-hex bridge does not affect any other hex in that bridge, unless more than half its total hexes have been destroyed; in this case, the entire bridge collapses. If the collapse of the final hex that destroys the whole bridge occurs due to movement, the entire bridge collapses immediately; if the final hex is destroyed by an attack, the bridge collapses at the end of the phase the attack occurred in (see *Collapse*, p. 73).

Under-Bridge Movement: A 'Mech may enter a bridge hex and be considered underneath the bridge if the 'Mech's LOS level (see p. 22) is equal to or less than the level of the bridge.

If a 'Mech cannot move on top of the bridge, or underneath it, the bridge is treated as a building hex for the purposes of movement (see *Urban Combat*, p. 66).

SKIDDING

If a running 'Mech makes a facing change in a paved hex at any point in the turn, and then attempts to enter a new hex in the same turn, before actually entering the new hex the 'Mech's controller must make a Piloting Skill Roll (see p. 53) to see if the 'Mech skids. The PSR is modified based on the total number of hexes moved in the turn so far (as shown on the Piloting Skill Roll Table, p. 54**,** under "Skidding Movement").

If the PSR is successful, the 'Mech does not skid, and may enter the new hex and continue its movement. If it fails, the 'Mech falls (see *Falling*, p. 56); it suffers normal falling damage and then skids.

See *Skidding*, page 68, for how to resolve the skid and the damage that results.

RECKLESS MOVEMENT

Applies to: Ice, Mud, Deep Snow, Swamp, Fog, Full-Moon Night, Moonless Night (see *The Battlefield*, pp. 59-65).

The increased MP cost imposed by the environmental conditions listed above represents the extra caution needed to avoid falling in such conditions. However, players who do not want to pay this MP cost can instead announce their intention to move recklessly before entering a hex affected by such a condition. When a 'Mech recklessly enters the hex, its controller must make a Piloting Skill Roll (see p. 53). Success means that the 'Mech enters the hex as normal. If the roll fails, the 'Mech immediately falls in the hex just entered (if this fall occurs on pavement, the 'Mech skids and its move is over; see *Skidding*, p. 68).

Jumping: 'Mechs expending Jumping MP ignore this rule.

Minimum Movement: A 'Mech using this rule (see p. 16) to enter a hex where a reckless movement MP modifier applies always triggers the PSR for moving recklessly.

OTHER ACTIONS

There are a few miscellaneous actions that also occur in or in close proximity to the Movement Phase.

HIDDEN 'MECHS

If a 'Mech attempts to enter a hex occupied by a hidden enemy 'Mech (jumping over it does not count), or ends its movement adjacent to such a hex, the hidden 'Mech is revealed. A hidden 'Mech revealed in these ways may immediately make a special point-blank weapon attack against the 'Mech that revealed it. See page 82 for the full details.

At the end of the Movement Phase, any hidden 'Mech within range of an enemy 'Mech's active probe and with line of sight to that enemy 'Mech is revealed.

LIFTING ITEMS (SIMPLIFIED)

A 'Mech may lift items using its hands. A 'Mech can lift and carry around cargo weighing up to 10 percent of the 'Mech's tonnage (this is increased to 20 percent if the 'Mech has active triple-strength myomer; see p. 117).

Lifting Restrictions: Only 'Mechs with functioning hand actuators may lift items.

A 'Mech carrying cargo cannot fire any arm or forward-firing torso-mounted weapons, and may not make club, punch, push, or physical weapon attacks. It may still charge, kick, and execute death from above attacks.

A 'Mech suffers no movement penalties when carrying cargo, and can still use Jumping MP.

Loading and Unloading: To pick up or drop items, a 'Mech must end its Movement Phase in the same hex as the item. One item is picked up (or unloaded in the same hex as the unloading 'Mech) in the End Phase of that turn.

A 'Mech cannot make any weapon or physical attacks in a turn it loads or unloads.

Dropping Cargo: During its movement, a 'Mech may drop all its cargo in the hex it is standing in (as opposed to properly unloading it, as described above) by expending 1 MP. Unlike regular loading/unloading, this does not affect its ability to make attacks.

Dropping an item may destroy it. Roll 1D6 for each item carried: on a result of 1–3, the item is destroyed, and on a result of 4–6, the item is undamaged and left in that hex.

If a 'Mech that is lifting items loses one or both hand actuators, it automatically drops all lifted items at the end of the phase in which the damage occurred. This does not require MP.

Combat: Any fall, or any damage dealt to an arm or front torso location, also damages that 'Mech's lifted cargo. For every such point of damage applied to the carrying 'Mech, a ton of cargo (or one item, whichever is more) is destroyed. The cargo does not reduce the damage.

Claws: A 'Mech with any claws must make a Piloting Skill Roll (see p. 53) with a +2 modifier when lifting items; if the object is very fragile (such as a human being), a further +1 modifier applies (at the gamemaster's discretion or mutual player agreement). If the PSR fails, the object is effectively destroyed.

TAG DESIGNATION

If using target acquisition gear (TAG; see p. 113), all 'Mechs making attempts to designate a target do so after the end of the Movement Phase, but before the beginning of the Weapon Attack Phase (Initiative order is still followed, however).

Arm flipping (see p. 24) or torso-twisting (see p. 25) can be made immediately prior to firing TAG. This prevents any further use of such in that turn's Weapon Attack Phase.

COMBAT

After all players complete the Movement Phase, 'Mechs engage in combat. BattleMechs use two forms of combat: weapon attacks and physical attacks. Weapon attacks are made using ranged armaments such as missiles, lasers, and autocannons. For physical attacks (see p. 34), each 'Mech has a variety of options, most of which rely on the 'Mech's own weight to inflict damage.

Hidden 'Mechs: If a player plans to attack using a hidden 'Mech (see p. 82), they must reveal that 'Mech and place it on the map at the beginning of the Weapon Attack Phase.

WHAT IS A TARGET?

The target of an attack can be almost anything, even if the result will not inflict damage. For example, by firing at an empty hex, a player can expend unwanted ammunition or heat up their 'Mech to activate triple-strength myomer. Regardless of the target, all effects of an attack must be taken into account (ammunition used, heat generated, multiple-target modifiers, etc.).

You can never deliberately target a friendly 'Mech. However, an area-effect attack (AE; see p. 96) damages everything in its blast radius, even friendly 'Mechs.

Clearing Woods or Jungle: 'Mechs can target woods or jungle hexes in order to transform them into clear hexes (see p. 61).

LINE OF SIGHT

Normally an attacker needs a clear line of sight (LOS) to its target. A straight line running from the center of the attacker's hex to the center of the target's hex defines the LOS between two 'Mechs. Any hexes through which this line passes lie along the LOS, even if the line barely touches the corner of a hex. However, the hexes containing the attacking and target 'Mechs are *not* counted (see *Water Hexes*, p. 23, for the sole exception to this rule).

LOS is always mutual: if you can see a target, that target can see you (unless using the optional *Hidden 'Mechs* rules; see p. 82). You can check for LOS before declaring an attack.

Adjacent 'Mechs: 'Mechs in adjacent hexes always have LOS to each other, unless one 'Mech is completely underwater and a 'Mech in an adjacent hex is not (see *Water Hexes*, p. 23). Likewise, if both 'Mechs are in adjacent building hexes but at different levels, LOS may not exist (see *Combat within Buildings*, p. 72).

Indirect Fire: If a valid LOS does *not* exist to the target, LRMs and Thunderbolts may be fired indirectly (see *Indirect Fire*, p. 30).

LOS on the Border of Two Hexes: If the LOS falls exactly on the line separating two hexes, the target 'Mech's controller decides which of the two hexes the LOS passes through (see the Line of Sight Between Two Hexes Diagram at right). If a target is not player controlled (for example, a building or a hex), randomly determine which of the two hexes the LOS lies along. In both cases, this LOS is used for all attacks between those two units for the rest of that turn. This choice is also used when determining attack direction (see *Attack Direction*, p. 32).

LEVELS AND HEIGHT

Apply the following rules for the purposes of terrain and LOS.

• LINE OF SIGHT BETWEEN TWO HEXES DIAGRAM •

LOS Level: A 'Mech's current height, plus the level of its underlying hex, is known as its LOS level.

'Mech Heights: Standing 'Mechs are 2 levels in height; prone 'Mechs are 1 level in height.

TERRAIN HEIGHT AND DEPTH

A hex's level is marked on the map. Hexes with levels higher than 0 represent hills. Hexes with levels lower than 0 represent sinkholes.

Buildings: The level listed in a building hex is how many levels the building rises above the hex it is in. For example, a Level 2 building on a Level 4 hex makes the hex Level 6 for LOS purposes. A 'Mech standing there would thus have a LOS level of 8.

Depths & Sublevels: Ground hexes lower than Level 0 are marked as Sublevels, while water is marked in Depths. Both are treated as negative height. For example, a hex at Depth 2 or Sublevel 2 is at Level –2. Depth 0 is the same as Level 0 for LOS purposes.

Hexes as Targets: If firing directly at a hex (as opposed to a 'Mech or building within it), the hex adjacent to it along the LOS from the attacker is considered to be 1 level lower than it actually is (i.e. a Level 2 hill adjacent to it won't block LOS to it).

Woods/Jungle: Woods and jungles rise 2 levels above the level of the underlying hex (for example, a jungle on a Level 2 hex would rise 2 levels above that, reaching Level 4). 'Mechs in such hexes are on the underlying terrain, not on top of the trees.

INTERVENING TERRAIN

Any terrain that might interfere with LOS is said to be intervening terrain. This isn't necessarily all-or-nothing: woods or jungle only block LOS if there's more than one hex of it along the LOS. Only check the hexes along the LOS *between* the attacker and target, not the hexes they occupy (see *Water Hexes* on p. 23 for a notable exception).

Terrain along the LOS between attacker and target intervenes if:

- Its level is equal to or higher than the LOS level of both 'Mechs; or
- Its level is equal to or higher than the attacker's LOS level and adjacent to it; or
- Its level is equal to or higher than the target's LOS level and adjacent to it

EFFECTS OF INTERVENING TERRAIN

Again, intervening terrain does not include the hexes either the attacker or target are standing in. Intervening terrain has the following effects on line of sight.

Buildings and Bridges: One intervening building hex blocks LOS; bridges, however, do not.

Woods/Jungle: Three or more points of intervening woods/jungle block LOS. Light woods/jungle is worth 1 point, and heavy woods/jungle is worth 2 points.

Hills: One intervening hill hex blocks LOS.

Partial Cover: While partial cover (see p. 26) conceals part of a 'Mech, it does not block LOS to that 'Mech.

Water: Intervening water blocks LOS unless both attacker and target are submerged and no other terrain intervenes.

OTHER 'MECHS AND LOS

'Mechs never intervene, and so 'Mechs besides the attacker and target have no effect on LOS.

The Line of Sight Diagram on the right illustrates many of the principles governing line of sight.

The BattleMech in Hex A wants to attack in this turn. It is standing in a Level 0 hex, and is therefore considered to have an LOS level of 2. Checking LOS for the BattleMech in Hex A to the other 'Mechs shown, we find the following conditions:

LOS to the 'Mech in Hex B passes through one light woods and one heavy woods hex, and so is blocked.

LOS to the 'Mech in Hex C passes through two light woods and a heavy woods hex, and so is blocked.

The 'Mech in Hex D is standing in a Level 1 hex, and therefore has an LOS level of 3. Even though the LOS passes through hexes 2 and 3, those woods are at Level 2 for purposes of LOS and so do not intervene. The woods in Hex 1, however, are in the LOS and are at Level 3 for purposes of LOS. Because these woods are equal to or higher than both attacker and target, they intervene. The woods in Hex 4 are equal to or higher than the attacker and adjacent to the attacker, and so they intervene as well; only two of these light woods hexes intervene, however, and so they do not block LOS.

Finally, the 'Mech in Hex E is in a Level 3 hex, within Depth 1 water, and so has an LOS level of 4 (2 for standing, +3 for the hill, –1 for Depth 1). LOS is blocked by Hex 5.

Use the diagram to practice finding LOS with the other 'Mechs. Try to determine how many targets each 'Mech can see, and compare your results to the correct results that follow: Hex B has 2 targets, Hex C has 2 targets, Hex D has 3 targets and Hex E has 0 targets.

• LINE OF SIGHT DIAGRAM •

WATER HEXES

A standing 'Mech in Depth 2 or deeper water (or a prone 'Mech in Depth 1 or deeper water) is submerged. Being submerged completely blocks LOS to and from the 'Mech in that hex, unless the 'Mech attempting to draw LOS to it is submerged as well.

See the Underwater Line of Sight Table, on page 65, for a summary of LOS involving water.

FIRING ARCS

If an attacker has LOS to its intended target, the attacking player then checks the firing arcs of his 'Mech's weapons to see which weapons can hit the target. There are four firing arcs as shown on the Firing Arcs Diagram: the forward arc (in yellow), left side arc (in blue), right side arc (also in blue), and rear arc (in red).

Note that the firing arcs extend from the attacker to the edge of the playing area, and there are no "blind spots".

Prone 'Mechs: Arcs do not change if the attacker is prone.

Split-Location Weapons: A weapon whose critical slots are split between a torso and arm location uses the torso firing arc.

FORWARD ARC

All weapons may fire into the forward arc unless they are rear-mounted. Arm-mounted weapons may fire into the forward arc plus the appropriate side arc.

LEFT SIDE ARC

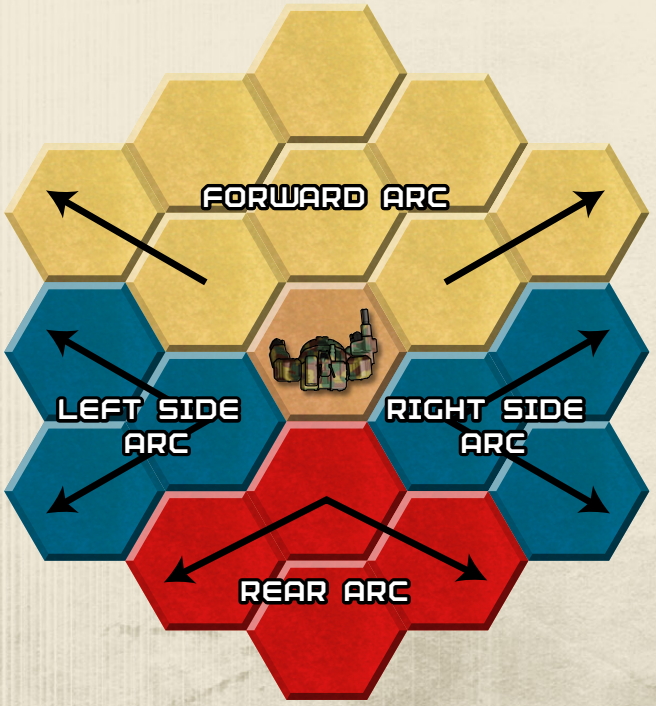
Weapons in a 'Mech's left arm may fire at targets in the left side arc (and also the forward arc).

RIGHT SIDE ARC

Weapons in a 'Mech's right arm may fire at targets in the right side arc (and also the forward arc).

REAR ARC

Only rear-mounted weapons may fire into the rear arc. Weapons which are rear-mounted will have an (R) beside their name on the record sheet's Critical Hit Table.



• FIRING ARCS DIAGRAM •

ARCS AND TORSO TWISTS

If a biped 'Mech makes a torso twist (see p. 25), its normal firing arcs for its arm- and torso-mounted weapons shift one hexside to the left or right (depending on the direction it twisted). However, arcs for leg-mounted weapons (and kick attacks) are unaffected.

If a 'Mech torso-twists, its firing arcs temporarily change, but its facing is still the same. This means that torso-twisting doesn't affect the direction of attacks made against the torso-twisted 'Mech (see *Attack Direction*, p. 32).

ARCS AND REVERSING (FLIPPING) ARMS

If a 'Mech flips its arms (see *Reversing (Flipping) Arms* below), it may fire any arm-mounted weapon into the rear firing arc instead of the usual firing arcs for those weapons.

ATTACK DECLARATION

Next, the player declares their attacks. For each attack, the following must be announced:

- Which specific weapon is being fired
- Which specific target it is being fired at

Each weapon can make one attack a turn (though some weapons can fire multiple shots per attack). If a weapon has a choice between multiple ammunition types or attack modes (such as rapid-firing), or the player wants to make a specialized attack with it (such as an aimed shot or indirect fire), the player must announce these choices at the same time as the attack.

All attacks by all players must be declared before any are resolved.

Leg-Mounted Weapons: Leg-mounted weapons may not fire through a hex that provides the firing 'Mech with partial cover, unless the partial cover is provided by water (see *Combat in Water*, p. 27).

Spotting: If a player declares an indirect fire attack (see p. 30), they must also declare which 'Mech is spotting for that attack. One 'Mech can spot for multiple attacks, but only if all those attacks are against a single target.

REVERSING (FLIPPING) ARMS

A 'Mech whose record sheet does not show hand and lower arm actuators in both arms on its Critical Hit Table and that does not have any weapons split between torso and arm locations can flip its arms. After flipping its arms, a 'Mech uses the rear firing arc for arm-mounted weapons, instead of the usual firing arcs.

Both arms must lack these actuators: a 'Mech with only one applicable arm cannot reverse either of its arms (and losing the required actuators through damage does not grant the ability). However, if a 'Mech that can reverse its arms loses one during a game, it can still reverse the remaining arm.

Reversing arms occurs during attack declaration. Both arms must be reversed (unless one is destroyed). A 'Mech cannot torso twist and reverse its arms in the same turn, or punch or make arm-mounted physical weapon attacks while its arms are reversed. In the End Phase, reversed arms automatically return to the front arc.

Prone 'Mechs: Prone 'Mechs may not reverse their arms.

EXPANDED ARM FLIPPING (OPTIONAL)

In standard rules, a 'Mech cannot both torso twist and reverse its arms. However, with this option, such an action is permitted. The controlling player must first announce whether a 'Mech is torso twisting or reversing its arms, and then make a Piloting Skill Roll (see p. 53). If the PSR is successful, the secondary action can occur. A failed roll does not result in a fall: instead, the second action (whether a torso twist or reversing arms) does not occur.

TORSO TWISTS

As part of each 'Mech's weapon attack declaration, a biped 'Mech can rotate its torso one hexside to the left or right while keeping its feet pointed straight ahead. This twist lasts for the remainder of the turn, affecting firing arcs for the Weapon Attack and Physical Attack Phases, but is not considered when determining hit locations.

The torso automatically returns to its forward position in the End Phase.

Four-Legged 'Mechs: Quad 'Mechs cannot torso twist.

FIRING WEAPONS

Players use their 'Mechs' armament to attempt to inflict damage on targets. Each weapon is always fired individually, even if multiple weapons of the same type are mounted in the same location.

Finding a Club: Instead of making any weapon or physical attacks that turn, a 'Mech may search rubble in the hex it occupies for a girder to use as a club. See *Club Attacks*, page 36, for more details.

STRATEGIC

G.A.T.O.R.

The attack process is not complicated—simply a matter of rolling 2D6 and trying to equal or exceed your modified Target Number. However, it's the "modified" part that can trip a player up: juggling the many potential modifiers can be imposing.

After you understand the *Target Number Modifiers* section below, you'll find that "GATOR" is a helpful mnemonic that reminds you what needs to be taken into account:

- **G** Gunnery Skill rating of the attacker (the base Target Number for the attack). Then add:
- **A** Attacker movement modifier
- **T** Target movement modifier
- **O** Other modifiers (potentially a big category, but typically only includes woods, partial cover, and heat)
- **R** Range modifiers

You start with your Gunnery Skill rating, then add all other applicable modifiers as indicated, and the result is your modified Target Number to roll against. In the Target Number Modifiers section that follows, modifiers are listed in GATOR order, beginning with the Attacker Movement Modifier.

If combined with the use of Movement Dice (see p. 13), attack resolution can be greatly sped up.

MAKING THE ATTACK

The base Target Number for a weapon attack is equal to the attacker's Gunnery Skill Rating (see *Skills*, p. 9).

The modified Target Number is the base Target Number plus all applicable modifiers covered in *Target Number Modifiers* below.

To make an attack, roll 2D6. If the result is equal to or greater than the modified Target Number, the attack hits.

If the modified Target Number is greater than 12, the shot automatically misses; rolling a 12 does not automatically hit. If a declared shot would automatically miss, the attacker can abort that shot, but another target cannot be chosen in its place that turn.

If the modified Target Number is 2 or less, the shot automatically hits. Because some weapons have additional effects based on the result of the attack roll, even automatic hits may still require the player to make a roll.

Resolution Order: The attacker chooses the order in which they make attack rolls for all of their 'Mech's declared attacks. From turn to turn, the attacker can change this order.

BASE TARGET NUMBER

GATOR

The base Target Number for a weapon attack is equal to the attacker's Gunnery Skill rating (see *Skills*, p. 9).

TARGET NUMBER MODIFIERS

The base Target Number may be modified by several factors (as showcased in the Strategic: G.A.T.O.R. sidebar; see left).

ATTACKER MOVEMENT MODIFIER

GATOR

An attacker that moves modifies its Target Number using the values in the Attack Modifiers Table (see p. 28). The modifier is based on the movement mode the attacker used that turn (Walking, Running, or Jumping), not the MP it expended or hexes it travelled.

If using Movement Dice (see p. 13), the color of the die indicates this modifier.

TARGET MOVEMENT MODIFIER

GATOR

A moving target is harder to hit, and so an attacker's Target Number is modified by its target's movement using the values in the Attack Modifiers Table (see p. 28). The target movement modifier (often abbreviated as "TMM") is based on the number of hexes traversed by the target this turn, not the number of Movement Points it spent.

If the target moved both backward and forward in the turn, base the TMM on the number of hexes moved from the hex in which the 'Mech last reversed its movement. For example, if the target moved backward three hexes and then forward two hexes, the target movement modifier would be based only on the final two hexes of movement, resulting in a TMM of 0.

If a target jumped this turn, raise its TMM by 1 in addition to the modifier for the number of hexes the target moved.

If using Movement Dice (see p. 13), the number on the die indicates this modifier.

Prone 'Mechs: Target movement modifiers apply for the whole turn, even if a 'Mech drops to the ground or falls that turn.

Skidding: Hexes skid in a turn count towards a 'Mech's target movement modifier for the turn (see *Skidding*, p. 68).

A JagerMech *is attacking with its AC/5s. The* JagerMech *used Running MP this turn, so it must add an attacker movement modifier of +2 to its Target Number. The target traveled five hexes, creating a +2 target movement modifier. The range to the target is two hexes, which is short range for the AC/5s. However, the AC/5 has a minimum range of 3. This means the Target Number of an attack at a range of three hexes would be modified by +1, while an attack made at two hexes—such as this attack—is modified by +2. The modified Target Number for the* JagerMech*'s AC/5s is 10: 4 (Gunnery Skill Rating) + 2 (Attacker Movement) + 2 (Target Movement) + 2 (Minimum Range) = 10.*

Meanwhile, an Atlas *is firing its LRMs. The range to its target is four hexes—short range for LRMs, but they also have a minimum range of 6. This means a +3 minimum range modifier applies to the attack. The* Atlas *did not move, and so no attacker movement modifier is applied. The target spent 5 Running MP, but only traveled two hexes, and since the target movement modifier counts only the number of hexes moved, no TMM applies. The modified Target Number for the LRM attack is 7: 4 (Gunnery Skill Rating) + 3 (Minimum Range Modifier) = 7.*

TERRAIN MODIFIERS

GATOR

Terrain can reduce the probability of a successful shot by forcing the attacker to account for intervening land features.

Death From Above: This attack (see p. 36) ignores terrain modifiers.

Light Woods/Jungle: Add a +1 modifier to an attack's Target Number if the target is in a light woods or light jungle hex. Apply a further +1 modifier per hex of light woods or light jungle intervening between the attacker and target (see *Intervening Terrain*, p. 22).

Heavy Woods/Jungle: Add a +2 modifier to an attack's Target Number if the target is in a heavy woods or heavy jungle hex. Apply a further +2 modifier per hex of heavy woods or heavy jungle intervening between the attacker and target (see *Intervening Terrain*, p. 22).

Partial Cover: Attacks against a target in partial cover (see below) receive a +1 Target Number modifier.

Water: A standing 'Mech in Depth 1 water receives partial cover, which provides a +1 Target Number modifier to attacks against it (see *Partial Cover Modifier* below).

PARTIAL COVER MODIFIER

GATOR

To receive partial cover, a target 'Mech must be standing adjacent to a hex one level higher than its current hex. This adjacent hex can be a hill or a building (but not a bridge), but must lie along the LOS between the attacker and target. For example, a standing 'Mech in a Level 0 hex adjacent to a Level 1 hex would receive partial cover if an attacker traces LOS through that Level 1 hex.

However, if the attacker has an LOS level higher than the target's LOS level, partial cover does not apply. In other words, an attacker firing downhill negates its target's partial cover (unless the partial cover is being provided by water or a building; see below).

The Line of Sight Diagram on page 23 illustrates some examples of partial cover.

The 'Mech in Hex C has partial cover from the 'Mechs in Hexes A and B. This is because it is adjacent to a hex that is one level below its own LOS level (see p. 13), along the LOS between the 'Mech and those two 'Mechs, and the LOS level of those two 'Mechs is equal to or less than the LOS level of the 'Mech. The 'Mech in Hex D has partial cover from the 'Mech in Hex A for the same reason.

For the 'Mech in Hex B, its target in Hex D would only receive partial cover if the 'Mech in Hex B declared its attacks first—LOS between the two passes exactly between the hexes that would allow the partial cover to intervene, meaning per LOS on the Border of Two Hexes *on page 22 that the target chooses the partial cover hex.*

Attacks against a target with partial cover apply a +1 Target Number modifier. In addition, if the hit location roll indicates a leg, the attack strikes the cover instead. In that event, if the partial cover is a building, the hit damages the building; otherwise, the attack does not deal damage.

Buildings: A standing 'Mech occupying a Level 1 building or a building hex one level below the roof receives partial cover, even if its attacker is at a higher level. If the hit location roll indicates a leg, the attack strikes the building instead, damaging it as normal.

Four-Legged 'Mechs: For quad 'Mechs, any hit to a leg—front or rear—constitutes a leg hit.

Prone 'Mechs: Prone 'Mechs cannot receive partial cover. Being prone and adjacent to a Level 1 hex along the LOS from an attacker instead means that the attacker cannot see the prone 'Mech at all (because prone 'Mechs are only one level high).

Water: A standing 'Mech in Depth 1 water normally receives partial cover. However, partial cover from water never applies to attacks that could not hit the legs in the first place (for example, if the attack uses the Punch Location Table Table and the target is a biped 'Mech). Otherwise, it applies even versus attackers that are at a higher level than it or making physical attacks. Effects that ignore partial cover and/or terrain modifiers negate the Target Number modifier provided by water, but not the leg protection.

If a fully submerged 'Mech attacks a 'Mech standing in Depth 1 water, the target still has partial cover, but an attack striking any location *except* for a leg strikes the cover.

Woods/Jungle: These terrain types do not provide partial cover.

HEAT MODIFIER

GATOR

The attacker's Target Number may suffer a modifier due to its heat build-up, as discussed in *Building up Heat* (p. 50). The Heat Data section of the record sheet summarizes these modifiers.

The heat modifier never applies to physical attacks.

WEAPON & EQUIPMENT MODIFIERS

GATOR

Some weapons and equipment provide modifiers to attacks. If this is the case, it will be noted in that item's description (see *Weapons and Equipment*, beginning on p. 96).

DAMAGE MODIFIERS

GATOR

The attacker's Target Number may suffer additional modifiers for damage to its arms or sensors, as discussed in *Critical Hits* (p. 45).

Lower Arm Actuators: 'Mechs whose record sheet did not come with lower arm actuators do not suffer actuator damage modifiers to weapon attacks, though the absent actuator still affects physical attacks.

IMMOBILE TARGET MODIFIER

GATOR

Attacks against an immobile target apply a –4 Target Number modifier.

The most common immobile targets are buildings; hexes; and 'Mechs shutdown, abandoned, missing all four limbs, or with an unconscious MechWarrior. Do not apply this modifier unless a target is specifically stated to be immobile, even if it has 0 MP.

Aimed Shots: This specialized attack can be declared against immobile targets. See page 30.

MULTIPLE TARGETS MODIFIER

GATOR

A 'Mech may engage more than one target in a turn. The only limit on the number of targets is how many weapons a 'Mech mounts.

During the attack declaration step, the attacker designates one target as the primary target. If any declared targets are in the attacker's forward arc, one must be the primary target. If the attacker is declaring attacks only against targets in the side and rear arcs, they may choose any of them as the primary target.

All targets other than the primary target, no matter how many, are secondary targets. Against secondary targets in the forward arc, apply a +1 Target Number modifier; against secondary targets in the side and rear arcs, the modifier is +2 instead.

These rules apply regardless of where the weapons are mounted. For example, a 'Mech with three medium lasers in its right arm fires one medium laser at a target in its front arc (designated as the primary target), and the second and third medium lasers at separate targets in its right side arc. The primary target receives no multiple targets modifier, while both secondary targets in the side arc each receive a +2 Target Number modifier.

Multiple Firing Arcs: Through torso twisting, a 'Mech with both upper-body and leg-mounted weapons may have two forward firing arcs at once. Regardless, a 'Mech may only have one primary target each turn.

Physical Attacks: The multiple targets modifier does not apply to physical attacks.

Stealth Armor: A 'Mech with an active stealth armor system cannot be attacked as a secondary target (see *Stealth Armor*, p. 115).

RANGE MODIFIER

GATOR

The farther away a target, the harder it is to hit. Count the hexes between attacker and target, taking the shortest path (start with the hex adjacent to the attacker, and include the target's hex). The total is the range to the target.

Each weapon's range is divided into range brackets: short, medium, and long. For example, a PPC has a range of 18 hexes. Its short-range bracket is "1-6", meaning that any attack made with a PPC against targets 6 hexes away or less is in the PPC's short range.

The range bracket a target is in determines the range modifier for an attack with that weapon against that target. A short-range shot does not modify the attack's Target Number. A medium-range shot adds a +2 Target Number modifier, while a long-range shot adds a +4 modifier.

Levels: Level differences are ignored when calculating range. A target one hex away but 99 levels higher than the attacker is still one hex away when determining range and range modifiers.

MINIMUM RANGE MODIFIER

GATOR

Some weapons, such as PPCs and long-range missiles (LRMs), are designed to be fired at long-range targets. If a target is too close, hitting it with such weapons is more difficult.

A weapon's minimum range, if any, appears on the Weapons and Equipment Tables beginning on page 120. If a target is within this range, use the following formula to determine the minimum range modifier: [Min. Range] – [Target Range] + 1. For example, if a 'Mech fires a PPC at a target three hexes away, the minimum range modifier would be +1. If the target is two hexes away the modifier is +2, and if one hex away the modifier is +3.

COMBAT IN WATER

A 'Mech standing in Depth 1 water may attack 'Mechs in water with its leg-mounted weapons. It may also attack 'Mechs that are not underwater, using weapons mounted in any other location.

Underwater 'Mechs can attack a standing 'Mech in Depth 1 water; the standing 'Mech has partial cover (see *Partial Cover Modifier*, p. 26), but instead of the normal procedure, an attack striking any location *except* for a leg is ignored.

See *Underwater Combat*, page 64, for the full details involving underwater combat.

Prone 'Mechs: A 'Mech prone in Depth 1 water is fully submerged.

• MINIMUM RANGE DIAGRAM (PPC EXAMPLE: BASE TARGET NUMBER + MINIMUM RANGE MODIFIER) •

ATTACK MODIFIERS TABLE

All Attacks: Weapons and Physical	Modifier
Attacker	
Movement (modifiers are cumulative)	
Stationary	None
Walked	+1
Ran	+2
Jumped	+3
Prone	+2 (does not apply to four-legged 'Mechs)
Skidding	+1 (does not apply to unintentional charges)
Terrain *(modifiers are cumulative)*	
Light Woods/Jungle	+1 per intervening hex; +1 if target in light woods/jungle
Heavy Woods/Jungle	+2 per intervening hex; +2 if target in heavy woods/jungle
Water*	
Depth 1	+1; see *Partial Cover Modifier*, p. 26
Depth 2	Underwater 'Mechs cannot target units that are not underwater (see *Terrain Modifiers*, p. 26).
Partial Cover	+1; see *Partial Cover Modifier*, p. 26
Target *(modifiers are cumulative)*	
Prone	–2 from adjacent hex; +1 from all others
Immobile	–4
Skidding	+2
Jumped	+1
Hexes Moved *(highest applicable modifier only)*	
Moved 0–2 hexes	0
Moved 3–4 hexes	+1
Moved 5–6 hexes	+2
Moved 7–9 hexes	+3
Moved 10–17 hexes	+4
Moved 18–24 hexes	+5
Moved 25+ hexes	+6
Stealth Armor	+1 at medium range, +2 at long range

Weapon Attacks Only	Modifier
Attacker	
'Mech Damage	
Sensor hit	+2
Shoulder hit	+4 for weapons in arm, disregard other damaged actuators in arm
Upper or lower arm actuator (each)	+1 for weapons in arm
Heat	
0–7	None
8–12	+1
13–16	+2
17–23	+3
24+	+4
Making Indirect Fire attack	+1 (+2 if the spotter also attacked this turn)
Spotting for Indirect Fire	+1

ATTACK MODIFIERS TABLE (CONTINUED)

Weapon Attacks Only (Continued)	Modifier
Range and Terrain	
Range	
Short	None
Medium	+2
Long	+4
Minimum range	[Minimum] – [Target Range] +1 (see *Minimum Range Modifier*, p. 27)
Each Intervening Hex/Level between Attacker and Target (as well as target's hex) in same multi-hex building	+1 per hex/level (maximum +3; see *Combat Within Buildings*, p. 72)
Target	
Secondary target in forward arc	+1
Secondary target in side or rear arc	+2
Weather Conditions	
Heavy Fog	+1 to all attacks of the pulse (P) and direct-fire energy (DE) types
Light	Modifiers reduced if the target has high heat (see p. 62)
Dusk/Dawn	+1
Full Moon Night	+2; ignore if the target is illuminated
Moonless Night	+3; ignore if the target is illuminated
Moderate/Heavy Rain	+1
Torrential Downpour	+2
Moderate/Heavy Snowfall	+1
Wind	
Moderate Gale	+1 to all attacks of the missile (M) type (see p. 97)
Strong Gale	+2 to all attacks of the missile (M) type, and +1 to all attacks of the direct-fire ballistic (DB) type (see pp. 96-97)
Storm	+3 to all attacks of the missile (M) type, and +2 to all attacks of the direct-fire ballistic (DB) type (see pp. 96-97)

Physical Attacks Only	
Attacker	
'Mech Damage	
Shoulder hit	No punching or physical weapon attack with arm; no clubbing attacks; +2 to pushing attack (each)
Upper or lower arm actuator hit (each)	+2 to punching and physical weapon attack with arm; half damage for punching attack with arm; +2 to clubbing attacks
Hand actuator hit	+1 to punching attack with arm; no clubbing attacks; no physical weapon attack with arm
Hip actuator hit	No kicking attacks
Upper or lower leg actuator hit (each)	+2 and half damage to kicking attack with that leg
Foot actuator hit	+1 to kicking attack with that leg
Other Modifiers	
Charging attack	Modify for relative Piloting Skills (see *Comparative Modifier*, p. 34)
Death from above attack	Modify for relative Piloting Skills (see *Comparative Modifier*, p. 34)
Rear kick (quad 'Mechs only)	+1
Unintentional charge	+3
Weather Conditions	
Moonless Night	+1; ignore if the target is illuminated

*See *Terrain Modifiers*, page 26, for exceptions.

PRONE 'MECHS

Prone 'Mechs may still participate in combat.

Water: A 'Mech prone in Depth 1 water is fully submerged.

FIRING WHILE PRONE

A prone 'Mech may fire some of its weapons, as long as neither of its arms has been destroyed.

The attacker chooses one arm: no weapons in that arm can fire that turn (this includes weapons split between that arm and the corresponding torso; which arm is chosen may be changed each turn). No leg-mounted weapons can be fired either. All other weapons may be fired, using the same arc as if it were standing and facing that direction. A prone 'Mech cannot torso twist.

Add a +2 Target Number modifier to weapon attacks made while prone, in addition to any other standard modifiers. For example, if a 'Mech runs (+2 modifier) and then drops prone in the same turn it makes a weapon attack, it receives a +4 Target Number modifier.

Four-Legged 'Mechs: Provided it still has all four legs, and has taken no critical hits to any hip actuator, a prone quad 'Mech fires as though it were standing (the +2 modifier for firing when down does not apply, and it may fire leg-mounted weapons).

One-Armed Prone Fire (Optional): Under this optional rule, a prone 'Mech with one arm destroyed may still prop itself up using its intact arm and fire any non-limb mounted weapons. Apply a +1 Target Number modifier (in addition to the standard prone modifier) to all shots made in this fashion.

ATTACKING PRONE 'MECHS

A prone 'Mech makes an easier target for opponents in adjacent hex, but a more difficult target at longer ranges. Apply a –2 Target Number modifier to any attack made against a prone 'Mech from an adjacent hex. If the attacker is not adjacent, apply a +1 modifier.

For example, if a 'Mech moves seven hexes and then drops prone, attacks against it by non-adjacent 'Mechs receive a +4 Target Number modifier (+3 for entering seven hexes, +1 for firing at non-adjacent prone 'Mech = 4); adjacent 'Mechs would receive a +1 Target Number modifier (+3 for entering seven hexes, –2 for firing at an adjacent prone 'Mech = 1).

SPECIALIZED ATTACKS

Several specialized attacks, such as indirect fire, have their own rules and Target Number modifiers, as outlined below.

AIMED SHOT

Players may announce that they are aiming for a specific hit location when declaring a weapon attack, but only against immobile targets (see *Immobile Target Modifier*, p. 27). The following weapon types cannot make an aimed shot: Area-Effect, Cluster, or Flak (see the Weapon and Equipment tables, beginning on p. 121, to find a weapon's type; see p. 96 for an explanation of each type). Additionally, indirect fire attacks and Rapid-Fire attacks firing more than one shot cannot be aimed shots.

Apply the –4 immobile Target Number modifier to the attack (unless aiming for the head; see below). If the attack hits, the attacker rolls 2D6: on a 6, 7, or 8, the shot hits the designated location. If not, the attacker rolls normally on the appropriate Hit Location Table (and so may hit the designated location after all).

This rule cannot be used with physical attacks.

Head Shots: If aiming at a 'Mech's head, the immobile target modifier is *not* applied, *and* an additional +3 modifier is added.

Partial Cover: If a target has partial cover, you can only aim for locations that are not behind cover. Do not apply the Target Number modifier for partial cover, and if a leg location is rolled when assigning a hit, re-roll until a non-leg hit location is rolled.

Targeting Computer: A targeting computer allows aimed shots against non-immobile targets. In this case, Pulse weapons also cannot be used, the standard –1 Target Number modifier for a targeting computer is ignored, and a +3 modifier is applied instead. Whether the target is immobile or not, aimed shots aided by a targeting computer in any way cannot target the head. See page 114 for the full details.

CLEARING WOODS OR JUNGLE

Weapon attacks can clear wooded and jungle hexes (see *Clearing Woods and Jungle*, p. 61).

INDIRECT FIRE

LRM and Thunderbolt launchers may be fired indirectly. Indirect fire allows a 'Mech without a direct line of sight to a target to attack that target. An attacker with valid LOS to a target cannot make an indirect fire attack against that target.

Another friendly 'Mech must have a valid line of sight to the target: this friendly 'Mech is referred to as the spotter. In order to choose to be a spotter, a 'Mech must not have charged or launched a Death From Above attack that turn; any other action is fine. A spotter can spot for any number of attackers against a single target, but can only spot to one target a turn.

An indirect fire attack applies the following modifiers:

- All standard modifiers for attacker and target movement;
- All standard modifiers for the spotter's movement;
- Terrain modifiers and partial cover based on line of sight from the spotter, not the firing 'Mech;
- +1 for indirect fire;
- Range modifier based on the range between the attacker and target, including minimum range modifiers;
- Finally, if a spotter makes any attacks during the Weapon Attack Phase of a turn that it also spots, apply a +1 Target Number modifier to those attacks, as well as an additional +1 modifier to the indirect fire attack.

Aimed Shots: No indirect fire attack may be an aimed shot.

Artemis FCS: No Artemis FCS bonuses apply to an indirect fire attack.

Narc Beacons: A target with an attached Narc homing pod (see p. 108) does not require a spotter to be the target of indirect fire from an attacker firing Narc-enabled missiles (see p. 107). If a spotter is not used, terrain modifiers do not apply to the attack, but the +2 modifier to the Cluster Hits Table roll that a Narc homing pod normally provides is lost (valid LOS from the attacker to the target still cannot exist). If a spotter is used, apply both the standard terrain modifiers and the +2 Cluster Hits Table modifier.

Partial Cover: Damage from indirect fire always strikes the target and not the partial cover, even if it hits a leg location; see *Partial Cover Modifier*, page 26. The exception is if the partial cover is provided by water: in this case, the indirect fire strikes the water and does no damage, unless it is a weapon of the AE (Area-Effect) type.

Semi-Guided Missiles: If their target has been designated by friendly TAG (see p. 113), these munitions ignore the target movement modifier. If fired indirectly, they also ignore all indirect fire, spotter, and terrain modifiers.

TAG: A 'Mech that hits a target with target acquisition gear (TAG; see p. 113) also spots the target for indirect fire; this is the 'Mech's one allowed spot for the turn (the reverse is not true: just spotting a target in no way benefits TAG-guided munitions). If a 'Mech uses TAG against a target and misses, the 'Mech can still attempt to spot any target for indirect fire in the normal fashion.

Note that spotting a target for indirect fire via TAG allows the spotting unit to ignore the special Target Number modifiers listed under the final bullet point above.

Torpedoes: Long-range torpedoes may be fired indirectly underwater, so long as both the torpedo launcher's location and target are underwater. The entire attack must take place submerged in the same body of water.

ROLLING TO HIT

Once the player has determined all Target Number modifiers for the attack, they make a roll to see if the attack is successful. For each weapon attack, the player rolls 2D6. If the result is equal to or greater than the modified Target Number, the attack succeeds.

Players choose the order they roll to hit for all of their 'Mech's declared attacks. From turn to turn, this order can differ at the player's discretion. However, you must resolve all attacks by a 'Mech against one target before moving on to attacks by the same 'Mech against another target.

A walking 'Mech with a Gunnery Skill of 4 fires all its forward-mounted weapons at one target, and two rear-mounted medium lasers at a second target. The attacker chooses the order to make their attack rolls in, but must resolve all fire against one target before moving to the next.

Its Ultra AC/10 has a modified Target Number of 11: 4 (Gunnery Skill) + 1 (attacker movement) + 2 (target movement) + 1 (partial cover) – 1 (targeting computer) + 4 (long range) = 11.

Its large pulse laser has a modified Target Number of 7: 4 (Gunnery Skill) + 1 (attacker movement) + 2 (target movement) + 1 (partial cover) – 2 (pulse weapon modifier) – 1 (targeting computer) + 2 (medium range) = 7.

Its LB 5-X AC has a modified Target Number of 9: 4 (Gunnery Skill) + 1 (attacker movement) + 2 (target movement) + 1 (partial cover) – 1 (LB-X firing cluster ammo) + 2 (medium range) = 9.

Its LRM-15 has a modified Target Number of 10: 4 (Gunnery Skill) + 1 (attacker movement) + 2 (target movement) + 1 (partial cover) + 2 (medium range) = 10.

Its two medium lasers fired at the secondary target each have modified Target Numbers of 9: 4 (Gunnery Skill) + 1 (attacker movement) + 1 (target movement) + 2 (secondary target in rear arc) – 1 (targeting computer) + 2 (medium range) = 9. Each medium laser is rolled separately.

AMMUNITION EXPENDITURE

Ballistic and missile weapons have limited ammunition. Weapons that require ammunition indicate the number of shots available for that weapon per ton of ammo carried in the Shots Per Ton column of the Weapons and Equipment Tables (see p. 121).

SHOTS

All weapons that require ammunition have their ammo stored in ammo bins. Each ammo bin provides a varying number of shots, depending on the weapon. For example, one ton spent on ammo provides an ammo bin containing twenty shots for an AC/5, or five shots for an AC/20. Missile ammunition works differently, in that a single shot for a missile launcher is always one firing of all the missile tubes of that launcher. For example, a ton of SRM 4 ammo provides an ammo bin with twenty-five shots (four missiles are fired per shot). A ton of LRM 20 ammo provides an ammo bin with six shots (twenty missiles are fired per shot). Put simply, each time a weapon that requires ammunition fires, regardless of its type or size or number of missile tubes, it uses one shot of that weapon's ammunition.

A weapon can draw ammo from any ammo bin that carries the exact ammunition for that weapon. The ammo need not be in the same location as the weapon on the Critical Hit Table of the 'Mech Record Sheet. For example, an LRM 15 in the left arm of a 'Mech can use LRM 15 ammo carried in any location, but cannot use LRM 5, 10, or 20 ammo. In addition, if multiple ammo slots of the correct type exist, a player can draw ammo from any slot desired in a turn, switching slots each turn at their discretion. For example, take a 'Mech with an LRM 15 and three LRM 15 ammo bins: two in the right arm and one in the left torso. The 'Mech fires the LRM 15 in each of three consecutive turns. Each turn, its controller can remove one shot from any one of the three LRM 15 ammo bins.

Similarly, multiple weapons of the exact same type (e.g., all LRM 5s, or all LRM 15s, but not all LRMs in general) can share ammunition bins. For example, a 'Mech with two AC/5s can have just one ton of AC/5 ammo to share between both, and firing both means that two shots are drawn from the single ammo bin.

As damage can cause ammunition to be lost or even to explode, it is important to track which bin each shot is fired from. For 'Mechs with multiple ammo bins for a single weapon type, do not assign all ammunition to a single "pool". For example, if your 'Mech has three tons of AC/5 ammo (20 shots per ton), that's not a single pool of 60 AC/5 shots. Instead, remember that you have three separate bins of 20 shots each, with each bin's ammo count recorded separately on the Critical Hit Table.

Ammunition Use Timing: Ammo expended is marked off when attack declarations are made, except for Streak missile launchers, which only mark off ammo after a successful to-hit roll.

Cluster Attacks: Though a cluster weapon (see p. 96) fires multiple projectiles, each firing still only uses one shot.

Rapid-Fire Attacks: Rapid-fire weapons (see p. 97) can fire multiple shots in a single attack. For example, a rotary AC/5 can fire between one and six shots in a single turn as its attack. One ton of ammo provides a rotary AC/5 with 20 shots. A player could fire it at a rate of one shot per turn for twenty turns, or fire it three times at a rate of six shots in three different turns (consuming 18 shots) or any combination thereof.

The ability to draw from any appropriate ammo bin also applies to rapid-fire weapons. For example, a 'Mech mounting a rotary AC/5 with three different ammo slots of the correct type can rapid-fire six shots in a single turn and then mark off all six shots in any combination across the three different ammo slots (two shots in each, all six shots in one, three shots in two, and so on).

HIT LOCATION

When an attack hits its target, the firing player must determine precisely where the attack struck. Hit location is determined by the attack direction and the target's facing.

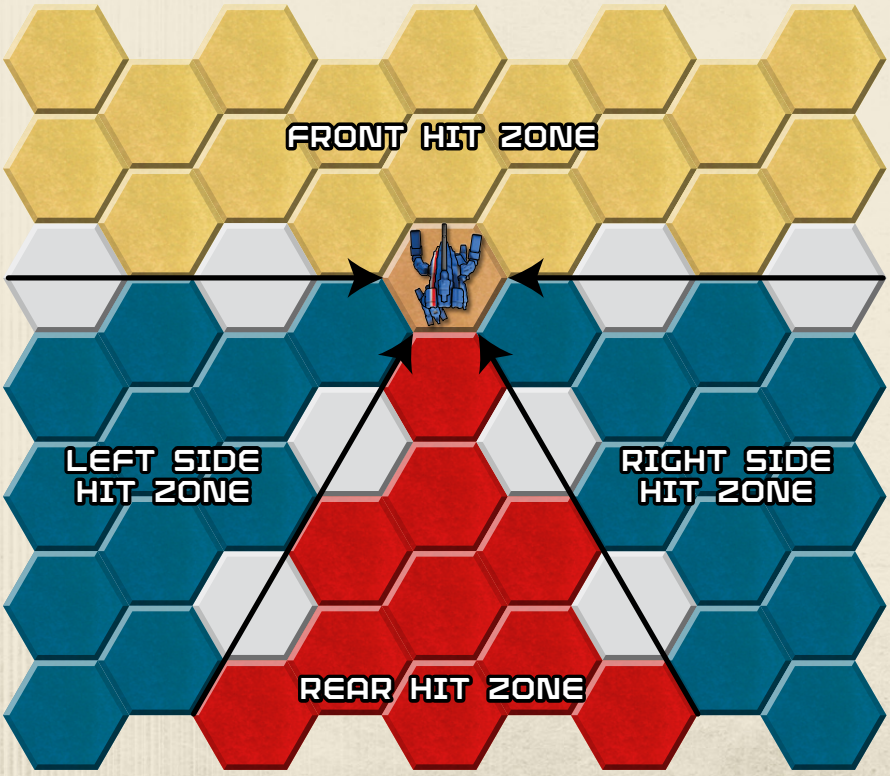
ATTACK DIRECTION

When an attack hits a 'Mech, it hits from the target's front, rear, left, or right side. Use the direction of a standing 'Mech's feet to determine its facing, disregarding any torso twists it has made.

Use the *Line of Sight* rules (see p. 22) to determine which hexside is crossed when determining attack direction. If the LOS crosses exactly between two hexsides, the hex picked when LOS was determined must be used here as well (see *LOS on the Border of Two Hexes*, p. 22).

Buildings: Attacks against buildings need not determine attack direction or hit location. Detailed explanations for assigning damage to buildings appear in *Urban Combat* (see p. 66).

Prone 'Mechs: If the target 'Mech is prone, use the hexside the top center of the 'Mech is pointing at as its facing. Any damage to a prone 'Mech from outside sources such as weapons fire is treated as if the 'Mech was standing with that facing.



• ATTACK DIRECTION DIAGRAM (WHITE HEXES = TARGET'S CHOICE) •

DETERMINING HIT LOCATION

To determine the location of a hit, the player rolls 2D6 and consults the appropriate column of the Hit Location Table (see p. 33); locations for four-legged 'Mechs appear in parentheses on the table.

As with resolving attack rolls, the attacker chooses the order in which they determine hit locations (and resolve damage to the target) for all of their 'Mech's announced attacks. From turn to turn, the attacker can change this order.

It is vital that every hit location roll be made one at a time. If you hit with multiple weapons in a single attack, it is tempting to just roll a bunch of dice and so try to resolve all hit locations at once. This is wrong, as the order in which attacks hit is very important, even if you hit with multiple of the same weapon. This is explained further in the *Damage* chapter (beginning on p. 41).

Cluster Attacks: After each hit by a cluster weapon, a roll is made on the Cluster Hits Table (see p. 33) to determine the number of projectiles in the shot that actually strike the target. Those that hit are broken up into a series of damage groupings, and then a separate hit location roll is made for each grouping (see *C: Cluster*, p. 96).

Critical Hit: A roll of 2 when determining hit location creates the chance for a critical hit in the torso location rolled, even if the armor remains intact in that location (see *Through-Armor Critical Hit*, under *Critical Hits*, p. 45).

Partial Cover: Remember that, as described in *Partial Cover Modifier* (see p. 26), if a 'Mech has partial cover and the hit location of an attack against that 'Mech indicates a leg, the attack strikes the cover instead (unless the attack came from underwater; see below). If the partial cover is a building, the hit damages the building.

Rapid-Fire Attacks: A separate hit allocation is made for each shot that hits (see p. 97 for details on Rapid-Fire weapons).

Water: If an underwater 'Mech attacks a standing 'Mech in Depth 1 water, the target still has partial cover (see *Partial Cover Modifier*, p. 26), but instead of the normal procedure, an attack striking any location *except* for a leg is ignored.

DAMAGE RESOLUTION

Only after all weapon attacks have been made is the damage caused as a result of those attacks applied. As such, damage dealt in a Weapon Attack Phase never affects attacks made that same phase (the attacks are all made before the gameplay effects of damage caused as a result are applied).

All Piloting Skill Rolls required due to weapon attacks are made at the end of the Weapon Attack Phase (i.e. after all attacks that phase), applying any modifiers resulting from damage inflicted that phase. See page 53 for details. Once this is done, players then proceed to the Physical Attack Phase.

See the *Damage* chapter, pages 41-49, for how to resolve damage.

CLUSTER HITS TABLE

Roll (2D6)	Weapon Size 2	3	4	5	6	7	9	10	12	15	20	30	40
2	1	1	1	1	2	2	3	3	4	5	6	10	12
3	1	1	2	2	2	2	3	3	4	5	6	10	12
4	1	1	2	2	3	3	4	4	5	6	9	12	18
5	1	2	2	3	3	4	5	6	8	9	12	18	24
6	1	2	2	3	4	4	5	6	8	9	12	18	24
7	1	2	3	3	4	4	5	6	8	9	12	18	24
8	2	2	3	3	4	4	5	6	8	9	12	18	24
9	2	2	3	4	5	6	7	8	10	12	16	24	32
10	2	3	3	4	5	6	7	8	10	12	16	24	32
11	2	3	4	5	6	7	9	10	12	15	20	30	40
12	2	3	4	5	6	7	9	10	12	15	20	30	40

'MECH HIT LOCATION TABLE

Roll (2D6)	Left Side	Biped (Four-legged) Front/Rear	Right Side
2*	Left Torso [critical]	Center Torso [critical]	Right Torso [critical]
3	Left Leg (Left Rear Leg)	Right Arm (Right Front Leg)	Right Leg (Right Rear Leg)
4	Left Arm (Left Front Leg)	Right Arm (Right Front Leg)	Right Arm (Right Front Leg)
5	Left Arm (Left Front Leg)	Right Leg (Right Rear Leg)	Right Arm (Right Front Leg)
6	Left Leg (Left Rear Leg)	Right Torso	Right Leg (Right Rear Leg)
7	Left Torso	Center Torso	Right Torso
8	Center Torso	Left Torso	Center Torso
9	Right Torso	Left Leg (Left Rear Leg)	Left Torso
10	Right Arm (Right Front Leg)	Left Arm (Left Front Leg)	Left Arm (Left Front Leg)
11	Right Leg (Right Rear Leg)	Left Arm (Left Front Leg)	Left Leg (Left Rear Leg)
12	Head	Head	Head

*A result of 2 may inflict a critical hit. Apply damage to the armor in that section in the normal manner, but the attacking player also rolls once on the Determining Critical Hits Table, page 46.

PHYSICAL ATTACKS

Physical attacks occur in the Physical Attack Phase. This phase occurs after the Weapon Attack Phase is over. As such, all weapon fire, as well as all damage and effects from it, is resolved before any physical attacks are declared and made in a turn.

'Mechs can make seven different types of physical attacks: charging (see p. 35), clubbing (see p. 36), death from above (DFA) (see p. 36), kicking (see p. 38), physical weapon attacks (see p. 38), punching (see p. 38), or pushing (see p. 40).

In order to make a physical attack, a 'Mech must be adjacent to its target (unless specified otherwise) and the target must be in a valid arc, as described in the rules for each attack.

Displacement: If a 'Mech's charge, DFA, or push attack would displace the target of another 'Mech's physical attack, the 'Mech with the lower Initiative (i.e. the first to move) resolves its physical attack first. If both attackers are on the same side, the controlling player(s) may choose which attack is resolved first. If no valid target exists, the attack automatically fails.

A charge, DFA, or push attack can displace a 'Mech off the playing area. In this case, off-board is not considered prohibited terrain, and the displaced 'Mech is considered destroyed.

Multiple Physical Attacks: A 'Mech may only make a single type of physical attack in a single turn. Even if a 'Mech mounts two physical weapons (identical or otherwise), it can only make a single physical weapon attack, unless the weapon's rules specify otherwise (such as with claws). However, when making a single punch attack, a player can punch with one or two arms (see *Punch Attacks*, p. 38).

A 'Mech can only be the target of one charge, DFA, or push attack in a turn. If one of these attacks is declared against a 'Mech, it cannot be targeted by any other attack of these types that turn.

MAKING A PHYSICAL ATTACK

The base Target Number for a physical attack is equal to the attacking 'Mech's Piloting Skill rating (see *Skills*, p. 9).

The modified Target Number equals the base Target Number plus the modifier for the specific physical attack as noted on the Physical Attack Modifiers Table below. All standard modifiers for weapon attacks apply (see pp. 28-29); the sole exceptions are heat and sensor modifiers, which never apply.

Physical attacks are not Piloting Skill Rolls: modifiers that only affect Piloting Skill Rolls do not affect physical attacks, and vice versa.

As with weapon attacks, if the modified Target Number is 2 or less, the physical attack automatically hits. If the modified Target Number is greater than 12, the physical attack automatically misses: in this case, the player can choose not to make the attack, thereby avoiding the possible need to make a Piloting Skill Roll for a failed attack.

PHYSICAL ATTACK MODIFIERS TABLE

Attack Type	Modifier
Charging	+0
Clubbing	–1
Death From Above (DFA)	+0*
Kicking	–2
Physical Weapon	See p. 38
Punching	+0
Pushing	–1

*All the normal attack modifiers apply, including the attacker's jumping movement, but the roll is not modified for terrain.

COMPARATIVE MODIFIER

Whenever a 'Mech charges or conducts a death from above attack on another 'Mech, subtract the target's Piloting Skill from that of the attacker's, and use the result as a modifier to the Target Number. This applies even if the target is unconscious.

CRITICAL DAMAGE

Critical damage to a 'Mech's arms or legs adds modifiers to the Target Numbers of physical attacks. See the Attack Modifiers Table, page 28, for a list of these modifiers.

DIFFERENT LEVELS

The rules for charging, clubbing, kicking, physical weapon, and punching attacks assume that the opposing 'Mechs are at the same level.

A 'Mech may make a physical attack against another 'Mech only if the level of the underlying hexes of both 'Mechs are within one level of each other. The Different Levels Table shows which types of physical attacks can be made in situations where the attacker and target are not on the same level as one another, and what Hit Location Table applies to that situation.

DIFFERENT LEVELS TABLE

Target is:	Allowed Physical Attack
Standing 'Mech 1 level higher	Charge, Club (Kick Table), DFA, Physical Weapon (Kick Table), Punch (Kick Table)
Standing 'Mech 1 level lower	Charge, Club (Punch Table), DFA, Kick (Punch Table), Physical Weapon (Punch Table)
Prone 'Mech 1 level higher	Club, DFA, Physical Weapon, Punch
Prone 'Mech 1 level lower	DFA

Note: A 'Mech can always make a death from above attack if it has the necessary Jumping MP, provided the target is valid.

PHYSICAL ATTACKS AND PRONE 'MECHS

Prone 'Mechs cannot make physical attacks against other 'Mechs.

Death from above and kick attacks may be made against a prone 'Mech. The attacker gains the –2 Target Number modifier for attacking an adjacent prone 'Mech.

A successful attack uses the Hit Location Table (see p. 33). For DFA attacks against prone 'Mechs, always use the Rear column.

Different Levels: A prone 'Mech one level higher than the attacking 'Mech can also be hit by punch, club, and physical weapon attacks. These attacks also use the Hit Location Table (unless noted otherwise for physical weapon attacks). A prone 'Mech one level lower may not be the target of a kick.

PHYSICAL ATTACKS AND WATER

A 'Mech standing in Depth 1 water may make any physical attack. However, a physical attack cannot be made against a fully-submerged 'Mech, unless the attack begins underwater as well.

For example, a 'Mech standing in Depth 1 water adjacent to a prone 'Mech in Depth 1 water can only make a kick attack, since the kick attack occurs completely underwater. The standing 'Mech cannot make a DFA against the prone 'Mech because a portion of the attack would take place outside the water.

Damage resulting from physical attacks is halved when applied to an underwater location (round down).

Partial Cover: For a standing biped 'Mech, Depth 1 water provides partial cover against any physical attack that does not roll on the Punch Location Table (bearing in mind that the attacker and target being at different levels can change what location table an attack uses; see the Different Levels Table on p. 34). For a standing quad 'Mech, Depth 1 water always provides partial cover.

If partial cover applies and the attacker is at the same LOS level (see p. 22) as the target, a hit that does not use the Kick Location Table and resolves to the legs is ignored; charges ignore this rule.

Depth 2+ Water: A 'Mech standing in Depth 2 (or deeper) water can make any physical attack, except for a death from above. See *Underwater Combat*, page 64, for the full details on combat when submerged.

CHARGE ATTACKS

Base Target Number: Piloting Skill (comparative modifier; see p. 34).

In order for a 'Mech to charge, it must not have jumped that turn. A charging 'Mech cannot make any weapon attacks that turn.

Charge attacks are declared in the Movement Phase, not the Physical Attack Phase. It is not necessary to move in a straight line to charge, or to use Running movement. All that is required is that the attacker has enough MP to enter the hex the target occupies, and can legally do so. The attacker stops one hex away from the target and only then declares the charge: when resolving any action against the charging 'Mech between that moment and the time the charge is finally resolved, treat the 'Mech as in that hex and having only moved that far. It does not actually finish its movement and enter the target hex until the Physical Attack Phase.

Like all other physical attacks, charges are resolved during the Physical Attack Phase. This means the attacker can only charge 'Mechs that have finished their movement. It also means a charging 'Mech cannot itself be the target of a charge or death from above attack, because the charging 'Mech's movement will not be finished until the end of the Physical Attack Phase.

If the attacking 'Mech falls during the Weapon Attack Phase or its pilot is unconscious at the start of the Physical Attack Phase, its charge automatically misses; resolve any fall normally at the end of the Weapon Attack Phase. An exception occurs if the 'Mech falls in a paved hex. In this case, at the end of the Weapon Attack Phase, the 'Mech skids a number of hexes equal to half the hexes it entered during the Movement Phase (round fractions up). When it skids into the hex of the 'Mech it was attempting to charge, it makes an unintentional charge (see *Skidding*, p. 68).

If the target falls during the Weapon Attack Phase, no attack is made and so the attacker does not move into the target's hex.

Multiple Attacks: A 'Mech may only be the target of one charge, death from above, or push attack in a given turn.

Shutdown Targets: When determining targets for a charge attack, 'Mechs that began the turn shutdown or with unconscious pilot are treated as though they already took their movement actions, regardless of what order their actions actually occur.

UMUs: A 'Mech can make a charge attack using Underwater Maneuvering Units.

CHARGE DAMAGE

If the attack succeeds, both 'Mechs take damage. Round all fractional damage up.

- **Damage to target:** Divide the charging 'Mech's weight by 10. Multiply this by the number of hexes moved (not MP spent) by the attacker in the Movement Phase. Do not count the hex the target is in. If the attacker moved both backward and forward that phase, only count the hexes moved after it last reversed its movement. If the attacker fell in the Movement Phase, no hexes it moved before the fall apply.
- **Damage to attacker:** 1 point of damage for every 10 tons the target weighs.

Divide all damage into 5-point Damage Value groupings (any leftover damage becomes its own grouping). The attacking player then rolls each grouping's hit location separately. For example, a 17-point charge forms three 5-point groupings and one 2-point grouping, and each grouping is a separate hit.

Buildings: A 'Mech that charges a building automatically fails its Piloting Skill Roll required to avoid damage when entering a building hex and so takes damage equal to the current CF of the building/10 (rounded up). This takes the place of the normal damage to the attacker caused by a charge.

Targets in Buildings: If a 'Mech charges a target in a building, the building absorbs damage as normal (see *Attacking 'Mechs Inside Buildings*, p. 71). The normal PSR to avoid taking damage when entering a building hex is required, with a +3 Target Number modifier in addition to the building modifier (see *Moving Into Buildings*, p. 67).

Unusual Targets: If a charge is made against any target (other than a building) with no tonnage—such as a hill, for example, due to a skid—calculate damage to the attacker using the attacker's tonnage rather than the target's.

A 65-ton JagerMech *moves 5 hexes and declares a charging attack against a 45-ton 'Mech (it has the required 1 MP left over to enter the target's hex). If the charging attack is successful, the target takes 33 points of damage (6.5 for the* JagerMech*'s tonnage multiplied by 5 for the number of hexes it moved, rounded up). The* JagerMech *itself suffers 5 damage (45 / 10, rounded up).*

LOCATION AFTER CHARGE

If the charge misses, the attacking 'Mech chooses to move into the hex to the right or left of the attacker's forward arc. If one of those hexes is prohibited terrain, the other hex must be chosen; if both are prohibited, the attacker does not move.

If the charge succeeds and the attacker was not destroyed, the attacker then advances into the target's hex. If the target 'Mech is not destroyed, it is displaced into the adjacent hex in the direction that the attacker charged it. If this displaces the target 'Mech into a hex two or more levels lower than its current hex, it automatically falls the full height (see *Falling*, p. 56). Regardless, if a 'Mech is already in the hex it is displaced into, see *Displacement*, on page 55.

If the target would be displaced into prohibited terrain, neither the attacker nor target moves (unless the target would be pushed off the playing area, which is allowed and destroys the target). All other effects occur, however, including any Piloting Skill Rolls required to avoid falling.

FALLS

After any successful charging attack, the attacking and target 'Mechs must make Piloting Skill Rolls with a +2 Target Number modifier, or fall in the hexes they currently occupy.

CLUB ATTACKS

Base Target Number: Piloting Skill –1.

To attack another 'Mech with a club, the target must be in the attacker's forward firing arc. The attacking 'Mech:

- cannot have fired any arm-mounted weapons in the same turn
- must have both its shoulder actuators undamaged
- must have two undamaged hand actuators

A club occupies both of the 'Mech's hands.

A successful attack with a club does 1 point of damage for every 5 tons that the attacking 'Mech weighs. Roll normally on the Hit Location Table.

Claws: A 'Mech with any claws applies a +2 Target Number modifier to its club attacks.

Missing Actuators: A 'Mech must have two undamaged hand actuators to use a club. Apply a +2 Target Number modifier to the attack for each damaged upper or lower arm actuator.

FINDING A CLUB

A 'Mech may not fire weapons or make physical attacks during the turn that it picks up a club.

Girder Clubs: Girders from rubbled medium, heavy, or hardened buildings may also be used as clubs. To search the rubble in the hex a 'Mech occupies for a suitable girder, the player rolls 2D6 during the Weapon Attack Phase. A result of 7 or greater is needed to find a girder in a rubbled medium building, 6 or greater for a rubbled heavy building, and 5 or greater for a rubbled hardened building. If a club is found, it may be picked up that same phase.

Limb Clubs: Any 'Mech arm or leg that has been blown off through critical hits or has fallen off as a result of side torso destruction is left in that hex and may be used as a club.

Tree Clubs: If the 'Mech is in a woods or jungle hex (or a hex that has been reduced to a rough hex by a chainsaw or dual saw), it may uproot a tree and use it as a club. Uprooted trees only last for one successful club attack.

DEATH FROM ABOVE ATTACKS

Base Target Number: Piloting Skill (comparative modifier; see p. 34).

A jump-capable 'Mech can leap onto its target, a risky maneuver that brings the full weight of the machine crashing down on the victim's head.

Death from above (DFA) attacks are declared in the Movement Phase, not the Physical Attack Phase. The attacker stops one hex away from the target and only then declares the DFA: when resolving any action against the 'Mech making the DFA attack between that moment and the time the DFA is finally resolved, treat the 'Mech as in that hex and having only moved that far. It does not actually finish its movement and enter the target hex until the Physical Attack Phase.

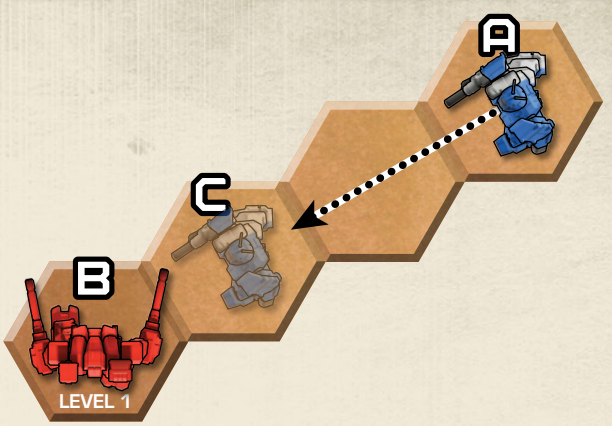
The attacker spends the usual Jumping MP needed to reach the target's hex, but must have enough MP available to clear any height requirement (the level of the hex the target is in, plus two if the target 'Mech is standing, as a standing 'Mech is always two levels high). For example, to make a DFA attack from a Level 0 hex against a 'Mech standing one hex away on a Level 3 hill would only cost 1 MP, but the attacking 'Mech would need at least 5 Jumping MP available to make the attack. See *Jumping Movement*, page 18.

When calculating the modified Target Number for a DFA, the attacker must always apply the +3 modifier for jumping movement: it is not already included. However, do not apply any terrain modifiers.

Like all other physical attacks, DFA attacks are resolved during the Physical Attack Phase. This means the attacker can only DFA a 'Mech that has finished its movement. It also means a 'Mech making a DFA attack cannot itself be the target of a DFA or charge, since its movement will not be finished until the Physical Attack Phase.

A 'Mech making a DFA cannot be the target of physical attacks, but may be the target of weapon attacks.

Multiple Attacks: A 'Mech may only be the target of one charge, death from above, or push attack in a given turn.



• DEATH FROM ABOVE DIAGRAM •

In the Death from Above Diagram at left, the 'Mech in Hex A is making a DFA attack against the 'Mech in Hex B. During the Weapon Attack Phase, the jumping 'Mech is considered to be in Hex C, with a level of 3 (the target hex's level +2). The 'Mech in Hex B may fire against the jumping 'Mech's front side with any weapons it can bring to bear at a range of 1. Other 'Mechs on the map can check for LOS and fire as though the jumping 'Mech were in Hex C with an LOS level of 5 (base level 3, +2 for standing).

Shutdown Targets: When determining targets for a DFA attack, 'Mechs that began the turn shutdown or with unconscious pilot are treated as though they already took their movement actions, regardless of what order their actions actually occur.

Stacking: A 'Mech executing a DFA attack does not count as in any hex for stacking purposes until it lands (see *Stacking*, p. 18).

UMUs: A 'Mech cannot make a DFA attack using UMUs (but can charge using them).

WEAPON ATTACK PHASE

The attacker cannot make weapon attacks in the same turn it executes a DFA.

During the Weapon Attack Phase, the attacker is adjacent to the target hex along the path that the attacker travels during the jump, and facing the target hex. If the path of the jump passes exactly between two hexes adjacent to the target, the attacker must choose which one they occupy. The attacker is in the air above its hex. Its level is thus two, plus the level of either the target's hex or the attacker's hex (whichever is higher).

If the attacking 'Mech fails a Piloting Skill Roll in the Weapon Attack Phase (or would automatically fall), instead of falling that phase the DFA is resolved in the Physical Attack Phase as if it missed (see *Falls After DFA*, at right). The DFA also misses if the attacking pilot is unconscious at the start of the Physical Attack Phase.

If a DFA's target is destroyed before the Physical Attack Phase, the DFA becomes a normal jump and the attacker lands as normal.

DFA DAMAGE TO TARGET

To determine the damage dealt to the target, divide the weight of the attacking 'Mech by 10 and multiply the result by 3 (round a fractional total up). For example, a *Jenner* weighing 35 tons inflicts 11 points of damage.

Damage from a DFA is not assigned to a single location. Instead, divide all damage from the attack into 5-point Damage Value groupings (leftover damage, if any, becomes its own grouping). The attacking player then rolls once on the Punch Location Table for each grouping (see p. 39). For example, a 12-point DFA forms two 5-point groupings and one 2-point grouping, and each grouping is assigned as a separate hit. Determine attack direction as though the attack had come from the attacking 'Mech's starting hex.

Prone 'Mechs: Resolve successful DFA attacks against a prone 'Mech on the rear column of the Hit Location Table (see p. 33).

DFA DAMAGE TO ATTACKER

If a DFA hits, divide the attacker's weight by 5, then divide the result into 5-point Damage Value groupings (as explained above). Consult the Front column of the Kick Location Table to find the hit location for each grouping (see p. 38) and apply this to the attacker.

LOCATION AFTER DFA

A DFA is completed in the Physical Attack Phase. In that phase, the attacker lands in the target's hex. If the DFA hits and the attacker was not destroyed, the target (if not destroyed) is displaced one hex in the direction opposite the attack. If this results in the target being pushed off the playing area, the target is destroyed.

If the DFA misses, the target chooses any legal adjacent hex and moves to it, even if immobile or prone. Unlike normal, this can be an occupied hex. If the target 'Mech is displaced into a hex two or more levels lower than its current hex, it automatically falls the full height (see *Falling*, p. 56). Regardless, if a 'Mech is already in the hex it is displaced into, see *Displacement*, on page 55.

Prohibited Terrain: If the target 'Mech would be displaced into prohibited terrain (besides off the playing area, which is allowed and destroys the target), another hex must be chosen as close as possible to the original hexside through which the target 'Mech would have moved. If two equally distant hexes are open, such as top-left and top-right, the controller can choose either one.

For example, if a target 'Mech would be displaced through the top hexside, but that is prohibited, its controller then looks at the top-left and top-right hexsides to see if those hexes are passable, then at the bottom-left and bottom-right hexsides, then finally at the bottom hexside.

If all the surrounding hexes contain impassable terrain, the target 'Mech cannot be displaced. In this case, if the DFA hits, the target is destroyed, but if the DFA misses, the attacker is destroyed.

FALLS AFTER DFA

After a successful death from above attack, the attacker makes a Piloting Skill Roll (see p. 53), with a +4 Target Number modifier. If the displacement resulting from a successful DFA did not already cause the target 'Mech to fall, then it must make a PSR as well (with a +2 modifier instead). If either 'Mech fails its PSR, it takes damage as from a 0-level fall.

On an unsuccessful attack the attacker automatically falls, treated as if it had fallen 2 levels. Roll on the Facing After Fall Table to determine facing (see p. 57), but the Hit Location for this damage is always the rear.

'MECH KICK LOCATION TABLE

Roll (1D6)	Biped Left Side	Biped Front/Rear	Biped Right Side
1–3	Left Leg	Right Leg	Right Leg
4–6	Left Leg	Left Leg	Right Leg

Roll (1D6)	Four-legged Left Side	Four-legged Front/Rear	Four-legged Right Side
1-3	Left Front Leg	Right Front Leg/Right Rear Leg	Right Front Leg
4-6	Left Rear Leg	Left Front Leg/Left Rear Leg	Right Rear Leg

KICK ATTACKS

Base Target Number: Piloting Skill –2.

A 'Mech cannot kick with a leg if it fired weapons mounted in that leg that turn. To make a kicking attack, both hip actuators must be undamaged, and the target must be in the kicking 'Mech's forward arc (based on its actual facing; ignore torso twists).

Kicks have a Damage Value of 1 point for every 5 tons of the attacking 'Mech's weight, assigned as a single damage grouping. Determine the damage location by rolling 1D6 and consulting the appropriate column of the Kick Location Table, based on whether the 'Mech is a biped or four-legged design.

If a 'Mech is kicked, it must make a Piloting Skill Roll (see p. 53). If a 'Mech attempts a kick attack and misses, it must make a PSR.

Four-Legged 'Mechs: A quad 'Mech can choose to kick into its rear arc instead (provided the hip actuators of both legs used in the kick are undamaged). Resolve the kick exactly like a standard kick, but with an additional +1 Target Number modifier.

Critical Damage: Reduce kick damage by half for each upper and lower leg actuator damaged on the attacking 'Mech (rounding down); this is cumulative. For example, an attacker with two leg actuators damaged reduces its kick damage to 1/4 its base value.

PHYSICAL WEAPON ATTACKS TABLE

Weapon Type	To-Hit Modifier	Damage Value**	Hit Location Table	Affected by TSM?
Claws	+1	1 per 7 tons	Punch	Yes
Flail	0	9†	Standard	No
Hatchet	–1	1 per 5 tons	Option‡	Yes
Lance	+1	1 per 5 tons	Standard	Yes
Mace	+1	1 per 4 tons	Option‡	Yes
Retractable Blade	–2*	1 per 10 tons	Option‡	Yes
Sword	–2	(1 per 10 tons) +1	Option‡	Yes
Talons	0	×1.5	As per attack	Kicks only
Vibroblade				
Activated	–2	See p. 109	Option‡	No
Deactivated	–2	(1 per 10 tons) +1††	Option‡	Yes

*Not affected by an absent or damaged hand actuator.

**Assign all damage as a single damage grouping. "1 per X tons" mean that the weapon does 1 point of damage for every X tons the 'Mech carrying it weighs (round fractional damage up); swords and deactivated vibroblades then add 1 point to this total.

†Not reduced by absent or missing actuators.

††Cannot be higher than the damage an activated vibroblade of the same type deals.

‡By default the weapon rolls on the regular Hit Location Table. However, when the player declares the physical weapon attack, they may also declare that it will use the Punch or Kick Hit Location Table instead. If so, apply a +4 Target Number modifier in addition to the weapon's standard modifier (the +4 modifier does not apply when forced to roll on a Punch/Kick Location Table due to attacks from different levels; see p. 34).

PHYSICAL WEAPON ATTACKS

Base Target Number: Piloting Skill, with modifier depending on weapon (see table at left).

'Mechs can be equipped with melee weapons, such as a hatchet or sword. A 'Mech cannot use a physical weapon in an arm if it fired weapons mounted in that arm that turn. The target of a physical weapon attack must be in the 'Mech's forward arc, or in the side arc corresponding to the arm in which the weapon is mounted.

All physical weapon attacks assign their damage as a single damage grouping. While a 'Mech can have multiple physical weapons, it can only attack with one of them each turn unless the weapon specifies otherwise (e.g. claws).

See *Physical Weapons*, on page 108, for more details.

Punch Attacks: If a 'Mech mounts a physical attack weapon in an arm other than a retractable blade which is retracted, it cannot make a punch attack with that arm as long as that physical attack weapon is intact.

Triple-Strength Myomer: A weapon listed on the table as being affected by triple-strength myomer (TSM) deals twice its normal damage when a 'Mech wielding it has active TSM (see p. 117).

PUNCH ATTACKS

Base Target Number: Piloting Skill.

A 'Mech cannot punch with an arm if it fired weapons mounted in that arm that turn. A 'Mech may punch with one or both arms each turn; a separate attack roll is made for each arm. All punch attacks must be made against targets in the attacking 'Mech's forward or side firing arcs. However, if a target is in the right or left arc, only the right or left arm, respectively, may punch that target.

A punch attack has a Damage Value of 1 for every 10 tons (or fraction of) that the attacker weighs, assigned as a single block of damage from each arm.

'MECH PUNCH LOCATION TABLE

BIPED			
Roll (1D6)	Left Side	Front/Rear	Right Side
1	Left Torso	Left Arm	Right Torso
2	Left Torso	Left Torso	Right Torso
3	Center Torso	Center Torso	Center Torso
4	Left Arm	Right Torso	Right Arm
5	Left Arm	Right Arm	Right Arm
6	Head	Head	Head

FOUR-LEGGED			
Roll (1D6)	Left Side	Front/Rear	Right Side
1	Left Torso	Left Front Leg/ Left Rear Leg	Right Torso
2	Left Torso	Left Torso	Right Torso
3	Center Torso	Center Torso	Center Torso
4	Left Front Leg	Right Torso	Right Front Leg
5	Left Rear Leg	Right Front Leg/ Right Rear Leg	Right Rear Leg
6	Head	Head	Head

A JagerMech *with Piloting Skill rating 5 punches a* Grasshopper, *standing in light woods, once, on its right side. Because the* JagerMech *has no lower arm actuator, the player adds a +2 Target Number modifier and reduces the normal damage by half (rounding down).*

Neither 'Mech moved, and so the modified Target Number is 9: 5 (Piloting Skill rating) + 2 (no lower arm actuator) + 1 (no hand actuator) + 1 (light woods) = 9. The player rolls a 9 and hits the target.

The JagerMech *weighs 65 tons, so its punch has a normal Damage Value of 7 (65 divided by 10, rounded up), but the missing actuator reduces this to 3 (7 divided by 2, rounded down). The attacking player rolls a 3 on the Punch Location Table, which means the attack hits the target's center torso.*

Reduce the damage by half for each upper or lower arm actuator damaged or not present, with these effects being cumulative (so if both arm actuators are missing or damaged, reduce the attack damage to one-quarter). Round fractions down, to a minimum of 1.

Determine the damage location by rolling 1D6 for each arm and consulting the appropriate column of the Punch Location Table twice, based on whether the target is a biped or four-legged 'Mech.

Multiple Targets: A 'Mech can make two punches at two different targets (including targets in different arcs) and ignores any multiple targets modifier.

Missing/Destroyed Actuators: A 'Mech cannot make a punch attack with an arm if the shoulder actuator in that arm has suffered critical damage.

'Mechs lacking a hand on the punching arm (whether by damage or by design) add a +1 Target Number modifier, as for a hand actuator critical hit. 'Mechs lacking a lower arm actuator on the punching arm (whether by damage or by design) must add a further +2 Target Number modifier, as for a lower arm actuator critical hit. In addition, as noted above, the punch inflicts only half the standard damage (round down).

PUSH ATTACKS

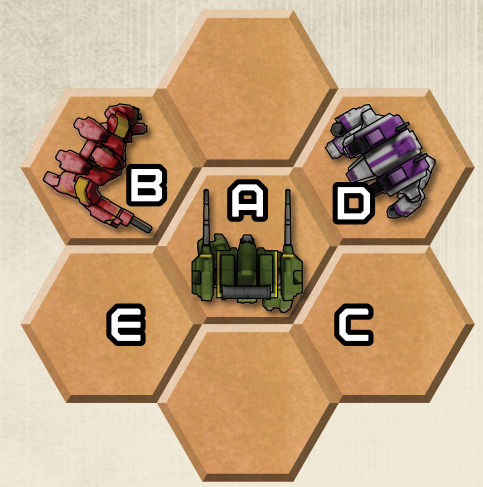
Base Target Number: Piloting Skill –1.

An attacker must have both arms to make a push attack, and cannot have made any arm-mounted weapon attacks that turn.

The target must be a standing 'Mech in the hex directly in front of the attacker (based on the orientation of the attacker's feet; a torso twist has no effect). The target cannot be performing a charge or DFA attack this turn, and must be at the same level as the attacker.

A successful push attack does not damage the target. Instead, it displaces the target into the adjacent hex in the direction that the attacker pushes it. The attacking 'Mech moves into the hex formerly occupied by its target (unlike a charge, this does not require additional MP expenditure). The defender must then make a Piloting Skill Roll (see p. 53) or fall in the hex it was forced into.

If the target 'Mech is displaced off the playing area, it is destroyed. If the target 'Mech is displaced into a hex two or more levels lower than its current hex, it automatically falls the full height (see *Falling*, p. 56). Regardless, if a 'Mech is already in the hex it is displaced into, see *Displacement*, on page 55.



• PUSHING DIAGRAM •

If in the Pushing Diagram above, the 'Mech in Hex A is successfully pushed by the 'Mech in Hex B, it moves into Hex C. If the 'Mech in Hex A is successfully pushed by a 'Mech in Hex D, it is forced into Hex E.

In both cases, the pilot of the target 'Mech must make a Piloting Skill Roll to remain standing, and its attacker advances into Hex A. The 'Mech in Hex A cannot push either of its opponents because neither of them lies directly in front of it.

Multiple Attacks: A 'Mech may only be the target of one charge, death from above, or push attack in a given turn.

If two 'Mechs attempt to push each other, resolve both attempts. If both attacks fail, nothing happens. If both attacks succeed, neither 'Mech moves, and both must make Piloting Skill Rolls or fall. If only one push attack succeeds, resolve it as usual.

Prohibited Terrain: If the target would be displaced into prohibited terrain (besides off the playing area, which is allowed and destroys the target), neither the attacker nor target moves. All other effects occur, including any PSRs required to avoid falling.

Shoulder Actuators: Apply a +2 Target Number modifier to push attacks for each damaged shoulder actuator the attacker has.

DAMAGE RESOLUTION

Only after all physical attacks have been made is the damage caused as a result of those attacks applied. As such, damage dealt in a Physical Attack Phase never affects attacks made that same phase (the attacks are all made before the gameplay effects of damage caused as a result are applied).

All Piloting Skill Rolls required due to physical attacks are made at the end of the Physical Attack Phase (i.e. after all attacks that phase), applying any modifiers resulting from damage inflicted that phase. See page 53 for details. Once this is done, players then proceed to the Heat Phase.

See the *Damage* chapter, pages 41-49, for how to resolve damage.

DAMAGE

This chapter explains how you apply damage to BattleMechs, and deal with the variety of special effects that can occur as a result of it.

Every weapon does a specific amount of damage, defined as Damage Value, which appears on record sheets as well as under the Damage Value column on the appropriate Weapons and Equipment tables, beginning on page 120. Follow the step-by-step procedure outlined in *Damage Resolution* below to determine the effects of damage.

Attacks: Remember that damage comes after attacks: *all* attacks by *all* 'Mechs in a phase must be declared before you start resolving any of them. Don't fall into the habit of bouncing back and forth between declaring and resolving for each weapon or each 'Mech.

Buildings: Hits against buildings are recorded differently from hits on 'Mechs. See *Urban Combat*, page 66, for details.

Damaging a MechWarrior: This is handled on page 58.

DAMAGE RESOLUTION

In short, damage from any source is first applied to the armor of the location damaged. Once all armor in a location is depleted, the internal structure of that location takes damage instead. Each time a location has its internal structure damaged, a critical hit in that location is possible (see *Critical Hits*, p. 45). Once the internal structure of a location reaches zero, that location is destroyed. After a location is destroyed, damage to that location transfers to the neighboring location.

Perhaps the most important thing to understand when applying damage is that every single hit is completely resolved—both the damage dealt and any critical hits inflicted—before moving on to the next hit. This includes single attacks that result in multiple hits, such as cluster weapons. For example, if an attack from a SRM-2 strikes a 'Mech, and both missiles from that attack hit, then the hit location and damage from the first missile is resolved, you check to see if any critical hits occur, any critical hits that do result are resolved (and if multiple critical hits occur, the attacker resolves each critical hit completely before moving on to the next), and only then does play move on to resolve the second missile from that attack.

Also note that the order in which damage is applied is important. If resolving multiple hits on a single 'Mech, be sure that everyone is aware of what order the hits are being resolved.

Resolution Order: The attacker chooses the order in which they resolve each of their 'Mech's successful attacks. From turn to turn, the attacker can change this order.

STEP-BY-STEP DAMAGE

The following steps offer a more thorough walkthrough of damage resolution.

To apply damage, begin with the amount of damage inflicted and its hit location, and start at Step 1. Answer each question yes or no, and follow the instructions.

If a head, leg, or torso location is destroyed, see separate sections for each of these events below for additional information.

Damage Transfer: The steps below assume you understand how to transfer damage. If not, in brief, once a location has been destroyed, all further damage transfers directly to the next location inward, if any (the Damage Transfer Diagram shows you which locations are inward; see p. 43). The damage is applied to the armor of the location transferred to, unless it came from an ammunition explosion (in which case it is applied to the internal structure instead).

Note that there is no damage transfer from the head or center torso locations. If a step below would call for transferring damage from one of these two locations inward, the attack is finished instead, and the remaining damage is lost.

See *Transferring Damage*, page 44, for more detail.

1. DOES THE LOCATION/FACING HAVE ARMOR?

Yes: Check off one armor circle on the Armor Diagram in the location (or facing, if a torso location) for every point of damage the hit deals, until all damage is applied or all armor in the location/facing is destroyed. Go to Step 2.

No: Go to Step 2, answering "Yes".

2. IS THERE DAMAGE REMAINING?

Yes: Check off one internal structure circle on the Internal Structure Diagram in that location for every point of damage remaining, until all damage is applied or all internal structure in the location is destroyed. Go to Step 3.

No: Attack is finished.

3. DOES ANY INTERNAL STRUCTURE REMAIN IN THE LOCATION?

Yes: Go to Step 4.

No: Go to Step 5.

4. AT THE START OF THIS PHASE, DID THE LOCATION HAVE ANY COMPONENTS THAT COULD TAKE A CRITICAL HIT?

Yes: Immediately roll once on the Determining Critical Hits Table, page 46. Apply any resulting critical hits to that location. Critical hits that cannot be applied are discarded. Go to Step 6.

No: Immediately roll once on the Determining Critical Hits Table, page 46. Apply any resulting critical hits to the next location inward (if any; see the Damage Transfer Diagram on p. 43). Go to Step 6.

5. ARE THERE EXPLOSIVE COMPONENTS IN THE LOCATION?

Yes: The location is destroyed. Immediately roll once on the Determining Critical Hits Table, page 46. Apply any resulting critical hits to the destroyed location. Only critical hits that strike explosive components are resolved; all others are discarded. Go to Step 6.

No: The location is destroyed. Go to Step 6.

6. IS THERE DAMAGE REMAINING?

Yes: Go to Step 7.

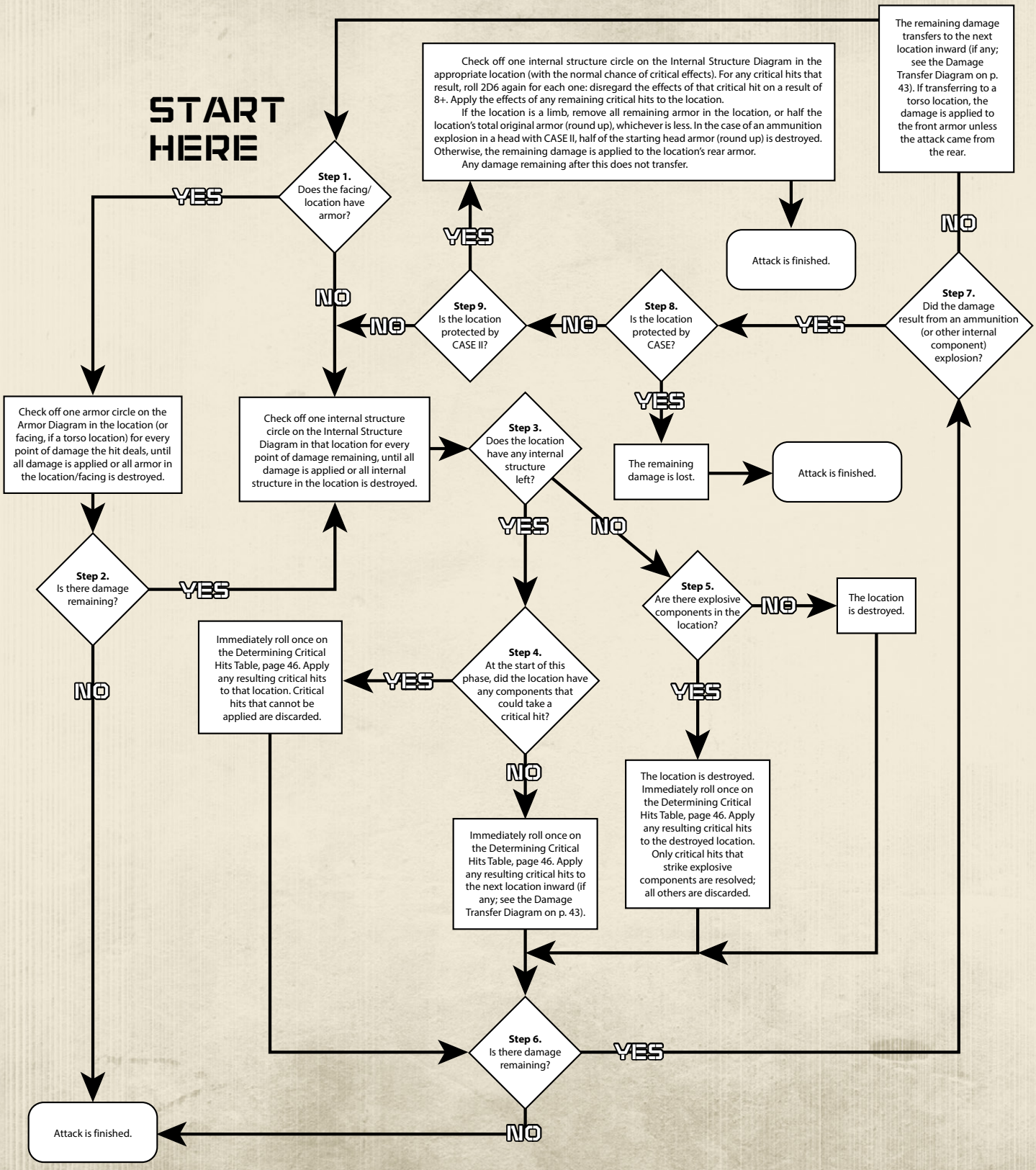
No: Attack is finished.

7. DID THE DAMAGE RESULT FROM AN AMMUNITION (OR OTHER INTERNAL COMPONENT) EXPLOSION?

Yes: Go to Step 8.

No: The remaining damage transfers to the next location inward (if any; see the Damage Transfer Diagram on p. 43). If transferring to a torso location, the damage is applied to the front armor unless the attack came from the rear. Go to Step 1.

• DAMAGE RESOLUTION FLOW CHART •



8. IS THE LOCATION PROTECTED BY CASE?

Yes: The remaining damage is lost. The attack is finished.

No: Go to Step 9.

9. IS THE LOCATION PROTECTED BY CASE II?

Yes: Check off one internal structure circle on the Internal Structure Diagram in the appropriate location (with the normal chance of critical effects). For any critical hits that result, roll 2D6 again for each one: disregard the effects of that critical hit on a result of 8+. Apply the effects of any remaining critical hits to the location.

If the location is a limb, remove all remaining armor in the location, or half the location's total original armor (round up), whichever is less. In the case of an ammunition explosion in a head with CASE II, half of the starting head armor (round up) is destroyed. Otherwise, the remaining damage is applied to the location's rear armor.

Any damage remaining after this does not transfer. The attack is finished.

No: The remaining damage transfers to the next location inward (if any; see the Damage Transfer Diagram at right), directly to the internal structure. Go to Step 2, answering "Yes".

DAMAGE TIMING

There is a very important distinction between resolving damage (the process described above) and applying its gameplay effects.

Damage dealt by attacks made in the Weapon Attack or Physical Attack Phases (including critical hit effects) is resolved as it happens, as explained above. *However, the gameplay effects of such damage are never applied until after all attacks in a phase have been both made and resolved*. This is because all declared attacks must be resolved, and is important because it means that damage dealt by attacks never affects any other attack made in the same phase.

Unlike attacks, the gameplay effects of damage from falls and displacement are applied immediately.

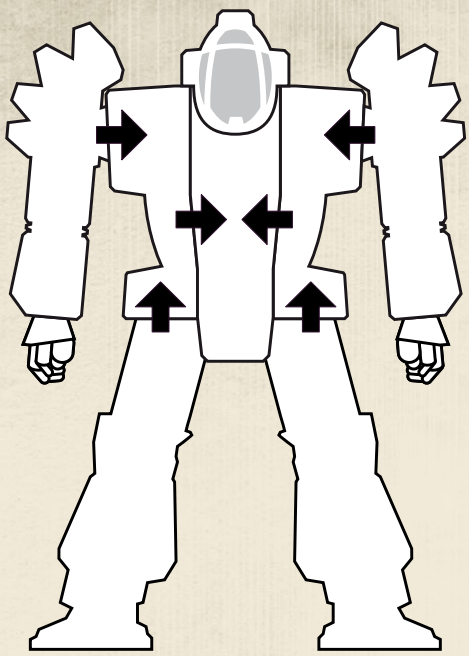
Piloting Skill Rolls: The gameplay effects of damage are always applied before any PSRs resulting from that damage. For example, if a foot actuator is destroyed in the Weapon Attack Phase, this forces a PSR. The standard +1 modifier for losing a foot actuator is applied to that PSR, as well as to any other PSRs called for that phase (and the rest of the game, of course).

HULL INTEGRITY AND BREACHES

Whenever an underwater location takes damage (such as the legs of a 'Mech standing in Depth 1 water), its controller rolls 2D6. On a result of 10 or greater, the 'Mech has lost integrity in that location, which fills with water. See page 65 for the effects of this.

LOCATION DESTRUCTION

If a location is destroyed, all components in that location are also destroyed, including all armor. If the same location takes another hit, or excess damage remains from the attack that destroyed the location, that damage transfers to (affects) the outer armor of the next location inward (see *Transferring Damage*, p. 44).



• DAMAGE TRANSFER DIAGRAM •

In addition, the destruction of the head, leg, or torso locations has further effects, detailed on pages 44-45.

Critical Hits: If a location is destroyed by the elimination of internal structure, and it contains one or more ammo slots (or other explosive items, such as a Gauss rifle), the controlling player must still check for critical hits due to damage to that location (see *Critical Hits*, p. 45).

Limb Clubs: An arm or leg which is blown off through critical hits or falls off as a result of side torso destruction remains in the hex in which it was lost and may be used as a club (see *Club Attacks*, p. 36).

In the midst of resolving damage in the Weapon Attack Phase, a 'Mech's right torso is destroyed.

As far as resolving damage goes, the torso is gone immediately. This means that the right arm also falls off immediately. If, when resolving any remaining hits that phase, any are scored against the 'Mech's right arm, they will transfer to the right torso (since, as far as damage resolution is concerned, the arm fell off the instant the torso was destroyed). And, since the right torso was also destroyed, the damage will transfer further inward, to the center torso (as per the Damage Transfer Diagram).

In terms of gameplay effects, all this destruction has absolutely no effect on any weapons located in the arm or torso that the target 'Mech had declared that it was firing that phase. This is because all declared attacks must be resolved. Only at the end of the phase are the arm and torso truly gone. Even if the 'Mech was blown to pieces, it still makes its declared attacks.

TRANSFERRING DAMAGE

Once a location has been destroyed, all further damage to that location transfers directly to the next location inward. The damage is applied to the armor of the location transferred to, unless it came from an ammunition explosion (see below).

Damage to a missing arm or leg transfers to the torso on the same side (so left leg or arm damage transfers to the left torso, while right arm or leg damage transfers to the right torso). Additional damage to a destroyed side torso location transfers to the center torso. Damage from a destroyed head or center torso does not transfer. The Damage Transfer Diagram on page 43 illustrates this principle, and the diagrams below explore it in more detail.

Damage dealt to the rear hit zone (see p. 32) that transfers inward transfers to the appropriate rear torso facing. For example, damage from the rear that hits a missing left leg is transferred to the left rear torso.

Ammunition Explosion: Damage from ammunition explosions transfers to the internal structure of the next location inward, bypassing the armor entirely.

Torso-Mounted Cockpit: If the head of a 'Mech with a torso-mounted cockpit is destroyed, excess damage to the head still does not transfer to other locations, and further strikes to the head have their location re-rolled.

HEAD DESTRUCTION

If a head has all its internal structure destroyed, the MechWarrior inside is killed and the 'Mech is destroyed. Remove the 'Mech from the map in the end of the phase in which it was destroyed.

Torso-Mounted Cockpit: The above assumes the 'Mech has a head-mounted cockpit. If using a torso-mounted cockpit, destruction of the head does not destroy the 'Mech or kill its pilot, though additional effects may occur (see *Torso-Mounted Cockpit*, p. 115).

LEG DESTRUCTION

When a standing 'Mech loses a leg, the 'Mech automatically falls and ends its movement for that turn. The absent leg applies a +5 modifier to all Piloting Skill Rolls: this replaces all other PSR modifiers generated by damage to that leg.

A 'Mech with one leg is reduced to 1 Walking MP (this overrides all MP increases) and cannot run. Despite all this, it can still use the *Minimum Movement* rule (see p. 16), and so can make one attempt to stand (see p. 19) in each subsequent turn. In an exception to the normal rules, only one Piloting Skill Roll is required for this attempt, even if damage or other factors means that the attempt would normally require two or more PSRs (though the modifiers from all such factors still apply).

If the 'Mech manages to stand it can jump in subsequent turns, but a PSR is required each time it lands.

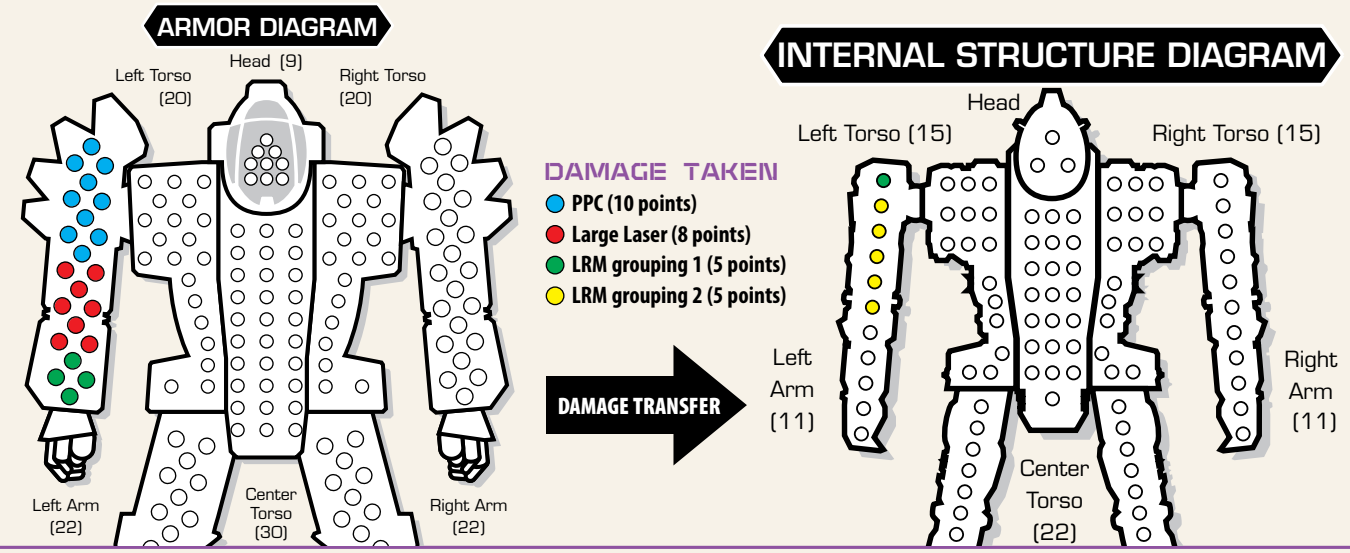
A 'Mech that loses both of its legs automatically falls as well. It is not considered immobile (see p. 18), but it has 0 MP available.

A Grasshopper*'s left arm is hit by a PPC (Damage Value 10), a large laser (Damage Value 8), and two 5-point groupings of long-range missiles (Damage Value 1 per missile hit, so 5 damage per grouping). Before this turn, the 'Mech had its full Armor Value of 22 in that arm.*

The PPC hit reduces the Armor Value by 10, so the Grasshopper*'s player fills in ten armor circles. The laser hit does 8 points of damage, and so the player fills in eight more circles, leaving four. The first missile grouping reduces the Armor Value by another 5 points, but since the* Grasshopper*'s remaining Armor Value is 4, that leaves 1 point of damage that the location's armor cannot absorb.*

The remaining 1 point of damage from the first missile grouping transfers to the arm's internal structure, and so the player fills in one circle on the Internal Structure Diagram, leaving ten circles out of the original eleven, and the attacker immediately makes a roll on the Determining Critical Hits Table (see p. 46), applying effects, if any. The last missile grouping reduces the arm's internal structure by another 5 points. The player fills in five more circles on the Internal Structure Diagram, once again immediately makes a roll on the Determining Critical Hits Table, and applies the effects, if any.

If the Grasshopper*'s left arm takes another 5 or more points of damage, the arm will be destroyed.*



It cannot change its facing or attempt to stand, though it can still attempt to prop itself up to fire if it has both arms.

Four-Legged 'Mechs: The effects of loss of legs are applied to a quad 'Mech as follows:

- **One destroyed leg:** Immediately falls and loses all movement and combat bonuses gained from being four-legged: it can no longer make lateral shifts, it loses its –2 modifier to Piloting Skill Rolls, it must make a successful PSR to stand after falling, and it suffers a +2 modifier to Target Numbers for firing when prone. In addition, a –1 MP penalty replaces any penalties associated with damaged leg actuators in the destroyed leg. If the 'Mech jumps, it still must make the usual PSR required for jumping and missing a leg, with a +5 modifier.
- **Two destroyed legs:** Functions with the same restrictions described above for a biped 'Mech that has lost one leg: it immediately falls, has only 1 MP, and so on.
- **Three destroyed legs:** Functions with the same restrictions described above for a biped 'Mech that has lost both legs: it immediately falls, has only 0 MP, and so on.
- **Four destroyed legs:** Automatically falls (if not already prone) and is immobile (see p. 18).

Minimum Movement: Although a one-legged 'Mech can no longer run, the one exception is that it can still use the *Minimum Movement* rule (which always counts as running; see p. 16).

TORSO DESTRUCTION

If a 'Mech's right or left torso has all of its internal structure destroyed, the corresponding arm is blown off immediately (see *Arm Blown Off*, p. 47, for the effects). Additional damage dealt in the same phase that would have struck that arm or torso automatically transfers to the center torso.

If the center torso is destroyed, the entire 'Mech is destroyed. Remove the 'Mech from the map in the end of the phase in which it was destroyed. If the center torso was destroyed by an ammunition explosion or Area-Effect type weapon, the MechWarrior is killed as well (unless they eject first; see *Ejection*, p. 81).

Four-Legged 'Mechs: If a quad 'Mech loses its right or left torso, the corresponding front leg is blown off immediately (see *Leg Blown Off*, p. 48). The rear leg is unaffected.

CRITICAL HITS

If a location takes one or more points of internal structure damage, but is not outright destroyed, the attacker checks for critical damage by rolling 2D6 and consulting the Determining Critical Hits Table (see p. 46). On an 8 or higher, the target 'Mech takes critical damage: the higher the roll, the greater the damage. Consult the 'Mech's Critical Hit Table to determine the effects (see *Applying Critical Hits*, p. 46).

If a location is destroyed, rather than just damaged, no check for critical hits in that location is made unless it contains one or more explosive slots. In that case, any resulting critical hits that strike explosive slots in that location are resolved as normal (see *Ammunition*, p. 47); hits that do not are always discarded.

The effects of critical hits, such as Piloting Skill Roll modifiers, are cumulative unless specifically stated otherwise, and last for the remainder of the game. Certain critical hits may force a PSR: if the text also applies a modifier to this roll, that modifier is in addition to any other modifiers unless stated otherwise.

Some effects can create a chance for a critical hit even if the location's internal structure was not damaged. These effects are detailed below.

Armor-Piercing Ammo: So long as it deals damage to armor, a hit with AP ammo (see p. 106) has a chance at causing a critical hit, even if the target's internal structure took no damage. If the armor damaged is laser-reflective armor (see p. 114), apply a +2 modifier to this critical hit check.

Armored Components: These prevent critical hits. See page 118.

Hardened Armor: As long as there is armor in a facing/location, apply a –2 modifier when rolling on the Determining Critical Hits Table due to points of damage being inflicted there, and the critical-enhancing effects of Armor-Piercing ammo (see p. 106) are negated.

Multiple Locations: If a single hit damages the internal structure of multiple locations, critical damage is resolved for each of those locations.

Primitive 'Mechs: When rolling on the Determining Critical Hits Table for a 'Mech of the Primitive tech base, apply a +2 modifier to the roll. A result over 12 is treated as 12.

Reinforced Structure: When rolling on the Determining Critical Hits Table for a 'Mech with reinforced structure (see p. 117), apply a –1 modifier to the roll.

Through-Armor Critical Hit: A roll of a 2 on the Hit Location Table may score a critical hit, even if internal structure was not damaged (though the attack must still have dealt at least 1 point of damage). This is known as a through-armor critical (TAC) hit; resolve it as any other critical hit chance.

The chance for a TAC hit is in addition to the normal check for critical hits resulting from damage to the internal structure, and applies even if a 'Mech has no armor left in that location. For example, a hit location result of 2 against a 'Mech with no torso armor left requires two rolls on the Determining Critical Hits Table, one for the TAC hit and one for damaging the internal structure.

If a TAC hit is scored on an already-destroyed left or right torso, the chance for a TAC transfers to the center torso along with the damage.

FLOATING CRITICALS (OPTIONAL)

Under standard rules, when the possibility of a through-armor critical hit arises from a roll of a 2 on the Hit Location Table, any critical hits are applied to the location initially rolled (center, right, or left torso, depending on facing). The following rule allows a possible critical hit to affect any location.

Using this option, when a player rolls a 2 on the Hit Location Table, roll the dice a second time to determine where the shot actually hits. Any critical hits inflicted are inflicted on that location. If another 2 is rolled, that result indicates that the shot struck the appropriate torso location, but does not grant another possible through-armor critical hit.

Partial Cover: If the target of the attack has partial cover and a leg location is rolled for a floating critical hit, re-roll until a non-leg location is rolled.

DETERMINING CRITICAL HITS TABLE

Roll (2D6)	Effect
2–7	No Critical Hit
8–9	Roll 1 Critical Hit Location
10–11	Roll 2 Critical Hit Locations
12	Head or Limb Blown Off / Roll 3 Critical Hit Locations*

*Roll 3 critical hit locations if the hit struck a torso location.

APPLYING CRITICAL HITS

When critical damage occurs, the attacker rolls for each critical hit to determine which slot in the damaged location the hit affects. The target player marks off this damage on their 'Mech's Critical Hit Table. It is possible that some locations can be blown off completely, even if they still have internal structure remaining. The exact procedure depends on the location damaged.

Head or Leg Hits: If the critical hit strikes the 'Mech's head or legs, roll 1D6, find the result on the Critical Hit Table, and mark off the damage. If the slot rolled is an inapplicable slot (see at right), roll the die again. If all slots in the location are inapplicable, then consult *Transferring Criticals* (see at right).

Torso or Arm Hits: The Critical Hit Table for these locations is divided into two blocks of six slots, with the first block labelled "1–3" and the second block "4–6". When assigning a critical hit to these locations, the attacker first rolls 1D6. This identifies which block of slots in that location takes the hit: on a result of 1–3, the hit is assigned to the first block of six slots, while on a result of 4–6, the hit is assigned to the second block. The attacker then rolls another 1D6. The result of this roll identifies the specific slot in that block that takes the hit. If the slot rolled is an inapplicable slot (see at right), roll both dice again (not just the last die).

If one of the blocks of six is filled entirely with inapplicable slots, simply ignore that block and only roll 1D6 when assigning a critical hit to that location, just as if you were assigning a leg or head critical. If all slots in the location are inapplicable, then consult *Transferring Criticals* (see at right).

In the Weapon Attack Phase, the internal structure of a Quickdraw*'s right torso is damaged. The attacker must immediately roll to determine critical hits before resolving any further weapons fire. The attacker rolls a 12, resulting in three critical hits. The right torso of the* Quickdraw *mounts two medium lasers, one of which was destroyed in a previous turn, and so one critical hit destroys the final medium laser. As all possible slots in the location were not hit before this phase, the two excess critical hits do not transfer and are lost.*

In the Physical Attack Phase of the same turn, the internal structure of the Quickdraw*'s right torso is damaged again. A 10 is rolled for determining critical hits, resulting in two critical hits. As all possible slots in the damaged location took critical hits before this phase, both critical hits transfer to the* Quickdraw*'s center torso.*

Multi-slot Components: Some items, such as double heat sinks, take up multiple slots on the Critical Hit Table. A single critical hit disables any item or structural component except the engine, gyro, or sensors, regardless of its number of slots. Critical hits on additional slots occupied by the item have no further effect, but soak up the hit: the critical is assigned to the slot, but no further effects occur.

Multiple Critical Hits: If multiple critical hits result from a roll on the Determining Critical Hits Table, resolve each critical hit completely before moving on to the next one.

Inapplicable Slots: If a critical hit roll would apply damage to a slot for which any of the following applies, it is an inapplicable slot and the player rolls again:

- the slot has already taken a critical hit
- the slot is empty
- the slot cannot take critical damage (such as endo-steel, CASE, TSM, or ferro-fibrous armor slots)

(On pregenerated record sheets, empty slots are labelled "Roll Again", and the words in slots that cannot take critical damage always appear fainter than the words in a slot filled with equipment that can, and are italicized.)

For a torso or arm hit, the player rolls both dice again, not just the last die. If all slots in the location are inapplicable, then consult *Transferring Criticals*, below.

A JagerMech *takes a critical hit to its left arm. The attacking player rolls the first die, and the result is 5. This means the critical hit affects a slot in the second half of the Critical Hit Table for the left arm (the block labelled "4–6"). The attacking player then rolls the second die and gets a 3. As that's an inapplicable slot (Ferro-Fibrous), the attacking player rolls both dice again. This time the result is a 2 (meaning the upper section) and a 6; the defending player notes the critical hit on that slot by crossing it out, knowing it just destroyed the Ultra AC/5.*

CRITICAL HIT TABLE

Left Arm

1-3	1.	Shoulder
	2.	Upper Arm Actuator
	3.	Ultra AC/5
	4.	Ultra AC/5
	5.	Ultra AC/5
	6.	~~Ultra AC/5~~
4-6	1.	Ultra AC/5
	2.	Autocannon/2
	3.	Ferro-Fibrous
	4.	Ferro-Fibrous
	5.	Ferro-Fibrous
	6.	Roll Again

TRANSFERRING CRITICALS

If a critical hit cannot be applied to a location, it might transfer to another location. This depends on the situation at the time of the hit.

If all possible slots in the damaged location had already taken critical hits in previous phases or turns, or are otherwise inapplicable slots (see above), the critical hit transfers to the next location per the Damage Transfer Diagram. Critical hits to the center torso and head never transfer.

If all possible slots in the damaged location were not hit in previous phases, any excess critical hits do not transfer and are lost.

Destroyed Locations: Critical hits that occur when a location is destroyed (rather than just damaged) never transfer.

CRITICAL HIT EFFECTS

Critical hit locations are arranged alphabetically by item; the location of the item on the 'Mech (head, leg, torso, arm) is noted in parentheses in this section.

AMMUNITION

If a critical hit destroys a slot carrying explosive ammunition, the ammo explodes (all ammunition is explosive unless specifically noted otherwise in *Weapons and Equipment* (see pp. 96-119). The MechWarrior takes 2 points of damage as a result of the feedback received through their neurohelmet. In addition, the 'Mech takes damage to its internal structure.

A critical hit to an ammo slot only explodes the ammo in that slot. Exploded missile ammo deals damage equal to the number of *missiles* remaining in the slot times their Damage Value. All other ammunition types deal damage equal to the number of *shots* remaining times their Damage Value. For example, one ton of exploding machine gun ammo deals 400 points of damage (200 shots x 2 damage), while one ton of SRM-2 ammo deals 200 points of damage (2 missiles per shot x 50 shots x 2 damage per missile). Apply the resulting damage to the Internal Structure Diagram (ammunition explosion damage starts the damage resolution process at Step 2, as described on p. 41).

Because an ammunition explosion damages the internal structure of the location where it explodes, a roll to determine critical hits is necessary; i.e. an ammo explosion could cause another ammo explosion, inflicting 4 points of damage on the MechWarrior.

CASE: CASE does not prevent any damage to the location that the ammunition explosion occurs in. However, if the explosion damage destroys that location, excess damage from the explosion has no effect and is deleted, rather than transferring inward.

CASE II: If the location is protected by CASE II, check off one internal structure circle on the Internal Structure Diagram in that location (with the normal chance of critical effects). For any critical hits that result from this damage, roll 2D6 again for each one: disregard the effects of that critical hit on a result of 8+. Apply the effects of any remaining critical hits to the location. Excess critical hits that cannot be applied are discarded.

If the location is a limb, remove all remaining armor in the location, or half the location's total original armor (round up), whichever is less. In the case of an ammunition explosion in the head when a small cockpit is used and CASE II is present, half of the starting head armor (round up) is destroyed. Otherwise, any remaining damage is applied to the location's rear armor.

Any remaining damage from the explosion does not transfer.

Empty Bins: If an empty ammunition slot takes a critical hit, the hit is still applied there, but no further damage or effects occur.

ARM BLOWN OFF (ARM)

This critical hit occurs when the player rolls a 12 on the Determining Critical Hits Table for an arm hit, and is automatic (neither player may choose to roll the three critical hits instead). The hit blows the arm off.

When an arm is blown off in this way, explosive components in that arm do not explode. Though the limb is destroyed, none of the armor or structure on the limb is lost (this is relevant when calculating whether the 'Mech will be required to make a Piloting Skill Roll for having taken 20+ points of damage that phase).

An arm blown off by a critical hit remains in the hex in which it was lost and may be used as a club (see *Club Attacks*, p. 36).

COCKPIT (HEAD)

A critical hit to the cockpit destroys that slot, kills the MechWarrior, and destroys the 'Mech. Remove the 'Mech from the map in the end of the phase in which it was destroyed.

Torso-Mounted Cockpit: If a 'Mech has a torso-mounted cockpit, the cockpit slot is located in the center torso instead of the head. The effects of its destruction remain the same.

ENGINE (TORSO)

'Mech engines have 3 points of shielding. Each critical hit to an engine slot destroys 1 point of shielding. As shielding is destroyed, the amount of heat escaping from the 'Mech's fusion drive increases.

The first hit increases the 'Mech's heat build-up by 5 points per turn. The second adds another 5 points of heat build-up per turn for a total of 10 points per turn. A 'Mech is destroyed if it suffers three engine hits (remember to count engine slots in the 'Mech's side torsos, if any): remove it from the map at the end of the phase in which it was destroyed.

Engine hits do not produce heat if the 'Mech is shut down.

ENGINE EXPLOSIONS (OPTIONAL)

Though the scientific reality of fusion engines prevents them from exploding, the dramatic effect of such an expensive piece of machinery blowing up in a huge ball of fire is extremely appealing.

If using this optional rule, when the fusion engine of a 'Mech takes four or more critical hits in the same phase, roll 2D6. On a result of 10 or higher, the engine explodes.

The explosion destroys the 'Mech and also starts a fire in the hex, regardless of the terrain (see *Effects of Fire*, p. 63). Any targets in adjacent hexes take damage equal to the engine's rating (which is the weight of the 'Mech multiplied by its unmodified Walking MP) divided by 10 (rounded to the nearest whole number; round .5 down). Targets two hexes away take damage equal to the engine's rating divided by 20. Targets three hexes away take damage equal to the engine's rating divided by 40.

Divide any damage into 5-point Damage Value groupings and randomly determine each location; determine the direction of the attack from the direction of the hex where the engine exploded.

Area-Effect: An engine explosion is treated as an Area-Effect (AE) type weapon (see p. 96).

Water: If the target is not in a water hex and an adjacent hex is a water hex, no damage is applied to any completely submerged 'Mech in that adjacent hex.

FOOT ACTUATOR (LEG)

This critical hit destroys the muscle (actuator) in the foot. For each foot actuator damaged, reduce the 'Mech's Walking MP by 1. Recalculate its Running MP accordingly by multiplying the new Walking MP by 1.5, rounding up.

PSR Modifiers: +1 each. The 'Mech's controller must make a PSR at the end of the phase in which the critical hit occurred.

Additionally, a PSR is required whenever the 'Mech jumps; the roll is made at the end of the 'Mech's movement.

Target Number Modifiers: +1 to all kick attacks.

GYRO (TORSO)

The gyroscope keeps the 'Mech upright and able to move. The first hit to a gyro damages it, and the second destroys it.

When a 'Mech's gyro is destroyed, the 'Mech automatically falls. 'Mechs with a destroyed gyro are always prone. They cannot stand up or leave their hex. However, they may change facing by one hexside per turn provided they have at least 1 MP available, and are not considered immobile.

PSR Modifiers: +3 if damaged. The 'Mech's controller must make a PSR when the first critical hit occurred, unless the 'Mech received a second gyro hit in the same phase (in which case it falls automatically). A damaged gyro forces a PSR every time the 'Mech runs or jumps, made at the end of the 'Mech's movement.

When a 'Mech's gyro is destroyed, the Piloting Skill Roll made to avoid damaging the MechWarrior in the fall (see p. 58) applies a +6 Target Number modifier for the destroyed gyro, rather than +3 for a damaged gyro.

Heavy-Duty Gyro: It takes three critical hits to destroy a heavy-duty gyro. The first critical hit does not force a Piloting Skill Roll, but applies a +1 modifier to all future PSRs. On the second critical hit, replace the above with the effects of the first critical hit to a standard gyro. The third critical hit destroys a heavy-duty gyro with all the usual effects for gyro destruction.

HAND ACTUATOR (ARM)

A critical hit to the hand actuator destroys the muscles controlling the 'Mech's wrist and hand. The 'Mech can no longer make physical weapon or clubbing attacks with that arm.

Target Number Modifiers: +1 to all punches made with that arm.

Missing Actuators: Some 'Mechs are designed without one or both hand actuators. The Target Number modifier for punch attacks given above still applies to such 'Mechs (i.e. whether the actuator is destroyed or merely absent is irrelevant).

HEAD BLOWN OFF (HEAD)

A hit blows off a 'Mech's head when the player rolls a 12 on the Determining Critical Hits Table for the head hit location (see *Head Destruction*, p. 44).

HEAT SINKS

A critical hit to a heat sink destroys it and reduces the 'Mech's ability to dissipate heat. For example, if a 'Mech with 16 single heat sinks has three of its heat sinks destroyed, it can now only dissipate 13 points of heat per turn.

HIP (LEG)

A critical hit to a hip freezes the affected leg in a straight position. The 'Mech cannot make kick attacks. Additionally, its Walking MP is cut in half (round up; apply before any leg or foot actuator damage MP reductions). Recalculate its Running MP accordingly by multiplying the new Walking MP by 1.5, rounding up.

After a hip critical hit, ignore any other critical hit modifiers from previous turns due to damage on that leg (even if this actually reduces the PSR penalty provided by the leg). However, other critical hits applied to that leg on this turn or later still apply.

A critical hit to the second hip reduces the 'Mech's MP to 0, but the 'Mech is not considered immobile.

PSR Modifiers: +2 per hip critical hit; this modifier overrides all other critical hit modifiers from that leg. The controlling player must make a PSR at the end of the phase in which the critical hit occurred.

Additionally, a PSR is required every time the 'Mech runs or jumps; the roll is made at the end of the 'Mech's movement.

Four-Legged 'Mechs: A quad 'Mech with at least one hip critical hit loses its ability to ignore the Target Number modifier for firing while prone (see p. 30). Four hip critical hits reduce its MP to 0 (but it is still not considered immobile).

JUMP JET/UMU (LEG/TORSO)

For each critical hit to a jump jet, reduce the 'Mech's Jumping MP by 1. For each critical hit to an underwater maneuvering unit (UMU), reduce the 'Mech's UMU MP by 1.

LEG BLOWN OFF (LEG)

This critical hit occurs when the player rolls a 12 on the Determining Critical Hits Table for a leg location hit, and is automatic—the player may not choose to roll the three critical hits instead. For the effects of losing a leg, see *Leg Destruction*, page 44.

When a leg is blown off in this way, explosive components in that leg do not explode. Though the limb is destroyed, none of the armor or structure on the limb is lost (this is relevant when calculating whether the 'Mech will be required to make a Piloting Skill Roll for having taken 20+ points of damage that phase).

A leg blown off by a critical hit is left in the hex it was lost in and may be used as a club (see *Club Attacks*, p. 36).

LIFE SUPPORT (HEAD)

Any critical hit knocks out this system permanently and leaves the pilot vulnerable to increased heat; the other critical slot can still take damage, but the hit has no additional effect. The MechWarrior takes 1 point of damage at the end of every Heat Phase that the 'Mech's Heat Scale is between 15 and 25; if the Heat Scale is 26 or higher, the MechWarrior takes 2 points of damage instead.

A life support critical hit also eliminates the 'Mech's internal air supply. If the 'Mech is submerged (in Depth 2 or deeper water, or prone in Depth 1 or deeper water) in the End Phase of any turn, the pilot takes 1 point of damage.

Torso-Mounted Cockpit: Warriors using a torso-mounted cockpit suffer more severe heat effects in the event of life support damage. If a 'Mech with a torso-mounted cockpit takes a life support critical hit, the MechWarrior suffers 1 point of damage at the end of every Heat Phase that the 'Mech's Heat Scale is between 1 and 14, and 2 points of damage at the end of every Heat Phase that the Heat Scale is 15 or higher.

LOWER ARM ACTUATOR (ARM)

This critical hit destroys the actuator in the 'Mech's lower arm. Damage from punches with this arm is halved (round down).

Target Number Modifiers:

+1 to all weapons firing from that arm.

+2 to any punches or physical weapon attacks with that arm.

+2 to all clubbing attacks.

Missing Actuators: Some 'Mechs are designed without one or both lower arm actuators. These 'Mechs do not suffer the weapon attack modifier for the missing actuators, though the modifiers for damage reduction and physical attacks still apply (see p. 40).

LOWER LEG ACTUATOR (LEG)

This critical hit destroys the muscle (actuator) in the lower leg. For each lower leg actuator critical hit, reduce the 'Mech's Walking MP by 1. Recalculate its Running MP accordingly by multiplying the new Walking MP by 1.5, rounding up.

Additionally, for each lower leg actuator critical hit, all kick attacks inflict half the standard damage (round down).

PSR Modifiers: +1 each. The controlling player must make a PSR at the end of the phase in which the critical hit occurred.

Additionally, a PSR is required every time the 'Mech jumps; the roll is made at the end of the 'Mech's movement.

Target Number Modifiers: +2 to all kick attacks.

SENSORS (HEAD)

Critical hits to a 'Mech's sensors make weapon attacks increasingly difficult. Critical hits to sensors do not affect other electronic equipment mounted by the 'Mech (such as ECM, C^3, and so on), or physical attacks.

Target Number Modifiers: +2 to all weapon attacks after the first hit. A second sensor hit makes it impossible for the 'Mech to fire its weapons. In both cases, physical attacks are unaffected.

Torso-Mounted Cockpit: A 'Mech using this system may survive two sensor critical hits and still perform weapon attacks.

Instead of the standard modifiers from sensor damage, the first sensor hit incurs a +2 Target Number modifier to all weapon attacks, while the second sensor hit increases this to a +4 modifier. If both sensor hits are in the head, the +4 modifier at the second sensor hit also applies to all physical attacks and Piloting Skill Rolls. Three sensor hits (or more) completely blind a 'Mech with a torso-mounted cockpit, making all weapon attacks impossible and applying a +4 modifier to all physical attacks and Piloting Skill Rolls.

SHOULDER (ARM)

A critical hit to this actuator freezes the shoulder joint. The 'Mech may not punch or make physical weapon attacks with that arm, nor may it make clubbing attacks.

Target Number Modifiers:

+4 to all weapons firing from that arm (overrides all other weapons fire modifiers from critical hits to that arm).

+2 to all pushing attacks.

UPPER ARM ACTUATOR (ARM)

This critical hit destroys the actuator in the 'Mech's upper arm. Damage from punches with the arm is halved (round down).

Target Number Modifiers:

+1 to all weapons firing from that arm.

+2 to all punches and physical weapon attacks with that arm.

+2 to all clubbing attacks.

UPPER LEG ACTUATOR (LEG)

This critical hit destroys the muscle (actuator) in the upper leg. For each upper leg actuator damaged, reduce the 'Mech's Walking MP by 1. Recalculate its Running MP accordingly by multiplying the new Walking MP by 1.5, rounding up.

Additionally, all kick attacks inflict half the standard damage (rounding down).

PSR Modifiers: +1 each. The controlling player must make a PSR at the end of the phase in which the critical hit occurred.

Additionally, a PSR is required every time the 'Mech jumps; the roll is made at the end of the 'Mech's movement.

Target Number Modifiers: +2 to all kick attacks.

WEAPONS AND EQUIPMENT

When a weapon or piece of equipment receives a critical hit, it is rendered useless for the remainder of the game. If the item takes up more than one critical slot, its remaining slots are not also marked off: each can still be hit by subsequent critical hits. This has no effect other than to absorb the critical hit.

For example, a PPC fills three critical slots. The PPC is disabled as soon as one of its three critical slots takes a hit, leaving its remaining two slots able to take critical hits but otherwise resulting in a useless weapon.

Explosive Components: Certain items or weapons, such as Gauss rifles, will create an ammunition explosion (see p. 47) when they receive their first critical hit. Whether an item created an explosion (and how much damage this causes) is indicated in each item's description in the Weapons and Equipment chapter. In all other respects, the critical hit is treated in the same manner as above.

DESTROYING A 'MECH

A 'Mech is considered destroyed and out of the game if its MechWarrior dies or the 'Mech suffers three engine hits (remember to count engine slots in the side torso if that torso is destroyed). Destruction of the cockpit or center torso also renders a 'Mech destroyed. Lastly, unless a 'Mech has a torso-mounted cockpit (see p. 115), destruction of the head destroys the 'Mech as well. Destroyed 'Mechs are removed from the map in the end of the phase in which they were destroyed, and have no further effect on game play.

Abandoned 'Mechs: If using the optional *Ejection* rules (see p. 81), a 'Mech whose pilot has ejected is abandoned, not destroyed. The hex the 'Mech is in continues to count as being occupied by a 'Mech. Stealth armor and any equipment listed in the *Electronics* section (see p. 109) continues to function. The 'Mech must be destroyed as normal to be removed from play.

Forced Withdrawal: If using the *Forced Withdrawal* rules (see p. 81), a 'Mech which can no longer move and has all its weapons destroyed is also considered destroyed.

MechWarrior Survival: Unless their 'Mech has a torso-mounted cockpit (see p. 115), a MechWarrior dies if the head or the head's cockpit slot is destroyed. A MechWarrior always dies if their 'Mech's center torso is destroyed by damage of the Area-Effect (AE) type or by an ammunition explosion (CASE of any kind does not prevent this). If the center torso is destroyed in any other fashion, the MechWarrior does not automatically die.

Even ejecting from a 'Mech can be dangerous, as the MechWarrior may suffer damage upon landing (see *Ejection*, p. 81).

Wreckage (Optional): If using this optional rule, when a 'Mech is destroyed it is removed from the field as normal, but the hex that it was in becomes a rough hex (see p. 60), in addition to any other terrain type or modification it already has (this includes the bottom of water hexes).

HEAT

Heat build-up is a BattleMech's greatest limiting factor. Though a 'Mech dissipates heat through its heat sinks, heat builds up when it moves or fires its weapons, and a high rate of activity can produce more heat than a 'Mech can dissipate. A warrior who pushes their 'Mech past its limits eventually must pay the price.

HEAT POINTS

The more heat points a 'Mech builds up, the greater its heat. A player tracks their 'Mech's heat points using the column of boxes on the 'Mech Record Sheet labeled "Heat Scale". The Heat Scale records heat levels from 0 to 30 heat points. The "Overflow" box is used to track heat levels above 30.

The Heat Point Table indicates the number of heat points generated by various actions and damage. It also shows the number of heat points a 'Mech can dissipate through its heat sinks and by standing in a water hex. As the 'Mech's heat reaches various levels on the Heat Scale, the 'Mech suffers the adverse effects listed for those levels.

BUILDING UP HEAT

Various activities build up heat at different rates. They can be broadly broken down into three categories: Movement, Weapons and Equipment, and Other.

MOVEMENT HEAT

Walking: Walking generates 1 heat point, no matter how many MP the 'Mech expends or how many hexes it moves.

Running: Running generates 2 heat points, no matter how many MP the 'Mech expends or how many hexes it moves.

Jumping: Standard jump jets generate 1 heat point per Jumping MP expended, with a minimum of 3 heat points. For example, when a 'Mech jumps 1, 2 or 3 hexes, it generates 3 heat points. If it jumps 4 hexes, it generates 4 heat points; if it jumps 6 hexes, it generates 6 heat points; and so on.

Improved jump jets instead generate 1 heat point per 2 hexes or portion thereof jumped, to a minimum of 3 heat points.

Underwater maneuvering units (UMUs) generate 1 heat point if used in a turn, no matter how many hexes the 'Mech moves. Mechanical jump boosters never generate heat.

Standing Up: Each attempt to stand creates 1 heat point, in addition to any heat generated by the movement mode the 'Mech is using that turn.

XXL Engines: A 'Mech with an XXL engine runs hotter than normal: standing still generates 2 heat points, Walking 4 heat points, and Running 6 heat points. The heat generated per hex jumped is doubled, with a minimum of 6 points per jump (the heat modifiers for improved jump jet use and jumping with an XXL engine cancel each other out).

WEAPONS AND EQUIPMENT HEAT

Most weapons, some equipment, and even some types of armor generate heat points when used. These sources are too many to summarize here: the Weapons and Equipment tables beginning on page 120 list the heat an item generates. Heat from such sources is never applied if the 'Mech is shut down.

Shutting Off Heat Sinks: A MechWarrior may wish their 'Mech to build up heat, particularly if using triple-strength myomer (see p. 117). To this end, a player may shut off as many of their heat sinks as desired in the End Phase. Shut-off heat sinks dissipate no heat and may only be switched back on during a subsequent End Phase.

Stealth Armor: When active, stealth armor creates 10 heat points a turn.

OTHER HEAT

Engine Critical Hits: The first engine hit to a 'Mech generates 5 points of heat per turn. The second hit raises this to 10 (total) points of heat per turn. This is not applied if the 'Mech is shut down.

Fire: During the Heat Phase, a 'Mech in a burning hex absorbs an additional 5 Heat Points. A 'Mech also absorbs 2 Heat Points for each burning hex it moved through during the Movement Phase. A 'Mech in a hex ignited during the Weapon Attack Phase of the turn is not affected by the fire until the next turn's Heat Phase (see *Fire and Smoke (Simplified)*, p. 63).

Heat: If the scenario is using Extreme Temperatures (see p. 62), for every 10 degrees C (or fraction thereof) higher than 50 degrees, add 1 heat point per turn.

Outside Heat Sources: Heat that is applied by external factors, such as heat-causing weapons, can also build up a 'Mech's heat. This includes fire and extreme heat (as covered above). These heat sources are cumulative, to a maximum of 15 heat points in a single turn.

HEAT POINT TABLE

Activity	Heat Points
Walking	+1 per turn
Running	+2 per turn
Jumping	+1 per hex (min. of 3 per turn)
Attempting to stand	+1 per attempt
Weapons fire	Per Weapons Tables, pp. 120-125
Equipment	Per entry in Weapons and Equipment, pp. 120-127
Heat sink	–1 per operational heat sink –2 per operational double heat sink –1 additional per operational single heat sink under water (maximum 6 points) –2 additional per operational double heat sink underwater (maximum 6 points)
First engine hit	+5 per turn
Second engine hit	+10 (total) per turn

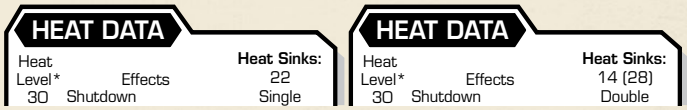
A Juliano *is struck by a plasma rifle and 5 inferno missiles, both of which cause heat. After rolling 1D6 to find the heat the plasma rifle hit causes (6 heat points), the* Juliano*'s controller tallies the heat generated: 6 for the plasma rifle + 10 for the infernos (2 heat points per missile x 5) = 16. However, outside heat sources can only inflict a total of 15 heat points in a turn, and so 1 point is wasted. These 15 heat points are added to the heat the* Juliano *generated by its own movement and attacks that turn.*

DISSIPATING HEAT

Heat sinks are the primary way in which a 'Mech rids itself of heat. However, there are a few others.

Heat Sink Types: Heat sinks may be standard or double. Standard heat sinks dissipate 1 heat point per turn, and double heat sinks dissipate 2 heat points per turn.

On a record sheet, if a 'Mech has double heat sinks, in the section listing a 'Mech's heat sinks, the number in parentheses is the amount of heat the 'Mech can dissipate in total. For example, a listing of 14 (28) means the 'Mech has 14 heat sinks, and dissipates up to 28 heat points each turn.



• HEAT DATA, SINGLE AND DOUBLE HEAT SINKS •

Coolant Pods: A coolant pod increases the cooling capacity of the Mech's active heat sinks by 1 point each, regardless of heat sink type (see p. 119).

Cold: If the scenario is using Extreme Temperatures (see p. 62), for every 10 degrees C (or fraction thereof) lower than –30 degrees, the 'Mech dissipates 1 heat point per turn.

Partial Wing: A partial wing (see p. 116) dissipates 3 heat points per turn, whether the 'Mech jumped or not.

Snow: Deep snow (see p. 61) dissipates 1 heat point per turn if at least one heat sink is mounted in the legs (if the 'Mech is prone, 3 heat points per turn are dissipated).

Water: Submerged heat sinks dissipate twice as much heat. A 'Mech standing in Depth 1 water has its leg-mounted heat sinks submerged. A 'Mech standing in Depth 2 or deeper water or prone in Depth 1 or deeper water has all of its heat sinks submerged (even those in the engine). No more than 6 additional heat points per turn can be dissipated due to water.

RECORDING HEAT BUILD-UP

During each turn's Heat Phase, each player adds up the heat points built up by their 'Mech, as well as any heat from outside sources (see *Outside Heat Sources*, p. 50). They then subtract the heat dissipated by heat sinks. The result may be positive or negative. Add this number to the current level of heat shown on the Heat Scale. If the result is negative, adjust the Heat Scale downward; if the result is positive, adjust it upward. The Heat Scale cannot drop below 0.

Overflow Heat: Mark any heat generated beyond 30 in the "Overflow" box on the record sheet.

EFFECTS OF HEAT

Excessive heat can injure a MechWarrior. It also causes a 'Mech to function less efficiently: the 'Mech moves more slowly, fires less accurately, and may shut down or even explode. Some of these effects are permanent; others disappear as the 'Mech cools.

Heat Timing: A 'Mech suffers heat effects after players have adjusted the heat level for the turn as described above. All heat effects resolve simultaneously, without affecting each other.

• HEAT DATA TABLE, WITH COLOR-CODED EFFECTS OVERLAY •

HEAT DATA

Heat Level*	Effects
30	Shutdown
28	Ammo Exp. avoid on 8+
26	Shutdown, avoid on 10+
25	–5 Movement Points
24	+4 Modifier to Fire
23	Ammo Exp. avoid on 6+
22	Shutdown, avoid on 8+
20	–4 Movement Points
19	Ammo Exp. avoid on 4+
18	Shutdown, avoid on 6+
17	+3 Modifier to Fire
15	–3 Movement Points
14	Shutdown, avoid on 4+
13	+2 Modifier to Fire
10	–2 Movement Points
8	+1 Modifier to Fire
5	–1 Movement Points

Movement Effect | Weapon Attack Effect
Shutdown Effect | Ammunition Effect

APPLYING HEAT EFFECTS

As usual, heat-based modifiers are cumulative with all other modifiers. However, heat-based modifiers of the same effect type are not cumulative with each other. For example, a 'Mech with a heat level of 28 has the following penalties: ammo explosion (avoid on 8+), shutdown (avoid on 10+), –5 MP, and a +4 Target Number modifier to weapon attacks. They do not also have, for example, –4 MP and –3 MP and –2 MP, etc. For a given effect type, you only ever use the highest applicable Heat Scale effect.

MOVEMENT

At 5, 10, 15, 20, and 25 heat points, subtract the number indicated from the 'Mech's Walking MP. For example, at anywhere from 5 to 9 heat points, subtract 1 from the 'Mech's Walking MP. Remember that Running MP is 1.5 times the current Walking MP; if the Walking MP is reduced, the player must also recalculate their 'Mech's Running MP, rounding fractions up.

Jumping: Reductions in Walking MP from heat build-up do not affect a 'Mech's Jumping (or UMU) MP.

A Hatamoto-Chi *begins a turn with a Heat Scale level of 4. During the turn, the 'Mech fires both its PPCs and walks (generating 21 heat points). It only has sixteen standard heat sinks working, which dissipate 16 heat points, leaving 5 to build up. During the Heat Phase, these 5 points are added to the 4 already on its Heat Scale, bringing the total to 9. In the next turn, the 'Mech must reduce its Walking MP by 1 (giving it a Walking MP of 3 and a Running MP of 5) and add +1 to its Target Number for weapon attacks (but not physical attacks).*

If the 'Mech repeats these actions in the next turn, the player must add 5 more heat points to the Heat Scale, bringing the total to 14. Now the player must make a 2D6 roll of 4 or higher to avoid shutdown. Even if they manage this, the Hatamoto-Chi*'s Walking MP is still reduced by 1 more, to 2, until its heat falls below 10 on the Heat Scale. At the same time, the 'Mech applies a +2 Target Number modifier to its weapon attacks.*

WEAPON ATTACKS

At 8, 13, 17, and 24 heat points, add the number indicated to the 'Mech's Target Number for weapon attacks. For example, at 8 heat points, add 1 to all weapon attack Target Numbers for as long as the heat remains at or above 8. As with movement, these effects are not cumulative, and disappear when heat build-up is reduced.

These penalties apply only to weapon attacks, not physical attacks.

SHUTDOWN

At 14, 18, 22, 26 and 30 heat points, a 'Mech attempts to shut down its power plant automatically as a safety procedure.

A conscious MechWarrior can override the power plant's safety shutdown procedure, as indicated by the "Avoid" Target Number listed with the effect, though shutdown cannot be avoided at 30 or more heat points. If their 'Mech's Heat Scale is at or above 14, the player rolls 2D6 once during the Heat Phase. If the roll result is equal to or greater than the highest Avoid Target Number corresponding to the 'Mech's heat level, the pilot avoids shutdown for that turn. If heat accumulation reaches two or more trigger levels in one turn, roll 2D6 only once, against the highest Avoid Target Number.

If a 'Mech shuts down involuntarily, a Piloting Skill Roll (see p. 53) is immediately required, with a +3 Target Number modifier. Regardless of how it shut down, any other PSR required while shut down fails automatically.

A shutdown 'Mech becomes immobile (see p. 18). Its equipment ceases to function except for heat sinks and life support, and it cannot attack or take any other action. Engine critical hits do not generate heat while a 'Mech is shut down. Outside heat sources (see p. 50) still create heat buildup, however.

A conscious MechWarrior can attempt to restart their power plant during in any subsequent Heat Phase. To do this, the player first records Heat Scale changes for the turn, and then rolls 2D6. If the result is equal to or greater than the highest current Avoid Target Number, the power plant restarts. A 'Mech may move and fire in the turn following the turn in which the power plant is restarted. When the heat drops below 14 on the Heat Scale, the power plant restarts automatically, even if the pilot is out of action.

Aimed Shots: Enemies can target a shutdown 'Mech with aimed shots (see p. 30).

Overflow Heat: A 'Mech's Heat Scale must be below 30 before a restart can occur.

Voluntary Shutdown: Players may elect to voluntarily shut down their 'Mech during any End Phase. They may then restart the power plant in any subsequent End Phase, provided they first pass any Shutdown Avoid checks due to heat (if needed).

AMMUNITION

If a 'Mech with unemptied bins of explosive ammunition reaches 19 points or higher on the Heat Scale, one of these bins may explode. To avoid an explosion, the 'Mech's controller must roll 2D6 once and meet or exceed the Target Number of the highest Ammo Explosion threshold reached; a 'Mech with 19-22 heat would need a 4+, 23-27 heat a 6+, and 28-30 heat an 8+ to avoid.

When an overheated 'Mech's ammo explodes, the ammunition critical slot with the most destructive ammo per shot explodes first. Ammo per shot is defined as the Damage Value done by one turn's worth of shots. For example, the ammo per shot of a machine gun slot has a Damage Value of 2, a LRM-15 a Damage Value of 15, and a SRM-6 a Damage Value of 12. When the 'Mech carries two ammo slots with equivalent per-shot Damage Values, the 'Mech's controlling player chooses which ammo explodes.

All of the appropriate ammo type in a single critical slot explodes. If more than one critical hit slot holds the appropriate ammo type, the one with the most shots remaining explodes. If two or more locations have an equal number of shots remaining, randomly determine which one explodes.

Resolve the explosion following the rules in *Critical Hit Effects*, page 45. An ammunition explosion has no effect on any 'Mech other than the one that suffered it.

Exploding/One-Shot Weapons: Exploding weapons and their ammunition (such as Gauss rifles), as well as all One-Shot (OS) type weapons, do not explode from overheating.

Inferno Missiles: A 'Mech carrying Inferno ammunition must roll an additional set of Heat Scale Avoid Rolls to determine whether or not its inferno ammo explodes (see p. 107).

MechWarrior Damage: Ammunition explosions automatically injure the MechWarrior, inflicting 2 points of damage and thus forcing two Consciousness Rolls (see p. 58).

DAMAGE TO MECHWARRIORS

If a 'Mech's life support system takes one or more critical hits, its MechWarrior takes 1 point of damage at the end of every Heat Phase that the 'Mech's Heat Scale is between 15 and 25; if the Heat Scale is 26 or higher, the MechWarrior takes 2 points of damage instead.

Torso-Mounted Cockpit: If a 'Mech with a torso-mounted cockpit takes a life support critical hit, the MechWarrior instead suffers 1 point of damage at the end of every Heat Phase that the 'Mech's Heat Scale is between 1 and 14, and 2 points of damage at the end of every Heat Phase that the Heat Scale is 15 or higher.

OTHER ACTIONS

While events such as movement or combat generally occur at fixed intervals, there are several important events that can occur at a variety of points in a *BattleTech* turn. Rules that can apply in multiple phases are covered in this chapter.

PILOTING SKILL ROLLS (PSR)

MechWarriors must make Piloting Skill Rolls (PSRs) under a variety of circumstances. A PSR is usually made to avoid falling; unless specified otherwise, failing a PSR means the 'Mech falls. All events that require a PSR are listed in the Piloting Skill Roll Table on page 54.

Four-Legged 'Mechs: A quad 'Mech adds a –2 modifier to all Piloting Skill Rolls made to avoid falls as long as none of its legs are destroyed.

Immobile 'Mechs and Unconscious Warriors: Immobile 'Mechs or 'Mechs with an unconscious warrior forced to make a Piloting Skill Roll—such as when shut down due to overheating—automatically fail the roll.

'Mech Damage: A 'Mech only makes one Piloting Skill Roll for taking 20 or more damage points in a single phase, regardless of how many points of damage is taken that phase.

Multiple Rolls: If a 'Mech is required to make multiple Piloting Skill Rolls at the same time (e.g. a 'Mech is kicked and takes 20 damage from that kick, both of which force a PSR), each PSR is rolled one at a time (though all modifiers are cumulative and apply to all rolls). If one of these PSRs fails, any remaining PSRs in that sequence are ignored.

Prone 'Mechs: A prone 'Mech ignores all Piloting Skill Rolls required to avoid falls, except when attempting to stand.

MAKING PILOTING SKILL ROLLS

The Piloting Skill Roll Table (see p. 54) lists the events that require a player to make a PSR for their MechWarrior. A PSR is required every time one of these events occurs (the exact moment in the phase when the PSR is made is explained under *Piloting Skill Roll Timing*, see p. 55). Piloting Skill Rolls use the MechWarrior's Piloting Skill as the base Target Number, modified by the following:

- all indicated modifiers for the event
- all modifiers applied by other events which have already occurred in the same phase
- all modifiers applied due to Preexisting Damage (as seen on the Piloting Skill Roll Table)

The result is the modified Target Number.

To make a Piloting Skill Roll, the player rolls 2D6. If the result is equal to or greater than the modified Target Number, the roll succeeds. If the result is less than the modified Target Number, the roll fails.

A player cannot choose to intentionally fail a Piloting Skill Roll.

EXPANDED DAMAGE MODIFIERS (OPTIONAL)

As an option, the Piloting Skill Roll modifier for taking 20+ points of damage in a single phase may be modified by the number of damage points taken. For every 20 points of damage a 'Mech takes, adds a +1 modifier. For example, a 'Mech that takes 40 to 59 points of damage receives a +2 modifier, a 'Mech that takes 60 to 79 points of damage receives a +3 modifier, and so on.

PILOTING SKILL ROLL TABLE

Cause of PSR	Modifier
Damage to 'Mech	
'Mech takes 20+ damage points in one phase	+1[1]
Involuntary fusion reactor shutdown	+3[2]
Leg/foot actuator destroyed	+1
Hip actuator destroyed	+2[3]
Gyro hit	+3
Gyro destroyed	Automatic fall[4]
Leg destroyed	Automatic fall[5]
Physical Attacks Against 'Mech	
'Mech was kicked	0
'Mech was pushed	0
'Mech was successfully charged/hit by death from above	+2
'Mech's Actions	
Missed kick	0
Made a successful charging attack	+2
Made a successful death from above attack	+4[6]
Entered Depth 1 water hex	–1
Entered Depth 2 water hex	0
Entered Depth 3+ water hex	+1
Attempted to stand	0
Entered sand hex	+1
Entered light jungle hex	+1
Entered heavy jungle hex	+2
Entered rubble hex	0
Running after facing change while on pavement	See *Skidding*, p. 68.
Jumped with damaged gyro or leg/foot/hip actuators	per Preexisting Damage, at right
Jumped with destroyed leg	per Preexisting Damage, at right
Ran with damaged hip or gyro	per Preexisting Damage, at right
Terrain Modifications	
Ice	+3[7]
Mud	+1[7]
Deep Snow	+1[7]
Swamp	+1[7]

[1]This is damage actually taken, after any damage modifiers are applied.

[2]Only during the phase that the reactor shuts down. If the MechWarrior must make a Piloting Skill Roll for a 'Mech with a shutdown reactor, the 'Mech automatically falls; in either case, if the 'Mech falls, the warrior automatically takes 1 point of damage (see *Falling Damage to a MechWarrior*, p. 57).

[3]This modifier can replace other modifiers. See *Hip (Leg)* on page 48 for details.

[4]The modifier for a destroyed gyro is +6 when making a Piloting Skill Roll to avoid damaging the MechWarrior during an automatic fall.

[5]The modifier for a destroyed leg is +5 when making a Piloting Skill Roll to avoid damaging the MechWarrior during an automatic fall.

Cause of PSR	Modifier
Skidding Movement	
Hexes moved in turn	
0–2	–1
3–4	0
5–7	+1
8–10	+2
11–17	+4
18–24	+5
25+	+6
Building Movement[8]	
'Mech entering light building hex	0
'Mech entering medium building hex	+1
'Mech entering heavy building hex	+2
'Mech entering hardened building hex	+5
Hexes moved in turn	
1–2	0
3–4	+1
5–6	+2
7–9	+3
10–17	+4
18–24	+5
25+	+6

Special Cases[9]	Modifier
MechWarrior trying to avoid damage when their 'Mech is falling	+1/every level above 1
Four-legged 'Mech with all legs	–2
Unintentional charge	+3
'Mech mounts AES in its legs	–2
'Mech mounts Hardened Armor	+1
'Mech mounts small or torso-mounted cockpit	+1

Preexisting Damage	
Per leg/foot actuator previously destroyed	+1
Per hip actuator previously destroyed	+2[10]
Gyro previously hit	+3
Leg previously destroyed	+5[11]

[6]Automatic fall if death from above attack is unsuccessful.

[7]This penalty applies to all PSRs made in this hex (including the PSR made when entering it).

[8]To avoid damage only; does not result in a fall if Piloting Skill Roll fails (see *Moving Into Buildings*, p. 67). Add an additional +1 modifier if 'Mech is charging or being charged (in addition to the +2 modifier normally required in that situation). A 'Mech displaced into a building automatically fails this PSR.

[9]These cases do not cause PSRs, but their modifiers apply to all PSRs made.

[10]Do not add modifiers for other damaged actuators in the leg.

[11]Ignore all other critical hit modifiers to that leg from previous phases (see p. 48).

PILOTING SKILL ROLL TIMING

Exactly when a Piloting Skill Roll is made depends on the phase of the turn and the action which triggers it.

Movement Phase: Piloting Skill Rolls required due to a movement action (entering water, trying to stand up, entering rubble, and so on) are made immediately following the action. As such, if a PSR is made to enter a hex, and the PSR fails, the 'Mech falls in the hex it entered.

Multiple rolls may be required during movement. For example, if a 'Mech wants to move through three hexes of Depth 1 water, the player must make a PSR when the 'Mech enters each water hex.

If a PSR is required in the Movement Phase due to weapon attacks (possibly due to attacks by hidden 'Mechs; see p. 82), the roll is also made immediately.

Attack Phases: Piloting Skill Rolls due to attacks made in the Weapon or Physical Attack phases are made at the end of the phase the attacks occurred in (i.e. after all attacks that phase). Apply all modifiers resulting from damage that phase.

Consciousness Rolls: If both a Consciousness Roll and a PSR are required at the same time, the Consciousness Roll comes first.

Displacement: Any PSR required due to displacement (see below) happens immediately, regardless of the phase.

MechWarrior Damage: Any PSR made to see if a MechWarrior is damaged by a fall (see p. 57) is made immediately, before applying any damage caused by the fall.

DISPLACEMENT

A 'Mech forced into another hex is displaced. This typically occurs due to charge, push, and death from above attacks.

If a 'Mech is displaced into a hex of the same level, or one or two levels higher, check if the newly-entered hex contains another 'Mech. If so, see *Domino Effect*, at right.

If a 'Mech is displaced downward, what happens depends on how many levels down the 'Mech is displaced, and if the newly entered hex is occupied by another 'Mech:

- If the hex the 'Mech is displaced into has no other 'Mech in it, and is one level lower, follow the normal rules called for by the action that caused the displacement.
- If the hex the 'Mech is displaced into contains another 'Mech, and that hex is one level lower, a domino effect occurs (see right).
- If the hex the 'Mech is displaced into has no other 'Mech in it, and is two or more levels lower, an automatic fall occurs (see *Falling*, p. 56).
- If the hex the 'Mech is displaced into contains another 'Mech, and that hex is two levels or more lower, then an accidental fall from above occurs (see p. 57).

Actions that cause a displacement often call for a Piloting Skill Roll by the 'Mech being displaced. Any fall that results is a 0-level fall that occurs in the hex the 'Mech is displaced into, unless the 'Mech is displaced down two or more levels. In that case, the PSR is ignored because the 'Mech automatically falls; this fall distance is calculated from the 'Mech's level before the fall as normal.

Buildings: If a 'Mech is displaced into a building hex, both are damaged as if the displaced 'Mech had voluntarily moved into it and failed its PSR (see *Moving Into Buildings*, p. 67). Remember that each level of a building hex is separate for stacking purposes.

During the Weapon Attack Phase, a 'Mech whose MechWarrior has a Piloting Skill of 5 takes 40 points of damage and loses two leg actuators. At the end of the phase, the player makes three Piloting Skill Rolls: one for taking 20 or more points of damage, and two more for losing two leg actuators. The modified Target Number for each of the three rolls is 8: 5 (Piloting Skill) + 1 (20+ points of damage) + 1 (damaged leg actuator) + 1 (damaged leg actuator).

During the Physical Attack Phase, the same 'Mech is kicked in the leg by two other 'Mechs, losing another actuator and taking 20 more points of damage. At the end of the phase, the player must make four more Piloting Skill Rolls: two for getting kicked twice, one for losing a leg actuator, and one for the 20 points of damage. The modified Target Number for each of the four rolls is 9: 5 (Piloting Skill) + 2 (existing actuator damage) + 1 (another damaged leg actuator) + 1 (20+ points of damage).

If resolving building collapses and displacement, see *Damage and Displacement* on page 73.

Destroyed 'Mechs: A 'Mech destroyed by an action that would also displace it (such as a charge) does not move or resolve any displacement.

Forced Off-Board: A 'Mech displaced into a partial hex or off the playing area is considered destroyed for the remainder of the scenario.

Prohibited Displacement: A 'Mech cannot be displaced into a hex three or more levels higher than the hex it was moved from. In such a case, the displacement cannot occur, and neither the target nor attacking 'Mech move, unless the action specifically states otherwise. All other effects of the displacing action occur.

Timing: Displacement (including any resulting PSRs and damage) is resolved immediately after the action that caused it, regardless of what phase it is. If any damage was inflicted by the action that caused the displacement, apply it immediately, before resolving displacement and even though the phase is not over.

If a physical attack might cause displacement, and in doing so interrupt another physical attack, the 'Mech with the lower Initiative (i.e. the first to move) resolves its physical attack first. If both attackers are on the same side, the controlling player(s) may determine which attack is resolved first. If no valid target exists, the attack automatically fails.

DOMINO EFFECT

If a 'Mech is forced into a hex occupied by another 'Mech, the 'Mech already in the hex (known as "the target 'Mech") is forced out of that hex. This is called a domino effect. The target 'Mech immediately makes a Piloting Skill Roll.

If the target 'Mech fails its PSR, it is displaced into the hex in the direction opposite the hexside where the intruding 'Mech entered the hex. The target 'Mech then falls in that hex.

Dodging a Domino Effect: If the target 'Mech's PSR is instead successful, and the domino effect also originates from one of the target's four side hexes, then the target can dodge the domino effect by immediately moving one hex directly forward or backward, so long as all of the following are true:

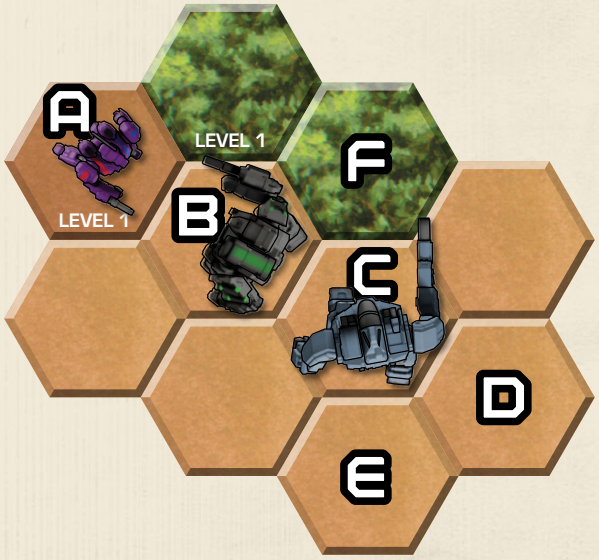
- it is mobile
- it is standing
- it has enough MP unspent that turn to move into the hex
- it did not jump that turn
- the hex being moved to does not already have a 'Mech in it

As usual, the 'Mech cannot move backward if it ran that turn.

If the target 'Mech's PSR is successful but it cannot dodge, it is still forced into the next hex as noted above, but does not fall.

Multiple Domino Effects: If a 'Mech is the target of a domino effect, and there is another 'Mech in the hex it is displaced into, another domino effect occurs. Such domino effects continue as long as 'Mechs remain in hexes adjacent to one another in the direction of the effect.

If there is no legal hex for the last 'Mech in a multiple domino effect chain to be displaced into, it is destroyed.



• DISPLACEMENT DIAGRAM •

In the Displacement Diagram above, the 'Mech in Hex A is displaced one level down into Hex B. The 'Mech already in Hex B makes an immediate Piloting Skill Roll. We'll assume it has no MP left. As such, whether it passes or fails it will in turn be forced into Hex C.

The 'Mech already in Hex C does have MP left, and the displaced 'Mech is coming from one of its side hexes, so the 'Mech in Hex C can try to dodge. First, its controller must make a PSR. If its roll fails, it is forced into Hex D and falls (and if another 'Mech was in that hex, the domino effect would continue).

If its PSR is successful (and it did not run that turn), it may move one hex directly backward, into Hex E, ending the domino effect. If the 'Mech had 2 or more MP left, it could instead move forward into the light woods in Hex F. If it had no MP left, however, it still would be forced into Hex D, but would not fall.

FALLING

If a 'Mech falls, it suffers damage and its MechWarrior may be injured as well.

LOCATION AFTER A FALL

When a 'Mech falls while moving from one hex to another, it falls in the hex it was moving into. If a fall occurs because of an attack or any other combat-related reason, the 'Mech falls in the hex it occupies.

Unless stated otherwise, 'Mech falls are 0-level falls; this includes 'Mechs that fall due to jumping. However, if a 'Mech is displaced downwards two or more levels (see *Displacement*, p. 55), subtract the level of the hex the 'Mech landed in from the level of the hex it was in immediately prior to the fall: this is the number of levels it fell.

Falling and Collisions: If a 'Mech falls two or more levels into a hex occupied by another 'Mech, the two 'Mechs may collide (see *Accidental Falls From Above*, p. 57).

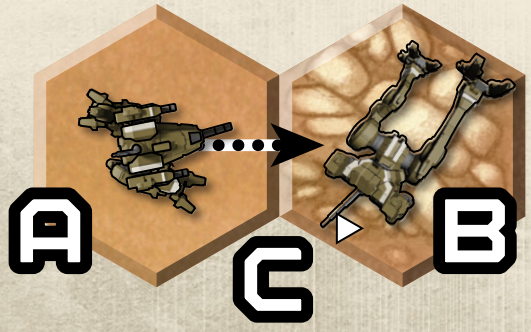
FACING AFTER A FALL

When a 'Mech falls, it takes damage and its facing may change. This facing change determines the Hit Location Table used when assigning damage from the fall. To determine the 'Mech's facing after the fall and the area of the 'Mech that takes damage, roll 1D6 and consult the Facing After Fall Table (see p. 57).

A fallen 'Mech is always prone and face down, regardless of which Hit Location their falling damage was applied to.

Damage Arc for Prone 'Mechs: If the target 'Mech is prone, use the hexside its head is pointing at as its facing. Any damage to a prone 'Mech from outside sources such as weapons fire is treated as if the 'Mech was standing with that facing.

In the Falling Diagram below, a player wants their 'Mech in Hex A to move into Hex B. As soon as the player enters the rubble hex, they must make a Piloting Skill Roll. The roll fails, and so the 'Mech's controller rolls 1D6 and consults the Facing After Fall Table. A result of 3 means the 'Mech is now facing two hexsides to the right (clockwise) of its original facing and takes the damage from the fall on its right side (i.e. it uses the right side column of the Hit Location Table). It lies prone and face down in the rubble hex, facing hexside C.



• FALLING DIAGRAM •

A prone Grasshopper *in a Level 1 hex attempts to stand. The MechWarrior fails the Piloting Skill Roll, and so the 'Mech falls again in the same hex. This means the 'Mech fell 0 levels. The player rolls a 1 on the Facing After Fall Table and finds that the 'Mech landed on its front, so the damage locations are rolled on the Front column of the Hit Location Table.*

The Grasshopper *suffers 7 points of damage (70 tons divided by 10 equals 7; the number of levels fallen plus 1 equals 1; 7 x 1 = 7). The player divides the 7 points into one 5-point grouping and one 2-point grouping, and then uses the Front column of the Hit Location Table to determine the exact location of the damage.*

FALLING DAMAGE TO A 'MECH

A fall deals damage equal to 1 point for every 10 tons that the 'Mech weighs (rounding up), multiplied by the number of levels plus 1 that the 'Mech fell. Divide the damage into 5-point Damage Value groupings (assigning any remaining points of damage to an undersized grouping), and then determine a hit location for each grouping. For example, a 'Mech that suffers 33 points of falling damage takes six 5-point hits and one 3-point hit, each assigned as a separate hit. Use the appropriate column of the Hit Location Table (see p. 33), as specified by the Facing After Fall Table (see below).

If the fall occurs during the Movement Phase, resolve the damage as it happens.

Water: If a 'Mech falls when standing in Depth 1 or greater water, it suffers normal falling damage divided by 2 (round down).

A 'Mech above the water's surface that falls into water suffers normal falling damage divided by 2 for hitting the water's surface, and also normal falling damage divided by 2 for the fall from the water's surface to the bottom of the water hex. Damage for hitting the water's surface equals tonnage/10 (round up) x (# of levels fallen + 1)/2). Normal damage for hitting the bottom of the water hex equals tonnage/10 (round up) x (depth of water hex + 1)/2. Damage is resolved separately; round any fractional final damage values down.

If a 'Mech takes damage in water, it may suffer a hull breach. See page 65.

FACING AFTER FALL TABLE

Roll (1D6)	New Facing	Hit Location
1	Same Direction	Front
2	1 Hexside Right	Right Side
3	2 Hexsides Right	Right Side
4	Opposite Direction	Rear
5	2 Hexsides Left	Left Side
6	1 Hexside Left	Left Side

The MechWarrior piloting the Grasshopper *in the previous example has a Piloting Skill of 5. It does not have any pre-existing damage, it did not take 20 points of damage this phase, and it fell 0 Levels, and so no modifiers apply to the Target Number for the Piloting Skill Roll. Its controller therefore need only roll 5 or better to avoid damaging the MechWarrior.*

FALLING DAMAGE TO A MECHWARRIOR

A player makes a second Piloting Skill Roll after every fall (often called a "seatbelt check"). All modifiers applied to the PSR that caused the fall are applied to this second roll, with an additional +1 modifier applied to the Target Number for every level above 1 fallen. If the roll succeeds, the MechWarrior avoids taking damage. If the roll fails, the MechWarrior takes 1 point of damage.

Automatic Damage: A MechWarrior automatically takes 1 point of damage if at the time of the fall their 'Mech is immobile (see p. 27) or if the modified Target Number for the PSR to avoid MechWarrior damage is greater than 12.

ACCIDENTAL FALLS FROM ABOVE

A 'Mech cannot intentionally fall onto another 'Mech. However, if a 'Mech is displaced into a hex two or more levels lower than its previous hex, and that hex already contains a 'Mech, then the displaced 'Mech might fall onto the target 'Mech. The displaced 'Mech makes an attack roll with a base Target Number of 7, modified by target movement and terrain.

FALLING 'MECH MISSES TARGET

If the attack roll fails, the falling 'Mech lands in an adjacent hex as close as possible to the hex from which it fell; if multiple such hexes exist, randomly determine which hex it lands in. Resolve this as a standard fall.

FALLING 'MECH HITS TARGET

If the attack roll is successful, divide the falling 'Mech's weight by 10 (round up), and multiply this by the number of levels fallen to the point of impact (the level of the underlying hex plus the level of the target 'Mech –1). Divide this damage by 2 (round down) if the target 'Mech is fully submerged. Then divide the damage into 5-point Damage Value groupings (assigning any remaining points of damage to an undersized grouping), and assign a hit location for each grouping to the target 'Mech using the Punch Location Table.

Determine damage to the falling 'Mech as normal for a fall of that height. Roll for facing as normal on the Facing After Fall Table (see left), but the 'Mech always lands on its back and all damage is applied to the rear.

The target 'Mech is immediately displaced to a random neighboring legal hex (see *Displacement*, p. 55), and must make a Piloting Skill Roll (see p. 53) with a +2 modifier or fall in the hex it was displaced into.

FORBIDDEN DISPLACEMENT

If there is no legal hex for the falling or target 'Mech to be placed in, that 'Mech is destroyed.

DAMAGING A MECHWARRIOR

Three types of damage to a 'Mech can also damage the MechWarrior inside: head hits, falling, and internal ammunition explosions. In addition, excessive heat buildup can harm the MechWarrior if the 'Mech's life support system suffers a critical hit.

A MechWarrior can withstand up to 5 points of damage. The sixth point of damage kills the warrior.

Ammunition Explosions: An ammunition explosion causes 2 points of damage to the MechWarrior as a result of severe electric shock they receive through the neurohelmet.

Falling: If a 'Mech falls, its controller must immediately make a Piloting Skill Roll for the MechWarrior, before applying any falling damage to the 'Mech. All modifiers applied to the PSR that caused the fall are applied to this second roll, with an additional +1 modifier applied to the Target Number for every level above 1 fallen as well. If the roll fails, the warrior takes 1 point of damage.

If the 'Mech is immobile (see p. 27), or if the modified Target Number for the PSR to avoid MechWarrior damage is greater than 12, such a fall automatically damages the warrior.

Head Hits: The MechWarrior takes 1 point of damage whenever the 'Mech's head takes a hit dealing 1 or more points of damage, even if the hit does not penetrate the 'Mech's armor (unless the 'Mech has a torso-mounted cockpit; see p. 115).

Heat: When a 'Mech's life support system takes a critical hit, the MechWarrior takes 1 point of damage at the end of every Heat Phase that the 'Mech's Heat Scale is between 15 and 25; if the Heat Scale is 26 or higher, the MechWarrior takes 2 points of damage instead.

If a 'Mech with a torso-mounted cockpit takes a life support critical hit, the danger from heat is greater. If a 'Mech with a torso-mounted cockpit suffers a life support critical hit, the MechWarrior suffers 1 point of damage at the end of every Heat Phase that the 'Mech's Heat Scale is between 1 and 14, and 2 points of damage at the end of every Heat Phase that the Heat Scale is 15 or higher.

CONSCIOUSNESS ROLLS

MechWarriors can survive up to 5 points of damage, but they may be knocked unconscious long before taking that much. Every time a warrior takes a point of damage, its controller must immediately make a Consciousness Roll. Roll 2D6 and consult the Consciousness Table below (the Warrior Data section of the 'Mech Record Sheet also contains the information on this table). If the roll result is equal to or greater than the MechWarrior's consciousness number, the warrior remains conscious. If the result is less than the consciousness number, the warrior is immediately knocked unconscious.

WARRIOR DATA

Name: _____

Gunnery Skill: ___ Piloting Skill: ___

Hits Taken	1	2	3	4	5	6
Consciousness#	3	5	7	10	11	Dead

• MECHWARRIOR DATA •

The player makes this roll for every point of damage taken; for example, in the case of an ammunition explosion in a 'Mech, that MechWarrior will need to make two consecutive Consciousness Rolls. However, if a MechWarrior is knocked unconscious, no further Consciousness Rolls need be made that turn.

A 'Mech with an unconscious warrior becomes immobile (see p. 18). The 'Mech cannot move, fire, or take any other action, though its player can still assign it a movement selection (see p. 10).

A warrior that takes 6 points of damage is dead. Remove the MechWarrior's 'Mech from the map in the end of the phase in which they were killed.

Equipment: If a MechWarrior is unconscious, any piece of equipment that can be activated or deactivated in the End Phase (see p. 12) continues to function (provided it started the turn activated), as does CASE.

Piloting Skill Rolls: If a MechWarrior is forced to make both a Consciousness Roll and a PSR at the same time, the Consciousness Roll always comes first.

All PSRs required of an unconscious MechWarrior automatically fail.

Recovering Consciousness: During the End Phase of each turn after the turn in which the MechWarrior lost consciousness, the player rolls 2D6. If the result is equal to or greater than the consciousness number for the warrior's current total damage points, the warrior regains consciousness. The player need not roll again to determine unconsciousness until the warrior takes new damage.

For example, a MechWarrior with 3 points of damage must make a single 2D6 roll in the End Phase to wake up. A roll of 7+ means the warrior regains consciousness that phase.

DUMPING AMMUNITION

During the End Phase, a player can declare that their 'Mech will dump ammunition in the next turn, meaning the MechWarrior will open the ammo loading doors on the back of the 'Mech and allow ammunition to feed out.

Ammunition must be dumped by critical hit slot (i.e. you cannot dump only some of the ammo in a slot; it must be all or nothing). Any or all slots may be dumped as a single action.

When a player announces that their 'Mech will dump ammunition, that ammunition is immediately unavailable for use. However, the actual dumping of the ammunition is not carried out until the next End Phase. For that one turn, the ammunition remains onboard in its normal location and critical hit slot, and can still suffer critical hits or the effect of heat buildup.

A 'Mech dumping ammunition cannot run or jump in that turn, but can still fire. Any hit against the dumping 'Mech on any rear torso location inflicts normal damage, but it also causes all dumping ammunition that can explode to do so (damage is applied to the location the ammunition is stored in, as normal).

Ammunition dumped in a hex cannot be exploded or used for any type of attack.

Underwater: 'Mechs may not declare their intention to dump ammunition when submerged. If a 'Mech submerges during the turn its ammunition is being dumped, all three torso locations automatically flood (see *Hull Integrity and Breaches*, p. 65).

TAG DESIGNATION

If using target acquisition gear (TAG; see p. 113), all 'Mechs making attempts to designate a target do so after the end of the Movement Phase, but before the beginning of the Weapon Attack Phase (Initiative order is still followed, however).

THE BATTLEFIELD

Warfare in the Inner Sphere occurs on hundreds of worlds and in a wide variety of environments, from the gentle rolling hills of Bountiful Harvest to the frozen, broken wastelands of Misery, from the shattered cities of Alexandria to the planet-wide dust storms of Mars. This chapter will cover some of the most common environments in which the typical warrior can expect to fight.

PLANETARY CONDITIONS

"Planetary conditions" is a general term used to describe all terrain and weather types/conditions that can be used in a game. Conditions not printed on a map can easily be represented via counters and the like, so long as their representation is understood by all players prior to the game.

Players should be aware that some conditions and terrain types introduce a number of additional modifiers and sub-rules (such as tracking the spread of fires or range reductions for battle underwater) which may complicate and slow down basic gameplay. When choosing maps and constructing scenarios, players should consider whether the increased novelty and variety that comes with certain conditions balances out the time required to manage them.

TERRAIN TYPES

The following details a variety of terrain types reasonably common in standard Terran environments. Many of the rules for these terrain types also appear elsewhere in this book, as terrain affects a game in many different ways. However, they have been summarized here for easy reference.

A terrain type's MP Modifier specifies how much additional MP is required to enter that hex, on top of the standard 1 MP required to enter any hex. For example, it costs 3 MP to enter a light jungle hex: the standard 1 MP, and the +2 MP added by light jungle.

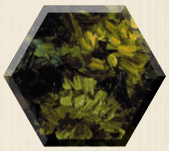
If a terrain type lists "None" for its PSR modifier, a Piloting Skill Roll is not called for when entering that terrain. In any other case, a PSR is required when entering a hex of that type, with the modifier given (which can be 0) applied to that roll in addition to the standard PSR modifiers.

CLEAR

Line of Sight (LOS): No effect.
MP Modifier: None. It costs only 1 MP to enter a clear hex.
PSR Modifier: No PSR.
Attack Target Number Modifier: None.
Special: If a hex is not marked as containing another terrain type, assume it is clear.

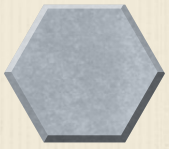
JUNGLE, LIGHT

Line of Sight (LOS): Three or more points of intervening woods/jungle block LOS. Light woods/jungle is worth 1 point.
MP Modifier: +2.
PSR Modifier: +1.
Attack Target Number Modifier: +1 if the target occupies such a hex, and +1 for each additional hex in between the attacker and target.



JUNGLE, HEAVY

Line of Sight (LOS): Three or more points of intervening woods/jungle block LOS. Heavy woods/jungle is worth 2 points.
MP Modifier: +3.
PSR Modifier: +2.
Attack Target Number Modifier: +2 if the target occupies such a hex, and +2 for each additional hex in between the attacker and target.

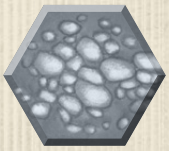


PAVEMENT

Line of Sight (LOS): No effect.
MP Modifier: None. For the MP bonus that roads sometimes bestow, see *Road Movement*, page 20.
PSR Modifier: No PSR (but see below).
Attack Target Number Modifier: None.
Special: Pavement includes roads. 'Mechs may skid on pavement (see *Skidding*, p. 68).

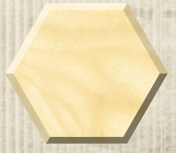
ROUGH

Line of Sight (LOS): No effect.
MP Modifier: +1.
PSR Modifier: No PSR.
Attack Target Number Modifier: None.



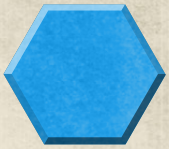
RUBBLE

Line of Sight (LOS): No effect.
MP Modifier: +1.
PSR Modifier: 0.
Attack Target Number Modifier: None.



SAND

Line of Sight (LOS): No effect.
MP Modifier: 0.
PSR Modifier: +1.
Attack Target Number Modifier: None.



WATER (DEPTH 0)

Line of Sight (LOS): No effect.
MP Modifier: None.
PSR Modifier: No PSR.
Attack Target Number Modifier: None.

WATER (DEPTH 1)

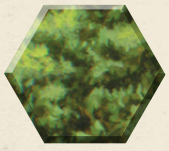
Line of Sight (LOS): No effect.
MP Modifier: +1; level change costs are not included. See *Movement in Water*, page 19.
PSR Modifier: –1.
Attack Target Number Modifier: None, but it provides partial cover to 'Mechs standing within it (see *Partial Cover*, p. 26).

WATER (DEPTH 2)

Line of Sight (LOS): No effect.
MP Modifier: +3; level change costs are not included. See *Movement in Water*, page 19.
PSR Modifier: 0.
Attack Target Number Modifier: None, but see *Attacking Underwater*, page 64.

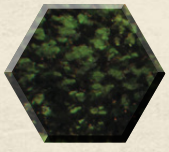
WATER (DEPTH 3+)

Line of Sight (LOS): No effect.
MP Modifier: +3; level change costs are not included. See *Movement in Water*, page 19.
PSR Modifier: +1.
Attack Target Number Modifier: None, but see *Attacking Underwater*, page 64.



WOODS, LIGHT

Line of Sight (LOS): Three or more points of intervening woods/jungle block LOS. Light woods/jungle is worth 1 point.
MP Modifier: +1.
PSR Modifier: No PSR.
Attack Target Number Modifier: +1 if the target occupies such a hex, and +1 for each additional hex in between the attacker and target.



WOODS, HEAVY

Line of Sight (LOS): Three or more points of intervening woods/jungle block LOS. Heavy woods/jungle is worth 2 points.
MP Modifier: +2.
PSR Modifier: No PSR.
Attack Target Number Modifier: +2 if the target occupies such a hex, and +2 for each additional hex in between the attacker and target.

TERRAIN MODIFICATIONS

The following are not terrain types of their own, but modifications to existing terrain, which may be overlaid on most of the terrain types above. For example, a heavy woods hex may also be shrouded in deep snow, and smoke may drift over any hex. All effects applied by the terrain modification stack with those of the terrain modified.

ICE (SIMPLIFIED)

If applied to water hexes, the hex is presumed to be frozen solid, and cannot be broken or melted through during the battle.

Cannot be used with Extreme Temperatures (see p. 62) of above 30 degrees Celsius.

MP Modifier: A hex covered in ice adds +1 MP cost per hex (but see *Reckless Movement*, p. 20).

PSRs: All PSRs made in a hex with ice (including the PSR made when entering one, if Reckless Movement was used) apply a +3 Target Number modifier.

MUD (SIMPLIFIED)

Cannot be used with Extreme Temperatures (see p. 62) of below –30 degrees Celsius.

MP Modifier: A hex covered in mud adds +1 MP cost per hex (but see *Reckless Movement*, p. 20).

PSRs: All PSRs made in a hex with mud (including the PSR made when entering one, if Reckless Movement was used) apply a +1 Target Number modifier.

SMOKE, LIGHT

Light smoke acts as light woods for the purpose of line of sight and weapons fire modifiers. See *Fire and Smoke (Simplified)*, page 63.

SMOKE, HEAVY

Heavy smoke acts as heavy woods for the purpose of line of sight and weapons fire modifiers. See *Fire and Smoke (Simplified)*, page 63.

SNOW, DEEP

Deep snow may not be added to water hexes. Cannot be used with Extreme Temperatures (see p. 62) of above 30 degrees Celsius.

MP Modifier: A hex covered in deep snow adds +1 MP cost per hex (but see *Reckless Movement*, p. 20).

PSRs: All PSRs made in a hex with deep snow (including the PSR made when entering one, if Reckless Movement was used) apply a +1 Target Number modifier.

Heat: Deep snow dissipates 1 heat point per turn if at least one heat sink is mounted in the legs (if the 'Mech is prone, 3 heat points per turn are dissipated).

SWAMP (SIMPLIFIED)

Cannot be used with Extreme Temperatures (see p. 62) of –30 degrees Celsius and below.

MP Modifier: A swamp hex adds a +1 MP cost per hex (but see *Reckless Movement*, p. 20).

PSRs: All PSRs made in a hex with a swamp (including the PSR made when entering one, if Reckless Movement was used) apply a +1 Target Number modifier.

CLEARING WOODS OR JUNGLE

'Mechs can declare weapon attacks to clear woods or jungle hexes. The Terrain Factor (TF) of a hex determines the Damage Value the hex can take before being reduced from heavy to light, or from light to rough terrain (you cannot reduce a woods or jungle hex to clear terrain). A hex's type does not change until its TF falls below the value of the reduced hex shown on the Terrain Factor column of the Terrain Factor and Conversion Table (see below).

For example, a heavy woods hex that takes 35 points of damage remains a heavy woods hex; if the same hex takes 6 more points of damage it is automatically reduced to light woods. A heavy woods hex can be changed to a rough hex in a single Weapon Attack Phase, provided enough damage is inflicted.

Finally, clearing woods or jungle requires a declared attack against that terrain's TF. If an attack against a 'Mech passes through or into such a hex, the attack does not reduce that hex's TF.

Area-Effect Weapons: Weapons of the area-effect (AE) type always damage any woods or jungle hexes in their blast area, even if the attack was not declared as an attempt to do so. Double the Damage Value of these weapons against such hexes.

Cluster Weapons: For weapons that use the Cluster Hits Table, use the weapon's full Damage Value (i.e., do not roll on the table).

Flechette Ammunition: This ammunition (see p. 107) deals double damage to woods and jungle hexes.

LOS: When trying to clear a woods or jungle hex, treat the hex as a standing 'Mech. If this "'Mech" would have partial cover from the attack (see p. 26), than the attacker cannot attempt to clear the woods/jungle.

Plasma Weapons: Versus woods or jungle, plasma rifles deal (10+2D6) × 2 damage, while plasma cannons deal 3D6 × 2 damage.

TERRAIN FACTOR AND CONVERSION TABLE

Terrain Factor	Former Terrain	New Terrain
Heavy Jungle: 100	Heavy Jungle	Light Jungle
Light Jungle: 60	Light Jungle	Rough
Heavy Woods: 90	Heavy Woods	Light Woods
Light Woods: 50	Light Woods	Rough

WEATHER CONDITIONS

Weather conditions represent global situations that apply to an entire playing area.

The following entries describe rules for weather conditions. They are organized by general category, with varying degrees of conditions for each type. Note that many of these conditions can be combined at once. For example, players might agree to combine the rules for a Strong Gale or Storm, Heavy Snowfall, and Extreme Temperatures to simulate combat in a fierce blizzard.

EXTREME TEMPERATURES

If a scenario takes place under such conditions, modify movement and combat as follows.

Combat in temperatures between 30 and –30 degrees Celsius has no game effect. However, fighting in higher or lower temperatures can affect some terrain (see p. 61), the chance of fires starting (see p. 64), as well as how 'Mechs dissipate heat.

For each 10 degrees C (or fraction thereof) higher than 50 degrees, add 1 heat point to the 'Mech's heat buildup each turn. For every 10 degrees C (or fraction thereof) less than –30 degrees, the 'Mech dissipates 1 heat point per turn.

FOG

LIGHT FOG

MP Modifier: Apply a +1 MP cost per hex (but see *Reckless Movement*, p. 20).

HEAVY FOG

MP Modifier: Apply a +2 MP cost per hex (but see *Reckless Movement*, p. 20).

Attack Target Number Modifier: +1 to all attacks of the Pulse (P) and Direct-Fire Energy (DE) types (see p. 97).

LIGHT

Low-light conditions can present a host of battlefield complications.

Searchlights: Searchlights illuminate targets, thus offsetting many of the penalties of fighting in a low-light environment. Mounted searchlights have a range of 30 hexes (half underwater).

A searchlight illuminates all targets in a chosen hex that the 'Mech with the searchlight could normally trace LOS to, in the arc where the searchlight is mounted, during any Weapon or Physical Attack Phase. A searchlight also illuminates all intervening hexes (and hence all 'Mechs in those hexes) between the target hex illuminated and the searchlight-equipped attacker in the same manner. Finally, a 'Mech with a searchlight illuminates itself.

For simplicity's sake, all searchlight beams are two levels high. A searchlight is presumed to be aiming at ground level, but a player may specify any level from 1 to 29 at, above, or below the level the 'Mech is at (and within LOS). The searchlight then illuminates that level and the one immediately above it. For example, a searchlight could specify that it is aiming up to level 29; if it did so, levels 29 and 30 would be illuminated.

If all players agree—or the scenario designates this—'Mechs without searchlights can be given a mounted searchlight for the duration of the scenario. Each time a searchlight-equipped 'Mech takes a hit in any torso location (Front or Rear), the attacker must roll 2D6. A result of 7+ means the searchlight is destroyed in addition to the normal effects of the attack.

Finally, a 'Mech's controller can turn its searchlight off or on during any End Phase.

DUSK/DAWN

Attack Target Number Modifier: +1 to all weapon attacks, unless the target 'Mech's heat scale is at 25 heat points or higher.

FULL MOON NIGHT

MP Modifier: +1 MP cost per hex (but see *Reckless Movement*, p. 20). A 'Mech with a searchlight ignores this penalty.

Attack Target Number Modifier: +2 to all weapon attacks. Reduce this by 1 for every 20 points of heat on the target 'Mech's heat scale (minimum of 0). Ignore this modifier if the target is illuminated.

MOONLESS NIGHT

MP Modifier: +2 MP cost per hex (but see *Reckless Movement*, p. 20). A 'Mech with a searchlight ignores this penalty.

Attack Target Number Modifier: +3 to all weapon attacks, and +1 to all physical attacks. Reduce each modifier by 1 for every 15 points of heat on the target 'Mech's heat scale (minimum of 0 in both). Ignore these modifiers if the target is illuminated.

RAIN

MODERATE RAINFALL

Attack Target Number Modifier: +1 to all weapon attacks.

Mud: Moderate rainfall automatically includes all modifiers and effects of mud (see p. 61) in any clear, Depth 0 water hex on the playing area.

HEAVY RAINFALL

Attack Target Number Modifier: +1 to all weapon attacks.

PSRs: Apply a +1 modifier to all Piloting Skill Rolls.

Mud: Heavy rainfall automatically includes all modifiers and effects for mud (see p. 61); mud occurs in every hex across the playing area (except for water hexes of Depth 1+, standard roads, and building hexes).

TORRENTIAL DOWNPOUR

Attack Target Number Modifier: +2 to all weapon attacks.

PSRs: Apply a +2 modifier to all Piloting Skill Rolls.

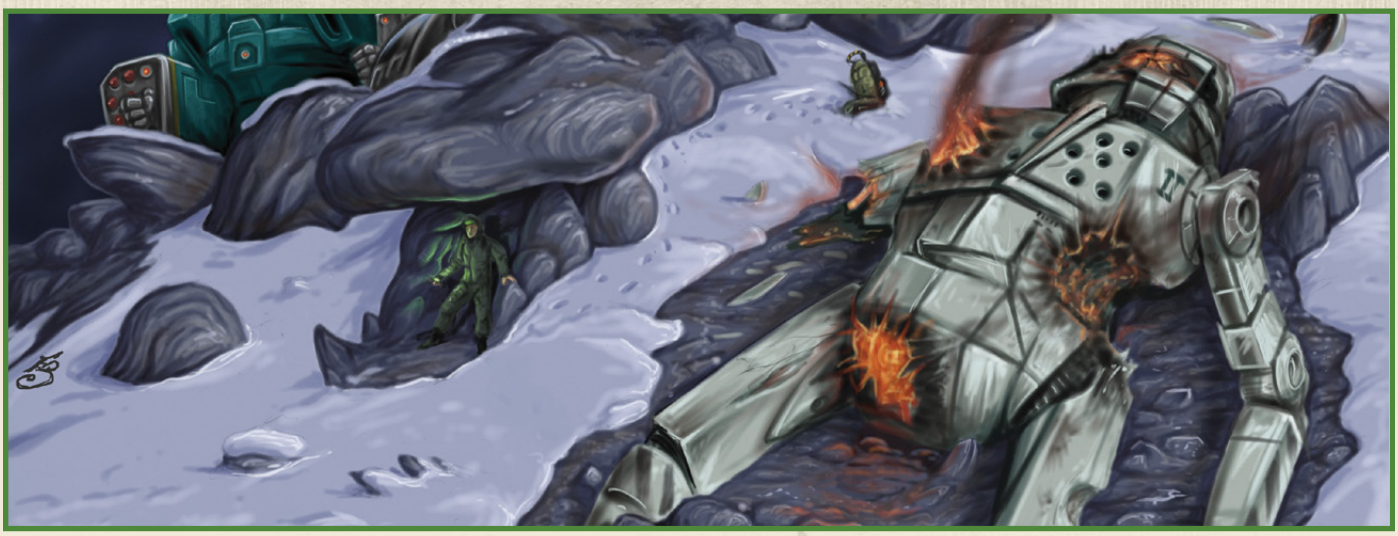
Mud: A torrential downpour automatically includes all modifiers and effects of mud (see p. 61); mud occurs in every hex across the playing area except for roads, building hexes, and water hexes of Depth 1+. Additionally, all Depth 0 water hexes become swamp hexes (see p. 61).

SNOW

MODERATE SNOWFALL

Attack Target Number Modifier: +1 to all weapon attacks.

Deep Snow: Moderate snowfall eventually covers the battlefield in snow. At the start of Turn 20 of a scenario, every hex is covered in deep snow (see p. 61). This includes the tops of buildings, but not water hexes.



HEAVY SNOWFALL

Attack Target Number Modifier: +1 to all weapon attacks.

PSRs: Apply a +1 modifier to all Piloting Skill Rolls.

Deep Snow: Heavy snowfall eventually covers the battlefield in snow. At the start of Turn 15 of a scenario, every hex is covered in deep snow (see p. 61). This includes the tops of buildings, but not water hexes.

WIND

MODERATE GALE

Attack Target Number Modifier: +1 to all attacks of the missile (M) type (see *M: Missile*, p. 97).

Fire: If using the simplified Fire and Smoke rules, fire spreads more easily in these conditions (see *Determining Fire Spread*, p. 64).

STRONG GALE (SIMPLIFIED)

Attack Target Number Modifier: +2 to all attacks of the missile (M) type, and +1 to all attacks of the direct-fire ballistic (DB) type (see *DB: Direct-Fire Ballistic*, p. 97).

PSRs: Apply a +1 modifier to all Piloting Skill Rolls.

Fire: If using the simplified Fire and Smoke rules, fire spreads more easily in these conditions (see *Determining Fire Spread*, p. 64).

STORM (SIMPLIFIED)

Attack Target Number Modifier: +3 to all attacks of the missile (M) type, and +2 to all attacks of the direct-fire ballistic (DB) type (see *DB: Direct-Fire Ballistic*, p. 97).

PSRs: Apply a +3 modifier to all Piloting Skill Rolls.

Fire: If using the simplified Fire and Smoke rules, fire spreads more easily in these conditions (see *Determining Fire Spread*, p. 64).

FIRE AND SMOKE (SIMPLIFIED)

Players may optionally use the following rules to introduce the effects of fire to the battlefield. Place a fire counter (a pre-made counter, a penny or whatever else players agree to use) on any hex that is set on fire during the game.

WIND DIRECTION

Games in which fire and smoke are present require players to first determine the battlefield's wind direction. At the start of the game, declare the same one side of every hex to be Direction 1, numbering the remaining hexsides 2 through 6 clockwise. Roll one die. The wind will blow in the direction shown on the die for the entire game. This single-hex direction is considered "downwind."

STARTING A FIRE

The ease or difficultly of starting a fire depends on whether or not it is intentionally lit.

Weather Conditions: If any rain or snow weather conditions are present on the battlefield, fires cannot be started. Fire also cannot be started in hexes with Deep Snow or Ice.

ACCIDENTAL FIRES

Any 'Mech attempting to clear a woods or jungle hex (see *Clearing Woods or Jungle*, p. 61) runs the risk of accidentally setting the woods on fire. To represent this risk, the player rolls 2D6 before each clearing attempt. On a result of 5 or less, the woods take damage as usual, but have also been set afire.

INTENTIONAL FIRES

Players who intend to start fires may declare that their 'Mech will fire its weapons at any woods, jungle, or building hex. On a successful attack against the hex, the player rolls 2D6 and consults the Fire Table (p. 64). Modifiers are applied to the Target Number, not the roll; any result equal to or higher than the Target Number means a fire has started.

EFFECTS OF FIRE

Any hex that is set on fire during the game must be marked as such with a fire counter. Once started, a fire continues to burn for the rest of the game (see *Extinguishing Fire*, p. 64).

During the Heat Phase, a 'Mech occupying a burning hex adds 5 Heat Points. A 'Mech also absorbs 2 Heat Points for each burning hex that it moved through during the Movement Phase. A 'Mech occupying

FIRE TABLE

Weapon Type	Target Number
Plasma Weapon or Inferno	Automatic
Flamer	4+
Direct-Fire Energy or Pulse Weapon	7+
Missile or Direct-Fire Ballistic	9+

Environment*	Modifier
Woods	+0
Jungle	+1
Building: Light	–2
Building: Medium	–1
Building: Heavy	+0
Building: Hardened	+1
Extreme Temperatures	**

*Environments not listed cannot be set on fire.

**For every 10 degrees above 30 degrees Celsius, apply a –1 modifier. For every 10 degrees below –30 degrees Celsius, apply a +1 modifier.

a hex ignited during the Weapon Attack Phase of the turn will not be affected by the fire until the Heat Phase of the following turn.

Buildings: Every End Phase, a building that is on fire loses 2 CF (regardless of its size or type).

ILLUMINATION

If Light-based weather conditions are in play (see p. 62), any hex that is on fire is illuminated, along with all hexes adjacent to it.

SMOKE

In the End Phase of a turn in which a fire was started, any hex on fire fills the first two hexes downwind from it with Smoke (see p. 61). However, a hex that is on fire cannot have smoke as well.

Light vs. Heavy: Smoke generated in this fashion is always light smoke, unless a heavy or hardened building is burning, in which case it is heavy smoke.

SPREADING FIRE

Fires on the battlefield can spread from hex to hex in the direction of the wind through woods, jungle, and some building hexes, but they cannot spread into other terrain.

DETERMINING FIRE SPREAD

During the End Phase of every turn, check to see if any fires currently on the map spread to additional hexes. A fire hex can spread if it has a neighboring hex downwind that can burn (a woods, jungle, or light or medium building hex) and is not currently on fire. For each such fire hex, roll 2D6. If the result is a 12, the fire spreads downwind one hex.

Buildings: Heavy and hardened buildings cannot catch fire in this fashion.

Smoke: For simplicity's sake, smoke does not spread on its own under these rules.

Wind: If using the Moderate Gale, Strong Gale, or Storm wind conditions (see p. 63), on a roll of a 12 a fire hex spreads two hexes downwind, rather than just one.

EXTINGUISHING FIRE

A fire will continue to burn for the rest of the game unless extinguished. Most 'Mechs lack the equipment needed to put out a fire, making extinguishing one in battle a tricky proposition. If 40 or more points of damage from weapons of the area-effect Type are dealt to a hex on fire in a single phase, the fire is extinguished.

Smoke: Remove any smoke produced by extinguished fire hexes during the End Phase of the turn the fire was extinguished.

UNDERWATER COMBAT

To be considered underwater (i.e. fully submerged), a 'Mech must be standing in a Depth 2 or deeper water hex, or lying prone in a Depth 1 or deeper water hex. A 'Mech standing in a Depth 1 water hex has only its legs submerged (i.e. is partially submerged).

ATTACKING UNDERWATER

A 'Mech standing in Depth 1 water may make underwater attacks with leg-mounted weapons, as well as attacks from weapons mounted in any other location against 'Mechs not fully submerged. Underwater 'Mechs can attack a standing 'Mech in Depth 1 water; the target has partial cover (see p. 26), and an attack striking any location except for a leg strikes the cover.

A 'Mech underwater cannot normally fire on or take fire from other 'Mechs unless those 'Mechs are also underwater. However, other 'Mechs can attack a standing 'Mech in Depth 1 water; the target has partial cover, and an attack striking any location except a leg deals no damage.

Calculate line of sight normally, treating depth as negative numbers. For example, Depth 1 is at Level –1 and so is two levels below a Level 1 hill. Depth 2 or deeper water completely blocks LOS to and from the 'Mech standing in that hex, while LOS is blocked to a prone 'Mech in Depth 1 or deeper water.

'Mechs without a clear line of sight cannot attack one another. The Underwater Line of Sight Table (see p. 65) summarizes which 'Mechs can fire at each other, and with what modifications. Only lasers and PPCs may be used in an underwater attack; torpedo launchers (see p. 104) are the sole exception (see the Underwater Range Table, p. 65).

Underwater Ranges: Water greatly reduces the ranges of all energy weapons. When a submerged 'Mech makes an attack, use the weapon ranges shown on the Underwater Range Table to determine range modifiers. If a weapon listed in the Weapons and Equipment Tables does not appear on the Underwater Range Table, it cannot be used underwater; see *Torpedo (SRT, LRT, NLRT) Launcher*, page 104, for the sole exception.

UNDERWATER LINE OF SIGHT TABLE

Attacker is:	Target is:		
	Underwater[1]	*Ground*[2]	*'Mech at Depth 1*
Underwater[1]	Yes	No	Yes[4]
Ground[2]	No	Yes	Yes[3]
'Mech at Depth 1	Yes[6]	Yes[5]	Yes[7]

[1]Includes standing 'Mechs at Depth 2+ and prone 'Mechs at Depth 1+.
[2]Includes all 'Mechs at Level 0+.
[3]At +1 for partial cover; use the appropriate column of the 'Mech Hit Location Table, with any leg hits striking the water and inflicting no damage.
[4]At +1 for partial cover; use the appropriate column of the 'Mech Hit Location Table, with any location hit but leg hits striking the water and inflicting no damage.
[5]Can only fire torso, arm or head weapons.
[6]Can only fire leg weapons.
[7]Leg weapons can only hit legs; all other weapons can only hit non-submerged locations (see *Water*, under *Partial Cover Modifier*, p. 26).

Physical Attacks: Any physical attack that occurs underwater inflicts half its standard damage (round down). A 'Mech standing in Depth 2 (or deeper) water can make any physical attack, except for a death from above.

For rules on physical attacks in Depth 1 water, see *Physical Attacks and Water*, page 35.

HULL INTEGRITY AND BREACHES

When an underwater location takes damage (such as a leg in Depth 1 water), the 'Mech's controller rolls 2D6: this is often called a hull breach check. On a result of 10 or greater, the 'Mech has lost integrity in that location, which fills with water.

Only roll for flooding when an underwater 'Mech takes damage, not for a damaged 'Mech that later enters the water (a 'Mech above the surface of a water hex that falls into a water hex does not check for hull integrity until it strikes the bottom of the water hex). However, if all armor in a location (or facing, if a torso location) is destroyed, that location is automatically breached if submerged at that time or later.

All components in a breached location become nonfunctional. If the breached location contains engine slots, the engine functions as if it took as many critical hits as engine critical slots in that location; if the breached location contains the cockpit, the 'Mech is considered destroyed and its MechWarrior killed. If a side torso is flooded, the corresponding arm (front leg, for quad 'Mechs) and its components become nonfunctional as well. Even if a 'Mech exits the water, all limbs and components rendered non-functional due to flooded locations remain non-functional. An arm rendered non-functional in this fashion counts as destroyed for the purposes of firing while prone (see p. 30).

These effects are not critical hits, and will not cause ammunition to explode. Components in the breached location can still take critical hits, even though the component is out of commission.

UNDERWATER RANGE TABLE

Weapon	Short	Medium	Long
Small Laser	1	2	—
Medium Laser	1–2	3–4	5–6
Large Laser	1–3	4–6	7–9
Light PPC*	1–4	5–7	8–10
PPC*	1–4	5–7	8–10
Heavy PPC*	1–4	5–7	8–10
Snub-nose PPC	1–6	7–8	9
ER Micro Laser	1	2	—
ER Small Laser (Clan)	1	2	3–4
ER Small Laser (IS)	1	2	3
ER Medium Laser (Clan)	1–3	4–7	8–10
ER Medium Laser (IS)	1–3	4–5	6–8
ER Large Laser (Clan)	1–5	6–10	11–16
ER Large Laser (IS)	1–3	4–9	10–12
ER PPC (Clan and IS)	1–4	5–10	11–16
Micro Pulse Laser	1	2	—
Small Pulse Laser (Clan)	1	2	3–4
Small Pulse Laser (IS)	1	2	—
Medium Pulse Laser (Clan)	1–3	4–5	6–8
Medium Pulse Laser (IS)	1–2	3	4
Large Pulse Laser (Clan)	1–4	5–10	11–14
Large Pulse Laser (IS)	1–2	3–5	6–7
Small Heavy Laser	1	2	—
Medium Heavy Laser	1–2	3–4	5–6
Large Heavy Laser	1–3	4–6	7–9
Binary Laser (Blazer) Cannon	1–3	4–6	7–9
Chemical Laser, Small	1	2	—
Chemical Laser, Medium	1–2	3–4	5–6
Chemical Laser, Large	1–3	4–6	7–9
ER Pulse Laser, Small	1	2	3–4
ER Pulse Laser, Medium	1–3	4–6	7–9
ER Pulse Laser, Large	1–4	5–10	11-16
Improved Heavy Laser, Small	1	2	—
Improved Heavy Laser, Medium	1–2	3–4	5–6
Improved Heavy Laser, Large	1–3	4–6	7–9
Re-engineered Laser, Small	1	2	—
Re-engineered Laser, Medium	1–2	3–4	5–6
Re-engineered Laser, Large	1–3	4–6	7–9
Variable-Speed Laser, Small	1	2	3
Variable-Speed Laser, Medium	1	2–3	4–6
Variable-Speed Laser, Large	1–2	3–5	6–9
X-Pulse Laser, Small	1	2	3
X-Pulse Laser, Medium	1–2	3–4	5–6
X-Pulse Laser, Large	1–3	4–6	7–9
Laser Anti-Missile System	—	—	—

* Minimum range 3.

URBAN COMBAT

Several published mapsets have buildings printed directly on the mapsheets. Players can also use counters to mark buildings—any type will do, but the building's type and level should be clearly marked, and the counter should fit appropriately within the hex or hexes it will occupy on the mapsheet. If any counters are to represent a single, continuous building, this must be noted clearly for all players.

Rules for rarer and more complicated structures (from railway stations all the way up to the legendary Castle Brian Star League megafortress) can be found in *Tactical Operations*.

BUILDING TYPES

BattleTech divides buildings into four types: light, medium, heavy, and hardened. Each building will also have two values: its level, and its Construction Factor. Type is a permanent attribute, while Construction Factor and level may fluctuate based on damage inflicted to the building.

Default Values: If a scenario does not specify a building's type, treat it as medium, Level 2.

CONSTRUCTION FACTOR

The Construction Factor (CF) determines how a building's physical structure affects play. This number applies to a hex as a whole; each level of a building hex does not have its own CF.

The CF is the number of damage points a building hex can take before being reduced to rubble. This damage, regardless of when it occurs, is tracked across a game. The CF also represents the number of tons of additional weight each level of a building hex can support without collapsing.

The range of possible CF values for each building type appears on the Building Modifiers Table, page 67.

Current CF: The CF of a building is always defined at the instant an action takes place. For example, if two 'Mechs enter the same building hex at different times in the same Movement Phase, the CF as far as both 'Mechs are concerned will be different, as the first 'Mech will have damaged the building hex.

Default Values: If the scenario does not specify a building's CF, assume that a light building hex has CF 15, a medium building hex has CF 40, a heavy building hex has CF 90, and a hardened building hex has CF 120.

Multiple-Hex Buildings: For buildings that cover more than one hex, each hex has its own CF value. The destruction of a single hex in a multi-hex building does not affect any other hex within the same structure, unless more than half the total hexes have been destroyed. For more information, see *Collapse* (see p. 73).

BUILDING LEVELS

Treat building levels exactly like other terrain levels for line of sight and movement, with each level of a building measuring roughly six meters high. These levels do not represent a specific number of floors, but rather are an abstraction of the building's interior.

STRATEGIC

SIMPLIFIED BUILDINGS

Urban combat can be fun, but it also introduces additional Piloting Skill Rolls and the need to track damage to buildings. For those in search of simpler, quicker games, a common approach is to treat all buildings as hills of an equivalent level that can be neither entered nor destroyed.

Building levels affect movement in the same way as all other terrain levels. In other words, a 'Mech can move to the roof of a building hex, and while staying on roofs change levels as though each building were a hill (provided the 'Mech does not change more than its maximum allowed level changes in a single hex and the CF of each hex can support the 'Mech).

BASEMENTS

A building's description—or the scenario—states whether or not it has a basement. Otherwise, players determine if a building has a basement—and the effect of a 'Mech falling into it—by rolling 2D6 and consulting the Basements Table below whenever a 'Mech enters a building hex at Level 0 for the first time in a scenario. Only one basement roll is made for each building hex per scenario.

A 'Mech automatically falls through the floor into a basement if it is on Level 0 and its tonnage is greater than the building hex's CF at the start of the current Movement Phase. Use the Facing After Fall Table (see p. 57) to determine the 'Mech's resulting facing and which facing to apply the fall damage to. A collapsed basement creates a sublevel below the ground floor of the building hex, with a depth equal to the number of basement levels (1 or 2).

A collapsed basement will not automatically collapse the building hex but the building hex will take damage equal to the tonnage of the falling 'Mech divided by 10 (round fractions up). Double the damage for a two-level basement.

BASEMENTS TABLE

Roll (2D6)	Effect
2	Double basement. The 'Mech falls 2 levels.
3-4	Basement. The 'Mech falls 1 level.
5-8	No basement.
9	Small basement. No effect on 'Mechs.
10-11	Basement. The 'Mech falls 1 level.
12	Double basement. The 'Mech falls 2 levels.

MOVEMENT EFFECTS

'Mechs can move into, onto, or through buildings. The rules in this section cover entering building hexes below the level of the roof; i.e. movement actually inside the building in a hex, as opposed to entering a hex on the roof of the building. The Building Modifiers Table below summarizes movement costs and modifiers for each type of building.

BUILDING MODIFIERS TABLE

Building Type	Original CF	MP Cost Per Hex	Piloting Skill Modifier
Light	1–15	+1	0
Medium	16–40	+2	+1
Heavy	41–90	+3	+2
Hardened	91–150	+4	+5
Cost to Enter Any Hex		*1*	

BUILDING MOVEMENT MODIFIERS TABLE

Hexes Moved In Turn	Piloting Skill Modifier	Hexes Moved In Turn	Piloting Skill Modifier
1–2	0	10–17	+4
3–4	+1	18–24	+5
5–6	+2	25+	+6
7–9	+3		

MOVING INTO BUILDINGS

'Mechs can enter a building hex. By default this is done at the building's ground level (i.e. the level of the underlying hex the building is in). A 'Mech may only enter a building hex at a level higher than the building's ground level if entering from an adjacent hex with a higher level (e.g. if a 'Mech walks from a Level 3 hill into a neighboring building hex on a Level 1 hill, it would enter the building at Level 2).

When a 'Mech voluntarily enters a building hex (not including moving onto the roof), its controller must make a Piloting Skill Roll (see p. 53). Modify the roll using the table above, based on both the building type (on the Building Modifiers Table) and the number of hexes moved at the time of the PSR (on the Building Movement Modifiers Table). If the roll fails, the 'Mech takes damage equal to the building hex's current CF divided by 10 (round up). This damage hits from the direction the 'Mech moved into the hex (e.g. a quad 'Mech shifting left into a building takes damage to its left side). A failed roll does not cause the 'Mech to fall.

Regardless of the result of this Piloting Skill Roll, the hex suffers damage equal to the 'Mech's tonnage divided by 10 (rounded up).

Once inside a building, a 'Mech may not change levels, except as noted under *Basements*, page 66.

Collapse: If a 'Mech's tonnage on any single level of a building hex exceeds the hex's current CF (except Level 0, the ground floor; see *Basements*, p. 66), the entire building hex immediately collapses. See *Collapse*, on page 73.

Falling: If a 'Mech falls while in a building hex, damage is resolved to both the 'Mech and the building as if the 'Mech had voluntarily entered the building hex and failed its PSR.

Involuntary Entry: 'Mechs that involuntarily enter a building hex (after being displaced or via a skid) use the *Displacement* or *Collisions With Stationary Objects* rules to resolve such movement (see pp. 55 and 70, respectively).

Jumping: A 'Mech cannot use Jumping MP to move into a building, though it can jump onto a building roof. However, it can use Jumping MP to leave a building hex, provided it does not enter another building hex that Movement Phase other than by its roof.

Stacking: The stacking limit (see p. 18) is applied to each level of a building hex separately, rather than the hex as a whole. For example, a five-level building hex could have a 'Mech in each level of the building, even though they would all be in the same hex (normally forbidden).

SKIDDING

If a 'Mech using running movement makes a facing change in a paved hex at any point in the turn, and then attempts to enter a new hex in the same turn, before that 'Mech actually enters the hex the player must make a Piloting Skill Roll (see p. 53) to see if the 'Mech skids. The roll is modified based on the total number of hexes moved in the turn so far, using the Skid Modifiers Table (see below).

If the roll succeeds, the 'Mech does not skid and can enter the hex and continue its movement; multiple rolls could thus occur in a turn. If it fails, the 'Mech falls in its hex (however, do not resolve the fall damage yet as that is applied later; see *Damage*, below) and begins to skid.

A 'Mech skids for a number of hexes equal to the number of hexes it has moved in the turn so far (not MP spent), divided by 2 (round up). The skid takes the 'Mech in the direction it was travelling before making the facing change that caused it to skid. If a 'Mech skids off the playing area, it is considered destroyed for the remainder of the scenario.

After a skid, the 'Mech's movement ends, even if it had MP remaining.

Combat Effects: During a turn in which a 'Mech skids, add a +2 Target Number modifier for all weapons fire and physical attacks made against it, and a +1 Target Number modifier for all weapons fire and physical attacks (except unintentional charges) made by it. Hexes skid in a turn count towards a 'Mech's target movement modifier for the turn.

Damage: Skidding damage is calculated at the end of the skid (except in the case of a potential collision, which interrupts this process; see p. 70). It is always applied in 5-point Damage Value groupings.

First calculate the 'Mech's fall damage (see *Falling*, p. 56). Then calculate its skid damage, which is one-half its normal falling damage, rounded up: multiply this result by the number of hexes it skid. Add together the damage from the fall and all hexes skid into one "attack". For example, a 65-ton 'Mech that falls (7 damage) and skids 2 hexes (4 damage per hex) must resolve 15 damage (in 5-point groupings), not 7 damage, 4 damage, and 4 damage.

Use the column of the Hit Location Table determined by the Facing After Fall Table to apply this damage.

SKID MODIFIERS TABLE

Hexes Moved	Piloting Skill Modifier
0-2	–1
3-4	0
5-7	+1
8-10	+2
11-17	+4
18-24	+5
25+	+6

SKIDDING DIAGRAM 1: INITIAL MOVEMENT

In Skidding Diagram 1 on page 69, a player wants their Phoenix Hawk *PHX-S3 in Hex A to run to Hex H. It has a Piloting Skill of 5. As part of its move, the 'Mech runs to Hex C and makes a facing change toward Hex D. No Piloting Skill Roll is required.*

However, when the Phoenix Hawk *attempts to enter Hex D, the player must make a PSR before entering the hex, because the 'Mech will be running as it moves into a new hex after making a facing change on pavement.*

So far, the Phoenix Hawk *has moved 2 hexes, and so per the Skid Modifiers Table it applies a –1 modifier to its PSR. With a Piloting Skill of 5, the player needs to roll a 4 or higher to avoid skidding. The player rolls a 10, and the 'Mech continues to run toward Hex H.*

The Phoenix Hawk *makes another facing change in Hex E toward Hex F. In order to move safely from Hex E to Hex F while running on pavement, the player must make another PSR, this time modified by 0 because the 'Mech has moved 4 hexes. The Target Number is 5 (5 + 0).*

SKIDDING DIAGRAM 1: SKIDDING

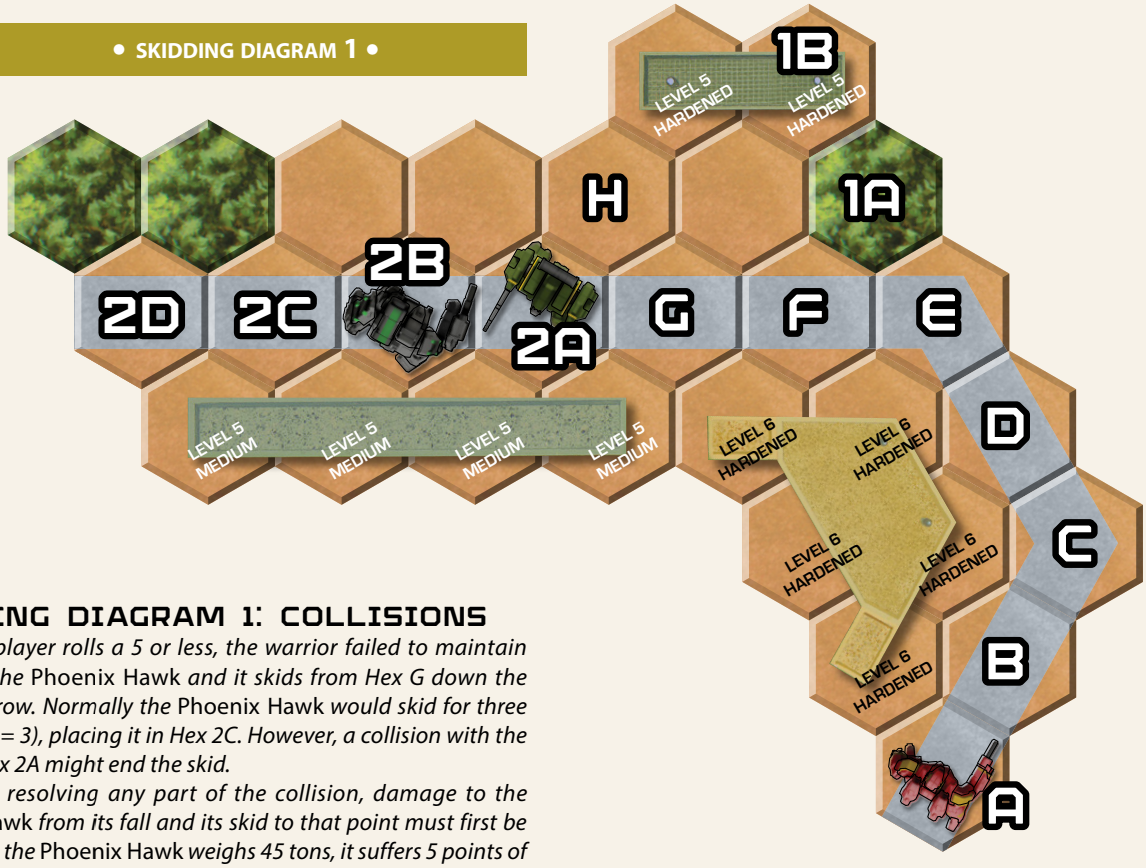
If the player fails the PSR (rolls a 4 or less), the Phoenix Hawk *skids down the 1A-1B hex row. Normally it would skid for two hexes (4 / 2 = 2), placing it in Hex 1B. However, the light woods in Hex 1A must be taken into account. The Movement Point Cost Per Hex for light woods is 1, so that is subtracted from the total amount of hexes to be skid, meaning the 'Mech only skids 1 hex into 1A.*

The 45-ton Phoenix Hawk *suffers 5 points of damage falling into hex E where it failed its Piloting Skill Roll (45 tons divided by 10 is 4.5, rounded up to 5) and 3 points of damage per hex it enters during the skid (half the falling damage of 4.5, rounded up) for a total of 8 (5 + 3 = 8); it only skid 1 hex and the woods don't deal additional damage. Once the skid is finished, the player rolls on the Facing After Fall Table (see p. 57) to determine the Hit Location Table column (see p. 33) where the falling/skid damage will be applied: one 5-point Damage Value grouping and one 3-point grouping.*

If the light woods had not been in Hex 1A, the Phoenix Hawk *would have skid into the building Hex of 1B, which would have required its controller to resolve the damage to both the 'Mech and the building hex as though the 'Mech had charged the building (see* Charge Damage, *p. 35).*

If the player instead passes the PSR (rolls a 5 or greater), then the Phoenix Hawk *continues to run towards Hex H. The 'Mech makes another facing change in Hex G towards Hex H. Once more, in order to move safely from Hex G to Hex H while running on pavement (even though the hex the 'Mech is moving into is not pavement), the player must make another Piloting Skill Roll, this time modified by +1 because the 'Mech moved 6 hexes. The Target Number is 6 (5 + 1).*

• SKIDDING DIAGRAM 1 •



SKIDDING DIAGRAM 1: COLLISIONS

If the player rolls a 5 or less, the warrior failed to maintain control of the Phoenix Hawk *and it skids from Hex G down the 2A-2D hex row. Normally the* Phoenix Hawk *would skid for three hexes (6 / 2 = 3), placing it in Hex 2C. However, a collision with the 'Mech in Hex 2A might end the skid.*

Before resolving any part of the collision, damage to the Phoenix Hawk *from its fall and its skid to that point must first be resolved. As the* Phoenix Hawk *weighs 45 tons, it suffers 5 points of damage falling into Hex G, where it failed its Piloting Skill Roll (45 tons divided by 10 is 4.5, rounded up to 5. A* Phoenix Hawk *would take 3 points of damage per hex skid (half the falling damage of 4.5, rounded up), but since a collision scenario began, the* Phoenix Hawk *interrupts its skid in Hex G to resolve it, and so hasn't skid any hexes yet. The player rolls on the Facing After Fall Table (see p. 57) to determine its facing and the Hit Location Table column (see p. 33) the damage is applied to. Assuming the skidding* Phoenix Hawk *survives, only then is the potential collision resolved.*

The 'Mech in Hex 2A has not yet moved this phase. The Target Number for the resulting unintentional charge attack is 11: 5 (base Target Number) + 2 (attacker ran) + 0 (no target movement modifier, as the target has not yet moved this phase) + 1 (skidding) + 3 (unintentional charge). The Phoenix Hawk*'s controller rolls an 11! But because the target 'Mech has not yet moved, it can still avoid the successful charge by making a Piloting Skill Roll (in this case, needing a result of 4 or better). The enemy 'Mech's controller, however, rolls a 3, which means it failed to dodge.*

As the falling and skidding damage happens first, this might result in the Phoenix Hawk *being destroyed before the collision actually occurs. If so, then the target 'Mech would still be able to move normally this turn. If not, the target 'Mech cannot move again later in the Movement Phase.*

The Phoenix Hawk*'s charge deals 27 points of damage: 4.5 (45 tons divided by 10) × 6 (the number of hexes the* Phoenix Hawk *entered during the turn, not counting skidding hexes). Its controller then rolls six hit locations: five 5-point Damage Value groupings and one 2-point grouping, all applied to the front side of the Kick Location Table. The* Phoenix Hawk *also takes damage from the successful charge, equal to 1 point of damage for every 10 tons that the target weighs (round fractions up), applied in 5-point Damage Value groupings. The attack comes from the right hexside of Hex 2A, since that's the hex facing the* Phoenix Hawk *collided with, and is applied to the* Phoenix Hawk *based on the facing it rolled when it fell.*

At the end of the successful charge, the Phoenix Hawk *ends its move in Hex 2A (even though it would normally have skid for two more hexes, its unintentional charge automatically ends its skid and its movement in general). The target 'Mech is moved into Hex 2B. The* Phoenix Hawk *is already prone, and so its controller need not make a Piloting Skill Roll to avoid a fall, but the target 'Mech must make the standard PSR to avoid a fall after impact; if it fails, the target 'Mech will fall into Hex 2B.*

Finally, the displacement of the original 'Mech in Hex 2A to Hex 2B automatically causes a domino effect to the 'Mech already in Hex 2B. If the 'Mech already in 2B had not yet moved this turn, it could attempt to get out of the way (see Domino Effect, *p. 55); if it had already moved, it will be moved to Hex 2C, after which the 'Mechs now in Hexes 2B and 2C each must make a PSR to avoid falling.*

As the Phoenix Hawk *skid that turn, once all effects are resolved its movement would be over, regardless of if it still had MP left.*

INTERRUPTING A SKID

Accidental Falls: If the level of the hex to be entered is two or more lower than the level of the skidding 'Mech's current hex, the 'Mech does not take skid damage from skidding into that hex, but automatically falls in that hex (see p. 56). The damage from the accidental fall is combined with the damage from the initial fall that started the skid plus any hexes skid and is all resolved together; make only one roll on the Facing After Fall Table to resolve both falls (though you still make two PSRs to avoid MechWarrior damage from the falls). The 'Mech's skid ends in the hex fallen into.

If there is already a 'Mech in the hex to be fallen into, instead see *Fall From Above*, under *Collisions With 'Mechs* at right.

Collisions: Collisions interrupt a skid and may completely stop it, but are dealt with separately (see *Collisions—General* at right).

Damage: Damage can be applied at various stages during a skid, and is resolved immediately when applied. If a skidding 'Mech is destroyed at any point in its skid, its skid ends immediately and no displacement (if any) is resolved.

Terrain: If a skidding 'Mech moves into a terrain hex (not including buildings, level changes, or water), other than clear and/or paved hexes, then that terrain reduces the distance it skids. For each such hex entered, subtract the Movement Points required to enter that terrain type from the remaining hexes to be skid, as shown on the Movement Costs Table, page 17. The terrain applies no additional damage. If the 'Mech would enter a hex that would reduce its remaining hexes to move to 0 (or less), it still enters that hex, ending its movement there.

For example, if a 'Mech skidding through six hexes enters light woods for its third hex, it would subtract two from the hexes left to skid (2 MP for a light woods hex), leaving it two hexes left to skid. If the next hex it entered were a heavy woods hex, that would subtract three hexes (3 MP for a heavy woods hex), which it doesn't have left, but it would skid into that hex regardless, ending its skid there.

Additionally, if the level of the hex to be entered is one or more higher than the level of the skidding 'Mech's current hex, the skidding 'Mech automatically crashes into it and stops (see *Levels*, under *Collisions with Stationary Objects*, p. 71).

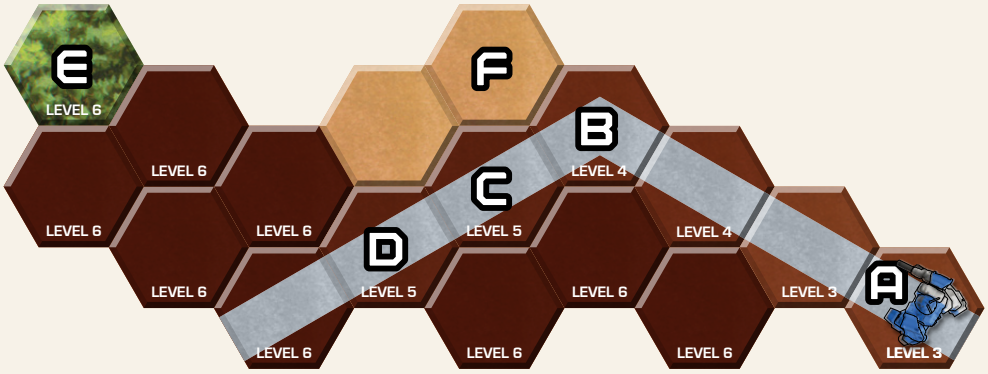
Water: 'Mechs automatically end their skid if they enter Depth 1 or deeper water, taking normal skidding or falling damage for entering that hex (based on how they entered it) and so risking a hull breach (see p. 65).

COLLISIONS—GENERAL

If an obstacle (e.g. a building, a hill, or a 'Mech) lies in the path of the skid, the skidding 'Mech may collide with it.

If a roll is to be made to see if a collision might occur (or one is automatically about to occur), the normal skidding damage resolution process is interrupted. When the interruption occurs, determine the skidding 'Mech's damage from its initial fall plus any skidding damage from the number of hexes skid up to and including where the skid was interrupted (if any), and resolve that together. Only then (if the skidding 'Mech survives) is the collision scenario resolved. If a 'Mech continues to skid after resolving it, the additional skidding damage is resolved afterwards. Each time a skid is interrupted afterwards with another collision scenario, apply all skidding damage to that point that has not yet been resolved.

For example, if a 'Mech falls and skids for five hexes, but runs into a building hex in the second hex of its skid—which it destroys and continues to skid—the damage from the fall and two hexes of skid would be applied together first, then the collision damage, then the damage from the collapsing building, and finally the damage from the additional hexes skid would be resolved.



• SKIDDING DIAGRAM 2 •

SKIDDING DIAGRAM 2: SKIDDING

In Skidding Diagram 2, Player 1 wants his 55-ton Griffin C *in Hex A to end its movement in Hex D. To spend the required 8 MP, this 'Mech must run (it has a Walking MP of 6 and a Running MP of 9). The* Griffin *runs to Hex B. When it attempts to enter Hex C, its controller must make a Piloting Skill Roll (due to the road) against a Target Number of 4 (4 for the MechWarrior's Piloting Skill, + 0 because the 'Mech has only moved 3 hexes at this point).*

If the roll is successful, the 'Mech can move to Hex D. There, the controlling player—thinking ahead to where he wants his 'Mech to be in the next turn—will spend the Griffin's *remaining 1 MP to turn to Hexside 6, in preparation for moving to the light trees in Hex E.*

If the roll in Hex B fails, the Griffin *skids into Hex F. Hex B has a level of 4, while Hex F has a level of 0. This four-level drop means that the controlling player must consult the* Falling *rules (see p. 56) to determine damage and results. In this case, the 'Mech takes 36 points of damage (6 for the initial fall—there is no damage for the 1-hex skid as the 'Mech skids 'into air' when it enters Hex F—and 30 for the accidental fall from the level change). The player assigns the damage in 5-point Damage Value groupings, using the appropriate column of the table once the facing has been determined (see* Facing After a Fall, *p. 56).*

Had any other 'Mech been in Hex F when the Griffin *fell off the cliff, a collision might have occurred, which would have required the players to use the* Accidental Falls From Above *rule (see p. 57).*

Collision damage is applied in 5-point Damage Value groupings. For the purposes of calculating attack direction, damage to a skidding 'Mech from colliding with something is treated as if it came from the hex facing that the skidding 'Mech collided with (i.e. it always comes from the opposite direction that a 'Mech is skidding in). Unless specified otherwise, the Hit Location Table column used to assign damage to a skidding 'Mech is based on that 'Mech's facing relative to this attack direction, just like any other attack.

When calculating charge damage from a collision with a building or 'Mech, use only the distance the skidding 'Mech moved this turn prior to its skid: don't count the hexes it skid.

Multiple collisions can occur in a single skid.

COLLISIONS WITH STATIONARY OBJECTS

Empty Buildings: A 'Mech that skids into an empty building hex moves into that hex, interrupts its skid there, and then resolves a collision. The hex takes damage as if the skidding 'Mech had charged it (see *Charge Damage*, p. 35). The 'Mech takes damage equal to the building hex's current CF divided by 10 (round up). Check for a basement (see p. 66); if there is a basement, the fall (if any) occurs first, before a building collapse (if the building takes enough damage to collapse). If the building hex is not destroyed or if the 'Mech fell into a basement, the 'Mech's skid ends in that building hex.

If the building hex is destroyed and the 'Mech did not fall into a basement, and the skidding 'Mech still has hexes left to skid, look at the MP Cost column on the Building Modifiers Table (see p. 67), and subtract that number of MPs for the type of building hex destroyed from the remaining hexes to be skid. After subtracting, if there are still hexes to be skid, the skid continues.

For example, if a 'Mech skid into a Medium Building hex with no basement, destroyed that building hex, and still had four hexes to skid, the 'Mech's controller subtracts 2 MP (after looking at the MP Cost column of the Building Modifiers Table), leaving the 'Mech to skid two more hexes. However, if the building hex was not destroyed then the skid ends, with the skidding 'Mech now in the building hex.

Occupied Buildings: If a 'Mech would skid into a building hex occupied by another 'Mech, resolve the scenario as a collision with a 'Mech (see below), though the building absorbs damage as normal (see *Attacking 'Mechs Inside Buildings*, p. 71). If the 'Mech already in the building is displaced or destroyed as a result of this collision, the skidding 'Mech then moves into the building hex and a collision with the now-empty building is resolved.

Levels: If the level of the hex to be entered is higher than the skidding 'Mech's current hex, the 'Mech instead stops in its current hex, interrupts its skid there, and then suffers a collision. The 'Mech takes 1 point of damage for every 20 tons it weighs (round fractions up) and then its skid ends.

COLLISIONS WITH 'MECHS

If there is another 'Mech in the path of a skid, the skidding 'Mech stops in the hex immediately prior to it, interrupts its skid there, and then checks to see if it collides with the other 'Mech. Resolve this as a charge, applying the +3 modifier for an unintentional charge but not the comparative modifier for differences in Piloting Skill between attacker and target. If the unintentional charge misses, the skidding 'Mech misses the target 'Mech and continues its skid.

If the unintentional charge succeeds, and the target does not dodge (see below) or does dodge but fails to leave its hex, the skidding 'Mech resolves damage as per a charge (see *Charge Damage*, p. 35). Damage to the target 'Mech is rolled on the Kick Location Table on page 38 (unless the target is prone, in which case use the 'Mech Hit Location Table on p. 33). This is resolved immediately and, unlike a standard charge, can affect a 'Mech that has not yet moved.

After the charge is resolved, if the target 'Mech was destroyed then the skid continues. If the target 'Mech was not destroyed, adjust the position of the 'Mechs involved per the rules for a successful charge given under *Location After Charge* (see p. 36); the skidding 'Mech takes an additional hex of skidding damage for entering the target's hex. The skid then ends.

Dodging a Collision: A 'Mech that has not yet moved during the current Movement Phase can attempt to dodge a successful unintentional charge.

The target 'Mech's controller must make a Piloting Skill Roll. Success means the target 'Mech may make its movement immediately. Failure means the target remains in its hex (this failure does not cause a fall). Whether successful or not, the dodging 'Mech cannot move again later in the Movement Phase.

Fall From Above: If a skidding 'Mech will accidentally fall into a hex occupied by another 'Mech, the skid is interrupted in the hex prior to the fall. Then, instead of an unintentional charge, you resolve an accidental fall from above (see p. 57); no skidding damage occurs for moving into the hex fallen into. Afterward, the unit's skid ends.

COMBAT EFFECTS

Combat in and around buildings may damage the buildings and the 'Mechs inside. Players may attack 'Mechs inside buildings, or the buildings themselves.

Art vs. Rules: For LOS purposes, a building always fills its hex, regardless of what the art may show on the playing map.

ATTACKING BUILDINGS

Building hexes are immobile targets (–4 Target Number modifier). However, weapon attacks targeted at a building hex from adjacent hexes always hit, as do all physical attacks (though players must still roll for weapon effects, such as jamming if firing an Ultra autocannon at double-rate, for example). Always use the full Damage Value for cluster weapons that strike an adjacent building hex; i.e. cluster type weapons do not use the Cluster Hits Table in this case.

When a building hex suffers damage, subtract the weapon's Damage Value from the building hex's current CF. When the cumulative damage equals or exceeds the building hex's CF, the hex becomes rubble for the rest of the game. If it is part of a multi-hex building, this may or may not result in the rest of the building hexes collapsing (see *Collapse*, p. 73).

ATTACKING 'MECHS INSIDE BUILDINGS

'Mechs inside a building hex may be attacked. An attacker must have line of sight to the building hex, as well as to the target 'Mech. One intervening building hex blocks LOS but the building hex a 'Mech is in does not count (since it is not intervening terrain).

Buildings protect against attacks. A building reduces the damage of each hit on a 'Mech inside it by its current CF / 10 (round up; treat each Damage Value grouping as a separate hit). The building hex takes that amount of damage from the hit. Resolve damage to the building's CF at the end of the Weapon Attack Phase.

Shots fired at a 'Mech inside a building hex that miss do not damage the building (neither does any damage not applied to a 'Mech from a successful cluster attack). The exception is if the attacker is adjacent to the building, in which case all missed shots automatically strike the building.

AE Damage: Area-effect attacks ignore the above damage reduction rules (see *Area-Effect*, p. 96).

Partial Cover: A standing 'Mech occupying a Level 1 building or a building hex only one level below the roof automatically receives partial cover, even if its attacker is at a higher level. If the hit location roll indicates a leg, the attack strikes the building instead, damaging it as normal.

Physical Attacks: Physical attacks (other than charges) cannot be made against targets in a building hex, unless the attacker is in the same multi-hex building, on the same level, and is in a hex adjacent to the target (see *Combat within Buildings*, at right).

COMBAT WITHIN BUILDINGS

'Mechs in the same building—whether in the same building hex but on different levels, or in different hexes of a multi-hex building—can attack one another. Use the standard attack procedures, with the following modifications:

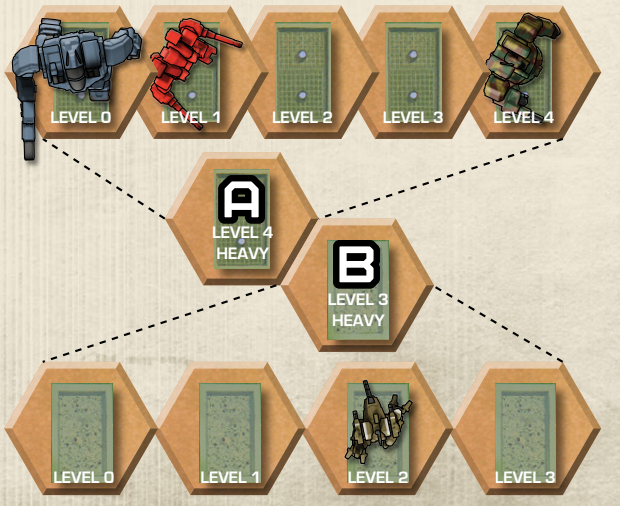
- Line of sight is blocked if more than two building hexes and/or levels intervene between the attacker and target hex.
- Apply an automatic +1 Target Number modifier if the target 'Mech is inside a building hex (i.e. not on a roof) and another +1 for each intervening building hex and/or level.
- Always use the Front/Rear column of the standard Hit Location Table to resolve hits (see p. 33). If the target is in the same hex as the attacker, treat it as if it were in the attacker's forward arc.

Rooftops: A 'Mech on a building's roof is not considered inside a building. Nonetheless, if a 'Mech inside a building is firing at another 'Mech on the roof of the same building (or vice versa), use the line of sight and Target Number modifiers given in the bullet points above.

Attacks by a 'Mech on a roof against a 'Mech inside another building use the rules for *Attacking 'Mechs Inside Buildings*.

Several opponents make successful attacks against a Hunchback *inside a medium building hex, hitting it with an AC/10, a small laser, a large laser, and nine missiles from an LRM-15. The building hex has a current CF of 38. Each attack's damage is thus reduced by 4 (38 ÷10 = 3.8, rounded up to 4).*

As such, the AC/10 inflicts 6 points of damage on the Hunchback. *The small laser inflicts no damage, while the large laser inflicts 4 damage points. The LRM damage is divided into a 5-point grouping and a 4-point grouping. One point of the 5-point grouping reaches the* Hunchback, *while the 4-point grouping is entirely absorbed by the building (the six missiles from the LRM-15 that did not strike the target do not damage the building hex). The* Hunchback *takes a total of 11 points of damage, while the building hex's CF is reduced from 38 to 19 (38 – 4 – 3 – 4 – 4 – 4 = 19) at the end of the Weapon Attack Phase.*



• BUILDING COMBAT DIAGRAM •

In the Building Combat Diagram at left, Player 1 has a BattleMaster *(Gunnery 4) on the hunt and has run to enter Building Hex A at Level 0. Player 2 has three 'Mechs: a* Wolverine *that did not move this turn at Level 4 (the roof) in Hex A, a* Warhammer *that moved 5 hexes to Level 1 of Hex A from a neighboring building, and a* Locust *on Level 2 in Hex C. The building has a CF of 60 on every level.*

During the Weapon Attack Phase, Player 1 looks for a target for his BattleMaster's *machine guns. He sees that three building hex levels are between the* BattleMaster *(Level 0, Hex A) and the* Wolverine *(Level 4, Hex A). More than two hexes and/or levels of intervening building between attacker and target means line of sight is not valid, and so Player 1 cannot attack the* Wolverine.

The Target Number to attack the Warhammer *is 10 (4 Gunnery, + 2 because the* BattleMaster *ran, + 2 for the target movement modifier, + 1 for combat within buildings, and + 1 for the intervening level). That's rather high, and so the* BattleMaster's *player looks for an alternate target.*

The Target Number to attack the Locust *is 8 (4 Gunnery, + 2 because the* BattleMaster *ran, + 1 for combat within buildings, and + 1 for the intervening level), making the* Locust *the best target. However, even if the attack strikes the* Locust, *the 2 points of damage from the machine gun will not damage it. This is because the building has a CF of 60: as a building absorbs 10% of its current CF from every attack directed against 'Mechs inside it, this means the* BattleMaster *would have to do at least 7 points of damage to hurt the target* Locust. *All the* BattleMaster *would accomplish is reducing the building's CF on Level 2 by 2 points (the damage of the machine gun).*

COLLAPSE

A building hex will collapse if it takes total damage equal to or greater than its CF, or if the tonnage of a 'Mech occupying any one level of a building hex above the ground floor exceeds the current CF of that building hex; regardless of whether a building has a basement or not, this never includes Level 0 of a building hex (see *Basements*, p. 66).

Because CF is tracked per hex of a multi-hex building, the collapse of one hex will not affect any other hex in that building, unless it brings the total of collapsed hexes to more than half, in which case the entire building collapses.

When a building hex (or a multi-hex building) collapses because its maximum weight limit has been exceeded, or because of damage from 'Mech movement, it collapses immediately. If the collapse is caused by damage from attacks, the collapse occurs at the end of the phase.

When a building hex collapses, any 'Mech in it takes damage. The base damage is equal to the hex's CF at the start of the phase divided by 10. If there are building levels above the 'Mech, add to this the base damage multiplied by the number of building levels above that 'Mech. Round the final result up. 'Mechs on the top level or roof apply no multiplier (i.e. the roof does not count as a level above for this purpose).

Divide this damage into 5-point Damage Value groupings, with any remaining points assigned to a final grouping. Roll the hit location for each grouping. For 'Mechs on the roof of a building hex, use the Front column of the Hit Location Table; for 'Mechs inside a building hex, use the Front column of the Punch Hit Location Table.

Basements and Collapse: When a building hex collapses, the resulting rubble will fill in any basement the building might have had (i.e. do not roll for a basement). A building with a basement becomes a rubble hex equal to the level of the terrain underneath the building, not a sublevel rubble hex. If the collapse occurs as a 'Mech moves into the building hex at Level 0 for the first time in the scenario, its controller need not make a basement roll (see *Basements*, p. 66).

If a 'Mech occupies a basement when the building hex above it collapses, that 'Mech is destroyed.

Falling: 'Mechs occupying a building level higher than 0 (including the roof) suffer standard falling damage (see *Falling Damage to a 'Mech*, p. 57) in addition to the collapse damage.

DAMAGE AND DISPLACEMENT

If the 'Mech fails its Piloting Skill Roll for entering a hex, damage from moving into the hex is applied first. Then all damage from the collapsing building hex or hexes is resolved, for all 'Mechs involved, followed by falling damage (if any; see *Falling Damage To A 'Mech*, p. 57) and then displacement (accidental falls from above, domino effect and so on; see *Displacement*, p. 55). With displacement, 'Mechs on the lowest levels are resolved first, then 'Mechs in higher levels.

If enough 'Mechs are occupying different levels of a multilevel building hex when it collapses, and those 'Mechs survive the collapse, some 'Mechs may be displaced multiple times. If a 'Mech cannot be displaced (for example, all the adjacent hexes are prohibited terrain), the 'Mech is destroyed.

COLLAPSE: MOVEMENT

In the Building Collapse Diagram on page 74, Player 1 wants the following to happen: a Locust *to walk inside a Level 2 building from Hex A to Hex B and a* Stinger *to jump to Hex B. (Because a 'Mech cannot enter a building hex using Jumping MP, the* Stinger *must land on the roof, and so it will end up at Level 3). Player 1 lost the Initiative, and so Player 2 plans to move his* Archer *last, walking from Hex C through Hex D to Level 0 of Hex E.*

The building itself has taken significant damage. Each hex with an X represents a collapsed building hex that is now rubble, while hexes D and E each only have 7 CF remaining. Hex B has taken 1 point of damage, leaving it with a CF of 39.

As the Locust *enters Hex B, Player 1 must make a Piloting Skill Roll to avoid damage against a Target Number of 7: 5 (Piloting Skill) + 1 (four hexes moved, including the building hex the 'Mech is attempting to enter) + 1 (Medium Building). Whether the roll is successful or not, Hex B takes 2 points of damage (20 for the tonnage of the* Locust *divided by 10), leaving the building hex with a current CF of 37.*

After the Locust *moves, the* Stinger *jumps to the building's roof in Hex B. No Piloting Skill Roll is required to enter the hex, nor is the hex damaged.*

The combined weight of the Locust *(20 tons) and* Stinger *(20 tons) is 40 tons, more than the building's current CF of 37. However, because these 'Mechs are located on two different levels in the hex, their combined weight does not collapse the hex.*

Player 2 can now make his planned move. His Archer *first walks from Hex C to Hex D, ending up at Level 0 inside the building in Hex D. As with the* Locust*, Player 2 must make a Piloting Skill Roll to avoid damage upon entering the building hex, against a Target Number of 6: 5 (Piloting Skill) + 1 (Medium Building). Regardless of the roll's success or failure, the building hex takes 7 points of damage (70 for the tonnage of the* Archer *divided by 10 = 7).*

COLLAPSE: HEX COLLAPSE

Hex D has a current CF of 7, and so the 7 points of damage caused by the Archer *collapses it. This collapse must be immediately resolved.*

Because the building collapses as the Archer *is entering the hex, no roll is needed to determine if the building hex has a basement. The collapse of Hex D does not exceed more than half the total hexes of the multi-hex building. The entire building spans eighteen hexes, of which Hex D is the ninth to collapse; the rest of the building hexes remain unaffected.*

Player 2 applies damage from the collapse after any damage received from entering the hex, if the Archer *failed its Piloting Skill Roll. At the beginning of the Movement Phase, Hex D's CF was 7; divided by 10, this gives a base damage of 0.7. The building had two levels above the* Archer*, and so the 'Mech takes 3 points of damage (base damage 0.7 x 3 = 2.1 (rounded up to 3). Player 2 assigns the 3 damage points using the Front/Back column of the Punch Hit Location Table.*

Having applied all appropriate damage, Player 2 can continue the Archer*'s movement into Hex E. As before, he must make a Piloting Skill Roll to avoid damage upon entering this building hex, against a Target Number of 6. If this PSR fails, damage is taken immediately. Whether the PSR succeeds or not, Hex E takes 7 points of damage (70 for the tonnage of the* Archer *divided by 10 = 7). Like Hex D, Hex E has a current CF of 7, and so the* Archer*'s entrance collapses it. The collapse occurs as the* Archer *entered the hex, meaning no basement roll is required.*

CONTINUED ON P. 74

COLLAPSE: BUILDING COLLAPSE

The collapse of hex E means that ten of the building's eighteen hexes have collapsed. Ten is more than half, and so the rest of the building hexes automatically collapse as well. These events must be resolved immediately.

First, the players apply damage inflicted by the collapsing building. In the case of the Archer, *at the beginning of the Movement Phase, Hex E had a CF of 7. Divided by 10, this gives a base damage of 0.7. The building had two levels above the Archer, and so the 'Mech takes 3 points of damage (base damage 0.7 x 3 = 2.1, rounded up to 3). Player 2 assigns the damage using the Front/Back column of the Punch Hit Location Table.*

Player 1's 'Mechs, all in Hex B, fare somewhat worse. At the beginning of the Movement Phase, Hex B had a CF of 37; divided by 10, which gives a base damage of 3.7. Luckily, the Stinger *is on the roof, meaning that no building levels are above it. It therefore takes 4 points of damage (3.7, rounded up), assigned against the Front/Back column of the Hit Location Table. The* Locust *is at Level 2 in Hex B. With only 1 level above the 'Mech, it also takes 4 points of damage, applied against the Front/Back column of the Punch Hit Location Table.*

COLLAPSE: FALLING AND DISPLACEMENT

Next, the players resolve falling damage. The Locust *falls two levels and so takes 6 points of damage: 20 tons divided by 10, multiplied by the number of levels fallen +1 (2 levels +1 = 3). Player 1 rolls on the Facing After Fall Table (see p. 57) and gets a 1 (same direction, front). He assigns damage to the front of the* Locust *in one 5-point grouping and one 1-point grouping, using the Front/Back column of the Hit Location Table. The* Locust *ends its fall prone at Level 0 in Hex B (now a rubble hex).*

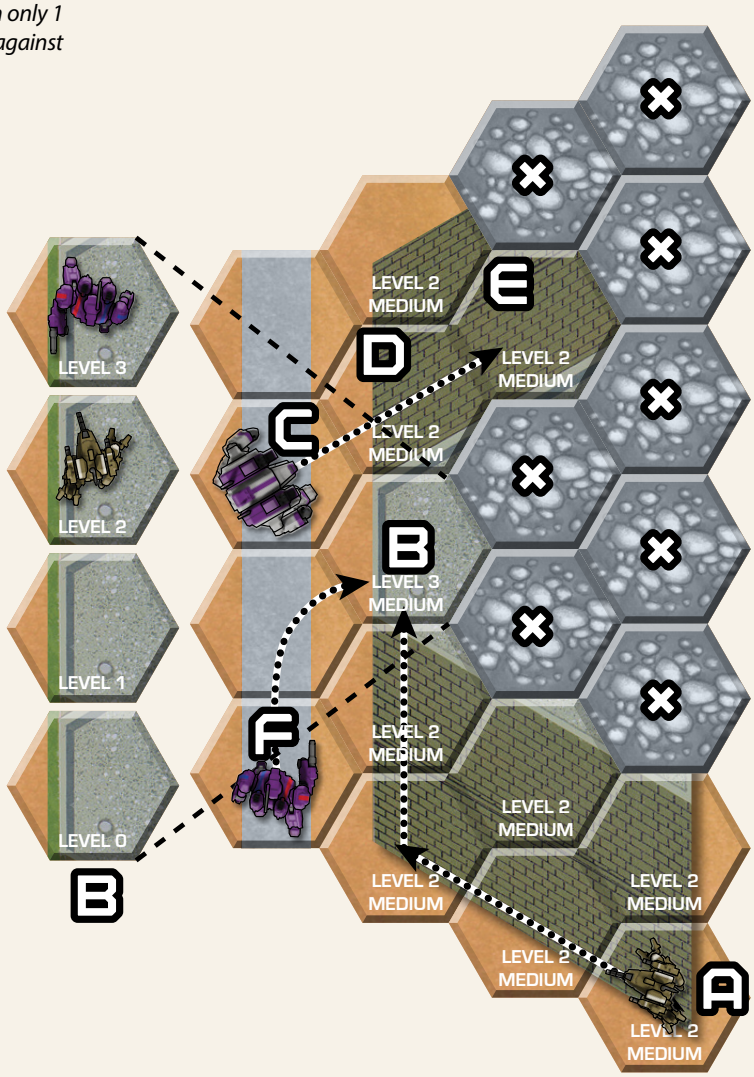
Player 1 next resolves the Stinger*'s fall. The* Stinger *was at Level 3, more than two levels higher than the Level 0 hex it's falling into. Also, the hex contains another 'Mech (the* Locust*), and so Player 1 must use the* Accidental Falls From Above *rule (see p. 57). The Target Number is 6 (7 for the base Target Number, + 1 for the target movement modifier, – 2 (target is prone and adjacent)).*

If the roll result equals or exceeds 6, the Stinger *strikes the* Locust *and inflicts 6 points of damage (the weight of the falling 'Mech divided by 10, x 3 for levels fallen) on the* Locust*'s rear, using the Front/Back column of the Punch Location Table. Player 1 then randomly determines an adjacent hex into which the* Locust *is displaced. If the result is less than 6, the* Stinger *does not strike the* Locust*, but instead is displaced into a randomly determined hex adjacent to Hex B. (If a building were still standing in that hex, the* Stinger *would take additional damage, but since every adjacent building hex is rubble, the 'Mech is off the hook.)*

Regardless of whether or not the Stinger *struck the* Locust*, the* Stinger *takes the same 8 points of falling damage: 20 tons (weight of the* Stinger*) divided by 10, multiplied by the number of levels fallen, + 1 (3 levels + 1 = 4). Player 1 assigns the damage to the back of the* Stinger *(per the rules for damage after an Accidental Fall From Above, see p. 57) in one 5-point grouping and one 3-point grouping, using the Front/Back column of the Hit Location Table. The* Stinger *ends the fall prone at Level 0 with a new facing. Its exact hex placement depends on whether the* Stinger *struck the* Locust *or not.*

If the Locust *were struck and randomly displaced into a hex that contained another 'Mech—or if the* Stinger *missed and were randomly displaced into such a hex—and if the entrance of either 'Mech into that hex would violate the stacking limit, a domino effect would occur, requiring immediate resolution.*

Had the Stinger *instead been any 'Mech massing 40 or more tons, Hex B would have collapsed when the* Stinger *made its jump, as the weight of the heavier 'Mech would be greater than the building's current CF after the damage inflicted by the* Locust*. In this scenario, when the* Archer *moved into Hex D, that hex's collapse would have demolished the rest of the hexes in the multi-hex building before the* Archer *could continue its movement into Hex E, meaning that the* Archer *would have moved into a rubble hex.*



• BUILDING COLLAPSE DIAGRAM •

SPECIAL CASE RULES

The *Special Case Rules* chapter offers detailed rules to resolve specific but less common situations that players may want to play out as part of their game; these are all often depicted in *BattleTech* fiction, for example. Players can use these rules to simulate the effects of artillery, minefields, hidden units, and other situations.

All special case rules are optional: the use of any of these rules in a game must be agreed upon by all players before play begins.

BATTLEFIELD SUPPORT

Artillery, minefields, and aerospace fighters—and their effect on the battlefield—are pivotal elements of *BattleTech* fiction. The following rules bring these types of support elements into any game.

BEFORE PLAY BEGINS

Players pay for Battlefield Support using Battlefield Support Points. If a scenario does not provide details for Battlefield Support, then the players must determine the amount of Battlefield Support Points (BSPs) they will have access to before the game begins (note the Extraction Raid, Objective Raid, Diversionary Raid and Planetary Assault terminology is flavor only; players simply pick which values they wish to use in what ever scenario/game they're playing):

- **Extraction Raid Support:** 5 BSPs
- **Objective Raid Support:** 12 BSPs
- **Diversionary Raid Support:** 20 BSPs
- **Planetary Assault Support:** 35 BSPs

Players may need to adjudicate some use of Battlefield Support Points. For example, if the Planetary Assault Support option is used, but one player spends their full allotment of BSPs on Light Density Minefields, the result would be 70 minefields on the playing area—potentially rather annoying unless the other player agrees to that style of play. As with any rules, players should understand and agree on the potential extremes before play begins.

Once the players have determined the amount of Battlefield Support Points they will have access to, they should secretly select and write down the specific Battlefield Support Types they want to have available during the scenario, as noted on the Battlefield Support Table. Players do not need to spend their full allotment of BSPs; unspent BSPs are lost once play begins, and may not be used to purchase Battlefield Support Types during the game. (Some additional secret notes may need to be written down based upon the specific support chosen; see details below.)

For example, if two players decide to use Objective Raid Support (12 BSPs), Player A may simply want three Sniper Artillery attacks (4 BSPs each), while Player B opts for one Aerospace Light Strike (2 BSP), an Aerospace Strafing attack (5 BSPs), a Thumper Artillery strike (3 BSPs), and four Light Density Minefields (2 BSPs).

One-Time Use: Each instance of a Battlefield Support Type is usable once only, whether it inflicts damage or not (see *Minefield Support*, p. 79, for the exception).

Playing Area: These rules are designed for use on a two-mapsheet playing area (31 hexes x 35 hexes). If players are using larger areas, they may wish to increase the BSPs provided at each tier. For every two maps added, add 50% to the BSPs provided.

DURING GAME PLAY

Use the following rules for each Battlefield Support Type during game play.

Attack Declaration: With the exception of Minefields (see p. 79), use the following rules when declaring an attack:

- Support Types can be used during the Weapon Attack Phase of any turn of the game.
- If a player wishes to use one or more Support Types, they must declare their use during the Attack Declaration portion of the Weapon Attack Phase of the turn they will be used (see *Artillery Support*, p. 77, for the exception). The player reveals their selection and use of that Support Type. If they purchased multiple instances of that same Support Type, they do not need to reveal the full number of that Support Type that they purchased but are not using.
- The controlling player declares the target hex for the attack; in the case of Strafing attacks, the player must designate multiple hexes (see *Strafing*, p. 76).
- A player can make a Battlefield Support attack at any time during the Weapon Attack Phase. As many can be made in a single turn as are available.

Attack Direction: If any Battlefield Support attack is successful, roll 1D6: on a result of 1–4, all Damage Groupings in the attack hit the front; on a result of 5–6, they all hit the rear. There are exceptions to this rule, such as artillery, scattering, and minefields: see below.

Area-Effect Weapon Type: Attacks of this Type (see p. 96) include the notation "[AE]" in their entry and on the Battlefield Support Table.

Damage Value Groupings: If an attack is successful, all Damage Value groupings are applied, but a separate hit location roll is made for each grouping. Each Damage Value grouping noted on the Battlefield Support Table consists of 5 damage. For example, the Heavy Bombing has six Damage Value groupings, so it would apply a total of 30 points of damage, in six Damage Value groupings of 5 damage each.

Target Number Modifiers: No modifiers are applied to the Target Numbers shown on the Battlefield Support Table (see p. 78). Players simply roll 2D6, comparing the result against the Target Number for that Support Type, to determine whether the attack is successful. (See *Artillery Support*, p. 77, for the exception.)

Water: If a target is submerged in water, it cannot be the target of any Battlefield Support attacks. If a bombing or artillery attack scatters onto a target submerged in water, or attempts to damage such a target in any way, it is ignored.

In the case of a 'Mech standing in Depth 1 water, any damage is applied using the Punch Hit Location Table (after rolling 1D6 to determine attack direction as normal).

OFFENSIVE AEROSPACE SUPPORT (STRIKES)

Light Strike Target Number: 5
Heavy Strike Target Number: 6

Successful Attack: If the attack is successful, make a separate roll on the appropriate Hit Location column for each Damage Value grouping.

Unsuccessful Attack: No effect.

OFFENSIVE AEROSPACE SUPPORT (BOMBING) [AE]

Light Bombing Target Number: 5
Heavy Bombing Target Number: 7

Successful Attack: If the attack is successful, make a separate roll on the appropriate Hit Location column for each Damage Value grouping.

If the hex targeted by a bombing is a building hex, and there is a 'Mech in that building hex, the full damage is applied simultaneously to both the building and the 'Mech.

Unsuccessful Attack: If the attack fails, it will scatter. Determine the Margin of Failure (MoF) between the modified Target Number and the die roll result. For each MoF point, the bombing attack will scatter one hex.

Next, roll 1D6 and compare it to the Scatter Diagram (see p. 77) to determine where the attack lands. If there is a 'Mech in that hex, even a friendly 'Mech, it is automatically hit; use the rules above for applying damage, but use a direction of attack as coming from the original targeted hex (i.e. do not roll the 1D6 to determine attack direction as for a normal attack).

OFFENSIVE AEROSPACE SUPPORT (STRAFING) [DE]

Strafing Target Number: 7

Upon the declaration of a strafing attack, a player must designate from one to five consecutive hexes, all in a straight line. When resolving the attack, the player makes a separate attack roll for every 'Mech in one of those five targeted hexes, against the Strafing Target Number of 7.

Successful Attack: If the attack is successful, make a separate roll on the appropriate Hit Location column for each Damage Value grouping. If more than one target was successfully struck, then the three Damage Value groupings of a Strafing attack hit each target; i.e. if the player successfully strikes four targets with a Strafing attack, all four targets would be hit by three Damage Value groupings, with the attacker resolving hit locations for each grouping. (Remember to first roll 1D6 to determine whether each attack strikes the front or rear of the 'Mech).

Unsuccessful Attack: If the attack fails in a hex, nothing occurs, but attacks in other hexes are unaffected.

DEFENSIVE AEROSPACE SUPPORT (LIGHT AIR COVER)

Light Strike Target Number: 3
Light Bombing Target Number: 4
Heavy Strike Target Number: 9
Strafing/Heavy Bombing Target Number: 11

This Battlefield Support Type is not an attack. Instead, it can potentially nullify an Offensive Aerospace Support attack.

After a player announces all Offensive Aerospace Support they wish to use in a given turn, an opponent may reveal their selection(s) of a Defensive Aerospace Support (Light Air Cover) to remove the attack(s). Before the attacker makes their roll, the defender rolls against a Target Number based upon the specific attack, as noted in this entry.

If a player reveals more than one Defensive Aerospace Support, they may choose to assign them all to a single attack, or if there are multiple attacks, they may split them up in any fashion. However, all such designations must be announced before any rolls are made.

Successful Roll: If the roll is successful, the opponent's Aerospace Support attack is completely nullified; no roll is made by the attacker, and their Aerospace Support selection is considered spent.

Unsuccessful Roll: If the roll fails, the attacking player resolves their attack as normal.

DEFENSIVE AEROSPACE SUPPORT (HEAVY AIR COVER)

Light Strike/Bombing Target Number: 9
Heavy Strike Target Number: 5
Strafing/Heavy Bombing Target Number: 6

This is not an attack. Instead, it can potentially nullify an Offensive Aerospace Support attack.

After a player announces all Offensive Aerospace Support they wish to use in a given turn, an opponent may reveal their selection(s) of a Defensive Aerospace Support (Heavy Air Cover) to remove the attack(s). Before the attacker makes their roll, the defender rolls against a Target Number based upon the specific attack, as noted directly above.

During the current turn, Chris reveals that he'll be attempting to make Light Bombing, Heavy Strike, and Strafing Aerospace Support attacks. With all of Chris' Aerospace Support attacks for this turn announced, Kevin reviews the battlefield. His 'Mechs are currently in bad condition and lined up perfectly for that strafing run; this could push him over the edge into defeat. He decides to reveal Defensive Aerospace Support: one Light Air Cover and one Heavy Air Cover.

Kevin knows he could assign both Support Types to try and take down the Strafing attack, but he's hurting enough that he needs to knock out two of the three attacks. He maximizes his chances against the Strafing attack by announcing he'll assign his Heavy Air Cover there, and then assigns his Light Air Cover to the Heavy Strike.

Kevin first rolls his Heavy Air Cover for the Strafing attack, needing a 6. He gets a 7! The Strafing attack is shot down with no further effects. Kevin's Target Number to take down the Heavy Strike with his Light Air Cover is 9; he rolls a 5, so the attack will still happen.

Now Chris rolls for the Heavy Strike, and gets a 7; this beats the Target Number by 1, so he'll be rolling for the four Damage Value groupings on the target. Ouch!

Finally, Chris only needs to roll a 5 to hit with his Light Bombing attack, but rolls a 2! The attack misses the intended target hex, because it's a bombing attack, it scatters (in this instance, three hexes: the Margin of Failure of the dice roll). Any target, friend or foe, in the hex it scatters to will automatically be attacked by the Light Bombing.

If a player reveals more than one Defensive Aerospace Support, they may choose to assign them all to a single attack, or if there are multiple attacks, they may split them up in any fashion. However, all such designations must be announced before any rolls are made.

Successful Roll: If the roll is successful, the opponent's Aerospace Support attack is completely nullified; no roll is made by the attacker and their Aerospace Support selection is considered spent.

Unsuccessful Roll: If the roll fails, the attacking player resolves their attack as normal.

ARTILLERY SUPPORT [AE]

Base Artillery Target Number: 8 (see below)

Pre-designated Hexes: Before scenario deployment begins, each side using Artillery Strikes secretly chooses five hexes on the playing area for each Artillery selection (hexes cannot be shared between Artillery Support selections). These are pre-plotted hexes. When the Artillery attack is made, if it is fired at one of the five pre-plotted hexes it automatically hits; i.e. no roll against a Target Number is made and no scatter occurs, the attack is simply announced and damage assigned (as noted below). There is still a one-turn delay between firing and the strike as normal (see below).

Note that this number of hexes is intended for a playing area composed of four maps. For every two additional maps used, add three additional pre-plotted target hexes per Artillery Support selection. However, regardless of the size of the playing area, no player can place more than five pre-plotted hexes per mapsheet (an area 17 hexes long by 15 hexes wide). These limits apply regardless of how much Artillery Support is in use by either side.

Targeting: When targeting a hex that has not been pre-designated, use the following rules.

During any turn's Weapon Attack Phase, the player must announce that an Artillery Support is incoming and secretly write down the target hex number. The artillery attack arrives during the Weapon Attack Phase of the *following* turn, at which time the player rolls to determine if the attack successfully strikes the targeted hex.

If a 'Mech chooses to spot the target hex for indirect fire (see p. 30) on both the turn the artillery attack was launched and the turn on which it arrives, apply a –2 Target Number modifier to the artillery attack.

The spotter can spot for any number of Artillery Support attacks (as well as indirect fire attacks) against a single hex, but can only spot to one hex a turn. If a spotter makes any attacks during the Weapon Attack Phase of a turn that it also spots, apply a +1 Target Number modifier to those attacks, as well as applying only a –1 (in place of the –2) modifier for targeting to the Artillery Support attack.

Successful Attack: If the attack is successful, the first number of Damage Value groupings in the Damage Value Grouping column on the Battlefield Support Table is applied to targets in the target hex. The second number of groupings is then applied to all targets in all adjacent hexes. If a third value is listed, it is applied to all targets in the adjacent hexes to those; see the Radius Diagram, at right.

In the case of the first target hex, make a separate roll on the appropriate column for each Damage Value grouping, after rolling the 1D6 to determine whether the attack will strike the front or rear of the 'Mech.

For the second (and third) radius of hexes, the direction of the attack for determining hit locations is based upon the central target hex; then roll for each Damage Value grouping.

If a building hex is damaged by artillery, and there are 'Mechs in the building, the full damage is applied simultaneously to both the building as well as the units.

• SCATTER DIAGRAM •

• RADIUS DIAGRAM •

Unsuccessful Attack: If the attack fails, it scatters. Determine the Margin of Failure (MoF) between the Target Number and the die roll result. The artillery attack will scatter one hex for each MoF point. Next, roll 1D6 and compare it to the Scatter Diagram (see p. 77) to determine where the shot lands. Any 'Mechs in any of those hexes, including friendly 'Mechs, as well as terrain and buildings are automatically hit. Use the rules above for applying damage, with the direction of attack for the new target hex coming from the original targeted hex (i.e. once again, do not roll the 1D6 for a normal attack).

Variant Artillery—Copperhead: Copperhead is an alternate Artillery Support attack, using the following rules:

- **BSPs:** The player designates whether each Artillery Support selection is a Thumper, Sniper, or Long Tom Copperhead Artillery Support attack. BSP costs are identical to the standard attacks.
- **Damage Value Groupings:** Copperhead Artillery munitions function as Arrow IV homing missiles (see p. 97), including their ability to home in on TAG-designated targets. The Damage Value Groupings for each type are as follows: Thumper: 1; Sniper: 2; Long Tom: 3.

BATTLEFIELD SUPPORT TABLE

Support Type	Target Number*	Damage Value Groupings**	Weapon Type	Battlefield Support Point (BSP) Cost
Offensive Aerospace Support				
Light Strike	5	2	N/A	2
Light Bombing†	5	3	AE	3
Heavy Strike	6	4	N/A	3
Heavy Bombing†	7	6	AE	4
Strafing	7	3	DE	5
Defensive Aerospace Support				
Light Air Cover		N/A	N/A	1
Light Strike	3			
Light Bombing	4			
Heavy Strike	9			
Strafing/Heavy Bombing	11			
Heavy Air Cover		N/A	N/A	2
Light Strike/Bombing	9			
Heavy Strike	5			
Strafing/Heavy Bombing	6			
Artillery Support†				
Thumper	8	3/1††	AE	3
Sniper	8	4/2††	AE	4
Long Tom	8	5/3/1††	AE	6
(Copperhead)	‡	‡	N/A	‡
(Illumination/Smoke)	8	‡‡	N/A	‡‡
Minefield Support				
Light Density	9	2	AE	0.5
Medium Density	8	4	AE	2
Heavy Density	7	6	AE	4

*No modifiers are applied to this Target Number, except for Artillery Support (see p. 77).
†Scatters if misses; see page 76.
‡See rules, page 78.

**All Damage Value groupings are five points of damage.
††Affects multiple hexes; see Artillery Support, page 77.
‡‡See rules, page 79.

Variant Artillery—Illumination/Smoke: Illumination and Smoke are alternate Artillery Support attacks, using the following rules:

- **BSPs:** The player designates whether each Artillery Support selection is a Thumper, Sniper, or Long Tom Illumination or Smoke Artillery Support attack. The BSPs for Illumination and Smoke are half the standard cost (rounding down).
- **Illumination:** Instead of damage, Illumination Artillery Support negates any Light-based Target Number modifiers (see p. 62) against targets within their area of effect—i.e. the target hex (and all adjacent hexes, if appropriate)—for the rest of the game.
- **Smoke:** Instead of damage, a Smoke Artillery Support fills the target hex (and all adjacent hexes, if appropriate) with heavy smoke (see p. 61) that rises 2 levels above the underlying terrain. This smoke remains in play for the rest of the game.

MINEFIELD SUPPORT [AE]

Light Density Target Number: 9
Medium Density Target Number: 8
Heavy Density Target Number: 7

Unlike all other Battlefield Support Types, minefields are not an attack declared during the Weapon Attack Phase. Instead, at the start of the game, prior to any unit deployment, a player who has selected one or more Minefield Support option writes down the hex number, corresponding to the map, which contains each minefield selection. The player can designate any full hex anywhere on the playing area, except water hexes.

The maximum number of minefield Damage Value groupings that can be placed in a single hex by a single player is 6. However, both sides can have a minefield in the same hex. In other words, a player could place a single Heavy Density minefield in a hex, or three Light Density fields, or a Light and a Medium Density field. In all those instances, two sets of each combination could be present in a hex, one for each side of the game.

During each Movement Phase, a minefield's controlling player secretly tracks if any 'Mechs enter a hex with a minefield. When a 'Mech—either enemy or friendly—enters a minefield hex, any player controlling a minefield in that hex immediately announces the presence of any minefields there and rolls against the listed Target Number(s). In all instances, each minefield is resolved separately.

Skidding (see p. 68) or displacement (see p. 55) can also trigger a minefield; if a 'Mech enters a minefield hex along the ground for any reason, the minefield's controlling player must immediately announce its presence and make an attack roll. 'Mechs expending Jumping MPs only trigger a minefield if they end their movement in a minefield hex.

Successful Attack: If the attack succeeds, roll on the front column of the Kick Location Table for each Damage Value grouping.

If a minefield is triggered and inflicts damage against a target, reduce the minefield's total Damage Value groupings by 1; if a minefield's Damage Value groupings are reduced to 0, it has no further effect in play.

Unsuccessful Attack: If the attack fails, unlike other Battlefield Support Types, the minefield stays active. Players may agree to place a counter to designate that the hex contains a minefield; or, if both players agree, they can decide to not place counters in such instances to enhance a "fog-of-war" aesthetic.

Detecting Minefields: A 'Mech with an active probe (see p. 109) can potentially reveal minefields. At any point during a 'Mech's movement, if it mounts an active probe and the effective radius of the probe covers any hex containing an enemy minefield, the controlling player of that minefield must make a roll. On a result of 10+, any enemy minefields in that hex are revealed.

Clearing Minefields: A player may use an LRM-20; Rocket Launcher 20; MRM-20, -30 or -40; ATM 12 (HE or standard) or ATM 9 (HE), or Artillery Support, to try and clear a minefield. The player must designate the attack to clear the minefield. When the attack hits the hex, the player rolls 2D6. On a result of 5 or better, the strike reduces the Damage Value groupings of all minefields in the hex, enemy and friendly, to 0.

For Artillery Support, any time the target hex of an artillery attack (not any adjacent hexes) contains a minefield, even if the artillery was not designated to clear the minefield, the controlling player should roll 2D6. On a result of 10+, the minefield is cleared.

In all instances, minefields cleared in this fashion do not explode.

Dropping 'Mechs: If a minefield is present in a hex in which a dropping 'Mech lands (see below), treat the dropping 'Mech as though it entered the hex along the ground.

DROPPING 'MECHS (SIMPLIFIED)

BattleMechs, being exceptionally versatile and resilient combat units, are capable of being dropped directly onto the battlefield, using special disposable equipment to plummet from DropShips high up in the atmosphere or even in orbit. The sudden appearance of many tons of 'Mech directly on top of an enemy position and seemingly out of nowhere has turned the tide of many battles.

Players may choose to keep some or all of their force in reserve at the start of the battle, if the scenario permits this or if all players agree.

DROP TIMES

How long it takes for a dropped force to arrive on the battlefield must be determined before the game begins, either by instructions given in the scenario or by the agreement of all players. The time prior to arrival may be a fixed rate (e.g., "all dropped 'Mechs arrive on Turn 5"), variable (e.g., "all dropped 'Mechs arrive at 1D6 turns into the battle"), or some combination of the two (e.g., "Alpha Lance arrives on Turn 2, but Beta Lance does not arrive for 1D6-1 turns after this"). If the players cannot or do not wish to choose, use the default of "forces arrive on Turn 3."

ARRIVAL

At the start of a turn in which dropped 'Mechs arrive on the field, the controller of each arriving 'Mech must secretly nominate a hex anywhere on the playing area in which the 'Mech will land. This can be done by writing the information on a piece of paper and turning it face down on the table. If a 'Mech is prohibited from entering a specific hex type, that hex cannot be chosen as the target, with one notable exception: a player can intentionally try to violate the stacking rules (see *Stacking*, p. 18).

At the end of the Movement Phase, after all non-dropping troops have moved, a Piloting Skill Roll (see p. 53) is made for each dropping 'Mech in the order the player controlling the 'Mechs chooses. If more than one player is dropping 'Mechs this turn, stagger the order by their initiative, as in standard movement.

Before each roll, the player must reveal the nominated target hex for the landing. A successful roll result indicates the 'Mech lands in the target hex at the end of the Movement Phase with any facing desired. If a roll fails, resolve that drop completely before moving on to the next 'Mech.

A dropping 'Mech may not move or make attacks in the turn that it landed, but may act normally on the following turn. Attacks made against a 'Mech on the turn it lands apply a +3 target movement modifier.

Flak: Type F (flak) attacks made against a 'Mech on the turn it lands apply a –2 Target Number modifier.

Minefields: If the 'Mech lands in a hex that contains a minefield (see p. 79), treat the dropping 'Mech as though it entered the hex along the ground.

Planetary Conditions: Any Piloting Skill Roll modifiers resulting from in-play Weather Conditions (see p. 62) are applied to the PSR above.

Stacking: Per the stacking rules (see p. 18), only one 'Mech can be in a single hex at the end of the Movement Phase. If a 'Mech violates the stacking rules when it lands use the *Displacement* rules to determine the effects (see p. 55); randomly determine the direction to displace any 'Mechs (if there are multiple 'Mechs, randomly determine the displacement direction for each one). This is not considered an accidental fall from above. If a 'Mech cannot be displaced (for example, all the adjacent hexes are prohibited terrain), the 'Mech is destroyed.

FAILED LANDINGS

A 'Mech that fails the Piloting Skill Roll for its drop will at the very least be damaged, and may even be destroyed.

Location: A 'Mech that fails its landing scatters 1D6 hexes for every Margin of Failure point. To determine the exact location, roll one die and compare it to the Scatter Diagram (see p. 77) to find the direction of the scatter. The dropping 'Mech will scatter in that direction a number of hexes as determined above.

If the 'Mech completely misses the playing area, it is considered destroyed for purposes of determining victory in the current scenario.

Damage: Damage for a failed landing is as though the 'Mech had fallen a number of levels equal to the roll's Margin of Failure (see *Falling*, p. 56). For example, if a 'Mech with a modified Piloting Skill target number of 6 or higher rolled a 3, the 'Mech would suffer damage as from a fall of 3 levels. A PSR for landing that fails by more than 7 means the 'Mech is automatically destroyed.

Stacking: Per the stacking rules (see p. 18), only one 'Mech can be in a single hex at the end of the Movement Phase. If a 'Mech violates the stacking rules when it scatters and enters a hex after a failed landing roll, use the *Displacement* rules to determine the effects (see p. 55); in this case, the movement direction of the dropping 'Mech for any displacement that might occur is traced from the hex that 'Mech originally aimed to land in. Unlike a successful landing, this is treated as an accidental fall from above (see p. 57).

EJECTION (SIMPLIFIED)

During the Movement Phase, if a MechWarrior is conscious then they can choose eject rather than take any other action that phase. An ejecting MechWarrior lands in the hex immediately behind their now-abandoned 'Mech. However, ejecting from a 'Mech can be dangerous. The pilot must make a Piloting Skill Roll to avoid taking damage, modified for the circumstances listed on the Ejection Modifiers Table. A pilot who fails this PSR takes damage equal to the roll's Margin of Failure divided by 2 (rounded down).

All equipment on an abandoned 'Mech continues to function, and the 'Mech still counts as occupying the hex it is in. However, the 'Mech is considered destroyed for its controller's movement and attack selection purposes.

Auto-Ejection: If an ammunition explosion occurs a MechWarrior will automatically eject, even if unconscious, unless the auto-eject system is disabled. This occurs the moment the explosion triggers, before any other damage or events are resolved, interrupting the normal ammo explosion process (though the pilot still takes the usual 2 points of damage due to ammo explosion feedback).

Players must note on a 'Mech's record sheet before each battle whether or not its auto-eject system is disabled; it is on by default, and can be turned on and off in the End Phase (see p. 12).

Full-Head Ejection System: See page 115.

Torso-Mounted Cockpit: This cannot be ejected from.

Water: Ejection cannot occur if the cockpit is submerged (unless it has a full-head ejection system).

Skin-of-the-Teeth Ejection (Optional): If players want more survivable pilots, use this additional option.

If a 'Mech has its auto-ejection system engaged, does not have a Full-Head Ejection System, and has its head destroyed, the auto-eject system activates. The MechWarrior automatically receives enough damage to bring them up to 5 points of damage. For example, an undamaged pilot and a pilot with 3 points of damage would both fill out their condition monitors to 5 points of damage. The pilot may still take damage during the ejection itself. These rules do not apply in the event of a critical hit to a cockpit slot (which outright kills the pilot).

EJECTION MODIFIERS TABLE

Landing Terrain	Modifier
Clear	–2
Water	–1
Deep Snow	–2
Rough	0
Rubble	0
Light Jungle/Woods	+2
Heavy Jungle/Woods	+3
Per Level of Building	+1

Planetary Conditions	Modifier
Heavy Snowfall/Strong Gale/ Torrential Downpour	+2
Storm	+3

Situation	Modifier
BattleMech Prone	+5
Pilot Unconscious	+3
Per Point of Head Internal Structure Damage	+1
Automatic Ejection	+1

FORCED WITHDRAWAL

Most forces will not fight to the last man. Instead, once they have taken appreciable amounts of damage, they will begin to retreat. The forced withdrawal rules help simulate this situation.

Under forced withdrawal, 'Mechs must retreat from the battlefield when rendered useless or in imminent danger of destruction (see *Crippling Damage*, below). A 'Mech making a forced withdrawal must always move toward its home map edge (even a single hex of movement in that direction satisfies this requirement). However, the 'Mech need not spend Running MP, and may retreat moving backward. If equipped with MASC, a supercharger, or some other system that increases speed at the risk of damage, a 'Mech need not engage that system when withdrawing.

Withdrawing 'Mechs may still attack an enemy 'Mech that closes within range of a weapon or physical attack.

CRIPPLING DAMAGE

Any 'Mech that suffers crippling damage must withdraw from the map. Crippling damage is defined as one or more of the following:

- Four or more points of damage to the MechWarrior
- The destruction of all sensor critical slots
- One gyro and one engine critical hit
- Two engine critical hits
- A side torso location is destroyed
- Internal structure damage in either three or more limbs or two or more torso locations (torso internal structure damage does not count towards crippling damage if that location still has front armor)
- The loss of all weapons. This is triggered if a 'Mech loses all weapons (through damage, jamming, or ammunition depletion) with a range greater than five hexes and if it can no longer cause more than 5 points of combined weapons damage. This does not apply if the 'Mech did not start with the ability to do 5 or more damage, or the ability to do damage at a range greater than five hexes; in this case the 'Mech can never trigger this condition.

If using these rules, when all of a 'Mech's weapons are lost (as defined above) and it can no longer move, the 'Mech is considered destroyed. This is in addition to the regular rules for determining destroyed 'Mechs, as found on page 49.

BROKEN MORALE (SIMPLIFIED—OPTIONAL)

Under this option, 'Mechs forced to withdraw apply a +3 Target Number modifier to all weapon and physical attacks, to reflect how badly shaken morale affects a MechWarrior.

HIDDEN 'MECHS

Players may hide 'Mechs on the map before play begins. Any player doing so writes down the number of each hex in which they hide a 'Mech. They must also designate its facing and if it is prone. A 'Mech remains hidden until it is revealed (placed on the map).

Hidden 'Mechs are not counted when making movement or fire selections. However, if a player plans to move a hidden 'Mech, they must reveal it at the start of the Movement Phase. If a player plans to attack with a hidden 'Mech (not including a point-blank shot; see below), they must reveal it at the start of the Weapon Attack Phase.

If a 'Mech attempts to enter a hex containing a hidden enemy 'Mech, it stops next to that hex instead (unless jumping; see below). However, if it is skidding, using the *Dropping 'Mechs* rule, or is being displaced into that hex, those rules take priority and the 'Mech does not stop. Either way, the hidden 'Mech is revealed. Additionally, a 'Mech that ends its move next to a hex containing a hidden enemy 'Mech reveals that 'Mech. Hidden 'Mechs revealed in these ways cannot move that phase.

Hidden 'Mechs may not attack, but may spot for indirect fire and/or act as part of a C^3 network.

Active Probes: At the end of the Movement Phase, any hidden 'Mech within range of an enemy 'Mech's active probe and with line of sight to that enemy 'Mech is revealed.

Area-Effect Attacks: Area-effect attacks automatically reveal any hidden 'Mech in the blast area, and damage them as normal. 'Mechs revealed by this act as normal in subsequent phases.

Buildings: If hiding a 'Mech in a building, its controller must note the level it is hidden on (this must be a level the 'Mech could normally access). A 'Mech that moves into or ends its move adjacent to a building hex reveals every hidden enemy 'Mech in that building, regardless of the level a hidden 'Mech is on.

Forbidden Hexes: A 'Mech can only be hidden in a hex/level it can legally enter and could normally access if not hidden. No 'Mech may hide in a clear, paved, or road/bridge hex. 'Mechs may only be hidden in water hexes if fully submerged.

Jumping: Jumping over hexes does not reveal 'Mechs hidden in them. If jumping into a hex would reveal a 'Mech hidden there, the jumping 'Mech still lands in that hex, causing a domino effect (see p. 55).

Minefields: If a 'Mech is hidden in a hex with minefields, the minefields attack if the 'Mech tries to walk or run out of the hex.

POINT-BLANK SHOTS

A hidden 'Mech revealed by enemy movement may immediately make a special point-blank weapon attack against the 'Mech that revealed it. It may first torso twist, and uses a Range of 1. Ignore Target Number modifiers for movement or terrain. Any damage dealt and PSRs required take effect immediately. If the target 'Mech is still capable of moving, has MP remaining, and isn't jumping, it may continue its move after the attack.

A 'Mech making a point-blank shot cannot perform any other action for the remainder of that turn.

Unavoidable Targets: If an enemy 'Mech skids, jumps, or is displaced into a hidden 'Mech, or drops onto one, any point-blank shot made occurs before any collision or displacement.

DESIGN QUIRKS

BattleTech fiction is filled with descriptions of designs whose unique quirks affect their abilities—the *Javelin* and its poor center of gravity that makes it prone to falls, the *Cyclops* and its advanced battle computer, and so on. There are also illustrations that fall far outside the norm, such as the *Stalker*, which plainly doesn't have arms, or the *Jenner*, with no torso that can be twisted, and so on.

Design quirks are a set of optional rules that allow players to bring this individuality to the tabletop. This section provides a series of positive and negative quirks, each with a numerical value that determines its relative strength or weakness.

It is strongly recommended that if a player modifies an existing 'Mech by assigning it a positive quirk, they should give it negative quirks of equal or greater value as well.

Strategic Operations: Quirks originally appeared in *Strategic Operations*. As such, some contain references to rules and unit and equipment types that do not appear in this book. These references have been left in to maintain compatibility with the wider *BattleTech* ruleset.

POSITIVE DESIGN QUIRKS

The following positive design quirks each enhance how the design performs during a game.

ACCURATE WEAPON (VARIES)

A weapon is more accurate than normal: all Target Numbers for that weapon receive a –1 modifier. The cost is 1 point per 5 points (or fraction thereof) of maximum damage the weapon can inflict in a single Damage Value grouping. If the "weapon" deals 0 damage (such as TAG), the cost is 2 points. More than one weapon can receive this quirk, but the cost for each must be paid.

ANTI-AIRCRAFT TARGETING (VARIES)

Some 'Mechs, like the *Rifleman*, have an advanced targeting system that can accurately target airborne units. This includes 'Mechs performing a combat drop (see p. 79), but not jumping 'Mechs. All attacks against such units while airborne (not grounded) receive a –2 Target Number modifier.

The cost is 1 point per 7 points (or fraction thereof) of maximum damage that all the weapons mounted on the 'Mech can inflict (excluding physical attack weapons). This bonus is only available when the 'Mech itself is on the ground.

BARREL FIST (1 POINT)

The cost for this quirk must be paid for each arm it applies to, and it can only be applied to arms with a lower arm actuator but no hand actuator. An arm with this quirk does not apply the +1 Target Number modifier for punching if it lacks a hand actuator.

BATTLE COMPUTER (5 POINTS)

The 'Mech is an advanced command unit equipped with a powerful tactical battle computer that allows for more effective command of a battle force; the *Cyclops* is a prime example. Each turn one or more such 'Mechs are on the battlefield with a conscious MechWarrior, their force receives a +2 modifier to all Initiative rolls.

This modifier is not cumulative with that of a Command 'Mech.

BATTLEFISTS (2 POINTS)

The Battlefists quirk applies only to 'Mechs that are equipped with hand actuators. This quirk reflects 'Mechs that feature hands which are better designed for melee combat as a consequence of their unique configuration (such as the *Awesome*'s mace-like fist, or the *Pillager*'s sharpened "knuckles"). This quirk is only paid for once, even if the 'Mech has two fists. If the 'Mech has two hands, and this quirk is only meant to apply to one of those hands, this must be specified (the cost remains the same).

A 'Mech with Battlefists receives a –1 Target Number modifier when making Punch attacks.

COMBAT COMPUTER (3 POINTS)

The 'Mech possesses an advanced combat computer, such as those installed in the *Stalker* and *Juliano*. The computer aids the MechWarrior in managing heat levels; each turn the 'Mech will generate four points of heat less than normal (but never less than zero).

COMMAND 'MECH (2 POINTS)

Some BattleMechs, such as the *Atlas*, *BattleMaster*, *Black Knight*, *King Crab*, *Marauder*, *Mongoose*, and *Wolverine* are designed as command 'Mechs. Each turn one or more such 'Mechs are on the battlefield with a conscious MechWarrior, their force receives a +1 modifier to all Initiative rolls.

This modifier is not cumulative with a Battle Computer.

COMPACT 'MECH (1 POINT)

This Quirk describes a 'Mech that is particularly slim in stature, such as the *Jackalope*. It enables the 'Mech to share a BattleMech cubicle with another Compact 'Mech. It also allows a 'Mech recovery vehicle to transport two Compact 'Mechs. In both cases, the combined tonnage of the two 'Mechs cannot exceed relevant tonnage limits.

The Compact 'Mech quirk only provides this bonus for transportation purposes; Compact 'Mechs cannot share the same space at the same time when undergoing maintenance or repairs.

This quirk can only be applied to 'Mechs that also possess the Narrow/Low Profile quirk, and is further limited to 'Mechs of medium weight (55 tons) or less.

COWL (4 POINTS)

BattleMechs such as the *Cyclops* have been outfitted with a protective cowl that provides an additional 3 points of head armor. However, attacks that originate along the row of hexes directly in front of the cowled 'Mech ignore this additional armor.

DIRECTIONAL TORSO MOUNT (2 OR 3 POINTS)

A well-known feature of the original *Goliath*, a Directional Torso Mount acts as a somewhat more restrictive BattleMech shoulder turret, allowing any weapons in the mount to shoot in either the front arc or the rear arc, depending on the mount's current facing. The mount's facing is set at the start of the game, and can be changed at the same time torso twists are made. However, unlike a torso twist, it does not reset at each End Phase: the arc chosen remains until deliberately changed. The mount rotates with any torso twist as normal.

Each time a location with a Directional Torso Mount takes a hit (Front or Rear), the player must roll 2D6. A result of 9+ means the mount is destroyed and its weapon locked in its current arc, in addition to the normal effects of the attack.

The 3-point version of this quirk is available only to quad 'Mechs. In this case, the mount operates as a full turret, capable of rotating a full 360 degrees.

No weapon with location placement restrictions (such as a heavy Gauss rifle) can be placed in a Directional Torso Mount.

DISTRACTING (1 POINT)

A 'Mech with the Distracting quirk is physically designed to intimidate (or confuse) its opponent with its appearance. This includes the fearsome visage of the *Atlas* BattleMech, but can also include 'Mechs cosmetically designed to appear more heavily armed or deadly than they truly are, such as the *Arbiter*.

A 'Mech with the Distracting quirk applies a +1 Special Source modifier to any Morale Checks its opponents must make when using the *Morale* rules (see pp. 179-180, *TO:AR*). If using the Demoralizer Special Piloting Ability from *Campaign Operations* (see p. 74, *CO*), the Distracting quirk reduces the Demoralizer's psychological attack Target Number modifier from +4 to +3.

This quirk may apply to a 'Mech in some periods but not others: for example, the *Mackie* would almost certainly be Distracting upon its debut, but would lose the quirk once the novelty wore off and newer designs were introduced.

EASY TO MAINTAIN (1 POINT)

Some 'Mechs, such as the *Thorn*, are easier to maintain and repair. All maintenance, repair, and replacement rolls made for a 'Mech with this quirk receive a –1 Target Number modifier.

This is an innate quality to the 'Mech, regardless of whether enough spare parts are available to actually fix it. As such, the *Vindicator* doesn't gain this quirk just because its spare parts are plentiful (and the *Assassin* has it despite the fact that its custom spare parts are rare).

EASY TO PILOT (2 POINTS)

Training 'Mechs such as the *Chameleon* and *Crockett* are designed to be easier for a rookie MechWarrior or pilot to operate.

A MechWarrior with a Piloting Skill higher than 3 will receive a –1 Target Number modifier for Piloting Skill Rolls required as a result of damage or terrain. More skilled MechWarriors receive no benefit.

EXTENDED TORSO TWIST (3 POINTS)

Unlike most 'Mechs, one with this quirk can twist its torso much further. When torso twisting, the 'Mech can change its facing by one or two hexsides. Quad 'Mechs normally cannot torso twist at all. However, a quad 'Mech with this quirk can perform a normal (not extended) torso twist.

FAST RELOAD (1 POINT)

BattleMechs such as the *Hatchetman* and *Enforcer* have weapons that use large removable ammunition magazines, allowing them to reload much faster than normal. This quirk can be assigned to individual weapons, locations on a 'Mech (such as the right torso), or the entire 'Mech. Affected weapons can reload in half the normal time.

Regardless of the option chosen, the cost for this quirk is the same.

FINE MANIPULATORS (0 POINTS)

Though 'Mechs generally rely on hand actuators for any manual work, some units, such as the *Jupiter*, feature manipulators otherwise more commonly found on vehicles for delicate work.

A 'Mech with this quirk may mount up to one Manipulator (see p. 245, *TM*) in each side torso and/or arm location. These do not follow the normal construction rules and are weight free; the exact number must be specified.

GOOD REPUTATION (0 POINTS)

Whether due to clever marketing, use by an iconic MechWarrior, or proven capabilities, some 'Mechs are extremely popular. Such 'Mechs are worth a premium on the open market, as buyer demand means that acquiring one at its base price is almost impossible. For each level of this quirk, the 'Mech is worth 10% more C-bills than normal. This quirk is not worth points because it is not, strictly speaking, a bonus to the 'Mech.

Note that in some eras and/or locations, such as during the late Third Succession War period, 'Mechs as a whole are so rare that all may be considered to have the 1 point version of this quirk. Furthermore, this quirk is not applicable to Clan machines, as the Clans do not really have a secondary 'Mech market and Inner Sphere customers will gladly pay inflated prices to acquire Clan 'Mechs regardless of their reputation.

HYPER-EXTENDING ACTUATORS (1 POINT)

The arm actuators of 'Mechs like the *Quickdraw* can bend much further than normal. Even if it has lower arm and/or hand actuators, it can still flip arms to fire all arm-mounted weapons into its rear arc.

IMPROVED COMMUNICATIONS (1 POINT)

The 'Mech has an exceptionally powerful communications suite. The 'Mech automatically ignores the first level of ghost targets (see p. 100, *TO:AR*) used against it. Additionally, it can always attempt a BattleMech Satellite uplink (see p. 162, *TO:AR*) without being forced to give up its movement and combat actions.

IMPROVED COOLING JACKET (1 POINT)

One weapon's design incorporates a highly effective cooling jacket. When fired, this weapon generates 1 point less heat (never less than 1 point). More than one weapon can have this positive quirk, but the cost for each must be paid.

IMPROVED LIFE SUPPORT (1 POINT)

When determining damage to the MechWarrior or pilot as a result of heat following a life support critical hit, treat the 'Mech's heat level as being 5 points lower (an example of this is the *Shadow Hawk*). This quirk also doubles the cockpit's standard life support time, from four days to eight.

IMPROVED SENSORS (3 POINTS)

A 'Mech with this quirk is treated as if it has an active probe (range 4 for Inner Sphere 'Mechs, range 5 for Clan 'Mechs). If it is equipped with an actual active probe, instead add 2 to that active probe's range.

IMPROVED TARGETING (3: SHORT, 4: MEDIUM, OR 5: LONG POINTS)

The 'Mech has advanced targeting capabilities in one range bracket. The quirk can be applied up to three times, but can be taken only once per range bracket. The cost of the quirk varies with the range bracket chosen. All ranged attack Target Numbers at the selected range bracket receive a –1 modifier.

A 'Mech with this quirk cannot take Improved Targeting for the Extreme range bracket, nor can this quirk be combined with Variable Range Targeting.

JETTISON-CAPABLE WEAPON (1 POINT)

A 'Mech can be equipped with mounted weapons that can be jettisoned in combat and—if recovered—remounted on that 'Mech with ease. This feature can be seen in early Star League-era attempts to develop handheld weapon mounts, such as on the *BattleMaster*, *Griffin*, *Wolverine*, and *Phoenix Hawk*.

The quirk is bought for a single weapon; a 'Mech can carry a maximum of two jettison-capable weapons, paying the quirk cost each time. It is assumed that any official variant replacing a jettison-capable weapon with another weapon in the same location also uses a jettison-capable weapon, though a gamemaster may rule

that variants based on field refits or low-tech conditions do not benefit. No Handheld Weapon (see pp. 127-128, *TO:AUE*) or item using pod technology may benefit from this quirk, though fixed weapons on an OmniMech may be jettison-capable.

A jettison-capable weapon may be dropped using the *Dumping Ammunition* rules (see p. 58), except no risk of hull breaches is created, and no risk of an explosion if a 'Mech jettisoning its weapons is struck from the rear. Jettisoning an ammunition-dependent weapon does not also jettison its ammunition. Treat any critical slots for a jettison-capable weapon as a Roll Again result if they are struck after the weapon is jettisoned.

In campaign play the location of a jettisoned weapon should be tracked, as it can be remounted on the 'Mech later using the same rules and time spent as an ammunition reload. A jettisoned weapon may be used by any 'Mech that is a variant of the 'Mech the weapon was originally designed for, as long as the variant features that exact weapon and in the exact same location.

MODULAR WEAPONS (1 POINT)

Though lacking the flexibility of an OmniMech, a 'Mech with Modular Weapons such as the *O-Bakemono* or *Mercury* can be repaired or customized more easily. Equipment can be replaced in half the normal time (though repairs in-place take the usual amount of time). When using the *Customization* rules (see p. 188, *CO*), half the time is required.

On some 'Mechs, only certain weapons or equipment are modular. In this case, the specific gear with the modular quality must be noted, but this does not reduce the quirk's cost.

MULTI-TRAC (2 POINTS)

A 'Mech with this quirk can track multiple targets and may attack any number of targets in its front and arm firing arcs in the same turn without adding the secondary-target modifier. Secondary targets in the rear arc are treated as normal.

NARROW/LOW PROFILE (3 POINTS)

Designs such as the *UrbanMech*, *Vulcan*, and *Lancelot* have a narrow or low profile that makes them harder to hit at range.

If the Margin of Success for a weapon attack made against a narrow/low profile 'Mech is 0 or 1, the hit is considered a glancing blow. For example, if you need to roll a 9 or better to hit the target, a result of 10 would be a glancing blow.

A glancing blow inflicts half the normal damage (rounded down); for weapons that roll on the Cluster Hits Table, instead apply a –4 modifier to the Cluster roll result (with a minimum result of 2). Additionally, apply a –2 modifier when rolling on the Determining Critical Hits Table any time a glancing blow yields the possibility of a critical hit; if using the *Advanced Determining Critical Hits* rule (see p. 83, *TO:AR*), apply a –4 modifier instead.

This quirk has no effect versus non-weapon attacks, such as falls or physical attacks. It also has no effect versus all-or-nothing weapon attacks, such as Streak missile launchers.

If using the *Linking Weapons* rule (see p. 83, *TO:AR*), the entire linked group is considered a glancing blow.

Damage Reduction: If also using the *Glancing Blow* rule (see p. 78, *TO:AR*), the effects stack (1/4 damage is dealt). Any subtractive damage reduction effects (such as ferro-lamellor armor) are applied after all other damage reduction effects.

NIMBLE JUMPER (3 POINTS)

Jump jets are brute-force thrusters designed only to transport a 'Mech directly from point A to point B. Some 'Mechs, however, such as the *Spider*, are especially nimble when jumping. When a 'Mech with this quirk jumps, it need not follow the most direct path to its target hex. Instead, it may deviate from that path by up to one hex of its controller's choice (assuming the 'Mech has the Jumping MP required, of course; additional heat for such a jump is generated as normal for the jump jet type in use). For example, a jump that would normally cover six hexes can instead cover up to seven.

If a 'Mech deviates with its jump in this fashion, attacks against it that turn receive an additional +1 target movement modifier (a total of +2 for jumping, instead of the normal +1; the extra hex travelled is also factored into its TMM). However, the deviating 'Mech's attacker movement modifier also increases, from +3 to +4.

This quirk can only be taken by light or medium 'Mechs, and only for those that have at least as many Jumping MP as they do Walking MP.

OVERHEAD ARMS (2 POINTS)

If it is standing and has line of sight to its target, a 'Mech with Overhead Arms treats its arm-mounted weapons as one level higher when determining the effects of terrain.

For example, a standing *Dasher* on Level 0 terrain would be Level 2 for LOS purposes as normal, but would treat attacks from its arm-mounted weapons as coming from Level 3. This does not allow the 'Mech to fire over Level 2 hexes directly in front of it (since the 'Mech must already have line of sight for the Overhead Arms to function), but it would ignore partial cover for targets at the same level (Level 2, in our example) unless the target was receiving it from water.

A 'Mech with Overhead Arms receives a +2 Target Number modifier to its punch and arm-mounted physical weapon attacks.

This quirk cannot be combined with the Low-Mounted Arms quirk, or taken by any 'Mech that either does not have arms, or lacks any form of direct-fire ranged weaponry in said arms (excepting OmniMechs).

PROTECTED ACTUATORS (1 POINT)

Armor protection around the actuators is more effective, making a 'Mech with this quirk more resistant to Leg and Swarm attacks by infantry. The Target Numbers for such attacks receive a +1 modifier.

REINFORCED LEGS (1 POINT)

Designed for executing the dreaded "Death From Above" attack, some 'Mechs (the *Highlander*, for example) suffer half the normal damage to the legs when performing Death From Above successfully.

RUGGED (1 OR 2 POINTS)

Some 'Mechs are extremely reliable, reducing the frequency with which they require maintenance. The cost is 1 point if the 'Mech can go for two times as long between maintenance periods as normal, and 2 points if it can go up to three times as long. See *Campaign Operations*, page 191, for details on maintenance.

SEARCHLIGHT (0 POINTS)

'Mechs such as the *Warhammer*, *Loki*, *Guillotine*, *Defiance* and *Rifleman* are equipped with mounted searchlights (see p. 62).

Each time a searchlight-equipped 'Mech takes a hit in any torso location (Front or Rear), the player must roll 2D6 to determine if the searchlight is destroyed. A result of 7+ means the searchlight is destroyed, in addition to the normal effects of the attack.

STABILIZED WEAPON (VARIES)

Some weapons, such as the center torso and head lasers of the *Mongoose*, are better cushioned against or otherwise compensated for the increased inaccuracy caused by moving at high speeds. If the 'Mech runs, all Target Numbers for that weapon receive a –1 modifier.

The cost is 1 point per 7 points (or fraction thereof) of maximum damage the weapon can inflict in a single Damage Value grouping. If the "weapon" deals 0 damage (such as TAG), the cost is 1 point. More than one weapon can receive this positive quirk, but the cost for each must be paid.

STABLE (2 POINTS)

'Mech designs such as the *Dragon* are exceptionally stable, receiving a –1 Target Number modifier when forced to make a Piloting Skill Roll as a result of a physical attack.

UBIQUITOUS (1 POINT)

Some 'Mechs, such as the *Archer*, *Wasp*, and *Locust*, have been in production for literally centuries, with thousands or even tens of thousands produced in that time across multiple nations. For other 'Mechs, the manufacturer has gone to exceptional lengths to ensure that parts are readily available, such as mass producing them or using parts from other, more common machines. Whatever the case, when attempting to locate replacement parts for these 'Mechs, apply a –2 Target Number modifier.

This quirk does not cross between the Clans and the Inner Sphere unless bought twice. For example, a 'Mech Ubiquitous amongst the Clans is not automatically Ubiquitous in the Inner Sphere, and vice versa.

VARIABLE RANGE TARGETING (VARIES)

The 'Mech's targeting system allows it to launch more accurate attacks at either long or short range, at the expense of reduced accuracy at other ranges. During a turn's End Phase, the controlling player must designate whether this improved targeting feature will be active at long or short range the next turn. All weapon attacks at the designated range receive a –1 Target Number modifier, but all weapon attacks at the other range receive a +1 Target Number modifier (medium range remains unmodified).

The cost is 1 point per 5 points (or fraction thereof) maximum damage that all the weapons mounted on the 'Mech can inflict (excluding physical attack weapons). This quirk cannot be combined with Improved Targeting.

VESTIGIAL HANDS (0 POINTS)

A 'Mech with vestigial hands lacks manipulators substantial enough for most game purposes, and does not have hand actuator critical hit slots. However, a 'Mech with vestigial hands may lift and drop items (see *Lifting Items (Simplified)*, p. 21) if a scenario objective requires it.

NEGATIVE DESIGN QUIRKS

The following negative design quirks each contain a rule detrimental to how the design performs during a game, whether during movement, in combat, when eliminating heat and so on.

Some of these quirks, such as Sensor Ghosts, are quite rare or even unheard of on standard examples of any BattleMech. These sorts of quirks are instead intended to allow players to add personality to worn and/or damaged machines.

AMMUNITION FEED PROBLEM (1 POINT)

The ammunition feed for one ammunition-using component has a tendency to jam at inconvenient moments. On an unmodified attack roll of 2 with such a weapon, roll 2D6. On a result of 10+, the weapon jams and cannot be fired again in this battle. On a roll of 12, the new round of ammunition will explode in the weapon for normal damage. Per normal rules, Gauss weapon ammunition will not explode, but the weapon itself does.

BAD REPUTATION (1 POINT)

While perfectly sound, this 'Mech has acquired a bad reputation (for example, the *Blackjack* during the Succession Wars). As a result, it is worth only half the normal resale value, unless 'Mechs are particularly rare for a given era and/or location (for example, in the late Third Succession War era, 'Mechs are remarkably hard to come by, and few would turn up their nose at even the most unpopular of machines).

Designs that are second-line or merely somewhat unpopular (such as the *Watchman* or *UrbanMech*) should likely not receive this quirk. Additionally, a 'Mech may have a Bad Reputation at one point in its career, but lose it later (such as the *Battle Hawk*).

The Clans do not really have the same sort of economic model when it comes to BattleMechs. Instead, if a Clan 'Mech has the Bad Reputation quirk and if the optional *Clan Honor* rules are in effect (see p. 273, *TW*), the MechWarrior begins the battle with 1 *dezgra* point. The 'Mech will still fetch its full value if ever sold on an Inner Sphere market.

Note that when buying a 'Mech with a Bad Reputation, players generally still must pay its full price.

COOLING SYSTEM FLAWS (3 POINTS)

A flaw in the design can result in the 'Mech generating excess heat. Whenever the 'Mech executes or receives a physical attack, falls, or is forced to make a Piloting Skill Roll because it received 20 points or more damage, roll 2D6. On a result of 10+ the 'Mech will generate 5 points more heat each turn for the rest of the battle. This can only occur once a battle.

CRAMPED COCKPIT (2 POINTS)

The poorly-designed cockpits of 'Mechs like the *Wolverine* and *Stinger* are very cramped. The 'Mech receives a +1 Piloting Skill Roll modifier. This quirk cannot be applied to 'Mechs with a Small Cockpit.

DIFFICULT EJECTION (1 POINT)

Certain designs have flaws in their emergency escape systems detrimental to a MechWarrior's health. If the MechWarrior fails a Piloting Skill Roll when ejecting, they suffer an additional point of damage (in addition to the standard damage taken).

DIFFICULT TO MAINTAIN (1 POINT)

Some 'Mechs are harder to maintain and repair. All repair or replacement rolls made for this 'Mech receive a +1 Target Number modifier.

EM INTERFERENCE (1 OR 2 POINTS)

An energy weapon is insufficiently shielded and causes interference with other delicate electronics. The turn after the weapon has been fired, the following equipment aboard the 'Mech will not function: any ECM, any Active Probe, Artemis IV FCS, Artemis V FCS, Blue Shield PFD, any C^3, Chameleon LPS, Cockpit Command Console, Electronic Warfare Equipment, MASC, MRM FCS, Narc (and iNarc), Null Signature System, Stealth Armor, Streak Launchers, Supercharger, Targeting Computer, and Void Signature System.

Alternatively, this quirk can be purchased for the 'Mech itself, in which case the problem occurs whenever any energy weapon on the 'Mech is fired. The cost for this version of the quirk is 2 points, but can only be purchased if the 'Mech has more than one energy weapon.

EXPOSED ACTUATORS (1 POINT)

The actuators on some designs (such as the *Commando)* are poorly protected and more vulnerable to the effects of Swarm and Leg attacks by infantry. The Target Numbers for such attacks receive a –1 modifier.

EXPOSED WEAPON LINKAGE (2 POINTS)

Some designs such as the *Cygnus* have the mechanics of a weapon dangerously exposed. When a location that holds such a weapon is hit, roll 2D6. On a roll of 10+, that weapon is unable to fire for the remainder of the game.

This quirk can be taken only once and only for a single weapon type, and affects all weapons of that type on the 'Mech (for example, all AC/20s, all ER PPCs).

HARD TO PILOT (2 POINTS)

This 'Mech is hard to operate; MechWarriors receive a +1 Target Number modifier to all Piloting Skill Rolls.

ILLEGAL DESIGN (0 POINTS)

'Mechs designed with this quirk do not follow the existing construction rules for some reason, achieving effects that are not normally allowed in game-play, and which can fail spectacularly at any moment. (Note: The total weight and critical space of all components must still fit within the design's tonnage and critical space.)

If employed, the opposing player may roll 2D6 to determine if the 'Mech with this quirk suffers a catastrophic failure once every 6 turns of combat. On a result of 6 or higher, such catastrophic failure occurs in 1D6 of the 'Mech's hit locations (using the Front/Back Hit Locations Table). A Critical Hit check is then made for each failure location determined, applying a +4 modifier to the roll result. On any modified roll result of 13+, the location suffers complete collapse if it is an arm, leg, head, wing, main gun, or turret. This will destroy all items within (and any explosive components checks should be made as appropriate for location destruction). If the affected location is none of the above, it suffers 4 critical hits, determined as normal.

Illegal 'Mechs suffer an additional +4 Target Number modifier to repair or replace any damaged or destroyed components, and double all repair and maintenance times for work performed on them. If an Illegal 'Mech repair fails, the item gains the Non-functional quirk as well.

Finally, because they are technically failures at the time of their production, Illegal 'Mechs should also receive the Obsolete quirk (see p. 88), with the year of their obsolescence determined to be the same as their year of introduction.

INACCURATE WEAPON (VARIES)

Being of poor design, a weapon is less accurate than normal and receives a +1 Target Number modifier. The value is 1 point per 5 points (or fraction thereof) of maximum damage the weapon can inflict in a single Damage Value grouping.

More than one weapon can receive this quirk.

LOW-MOUNTED ARMS (2 POINTS)

'Mechs equipped with extremely low-mounted arms—such as the *Dark Crow*—have mounted their arms far too low to take advantage of the 'Mech's height advantage. This quirk cannot be combined with the Overhead Arms quirk, nor can it be taken by any 'Mech that either does not have arms, or lacks any form of weaponry in its arms.

A 'Mech with Low-Mounted Arms fires its arm-mounted weapons as if they are actually mounted in the 'Mech's legs, even though these arms still rotate with the torso position. As a result, these weapons may be blocked from firing by partial cover such as low hills and buildings. By the same token, attacks against 'Mech with Low-Mounted Arms that pass over partial cover cannot strike the 'Mech's arms either, and instead will strike the covering terrain.

NO/MINIMAL ARMS (2 POINTS)

Some 'Mechs effectively have no arms, such as the *Stalker* and *Locust*, or arms that are totally ineffective in aiding a fallen 'Mech in regaining its feet. When making a Piloting Skill Roll to stand up, such 'Mechs receive a +2 Target Number modifier; this is not cumulative with the *Attempting to Stand* optional rules (see p. 22, *TO:AR*). In addition, the 'Mech cannot make physical attacks with its arms.

NO COOLING JACKET (2 POINTS)

A weapon's design excludes the standard cooling jacket. As a result, it generates 2 extra points of heat when fired.

NO EJECTION SYSTEM (2 POINTS)

Some 'Mechs—notably the *Spider*—lack an emergency escape system. The MechWarrior can never eject and thus cannot use the *Ejection (Simplified)* rules (see p. 81).

NON-FUNCTIONAL ITEM (5 POINTS)

A 'Mech with this quirk has one or more components or pieces of equipment that simply do not work. No amount of repairs or replacement will correct this issue; the 'Mech must be redesigned from the ground up to solve the problem.

A 'Mech can have multiple Non-functional quirks, but each must be assigned to the specifically non-operational components they possess.

NON-STANDARD PARTS (1 POINT)

Constructed with non-standard components—such as the *Clint*—it is more difficult to source parts for this 'Mech. When attempting to locate replacement parts, add +2 to the Target Number.

OBSOLETE (4 POINTS)

An Obsolete 'Mech is one built using technological and/or construction standards that have been completely abandoned, such as any Primitive 'Mech from the Star League era until the Jihad. Though still serviceable—and maybe even effective—an obsolete 'Mech is extremely difficult to maintain after its parts sources go silent, forcing technicians to resort to jury-rigging and hand-machining even the most basic structural parts and electronics to keep it battle worthy.

A 'Mech with the Obsolete quirk cannot have any variants in active production, and must note the year in which its production ceased as part of the quirk. After that, the 'Mech will become increasingly hard to maintain, and its resale price will drop (for anyone except antique collectors, that is). To reflect this, for every 15 years after production of an obsolete 'Mech ceased, any rolls to repair or replace equipment on the 'Mech—or to even locate said parts—will suffer a +1 Target Number increase (to a maximum of +5). In addition, for every 20 years after production of an obsolete 'Mech ceased, the resale cost (but not the purchase cost) will decrease by 10 percent (to a minimum of 50 percent).

A 'Mech that somehow comes back into production after becoming obsolete automatically negates this quirk 10 years after returning to production. Additionally, some 'Mechs may have parts manufactured for them even if the chassis themselves are out of production, and thus would not qualify for this quirk. Note that an obsolete 'Mech may still receive additional modifiers for other quirks such as Difficult to Maintain, Easy to Maintain, Non-Standard Parts, and Ubiquitous.

OVERSIZED (2 POINTS)

The Oversized quirk cannot be combined with the Narrow/Low Profile quirk, and is further limited to 'Mechs of heavy weight (60 tons) and larger. This quirk describes a 'Mech that has been built physically larger than the norm, such as the *Mackie* or *Scylla*.

Any Piloting Skill Rolls required of an Oversized 'Mech to avoid damage when passing through buildings suffer a +1 Target Number modifier. Furthermore, they find it harder to use covering terrain effectively: an Oversized 'Mech never receives the +1 Target Number modifier for partial cover when being attacked.

POOR COOLING JACKET (1 POINT)

A weapon has a meager or badly designed cooling jacket. As a result, it generates 1 more point of heat than normal when fired.

POOR LIFE SUPPORT (1 POINT)

When determining damage to the MechWarrior or pilot as a result of heat following a life support critical hit, treat the 'Mech's heat level as being 5 points higher. This quirk also reduces the cockpit's standard life support time from four days to two.

POOR PERFORMANCE (3 POINTS)

The 'Mech cannot reach maximum speed immediately. It must spend one turn expending MP equal to its Walking speed before it can use MP up to its Running speed the following turn.

POOR SEALING (2 POINTS)

A 'Mech with the Poor Sealing quirk suffers from imperfections in its structural and armor design that make it more susceptible to leaks and other failures in hostile environments. In any situation where a 'Mech must check for hull breaches and/or flooding (such as suffering damage while operating in vacuum, toxic environments and underwater), this quirk adds a +2 modifier to the roll when making a hull breach check (see p. 65). If the 'Mech would otherwise not be susceptible to hull breaches, consider it to have a base hull breach TN of 12 before applying this modifier.

In addition to the above, a 'Mech with Poor Sealing must also make a hull breach check when first entering a hostile environment (such as vacuum or toxic atmospheres, when entering water, or when becoming completely submerged in water). Failure will result in a flood to a random location that is exposed to the hostile environment (use the appropriate Front Locations Table to determine the affected area).

POOR TARGETING (2 POINTS)

The 'Mech has poor targeting capabilities in one range bracket. This quirk can be applied more than once, but can be taken only once per range bracket. All ranged attacks at the selected range bracket receive a +1 Target Number modifier.

A 'Mech cannot take Poor Targeting for Extreme Range.

POOR WORKMANSHIP (1 POINT)

Not all manufacturers are as dedicated to producing quality products. This 'Mech is one level of unit quality (see p. 189, *CO*) lower than normal. In addition, all rolls for critical damage are modified by +1, making critical hits more likely. Results of 13 are treated as 12 (except if using *Advanced Determining Critical Hits*, in which case do not modify down; see p. 72, *TO:AR*).

PROTOTYPE (2 POINTS)

The 'Mech is a prototype and still has some issues. Its systems are less resilient and lack the redundancy that will be incorporated into the production model. All rolls for critical damage are modified by +2, making critical hits much more likely. Results of 13 or over are treated as 12 (except if using *Advanced Determining Critical Hits*, in which case do not modify down; see p. 72, *TO:AR*).

Even if brand new, designs that have just entered service have all passed the prototype stage and so likely do not qualify for this quirk. 'Mechs found in the *Experimental Technical Readout* series of products would make good candidates.

RAMSHACKLE (3 POINTS)

Decades or even centuries of hard service and substandard maintenance have led to an unpredictable machine patched together with bailing wire and prayers.

The 'Mech's controller rolls 1D6 at the start of the battle. The result indicates what negative quirk, if any, applies to that 'Mech for the duration of the battle: Sensor Ghosts (1), EM Interference (entire 'Mech) (2), Hard to Pilot (3), Cooling System Flaws (4), Poor Performance (5), or no negative result (6).

SENSOR GHOSTS (3 POINTS)

The 'Mech's sensor suite projects ghost images, causing significant targeting problems. All ranged attacks receive a +1 Target Number modifier.

SLOW TRAVERSE (1 POINT)

The 'Mech has an underpowered turret traverse mechanism. Any turret (or 3-point Directional Torso Mount) on the 'Mech may only rotate one hexside in a turn.

STATIC AMMO FEED (1 POINT)

Weapons with the Static Ammo Feed quirk use a very crude or minimized form of ammunition feed that renders the weapon incapable of switching between multiple ammunition bins. It is most commonly seen in 'Mechs that feature exposed ammunition feeds, such as the *Blade* or the *Cygnus*, and thus often occurs in conjunction with the Exposed Weapon Linkage quirk.

The Static Ammo Feed quirk must be taken for each affected weapon type the 'Mech carries, though it can cover multiple weapons of that type at the same time. Thus, a 'Mech with two Autocannon/10s and two Autocannon/5s would need two Static Ammo Quirks to account for all four weapons having this issue, if desired—one for the AC/10s and the other for the AC/5s. This quick cannot be applied to weapons without switchable ammunition.

All ammunition carried for a weapon with Static Ammo Feed must be of the same type. This is true even if the weapon is of the Switchable Ammo (S) Type.

UNBALANCED (1 POINT)

The 'Mech's design (one notorious example is the *Javelin*) is unbalanced. Treat this quirk as a +1 Target Number modifier for Piloting Skill Rolls made whenever the 'Mech enters a hex requiring a PSR due to a building type or planetary condition (i.e. Terrain Type, Terrain Modification, and so on).

STRATEGIC

MASTER UNIT LIST

The Master Unit List is the free online database providing fans with a comprehensive list of official game units. Drawing on over thirty-five years of official *BattleTech* supplements, there you'll find more than 1,400 units and over 5,000 additional variants! And updates ensure you'll always be provided with the latest configurations.

Vast functionality has been incorporated to give you the tools necessary for building accurate and dynamic forces. The database can be sorted by *BattleTech* Eras spanning the history of the game's many rich time periods. You can also sort by the many *BattleTech* factions—the Star League, the Great Houses, the Clans, and of course Mercenaries and Pirates.

Even more tools are available at the free Master Unit List site. Beyond helpful walkthroughs to get you started, you'll find dice roll aids for cluster hits, force-balanced assignment tables, Battle Value skill calculations, and even tools to generate random planetary systems.

Visit WWW.MASTERUNITLIST.INFO today!

WEAK HEAD ARMOR (VARIES)

The value of this quirk is equal to the number of armor points effectively lost; for example, a 'Mech with 9 points of head armor but using a 4-point version of this quirk effectively has only 5 armor points.

WEAK LEGS (1 POINT)

This 'Mech's legs were never designed to take the stresses involved in physical attacks. If the 'Mech executes a Death From Above attack, roll on the Determining Critical Hits Table for each of its legs. If it is kicked in a leg, roll on the table for the leg that was kicked. In both cases, the critical check occurs even if the leg armor was not breached.

BATTLEMECH QUIRK LISTING

Technical Readout: Prototypes introduced the practice of including design quirks with each new unit, a process that has continued with subsequent Technical Readouts. However, earlier Technical Readouts lack quirk details. For players interested in a recommended list of quirks for older machines, the following should be of help.

Base Designs vs. Variants: Note that the following quirk list is intended to cover each 'Mech in the broadest possible sense. For the most part, each quirk assignment is meant to cover all variants of that machine, and all eras in which it appears. There are literally thousands of 'Mech variants in the *BattleTech* universe: attempting to assign quirks to every variant of every machine would require more space than is available. Players should thus feel free to modify these quirk assignments to deal with specific variants or eras.

Additionally, individual machines may be modified as usual to represent the trials and tribulations (or upgrades) it has undergone in the field, with the usual caveat that player-added positive quirks should be balanced by an equivalent point total in negative quirks as normal.

BATTLEMECH QUIRK TABLE

BattleMech	Quirks
Akuma	Easy To Maintain
Albatross	Easy To Maintain, Improved Targeting (Long)
Alfar	Non-Standard Parts
Annihilator	Easy To Maintain
Anubis	Extended Torso Twist; Difficult To Maintain, Exposed Actuators
Anvil	Extended Torso Twist, Improved Targeting (Medium)
Apollo	Fast Reload, Rugged (1)
Aquagladius	Difficult Ejection, Difficult To Maintain
Arbalest	Easy To Maintain
Arcas	Protected Actuators
Archangel	Command 'Mech, Improved Communications, Improved Sensors; Bad Reputation
Archer	Battlefists, Command 'Mech, Stable, Ubiquitous
Arctic Fox	Weak Head Armor (1)
Arctic Wolf	Improved Targeting (Short)
Arctic Wolf II	Improved Targeting (Short)
Argus	Easy To Maintain; No/Minimal Arms
Assassin	Easy To Maintain; Cramped Cockpit, Non-Standard Parts, Poor Life Support
Atlas	Battlefists, Command 'Mech, Distracting, Improved Comm.
Atlas II	Battlefists, Command 'Mech, Distracting
Avatar	Stabilized Weapons (CT Medium Lasers)
Awesome	Battlefists (LA)
Axman	Protected Actuators
Baboon (Howler)	Extended Torso Twist, Narrow/Low Profile, Searchlight
Balius	Rugged (1)
Bandersnatch	Stable, Ubiquitous
Banshee	Rugged (1); Bad Reputation (prior to 3025)
Banzai	Stabilized Weapons (RA Large Pulse Lasers); Difficult To Maintain, Non-Standard Parts, Prototype
Barghest	Directional Torso Mount (LT/RT)
Battle Cobra	Extended Torso Twist
Battle Hawk	Rugged (1); Bad Reputation
BattleAxe	Difficult To Maintain
BattleMaster	Command 'Mech, Jettison-Capable Weapon (PPC); Weak Head Armor (1)
Bear Cub	Easy To Pilot, Extended Torso Twist; Cramped Cockpit
Behemoth (Stone Rhino)	Barrel Fist (LA/RA), Protected Actuators; Oversized, Poor Performance, Weak Head Armor (1)
Beowulf	Command 'Mech, Extended Torso Twist; Exposed Actuators
Berserker	Distracting
Bishamon	Difficult To Maintain
Black Hawk (Nova)	Combat Computer, Narrow/Low/Profile
Black Hawk-KU	Easy To Maintain

BattleMech	Quirks
Black Knight	Command 'Mech
Black Lanner	Stable
Black Watch	Easy To Pilot
Blackjack	Bad Reputation
Blackjack (Omni)	Non-Standard Parts
Blade	Easy To Maintain, Modular Weapons; Exposed Weapon Linkage (RAC/5), Static Ammo Feed
Blitzkrieg	Directional Torso Mount (CT/LT); No/Minimal Arms
Blood Asp	Rugged (1)
Blood Kite	Easy To Maintain, Rugged (2)
Blood Reaper	Hyper-Extending Actuators, Rugged (1)
Bloodhound	Improved Sensors, Protected Actuators
Blue Flame	Directional Torso Mount (H/LT/RT); Bad Reputation
Bombard	Bad Reputation, Non-Standard Parts
Bombardier	Bad Reputation
Bowman	Fast Reload (Arrow IV), Stable; Bad Reputation (Clan)
Brahma	Stabilized Weapon (LAC/5)
Brigand	Nimble Jumper
Bruin	Rugged (1); Exposed Actuators
Buccaneer	Improved Targeting (Short); Bad Reputation
Burrock	Stable
Bushwacker	Narrow/Low Profile
Caesar	Hyper-Extending Actuators
Canis	Fast Reload
Cataphract	Protected Actuators
Catapult	No/Minimal Arms, Weak Head Armor (1)
Cauldron-Born (Ebon Jaguar)	Narrow/Low Profile
Centurion	Improved Targeting (Short); Non-Standard Parts
Cephalus	Improved Life Support; Bad Reputation (Clan), Difficult To Maintain, No/Minimal Arms
Cerberus	Barrel Fist (LA/RA), Improved Targeting (Long); Vestigial Hands
Cestus	Stabilized Weapon (Gauss Rifle)
Chameleon	Easy To Pilot
Champion	No/Minimal Arms
Charger	Barrel Fist (LA), Easy To Maintain; Bad Reputation
Chimera	Nimble Jumper
Cicada	No/Minimal Arms
Clint	Improved Targeting (Medium, Long); Exposed Actuators, Difficult To Maintain, Non-Standard Parts
Clint IIC	Improved Targeting (Medium, Long)
Cobra	Protected Actuators
Colossus	Stable; Low-Mounted Arms
Commando	Narrow/Low Profile; Exposed Actuators
Commando IIC	Narrow/Low Profile; Exposed Actuators

BATTLEMECH QUIRK TABLE

BattleMech	Quirks
Copperhead	Difficult To Maintain
Corvis	Modular Weapons
Cossack	Extended Torso Twist
Cougar	Ubiquitous
Coyotl	Extended Torso Twist
Crab	Easy To Maintain
Crimson Hawk	Protected Actuators
Crimson Langur	Easy To Maintain, Improved Communications, Protected Actuators
Crockett	Easy To Pilot; Poor Life Support
Cronus	Easy To Maintain
Crossbow	Modular Weapons (LRM 10s); Obsolete (2550-3070), Poor Targeting (Long)
Crossbow (Omni)	Improved Targeting (Short)
Crusader	Easy To Maintain, Rugged (1), Stabilized Weapon (Legs), Ubiquitous
Cudgel	Difficult To Maintain
Cuirass	Easy To Maintain
Cyclops	Battle Computer, Cowl; Difficult Ejection, Weak Head Armor (2)
Cygnus	Exposed Weapon Linkage, Static Ammo Feed
Daboku	Stabilized Weapon (Autocannons); Prototype
Daedalus	Difficult To Maintain
Daikyu	Extended Torso Twist
Daimyo	Extended Torso Twist, Narrow/Low Profile; Difficult To Maintain
Daishi (Dire Wolf)	Improved Targeting (Long); Difficult Ejection
Dark Crow	Low-Mounted Arms
Dart	Extended Torso Twist
Dasher (Fire Moth)	Narrow/Low Profile, Overhead Arms
Dasher II	Poor Performance
Defiance	Barrel Fist (LA/RA), Hyper-Extending Actuators, Searchlight, Stable
Deimos	Anti-Aircraft Targeting
Dervish	Easy To Maintain, Hyper-Extending Actuators
Deva	Improved Communications, Improved Sensors; Bad Reputation
Devastator	Hyper-Extending Actuators, Searchlight
Dragon	Narrow/Low Profile, Stable
Dragon Fire	Good Reputation (1); No/Minimal Arms
Dragonfly (Viper)	Improved Targeting (Long), Narrow/Low Profile
Dragoon	Easy To Maintain, Modular Weapons; Bad Reputation, Obsolete (2780)
Duan Gung	Protected Actuators; Non-Standard Parts
Eagle	Extended Torso Twist; Weak Head Armor (1)
Ebony	Improved Targeting (Medium)
Eidolon	Non-Standard Parts
Eisenfaust	Battlefists, Easy To Maintain, Protected Actuators
Emperor	Command 'Mech, Improved Cooling Jacket (LA/RA Large Laser)
Enfield	Extended Torso Twist
Enforcer	Barrel Fist (LA/RA), Fast Reload (AC/10), Improved Cooling Jacket (AC/10); Ammunition Feed Problem (AC/10)
Enforcer III	Barrel Fist (LA/RA), Fast Reload (UAC/10)
Epimetheus	Non-Standard Parts, Prototype
Excalibur	Vestigial Hands
Exterminator	Difficult To Maintain
Eyleuka	Exposed Actuators
Fafnir	Stabilized Weapon (LT/RT Heavy Gauss Rifles); No/Minimal Arms
Falcon	Narrow/Low Profile
Falcon Hawk	Good Reputation (1)
Falconer	Stable
Fennec	No/Minimal Arms
Fenris (Ice Ferret)	Improved Sensors
Fire Falcon	Improved Sensors
Fire Scorpion	Cowl
Fireball	Improved Communications, Narrow/Low Profile
Firebee	Poor Life Support, Weak Legs
Firefly	No/Minimal Arms
Firestarter	Ubiquitous; Exposed Weapon Linkage (LA/RA Flamers)
Firestarter (Omni)	Cowl; Difficult To Maintain
Flamberge	Improved Targeting (Medium)
Flashfire	Modular Weapons (Fluid Guns)
Flashman	Rugged (1)
Flea	Easy To Maintain, Improved Life Support; No/Minimal Arms
Galahad	Difficult To Maintain
Galahad (Glass Spider)	Anti-Aircraft Targeting, Extended Torso Twist, Multi-Trac
Gallant	Stabilized Weapons (RA Large Pulse Lasers)
Gallowglas	Good Reputation (1)
Garm	Extended Torso Twist
Ghost	Extended Torso Twist, Improved Communications, Improved Sensors
Gladiator	Cowl; Bad Reputation
Gladiator (Executioner)	Distracting
Goliath	Multi-Trac; Exposed Actuators
Goshawk (Vapor Eagle)	Nimble Jumper
Goshawk II	Improved Life Support, Nimble Jumper
Grand Crusader	Directional Torso Mount (LT/RT); Bad Reputation
Grand Crusader II	Easy To Maintain, Stable; Bad Reputation

BATTLEMECH QUIRK TABLE

BattleMech	Quirks
Grand Dragon	Extended Torso Twist, Narrow/Low Profile, Stable
Grand Titan	Multi-Trac; Difficult To Maintain
Grasshopper	Rugged (1)
Great Turtle	Improved Targeting (Short); Poor Performance
Great Wyrm	Stable
Grendel (Mongrel)	Rugged (1)
Griffin	Battlefists, Jettison-Capable Weapon (PPC), Rugged (1), Ubiquitous
Griffin IIC	Battlefists
Grigori	Improved Communications, Improved Sensors; Bad Reputation
Grim Reaper	Easy To Maintain, Protected Actuators
Grizzly	Difficult To Maintain, Exposed Actuators
Guillotine	Searchlight
Guillotine IIC	Searchlight
Gunslinger	Cowl
Gurkha	Bad Reputation, Exposed Actuators
Ha Otoko	Bad Reputation (Clan)
Hachiwara	Difficult To Maintain
Hammer	Extended Torso Twist, Improved Targeting (Long)
Hammerhands	Barrel Fist (LA/RA), Protected Actuators
Hankyu (Arctic Cheetah)	Protected Actuators
Hatamoto	Barrel Fist (LA), Easy To Maintain, Modular Weapons (Torso SRMs)
Hatchetman	Anti-Aircraft Targeting
Hauptmann	Easy To Maintain, Extended Torso Twist
Hector	Bad Reputation, Exposed Actuators, Obsolete (2420)
Helepolis	Reinforced Legs, Stable
Helios	Easy To Maintain
Hellfire	Difficult To Maintain
Hellhound (Conjurer)	Extended Torso Twist
Hellion	Weak Head Armor (1)
Hellspawn	Bad Reputation
Hellstar	Barrel Fist (LA/RA); Exposed Actuators
Hercules	Difficult To Maintain
Hermes	Improved Communications, Stable
Hermes II	Easy To Maintain, Improved Communications
Highlander	Command 'Mech, Cowl, Reinforced Legs; Difficult Ejection
Highlander IIC	Command 'Mech, Cowl, Reinforced Legs; Difficult Ejection
Hitman	Accurate Weapon (TAG), Stabilized Weapon (TAG)
Hollander	Reinforced Legs, Stabilized Weapon (Gauss Rifle); Unbalanced
Hollander II	Reinforced Legs, Stabilized Weapon (Gauss Rifle)
Hoplite	No/Minimal Arms
Hornet	Narrow/Low Profile; No/Minimal Arms
Hunchback	Battlefists
Hunchback IIC	Bad Reputation (Clan)
Huron Warrior	Improved Sensors
Hussar	Directional Torso Mount (CT), Improved Communications, Rugged (1)
Icarus II	Improved Targeting (Short)
Icestorm	Improved Sensors, Rugged (1)
Imp	Command 'Mech; Difficult To Maintain
Initiate	Hyper-Extending Actuators; Bad Reputation
Jackal	Improved Cooling Jacket (ER PPC); Weak Legs
Jackrabbit	Easy to Maintain, Bad Reputation
JagerMech	Anti-Aircraft Targeting
JagerMech III	Anti-Aircraft Targeting, Overhead Arms
Javelin	Unbalanced
Jenner	No/Minimal Arms
Jenner IIC	No/Minimal Arms
Jinggau	Narrow/Low Profile; Weak Head Armor (2)
Juggernaut	Improved Life Support; Difficult To Maintain
Jupiter	Fine Manipulators, Improved Sensors, Improved Targeting (Long)
Kabuto	Extended Torso Twist
Karhu	EM Interference
King Crab	Command 'Mech
Kingfisher	Rugged (1)
Kintaro	Rugged (1)
Kodiak	Battlefists, Distracting
Komodo	Extended Torso Twist, Narrow/Low Profile, Protected Actuators; Weak Head Armor (2)
Koschei	Improved Life Support; Weak Head Armor (1)
Koshi (Mist Lynx)	Improved Sensors, Narrow/Low Profile
Koto	EM Interference
Kraken (Bane)	Extended Torso Twist, Protected Actuators
Kuma	Accurate Weapon (Heavy Large Laser); Hard To Pilot
Kyudo	Easy To Pilot; Exposed Actuators
Lancelot	Anti-Aircraft Targeting, Narrow/Low Profile; Bad Reputation
Lao Hu	Good Reputation (1)
Legacy	Directional Torso Mount (LT/RT); Bad Reputation
Legionnaire	Easy To Maintain, Improved Targeting (Medium); Exposed Weapon Linkage
Lightray	Hyper-Extending Actuators; Bad Reputation, Exposed Actuators
Linebacker	Stable
Lineholder	Easy To Maintain
Lobo	Directional Torso Mount (AMS), Extended Torso Twist; Difficult To Maintain

BATTLEMECH QUIRK TABLE

BattleMech	Quirks
Locust	Narrow/Low Profile, Ubiquitous; Cramped Cockpit, No/Minimal Arms, Weak Legs
Locust IIC	Easy To Maintain, Narrow/Low Profile; No/Minimal Arms, Weak Legs
Loki (Hellbringer)	Searchlight
Longbow	Anti-Aircraft Targeting, Searchlight, Ubiquitous; No/Minimal Arms
Longshot	Difficult To Maintain
Lupus	Exposed Actuators
Lynx	Multi-Trac
Mackie	Easy To Maintain, Protected Actuators, Rugged (1); Oversized
Mad Cat (Timber Wolf)	Improved Targeting (Medium); Weak Head Armor (1)
Mad Cat Mk II	Improved Targeting (Medium)
Maelstrom	Good Reputation (1)
Malak	Improved Communications, Improved Sensors; Bad Reputation
Man O'War (Gargoyle)	Stabilized Weapons (LA/RA)
Mandrill	Narrow/Low Profile
Mangonel	Difficult To Maintain, No/Minimal Arms
Mantis	Non-Standard Parts
Marauder	Command 'Mech, Directional Torso Mount (RT), Hyper-Extending Actuators, Narrow/Low Profile; Exposed Weapon Linkage (AC/5)
Marauder II	Command 'Mech, Hyper-Extending Actuators, Narrow/Low Profile
Marauder IIC	Command 'Mech, Hyper-Extending Actuators; Exposed Actuators
Marshal	Easy To Maintain, Rugged (2)
Masakari (Warhawk)	Improved Targeting (Long)
Matador	Battlefists
Mauler	Stabilized Weapon (Autocannons)
Men Shen	Improved Communications, Improved Sensors; Exposed Actuators
Mercury	Easy To Maintain, Modular Weapons
Mercury II	Narrow/Low Profile, Stabilized Weapon (CT Medium Laser); Bad Reputation (Clan)
Merlin	Easy To Maintain, Rugged (1)
Mjolnir	Accurate Weapon (Mace), Easy To Maintain
Mongoose	Command 'Mech, Easy To Pilot, Stabilized Weapons (Small Laser & CT Medium Laser)
Mongoose II	Command 'Mech, Improved Communications
Morpheus	Protected Actuators; Difficult To Maintain
Morrigan	Stabilized Weapon (Heavy Large Laser); Weak Legs
Naga	Extended Torso Twist; Bad Reputation (Clan), No/Minimal Arms
Naginata	Cowl, Stable
Nexus	Bad Reputation
Nexus II	Extended Torso Twist; Bad Reputation
Night Gyr	Difficult To Maintain
Night Hawk	Easy To Maintain
Night Wolf	Directional Torso Mount (AMS), Hyper-Extending Actuators
Nightsky	Modular Weapon (Hatchet)
Nightstar	Command 'Mech, Good Reputation (1), Variable Range Targeting; Difficult To Maintain
Ninja-To	Stable
Nobori-nin (Huntsman)	Stable
No-Dachi	Hyper-Extending Actuators
Nova Cat	Extended Torso Twist; Exposed Actuators
Nyx	Narrow/Low Profile; Difficult To Maintain
O-Bakemono	Modular Weapons; No/Minimal Arms
Ocelot	Jettison-Capable Weapon (RA Medium Lasers)
Omen	Protected Actuators, Rugged (1)
Onager	Battlefists (RA); Exposed Actuators, Exposed Weapon Linkage (HAG/30)
Onslaught	Protected Actuators; Non-Standard Parts
Orion	Anti-Aircraft Targeting, Easy To Maintain, Rugged (1)
Orion IIC	Anti-Aircraft Targeting, Easy To Maintain, Rugged (1)
Orochi	No/Minimal Arms
Osiris	Directional Torso Mount (1 CT ER Medium Laser), Extended Torso Twist
Osprey	Directional Torso Mount (RT); No/Minimal Arms
Osteon	Extended Torso Twist, Multi-Trac, Protected Actuators; Bad Reputation (Clan)
Ostroc	Narrow/Low Profile
Ostscout	Improved Communications, Improved Sensors, Narrow/Low Profile, Rugged (1)
Ostsol	Improved Sensors, Narrow/Low Profile
Ostwar	Easy To Maintain
Owens	Improved Sensors
Pack Hunter	Reinforced Legs
Pack Hunter II	Stabilized Weapon (ER PPC)
Paladin	Difficult To Maintain, Non-Standard Parts
Panther	Improved Targeting (Short), Nimble Jumper
Parash	Jettison-Capable Weapon (Large Pulse Laser), Narrow/Low Profile
Pariah (Septicemia)	Extended Torso Twist, Multi-Trac; Bad Reputation (Clan)
Pathfinder	Improved Sensors
Patriot	Protected Actuators, Stable
Peacekeeper	Rugged (1)
Penetrator	Good Reputation (1), Stable

BATTLEMECH QUIRK TABLE

BattleMech	Quirks
Penthesilea	Rugged (1), Stabilized Weapons (CT ER Medium Lasers)
Peregrine (Horned Owl)	Stabilized Weapon (CT)
Perseus	Anti-Aircraft Targeting, Easy To Maintain
Phantom	Improved Sensors
Phoenix	Directional Torso Mount (SRMs); EM Interference
Phoenix Hawk	Command 'Mech, Improved Communications, Jettison-Capable Weapon (Large Laser), Ubiquitous
Phoenix Hawk IIC	Easy To Maintain, Fast Reload (torso weapons)
Pillager	Battlefists
Pinion	Bad Reputation (Clan)
Piranha	Improved Targeting (Short)
Porcupine	Distracting
Pouncer	Weak Head Armor (2)
Predator	Cramped Cockpit, Difficult Ejection
Prefect	Extended Torso Twist
Preta	Improved Communications, Improved Sensors; Bad Reputation, Exposed Actuators
Prometheus	Non-Standard Parts, Prototype
Prowler	Poor Life Support
Pulverizer	Accurate Weapon (Enhanced ER PPC), Easy To Maintain, Protected Actuators; Bad Reputation (Clan)
Puma (Adder)	Narrow/Low Profile
Quickdraw	Hyper-Extending Actuators; Exposed Actuators
Rabid Coyote	Improved Targeting (Short); Bad Reputation (Clan)
Raijin	Directional Torso Mount (LT/RT), Extended Torso Twist; Bad Reputation
Raijin II	Extended Torso Twist; Bad Reputation
Rakshasa	Easy To Maintain; Weak Head Armor (1)
Rampage	Vestigial Hand (RA); Bad Reputation, Obsolete (2780)
Raptor	Easy To Maintain
Raptor II	No/Minimal Arms, No Ejection System
Raven	No/Minimal Arms
Razorback	Narrow/Low Profile; No/Minimal Arms
Red Shift	Accurate Weapon (TAG), Improved Communications, Narrow/Low Profile; Bad Reputation
Rifleman	Anti-Aircraft Targeting, Improved Communications, Searchlight, Ubiquitous
Rifleman II	Anti-Aircraft Targeting, Improved Communications, Searchlight
Rifleman IIC	Anti-Aircraft Targeting, Improved Communications
Ronin	Non-Standard Parts
Rook	Rugged (1)
Ryoken (Stormcrow)	Stable
Ryoken II	Stable
Sagittaire	Improved Targeting (Short); Difficult To Maintain, Weak Head Armor (1)
Salamander	Difficult To Maintain
Sasquatch	Difficult To Maintain
Savage Coyote	Oversized
Scarabus	Rugged (1)
Scorpion	Bad Reputation, Hard To Pilot
Scylla	Oversized
Sentinel	Improved Communications; Ammunition Feed Problem (SRM 2)
Sentry	Easy To Maintain, Easy To Pilot, Rugged (1), Ubiquitous
Seraph	Improved Communications, Improved Sensors; Bad Reputation
Sha Yu	Improved Communications
Shadow Cat	Narrow/Low Profile
Shadow Cat II	Easy To Maintain, Narrow/Low Profile
Shadow Hawk	Battlefists, Improved Life Support, Rugged (1), Ubiquitous
Shadow Hawk IIC	Battlefists, Improved Life Support
Shen Yi	Weak Head Armor (1)
Shockwave	Battlefists, Easy To Maintain; Exposed Weapon Linkage (RAC/5)
Shogun	Barrel Fist (LA/RA); Difficult To Maintain, Non-Standard Parts
Shootist	Command 'Mech
Shugenja	Cowl, Hyper-Extending Actuators; Exposed Actuators
Silver Fox	Non-Standard Parts, Weak Head Armor (1)
Sirocco	Directional Torso Mount (All); Difficult To Maintain
Slagmaiden	Jettison-Capable Weapon (Shield); Exposed Actuators, Prototype
Sling	Cramped Cockpit
Snake	Improved Targeting (Short), Protected Actuators
Snow Fox	Narrow/Low Profile; Hard To Pilot
Solitaire	Improved Cooling Jacket (Heavy Large Laser), Stabilized Weapon (Heavy Large Laser); Exposed Actuators
Spartan	Improved Sensors
Spatha	Difficult To Maintain, Non-Standard Parts
Spector	Difficult To Maintain, Rugged (1)
Sphinx	Easy To Maintain, Rugged (1)
Spider	Easy To Maintain, Nimble Jumper; No Ejection System
Spirit	Jettison-Capable Weapon (SSRM 4)
Stag	Improved Communications, Improved Sensors, Stabilized Weapon (RA ER Large Laser); Bad Reputation (Clan)
Stalker	Combat Computer, Ubiquitous; No/Minimal Arms
Stalking Spider	Stable
Starslayer	Rugged (1)
Stealth	Improved Sensors

BATTLEMECH QUIRK TABLE

BattleMech	Quirks
Stiletto	Multi-Trac; Difficult Ejection
Stinger	Rugged (1), Ubiquitous; Cramped Cockpit
Stooping Hawk	Rugged (2)
Strider	Easy To Pilot
Striker	Easy To Maintain
Sun Cobra	Protected Actuators
Sunder	Extended Torso Twist
Supernova	Improved Targeting (Long)
Tai-sho	Improved Communications, Stabilized Weapon (UAC/10)
Talon	Rugged (1); No/Minimal Arms
Talos	Easy To Pilot
Tarantula	Easy To Pilot, Extended Torso Twist; Exposed Actuators
Targe	Extended Torso Twist; Bad Reputation, No/Minimal Arms
Tempest	Directional Torso Mount (H Medium Laser)
Templar	Easy To Maintain
Tessen	Bad Reputation
Thanatos	Stable
Thor (Summoner)	Improved Communications, Ubiquitous
Thorn	Easy To Maintain
Thresher	Improved Targeting (Medium)
Thug	Rugged (1)
Thunder	Cowl
Thunder Fox	Easy To Maintain
Thunder Hawk	Command 'Mech, Good Reputation (1), Variable Range Targeting; Difficult To Maintain
Thunder Stallion	Cramped Cockpit
Thunderbolt	Multi-Trac, Rugged (2), Ubiquitous
Ti Ts'ang	Hyper-Extending Actuators
Titan	Protected Actuators
Titan II	Protected Actuators
Toro	Cramped Cockpit
Toyama	Bad Reputation
Trebaruna	Command 'Mech
Trebuchet	Easy To Maintain, Fast Reload (torso weapons)
Tsunami	Difficult To Maintain
Tundra Wolf	Easy To Maintain; Weak Head Armor (1)
Turkina	Easy To Pilot
Uller (Kit Fox)	Narrow/Low Profile
UrbanMech	Extended Torso Twist, Narrow/Low Profile; No/Minimal Arms
UrbanMech IIC	Extended Torso Twist, Narrow/Low Profile; No/Minimal Arms
Ursus	Distracting, Protected Actuators; Cramped Cockpit
Ursus II	Distracting, Protected Actuators
Uziel	Cowl; Bad Reputation, Exposed Actuators
Valiant	Improved Targeting (Short), Protected Actuators; Weak Head Armor (1)
Valkyrie	Easy To Maintain, Improved Communications
Vanquisher	Stable; Bad Reputation, No/Minimal Arms
Venom	Nimble Jumper; Difficult Ejection
Verfolger	Hyper-Extending Actuators; Difficult To Maintain
Victor	Rugged (1)
Viking	Easy To Maintain, Multi-Trac; No/Minimal Arms
Vindicator	Rugged (1); Difficult Ejection
Viper (Black Python)	Improve Targeting (Short)
Vixen (Incubus)	Jettison-Capable Weapon (Large Pulse Laser)
Volkh	Non-Standard Parts
Von Rohrs (Hebi)	Bad Reputation
Vulcan	Narrow/Low Profile
Vulture (Mad Dog)	Improved Targeting (Medium)
Wakazashi	Difficult To Maintain
War Dog	Accurate Weapon (OS SSRM 2s), Fast Reload (OS SSRM 2s)
Warhammer	Rugged (2), Searchlight, Stable, Ubiquitous
Warhammer IIC	Searchlight, Stable
Warlord	Easy To Maintain
Wasp	Easy To Maintain, Extended Torso Twist, Ubiquitous
Watchman	Easy To Maintain, Easy To Pilot, Rugged (1)
Werewolf	Non-Standard Parts
White Flame	Directional Torso Mount (LT/RT); Bad Reputation, Exposed Actuators
Whitworth	Rugged (1); Weak Legs
Wight	Exposed Actuators, Weak Legs
Wildfire	Non-Standard Parts, Prototype
Wolf Trap (Tora)	Bad Reputation
Wolfhound	Easy To Maintain, Good Reputation (1)
Wolverine	Command 'Mech, Extended Torso Twist, Improved Communications, Jettison-Capable Weapon (AC/5), Protected Actuators, Ubiquitous; Cramped Cockpit
Wolverine II	Command 'Mech, Improved Communications, Protected Actuators
Woodsman	Easy To Maintain
Wraith	Ubiquitous; Difficult To Maintain
Wyvern	Rugged (1)
Wyvern IIC	Protected Actuators
Xanthos	Improved Life Support
Yao Lien	Non-Standard Parts
Yeoman	No/Minimal Arms
Ymir	Stable
Yu Huang	Command 'Mech, Good Reputation (1)
Zeus	Barrel Fist (RA), Easy To Maintain

WEAPONS AND EQUIPMENT

BattleMechs use a wide variety of weapon systems, ranging from autocannons to lasers to hatchets to missiles, as well as specialty electronics designed to hinder the enemy and assist allies on the battlefield.

The in-flux nature of the *BattleTech* universe makes a canonical list of available weapons and equipment difficult, since new items are continually introduced as the timeline progresses. The *BattleMech Manual* presents all the most common items as of the in-universe year 3145 that are available to BattleMechs and serve a purpose in 'Mech versus 'Mech combat. The Weapons and Equipment Tables, beginning on page 120, provide a simplified summary of each item.

BASIC INFORMATION

All items in this chapter are assumed to obey the following rules unless specifically stated otherwise:

Ammunition: If the item is a weapon requiring ammunition, a critical hit to its ammunition bin deals the weapon's damage times the number of shots left in that bin as an internal explosion to the affected location (see *Ammunition*, under *Critical Hit Effects*, p. 47).

If an ammunition type is stated as non-explosive, a critical hit to a bin containing that ammo still disables the bin for the rest of the game.

If a bin is empty and takes a critical hit, the slot still takes the critical hit, but but no ammunition explosion or effects occur.

Artillery & Mines: With the exception of Arrow IV Homing Missiles (see p. 97), these weapons are dealt with separately, under *Battlefield Support* (see p. 75).

Critical Hit Effects: A critical hit (see p. 45) to any slot of an item destroys that item. Critical hits to intact slots of an already-destroyed item are still absorbed, but have no further effect.

Heat Effects: If the item is a weapon requiring ammunition, overheating ammunition bins risk exploding. The 'Mech suffers the weapon's listed damage times the number of shots remaining unless the ammo is non-explosive (or the bin is empty). See *Ammunition*, under *Effects of Heat*, page 52.

CHAPTER CONTENTS

Items in this chapter are ordered as follows:

- Ranged Weapons (pp. 97-106)
- Special Munitions (pp. 106-108)
- Physical Weapons (pp. 108-109)
- Electronics (pp. 109-114)
- Structural Components (pp. 114-117)
- Miscellaneous Equipment (pp. 118-119)

WEAPON TYPES

Each weapon uses a series of abbreviations, given below, to detail any special effects it may have. Some types have no effect in 'Mech-only combat, but are still listed here in order to maintain compatibility across the wider *BattleTech* ruleset.

Weapons often feature multiple types at once. Additionally, some weapons (such as hyper-assault Gauss rifles) may take on different types at different times.

- **AE: Area-Effect.** AE attacks damage every target in their area of effect (this can be multiple hexes)—terrain, buildings, even friendly 'Mechs. In *BattleMech Manual* play, when determining AE damage to a building, double the normal damage and apply this once to the building's CF. Against 'Mechs, every 'Mech in the area of effect is damaged, regardless of the level it is at or if it is inside a building (i.e. no building reduces AE damage to 'Mechs inside it), unless the 'Mech is underwater.
- **AI: Anti-Infantry.** This has no additional effect versus 'Mechs.
- **C: Cluster.** Each shot in a cluster attack contains multiple projectiles, not all of which may hit the target. Cluster attacks use the Cluster Hits Table (see p. 33) to resolve their damage.

  The Damage Value for cluster attacks has two parts. For example, a silver bullet Gauss rifle's Damage Value is C1/15.

  The number to the right is the attack's weapon size. If a cluster attack hits, the attacker rolls 2D6 and consults the Cluster Hits Table. Weapon size tells you which table column to use: cross-reference this with the roll result. This gives the number of projectiles that hit the target (e.g. rolling a 7 on the Cluster Hits Table for an attack with a weapon size of 15 means that 9 projectiles hit). Multiply the number of projectiles that hit by the damage per projectile (1 damage each, unless stated otherwise; Missile weapons state the damage each missile deals). This is the attack's damage total.

  The number to the left is the attack's grouping size. This value ranges from 1 to 5. After a cluster attack hit, split its damage total into Damage Value groupings equal to the attack's grouping size. For example, an attack with a grouping size of 1 forms every point of damage dealt into its own Damage Value grouping, while an attack with a grouping size of 5 has every 5 points of damage form its own grouping. If at any point there's not enough damage to make a full damage grouping, form a single undersized damage grouping.

  The attacker rolls a separate hit location for every grouping. For example, for all standard SRM launchers (C2), every 2 points of Damage Value dealt is a separate hit with its own hit location roll. For all ATM, LRM, and MRM launchers (regardless of the launcher size or damage per projectile), the grouping size is 5 (C5): every 5 points of Damage Value dealt is a separate hit with its own hit location roll.

  Each grouping only ever deals one hit. For example, an LRM strike dealing 5 damage to the head is one hit to the head, not five.

- **DB: Direct-Fire Ballistic.** These weapons (except machine guns) can use a targeting computer when making attacks (see *Targeting Computer*, p. 114).
- **DE: Direct-Fire Energy.** These weapon types (except flamers) can use a targeting computer when making attacks (see *Targeting Computer*, p. 114).
- **E: Electronics.**
- **F: Flak.** Apply a –2 Target Number modifier to attacks made by a flak weapon against dropping 'Mechs (see p. 79). This is cumulative with all other modifiers provided by the weapon or its ammunition unless specified otherwise.
- **H: Heat-Causing.** This weapon may inflict additional heat, damage, or both (see also *Outside Heat Sources* in the *Heat* chapter, p. 50, for details).
- **M: Missile.** A target's anti-missile systems may reduce the number of missiles from a missile attack (see *Anti-Missile Systems*, p. 118). Most missiles are also Cluster weapons (see above).
- **OS: One-Shot.** One-shot weapons can only be fired once per game, and may not carry any additional ammunition. The single shot carried by an OS launcher may be of any munition type available to that launcher type. Despite using ammunition, OS weapons are not explosive unless specified as such.
- **P: Pulse.** Pulse weapons using a targeting computer receive the computer's Target Number bonus, but not when making any kind of Aimed Shot (see *Targeting Computer*, p. 114).
- **PB: Point-Blank.** Point-blank weapons can only be used against targets in the same or adjacent hexes.
- **R: Rapid-Fire.** A rapid-fire weapon may optionally fire multiple shots at a single target as its attack. The number after "R" in the Damage column of the appropriate Weapons and Equipment tables indicates the maximum number of shots that can be made in that attack (for example, an R2 weapon may fire one or two shots as its attack, while an R6 weapon may fire anywhere from one to six shots as its attack).

  When firing multiple shots in a single attack (i.e. "rapid-firing"), the weapon consumes ammo and creates heat for each shot fired; whether any hit is irrelevant. On a successful attack, the attacker rolls 2D6 and compares the result to the appropriate column of the Cluster Hits Table (see p. 33). The Weapon Size for the attack is the number of shots fired by the weapon that turn, and so this can change from turn to turn (for example, a Rotary AC/5 rapid-firing four shots uses the "4" column, but would use the "6" column if rapid-firing six shots, whereas a rapid-firing Ultra AC would always use the "2" column). The 2D6 result is the number of shots that strike the target.

  Once the controller determines the number of shots that hit, a different hit location roll is made for each shot. Each shot deals damage equal to the weapon's Damage Value.

  Each time a rapid-fire weapon fires more than one shot in a single turn, it may jam. This is determined by the attack's to-hit roll, which is made even if the attack would automatically hit. When firing two or three shots, a jam occurs on an attack roll of 2. If firing four to five shots, a jam occurs on a to-hit roll of 3 or less. If firing six to seven shots, a jam occurs on a to-hit roll of 4 or less, and so on. If the weapon jams, it is useless for the rest of the game (see *Rotary Autocannon*, p. 98, for the exception); the shots are still fired, however, and can still hit the target as normal.
- **S: Switchable Ammo.** This weapon can choose from multiple ammo types, as detailed in the *Special Munitions* sections for that weapon. The type being fired must be announced during attack declaration (see p. 24).
- **V: Variable Damage.** The Damage Value of weapons with variable damage declines at one or more range brackets. Such weapons always have three Damage Values: their values at short, medium, and long range (e.g. a weapon with Damage Values of V25/20/10 deals 25 damage points at short range, 20 at medium, and 10 at long).
- **X: Explosive.** Item explodes when damaged; see the individual item's description for the exact details.

ARTILLERY

ARROW IV HOMING MISSILES (SIMPLIFIED)

"Agincourt II: Electric Boogaloo."

An Arrow IV missile can be fired indirectly (similarly to an Artillery Support attack), or directly:

- **Indirect (only if target is more than 17 hexes away):** During any turn's Weapon Attack Phase, the player must announce that an Arrow IV Homing Missile is incoming and secretly write down the target hex number (make sure to mark off the ammo and allocate heat for the firing 'Mech). The Arrow IV attack arrives during the Weapon Attack Phase of the *following* turn.
- **Direct (only if target is 17 or less hexes away):** A direct-fire attack is made during the Weapon Attack Phase and has no time in flight: it arrives the same phase it is launched.

On the turn the homing missile arrives (or fired, if using Direct fire), it may attack any 'Mech successfully designated by friendly TAG within 8 hexes of the chosen hex. If there are multiple such 'Mechs, the firer chooses which to attack. If there are no such 'Mechs, it explodes harmlessly over the battlefield. (Undirected or misdirected missiles do not scatter.)

The firer then rolls 2D6. On a result of 4 or higher, the missile strikes the target, dealing 20 points of damage. This is treated as a hit from a Direct-Fire Ballistic weapon (i.e. in this instance it is *not* an AE type weapon), resolved against the 'Mech's facing relative to the 'Mech that fired the missile. On a result of 3 or less, however, the missile hits the hex occupied by the designated target instead, and inflicts 5 points of area-effect damage to all targets in the hex.

A friendly TAG may designate a target for any number of Arrow IV Homing Missiles per round from any number of attackers, with only one attack roll required to designate the target. However, a TAG-equipped 'Mech may not designate multiple targets for Arrow IV in a turn, even if it is equipped with multiple TAG units. If a 'Mech with an on-board Arrow IV launcher using homing missiles also mounts TAG, the 'Mech may designate a target and fire its homing missiles against it in the same turn.

An Arrow IV missile cannot make an Aimed Shot. It is unaffected by anti-missile systems of any kind. Nemesis Narc pods can result in an Arrow IV missile attacking a friendly target; see that item's description (p. 108) for details.

Tech Base: Clan and Inner Sphere

AUTOCANNONS

"Autocannon" (often abbreviated as "AC") is a general term for modern 'Mech-scale cannons, which have a wide variety of calibers, firing speeds, and munition types. Some fire a single massive round, others are shotgun-like weapons, and still others are high-speed Gatling-like cannons that spew out a large number of rounds in a short span of time.

Special Munitions: Some autocannons are able to employ special autocannon munitions, as detailed on page 106. Whether or not such munitions are available is noted for each autocannon type. Note that tech base restrictions still apply: a weapon may only select those munitions available to its tech base.

AUTOCANNON

"This is my BOOMSTICK!"
Special Munitions: Yes
Technology Base: Clan or Inner Sphere
Type: DB, S

LB-X AUTOCANNON (LB-X)

"Nothin' says lovin' like a face full of 'Mech-scale buckshot."
Special Munitions: Yes (cluster munitions only)
Technology Base: Clan or Inner Sphere
Type: DB, S

LIGHT AUTOCANNON (LAC)

"Lighter and sleeker, but no less effective. Well, maybe a bit."
Special Munitions: Yes
Technology Base: Inner Sphere
Type: DB, S

PROTOMECH AUTOCANNON (PAC)

"Not just for Protos anymore."
Special Munitions: Yes
Technology Base: Clan
Type: DB, S

ROTARY AUTOCANNON (RAC)

"It costs 72,000 C-bills to fire this marvel of Davion engineering for 10 seconds."

Game Rules: Rapid-fire, six shots maximum. Though a rapid-fire weapon, a RAC is an exception in that it can be unjammed during the battle.

Unjamming the weapon is started by declaring that action during the End Phase of a turn. On the next turn, the 'Mech unjamming its RAC must stand still or only use Walking MP, and cannot make weapon attacks, including spotting. At the end of the Weapon Attack Phase, the 'Mech's controller makes a Gunnery Skill Roll with a +3 Target Number modifier. Success means that the

weapon can be used again after that turn, while a failed roll means that the jam is not cleared, though the player may attempt to clear the weapon in a subsequent turn. Attempts can be made to clear multiple jammed RACs at once, though each individual weapon may only be the subject of a single clearing attempt per turn: a separate Gunnery Skill Roll is made for each attempt.

Critical Hit: If a rotary autocannon suffers a critical hit while jammed, an ammo explosion results with a Damage Value equal to a single shot for the weapon.

Special Munitions: No
Technology Base: Clan or Inner Sphere
Type: DB, R

ULTRA AUTOCANNON (UAC)

"Master Technician, I'd like to speak with you about my AC's firing circuitry..."

Game Rules: Rapid-fire, two shots maximum.
Special Munitions: No
Technology Base: Clan or Inner Sphere
Type: DB, R

FLAMERS

Flamers are a catch-all term for the wide variety of flamethrowers in use by the armed forces of the thirty-first century. Some tap into the 'Mech's own reactor for the ravening heat necessary, while others require ammunition for their devastating properties to work.

Ammunition: By default, flamers do not require ammunition. Those that do are noted as such.

External Heat: Every time a flamer is fired, before the attack roll is made, the player may announce they are applying heat to the target, instead of the weapon's normal Damage Value. The amount of heat applied is the same as the weapon's Damage Value, unless otherwise noted.

Enhanced Flamers (Optional): If using this optional rule, whenever a flamer of any kind is fired at a target that tracks heat (such as a 'Mech), instead of choosing whether to deal heat or damage to the target, a hit applies both.

Type: AI, DE, H

FLAMER

"Nice national park you got here. Be a shame if something were to happen to it."

Technology Base: Clan or Inner Sphere

ER FLAMER

"Less spread on this one, but I still wouldn't call it 'friendly fire.'"

External Heat: An ER flamer only deals 1 point of heat.
Technology Base: Clan or Inner Sphere

HEAVY FLAMER

"Remember, MechWarriors: only YOU can prevent infantry."

Ammunition: A heavy flamer requires ammunition. A critical hit to a heavy flamer ammunition slot causes the ammunition to explode for 5 damage times the number of shots remaining.

Critical Hit: Heavy flamer ammo explodes for 5 points per shot.
Technology Base: Clan or Inner Sphere

GAUSS RIFLES

A Gauss rifle uses a magnetized series of coils or rails, rather than chemicals or powder, to accelerate a ferrous slug to incredibly long ranges with almost negligible heat.

Ammunition: All Gauss weapon ammunition is non-explosive.

Critical Hit: The first critical hit to a Gauss rifle causes it to explode as if an ammunition critical slot was hit. The damage this causes varies by Gauss rifle type (see below), but such an explosion does not affect the rifle's remaining undamaged critical slots. Future critical hits to that rifle do not cause further rifle explosions.

GAUSS RIFLE

"This machine kills Clanners."

Critical Hit: The first critical hit to a Gauss rifle results in a 20-point ammunition explosion.

Technology Base: Clan or Inner Sphere
Type: DB, X

ANTI-PERSONNEL GAUSS RIFLE

"For when a machine gun just won't do."

Critical Hit: The first critical hit to an anti-personnel Gauss rifle results in a 3-point ammunition explosion.

Technology Base: Clan
Type: AI, DB, X

HEAVY GAUSS RIFLE

"Bigger is *better."*

Game Rules: A 'Mech expending MP and firing a heavy Gauss rifle in the same turn must make a Piloting Skill Roll (see p. 53) at the end of the Weapon Attack Phase, with a modifier based on the 'Mech's weight class: Assault –1, Heavy 0, Medium +1, Light +2.

Unlike other PSR effects, this is not cumulative; if a 'Mech fires two heavy Gauss rifles in the same turn, two rolls would be required, but only the single modifier would apply to each roll.

Critical Hit: The first critical hit to a heavy Gauss rifle results in a 25-point ammunition explosion. If the rifle is split between the center torso and side torso, CASE located in the side torso that contains the heavy Gauss rifle stops any transfer of damage to the center torso; if the rifle is struck in the center torso, however, a 25-point ammunition explosion applies to the center torso.

Technology Base: Inner Sphere
Type: DB, V, X

HYPER-ASSAULT GAUSS RIFLE

"You had me at 'hail of Gauss slugs.'"

Game Rules: When a hyper-assault Gauss rifle, or HAG, hits its target, roll on the appropriate Cluster Hits column, but apply a +2 modifier to this roll at short range and a –2 modifier at long range. Treat a result of less than 2 as 2, and a result of greater than 12 as 12.

When firing a HAG at a combat-dropping 'Mech (see pp. 79-80), its controller can choose to make a Flak attack: the attack gains the Flak type, in addition to the weapon's normal types, and cannot benefit from a targeting computer. Apply a –1 Target Number modifier to such attacks, in addition to the standard –2 Flak modifier (so –3 total).

Critical Hit: The first critical hit to a hyper-assault Gauss rifle results in an ammunition explosion, with the HAG-20 delivering a 10-point explosion, the HAG-30 a 15-point explosion, and the HAG-40 a 20-point explosion.

Technology Base: Clan
Type: C, DB, X

IMPROVED HEAVY GAUSS RIFLE

"Drag chute optional."

Game Rules: A 'Mech expending MP and firing an improved heavy Gauss rifle in the same turn must make a Piloting Skill Roll (see p. 53) at the end of the Weapon Attack Phase, with the following modifiers based on the 'Mech's weight class: Assault –1, Heavy 0, Medium +1, Light +2. Unlike other PSR effects, this is not cumulative; if a 'Mech fires two of these weapons in the same turn, two rolls would be required, but only the single modifier would apply to each roll.

Critical Hit: The first critical hit to an improved heavy Gauss rifle results in a 30-point ammunition explosion. If the rifle is split between the center torso and side torso, CASE located in the side torso that contains the rifle stops any transfer of damage to the center torso; if the rifle is struck in the center torso, however, a 30-point ammunition explosion applies to the center torso.

Technology Base: Inner Sphere
Type: DB, X

LIGHT GAUSS RIFLE

"This is undoubtedly the greatest weapon the Inner Sphere has ever seen." – Chief of FWLM Procurement

Critical Hit: The first critical hit to a light Gauss rifle results in a 16-point ammunition explosion.

Technology Base: Inner Sphere
Type: DB, X

MAGSHOT GAUSS RIFLE

"No utility, no problem!"

Critical Hit: The first critical hit to a Magshot Gauss rifle results in a 3-point ammunition explosion.

Technology Base: Inner Sphere
Type: DB, X

SILVER BULLET GAUSS RIFLE

"Nothing says 'Get out of my airspace!' like hypersonic buckshot."

Game Rules: Attack rolls for a silver bullet Gauss rifle apply a –1 Target Number modifier.

Critical Hit: The first critical hit to a silver bullet Gauss rifle results in a 20-point ammunition explosion.

Technology Base: Inner Sphere
Type: C, DB, F, X

LASERS

Designed to deliver a concentrated burst of extreme heat to a small area, a modern laser (its name originates from the ancient acronym "Light Amplification by Stimulated Emission of Radiation") is the most basic of energy weapons.

Ammunition: By default, lasers do not require ammunition. Those that do are noted as such.

LASER

"Red or blue: it's still half the battle."

Technology Base: Clan or Inner Sphere
Type: DE

BINARY LASER (BLAZER) CANNON

"So big. So impractical. I'm surprised it's not on every second Lyran 'Mech."

Technology Base: Inner Sphere
Type: DE

CHEMICAL LASER

"Why are there 'Do Not Eat' labels on the ammunition?"

Ammunition: Chemical lasers require ammunition, which follows all normal ammunition rules for critical hits and heat.

Technology Base: Clan
Type: DE

EXTENDED-RANGE (ER) LASER

"Reach out and touch someone."

Technology Base: Clan or Inner Sphere
Type: DE

EXTENDED-RANGE (ER) PULSE LASER

"Shoot farther, not smarter."

Game Rules: Attack rolls for ER pulse lasers apply a –1 Target Number modifier.

Technology Base: Clan
Type: AI (ER small pulse laser only), P

HEAVY LASER

"The closest thing to a laser freight train."

Game Rules: Attack rolls for heavy lasers apply a +1 Target Number modifier.

Technology Base: Clan
Type: DE

IMPROVED HEAVY LASER

"So… they replaced the accuracy problems with high explosives?"

Game Rules: Improved heavy lasers do not suffer the usual +1 Target Number modifier for standard heavy lasers. However, treat the first critical hit to an improved heavy laser as an ammunition explosion, with the small improved heavy laser causing a 3-point explosion, the medium causing a 5-point explosion, and the large causing an 8-point explosion.

Technology Base: Clan
Type: DE, X

PULSE LASER

"Hey! Who broke my large pulse laser's TarComp interface?"

Game Rules: Attack rolls for pulse lasers apply a –2 Target Number modifier.

Technology Base: Clan or Inner Sphere
Type: AI (small pulse laser and micro pulse laser only), P

RE-ENGINEERED (RE) LASER

"The laser so nice they engineered it twice."

Game Rules: Attack rolls for re-engineered lasers apply a –1 Target Number modifier. RE lasers also treat each armor circle provided by hardened, ferro-lamellor, and laser-reflective armors as 1 point of standard armor. Versus RE lasers, these armors provide no special protective bonuses of any kind.

Technology Base: Inner Sphere
Type: P

VARIABLE-SPEED PULSE (VSP) LASER

"Note: Your laser's stutter is intentional."

Game Rules: Just as the Damage Value of variable speed pulse lasers changed based on the range bracket the weapon is fired in, attack rolls made with such weapons benefit from various modifiers based on that range:

- Short range: –3 Target Number modifier
- Medium range: –2 Target Number modifier
- Long range: –1 Target Number modifier

Technology Base: Inner Sphere
Type: AI, P, V

X-PULSE LASER

"The 'X' makes it sound cool."

Game Rules: Attack rolls for X-pulse lasers apply a –2 Target Number modifier.

Technology Base: Inner Sphere
Type: AI (small X-pulse laser only), P

M-POD

"Shocking shrapnel shower shreds Savage Sorenson's Solaris streak!"

The short range and one-shot capability of this Solaris-designed weapon has prevented it from being widely used in a battlefield capacity.

Game Rules: Attacks with M-Pods apply a –1 Target Number modifier and the following range: 1 hex for short, 2 for medium, 3 for long. Use a Weapon Size of "15" on the Cluster Hits Table (see p. 33) at Short range, the "10" column at Medium range, and the "5" column at Long range. An M-Pod may attack targets in the same hex as the attacker.

Critical Hit: Unexpended M-Pods that take a critical hit explode for 5 points of damage as an ammunition explosion. Expended M-Pods can still be affected by a critical hit in the same way as empty ammo bins.

Technology Base: Inner Sphere
Type: C, OS, V, X

MACHINE GUNS

The machine gun (MG) has been favored as the quintessential anti-infantry weapon since the first crude models appeared in the nineteenth century.

Technology Base: Clan or Inner Sphere

Rapid-Fire Mode (Optional): Using this option, any machine gun may be modified to fire at a much higher rate, chewing up huge amounts of ammunition but greatly increasing the damage potential. The damage and heat inflicted and amount of ammo used depends on the type of machine gun used:

- For light machine guns, the damage and heat is 1D6–1 (to a minimum of 1).
- For standard machine guns, the damage and heat is 1D6.
- For heavy machine guns, the damage and heat is 1D6+1.

Each rapid-fire burst uses a number of rounds equal to the heat generated above x 3.

The controlling player must mark any machine guns to be used in rapid-fire mode on the record sheet at the beginning of the game. Such MGs can only be rapid-fired during the game. Machine guns cannot be fired in rapid-fire mode when part of a machine gun array (see p. 102); to use rapid-fire mode, the array must be shut off before the game begins, and left off throughout.

MACHINE GUN (MG)

"Damn, he's 91 meters out."

Type: AI, DB

HEAVY MACHINE GUN (HMG)

"We decided to replace that pesky gunpowder with a larger bullet."

Type: AI, DB

LIGHT MACHINE GUN (LMG)

"How does a 'light' machine gun weigh 500 kilos?"

Type: AI, DB

MACHINE GUN ARRAY (MGA)

"You see, it's kind of like a gatling gun made out of gatling guns…"

Game Rules: When a machine gun array hits a target, roll on the Cluster Hits Table to determine how many machine guns in the array hit. The Weapon Size used is the number of machine guns in the array (e.g., if two machine guns, roll on the "2" column, if 4, use the "4" column). The roll result is the number of hits that struck the target, with each hit's damage equal to the Damage Value of the machine gun type in the array.

Once the number of hits is determined, roll once on the appropriate Hit Location Table and apply and resolve all hits to that location one at a time. This means that if an MGA 4 successfully hits its target, the subsequent roll on the Cluster Hits Table indicates that three machine guns hit, and the hit location is determined to be the target 'Mech's head, then the weapon causes 3 points of MechWarrior damage and forces three Consciousness Rolls.

Like heat sinks, an MGA can be turned off during the End Phase of a turn. In any subsequent turn, the grouped machine guns can be fired individually. The MGA can be turned back on during the End Phase of any turn.

MISSILE LAUNCHERS

Ammo Explosions: A critical hit to a missile ammunition bin causes damage equal to the launcher size times the number of remaining shots times the damage inflicted per missile.

OS-Type launchers do not explode when hit.

Advanced Fire Control: Some launchers may utilize advanced fire control systems, such as Artemis. Those launchers that may do so will be noted, along with the type of fire control available. Note that tech base restrictions still apply; for example, a launcher type listed as being compatible with Artemis V still needs to be a Clan version of that launcher.

Indirect Fire: Some launchers may perform an indirect attack (see p. 30). Those that can will be noted.

Special Munitions: Some launchers are able to employ special missile munitions, as detailed on page 106. Those that can will be noted. Note that tech base restrictions still apply; a launcher may only select those munitions available to its tech base.

> *If a bin of SRM-6 ammunition that had already fired three shots was critically hit, then the total damage would be 6 (launcher size) x 12 (number of shots remaining in the bin) x 2 (damage per missile) = 144 total damage.*

ADVANCED TACTICAL MISSILE (ATM) LAUNCHER

"For the Clan Warrior who thinks LRMs don't have enough gimmicks."

Advanced Fire Control: ATMs have integral Artemis IV, which performs as its standard counterpart (i.e., adds +2 to its Cluster Hits Table roll; shots fired into or through an enemy ECM field lose this bonus), and requires no extra critical slots or tonnage.

Indirect Fire: No

Special Munitions: ATMs may only use Standard, Extended Ranged (ER), and High Explosive (HE) ATM munitions. These have no special rules, only varying range and damage values, and so all relevant information is found on the Clan Weapons Table (see p. 125). All three munition types benefit from the launcher's Artemis effect. Each requires its own ammo bin.

Regardless of the munition type used, damage from ATM hits is divided into 5-point damage groupings.

Technology Base: Clan

Type: C, M, S

ENHANCED LRM LAUNCHER (NLRM)

"You fool! You thought you had me cornered, but I have a smaller minimum range! And… backup weapons."

Advanced Fire Control: Yes (Artemis IV; see p. 110)

Indirect Fire: Yes

Special Munitions: Yes

Technology Base: Inner Sphere

Type: C, M, S

EXTENDED LRM LAUNCHER (ELRM)

"A kilometer's worth of pain. Just watch out for that minimum arming distance."

Game Rules: When attacking targets within their minimum range (10 hexes or less), ELRMs must use the Cluster Hits column for half their rated rack size (rounding up). For example, an ELRM-15 that hits within its minimum range would roll on the "8" column for Cluster Hits, rather than the 15 column (15 ÷ 2 = 7.5, rounded up to 8).

Advanced Fire Control: No
Indirect Fire: Yes
Special Munitions: No
Technology Base: Inner Sphere
Type: C, M

IMPROVED ONE SHOT (I-OS) MISSILE LAUNCHER

"After decades of research, we have finally discovered muzzle-loading LRMs." – LAW R&D

Game Rules: I-OS launchers can consist of any size MML, MRM, SRM, SRT, LRM, LRT, NLRM, or NLRT launcher, Thunderbolt launcher, or Narc and iNarc launchers, and use the same statistics as the launcher chosen, except they are also OS weapons (see p. 97).

A one-shot MML may have for its single shot any one munition type available to MML (not both types).

Advanced Fire Control: As per original launcher type
Indirect Fire: As per original launcher type
Special Munitions: As per original launcher type
Technology Base: Clan or Inner Sphere (as per original launcher type)
Type: As per original launcher type, plus OS

LONG RANGE MISSILE LAUNCHER (LRM)

"Locked, stocked, and sitting behind a nice tough hill."

Advanced Fire Control: Yes (Artemis IV and Artemis V; see p. 110)
Indirect Fire: Yes
Special Munitions: Yes
Technology Base: Clan or Inner Sphere
Type: C, M, S

MEDIUM RANGE MISSILE (MRM) LAUNCHER

"One giant leap backward. It's like the Succession Wars in missile form."

Game Rules: Attack rolls with MRMs apply a +1 Target Number modifier.

Advanced Fire Control: Yes (Apollo FCS; see p. 113)
Indirect Fire: No
Special Munitions: No
Technology Base: Inner Sphere
Type: C, M

MULTI-MISSILE LAUNCHER (MML)

"I've got Tandem-Charge SRMs, Infernos, Swarm-I, and Semi-Guided. Give me a target, and I'll frag it."

Game Rules: The multi-missile launcher can fire SRMs or LRMs, and each missile type retains its characteristic damage and ranges. Note that an MML-5 cannot draw on ammunition intended for a separate LRM-5 launcher.

Advanced Fire Control: Yes (Artemis IV; see p. 110)
Indirect Fire: Yes (LRMs only)
Special Munitions: Yes, as per missile type (SRM or LRM)
Technology Base: Inner Sphere
Type: C, M, S

ONE-SHOT (OS) MISSILE LAUNCHER

"The derringer of 'Mech weapons."

Game Rules: OS launchers can be any size MML, MRM, SRM, SRT, LRM, LRT, NLRM, or NLRT launcher, Thunderbolt launcher, or Narc and iNarc launchers, and use the same statistics as the launcher chosen, except they are also OS weapons (see p. 97).

A one-shot MML may have for its single shot any one munition type available to MML (not both types).

Advanced Fire Control: As per original launcher type
Indirect Fire: As per original launcher type
Special Munitions: As per original launcher type
Technology Base: Clan or Inner Sphere (as per original launcher type)
Type: As per original launcher type, plus OS

ROCKET LAUNCHER (RL)

"Surprisingly ineffective as a jump jet aid."

Game Rules: Attack rolls with rocket launchers apply a +1 Target Number modifier.

Advanced Fire Control: No
Indirect Fire: No
Special Munitions: No
Technology Base: Inner Sphere
Type: C, M, OS

SHORT RANGE MISSILE (SRM) LAUNCHER

"My Arctic Wolf *is the purest expression of "SRM is love, SRM is life". It needs more SRMs, though."*

Advanced Fire Control: Yes (Artemis IV and Artemis V; see p. 110)
Indirect Fire: No
Special Munitions: Yes
Technology Base: Clan or Inner Sphere
Type: C, M, S

STREAK LRM (SLRM) LAUNCHER

"As if the Clans needed an option to hit me with even more *missiles."*

Game Rules: Make an attack roll as normal. If successful, the Cluster Hits Table result is automatically 12 (no cluster roll is required); the player rolls to determine hit locations as per the regular cluster rules. If the roll fails, the launcher does not fire: no ammunition is used and no heat is generated. However, the attempted attack still counts as an attack for purposes such as physical attacks; for example, an attack made with a leg-mounted Streak launcher that does not hit still precludes making a kick with that leg that turn.

An attacker firing Streak missiles into or through a hostile Angel ECM bubble (see p. 113) must roll on the Cluster Hits Table (see p. 33) after a hit as though the launcher were a standard (non-Streak) model. Non-Angel ECM has no effect.

Advanced Fire Control: No
Indirect Fire: No
Special Munitions: No
Technology Base: Clan
Type: C, M

STREAK SRM (SSRM) LAUNCHER

"Locked on target!"

Game Rules: As per Streak LRMs (see above)
Advanced Fire Control: No
Indirect Fire: No
Special Munitions: No
Technology Base: Clan or Inner Sphere
Type: C, M

THUNDERBOLT MISSILE LAUNCHER

"The biggest bang for your buck since the Arrow IV."

Game Rules: Thunderbolt missiles are not cluster weapons, but deliver their damage to a single location instead. A Thunderbolt missile strike that occurs within the weapon's minimum range delivers only half the missile's rated damage, rounded down.

Anti-missile systems may engage an incoming Thunderbolt missile, using the standard rules for AMS versus a single missile (see p. 118).

Advanced Fire Control: No
Indirect Fire: Yes
Special Munitions: No
Technology Base: Inner Sphere
Type: M

TORPEDO (SRT, LRT, NLRT) LAUNCHER

"Flood the tubes! I've got an Atlas *to sink!"*

Game Rules: Torpedoes are both a launcher type and a munition type. Torpedo launchers can be any standard 'Mech-sized SRM, LRM, or NLRM launcher, and use the same statistics as the launcher type chosen.

Torpedoes may only be fired from launchers submerged in water of Depth 1 or deeper. The target must be at least partially submerged, and the attacker must trace LOS to the target entirely through water hexes of Depth 1 or deeper. If a torpedo hits a location that is not submerged, reroll the location.

See *Attacking Underwater*, page 64, for more details.

Advanced Fire Control: Yes (Artemis IV and Artemis V; see p. 110)
Indirect Fire: Yes (LRT and NLRT only)
Special Munitions: Artemis-equipped torpedoes only
Technology Base: Clan or Inner Sphere
Type: As per original launcher type

NARC MISSILE BEACONS

Rather than lobbing missiles, a Narc beacon fires large grappling munitions called "pods". The most common is the homing pod, which improves the accuracy of those missiles equipped with properly attenuated sensors. However, the more advanced iNarc system offers several specialty pod types, providing a user with a variety of tricks.

Buildings: Narc pods cannot be fired into or inside buildings, but may be fired at them.

Special Munitions: See page 107
Type: M, S

NARC MISSILE BEACON

"I call this one the 'Tattle-tale.'"

Game Rules: If a Narc missile beacon attack hits, a Narc pod is attached to the target 'Mech in the location struck, and are only destroyed if the location they are attached to is destroyed. In all following phases, any 'Mech attacking the target with Narc-equipped missiles (see p. 107) adds +2 to the result of their Cluster Hits Table roll. Additionally, once a Narc pod is attached to a target, all Narc-equipped missiles capable of indirect fire can be fired indirectly at the target without a spotter; all other standard modifiers for indirect fire apply (see p. 30). If used in this manner, the Narc-equipped missiles lose their +2 Cluster Hits Table modifier.

Narc homing pods do not function as long as they are within the area of effect of an active enemy ECM suite (see *ECM Suites*, p. 112), but the beacon itself is unaffected by ECM.

Critical Hit: Narc ammunition bins that suffer a critical hit explode for 2 points per pod (unless they are Explosive Pods; see p. 108).

Technology Base: Clan or Inner Sphere

iNARC MISSILE BEACON

"More pods, more problems."

Game Rules: The iNarc functions as a standard Narc missile beacon. However, iNarc pods are larger than standard Narc pods. As such, rather than making a physical attack in the Physical Attack Phase, a 'Mech may attempt to brush off iNarc pods which have been attached to it.

The 'Mech's controller can make up to two attempts to brush off pods each turn, one for each arm. A Piloting Skill Roll (see p. 53) is required for each attempt, with a +4 Target Number modifier as well as any modifiers normally applied to a punching attack (see *Punch Attacks*, p. 38). Arms that fired a weapon during the Weapon Attack Phase in the same turn are ineligible, as are arms mounting any physical attack weapon. A successful roll destroys one pod, chosen by the 'Mech's controller.

If the Piloting Skill Roll is unsuccessful, the 'Mech does not fall, but takes punching damage from the appropriate arm. The pod(s) stay attached. Roll 1D6 and consult the Front column of the Punch Location Table to determine the location of the damage. If the player declares that their 'Mech will make two attempts, both must be resolved, even if the first is successful (this could result in the first attempt being successful, yet the second attempt failing and still damaging the 'Mech).

Critical Hit: iNarc ammunition bins that suffer a critical hit explode for 3 points per pod (unless they are Explosive Pods; see p. 108).

Technology Base: Inner Sphere

PARTICLE PROJECTOR CANNONS

The particle projector cannon (PPC for short) is one of the most powerful non-ballistic weapons ever devised for the modern battlefield. Consisting of a magnetic accelerator, firing high-energy proton or ion bolts, PPCs have been used for centuries to flay armor through kinetic and thermal damage.

PPC

"'Azure whiplash?' No, too corny. 'Man-made lightning!' That's the ticket!" – Donal marketing exec

Technology Base: Clan or Inner Sphere
Type: DE

ER PPC

"The hand of Zeus."

Technology Base: Clan or Inner Sphere
Type: DE

HEAVY PPC

"This is the Lord's Light 3, the most powerful PPC in the Inner Sphere. It's my very favorite gun. I call her Doris."

Technology Base: Inner Sphere
Type: DE

LIGHT PPC

"Lighter package, same great range."

Technology Base: Inner Sphere
Type: DE

SNUB-NOSE PPC

"The thinking man's PPC."

Technology Base: Inner Sphere
Type: DE, V

PPC CAPACITOR

"Charge it!"

Game Rules: Instead of firing a PPC as normal, a 'Mech may charge its PPC capacitor (it cannot start a game charged, unless a specific scenario states otherwise). This requires one turn and builds up 5 points of heat. This charge can be held as long as needed, during which time the charged capacitor continues to generate 5 points of heat. When a charged PPC is fired, add 5 points of heat in addition to the heat the PPC normally generates.

For example, if a 'Mech with an ER PPC with a capacitor charges the weapon in Turn 1 but does not fire it until Turn 4, it would generate 5 points of heat from the charging capacitor for Turns 1 through 3, and 20 points in Turn 4 when it fires (15 for the ER PPC, plus 5 for the capacitor's charge release).

A PPC fired with an attached and charged PPC capacitor adds 5 points to its normal Damage Value. An attack roll result of 2 with a capacitor-charged PPC burns out the system before it can fire, inflicting a critical hit on the capacitor slot (in this instance, it does not explode; see below) as well as the first critical slot of the corresponding weapon, but inflicting no damage against the target.

Critical Hit: A PPC capacitor explodes if it or its attached PPC suffer a critical hit while charged, resulting in an internal explosion equal to the full combined maximum Damage Value of the PPC and its capacitor. The explosion also destroys all critical slots for both the PPC and its capacitor. However, if the PPC was declared as firing in the same Weapon Attack Phase, the capacitor does not count as charged: the critical hit only destroys the affected slot.

Technology Base: Clan or Inner Sphere
Type: X

PLASMA WEAPONS

Plasma weapons, rather than tapping a BattleMech's fusion reactor for their ammunition, instead incinerate a cartridge of plastic foam and use it as a projectile.

Game Rules: Versus woods, jungle or buildings, plasma rifles deal (10+2D6) × 2 damage, while plasma cannons deal 3D6 × 2 damage.

Ammunition: Non-explosive.
Type: DE, H

PLASMA CANNON

"We have perfected your 'Plasma Rifle', Spheroid. That will be one Assault 'Mech." – Clan Diamond Shark Merchant.

Game Rules: A plasma cannon has no Damage Value. However, a 'Mech hit by a Plasma Cannon must add 2D6 points of heat during that turn's Heat Phase (see *Outside Heat Sources*, p. 50).

Technology Base: Clan

PLASMA RIFLE

"For the pyromaniac who thinks they have everything."

Game Rules: In addition to its standard Damage Value of 10, a plasma rifle delivers 1D6 points of heat to 'Mechs during the Heat Phase (see *Outside Heat Sources*, p. 50).

Technology Base: Inner Sphere

RIFLES (CANNONS)

"Real cutting-edge stuff here. Might as well get a trebuchet."

Rendered obsolete by the modern autocannon, rifles are rarely seen in the Inner Sphere (usually in the most backward areas), while the Clans have abandoned them entirely.

Game Rules: Rifles of all sizes must subtract 3 points of damage (to a minimum of 0) for successful attacks against any 'Mech. For example, a light rifle (which deals 3 damage) is incapable of damaging 'Mechs. However, they still do their full damage against terrain and buildings.

Technology Base: Inner Sphere
Type: DB

SPECIAL MUNITIONS

If multiple munition types are available for a given weapon, what munition type the weapon is firing must be announced during attack declaration (see p. 24).

AUTOCANNON MUNITIONS

Ammunition Multiplier: Some autocannon munitions provide greater or fewer shots per ton than a ton of standard ammunition. Multiply the standard shots per ton for that weapon by the munition's ammunition multiplier to find the number of shots per ton that munition type provides (for reductions, round the final total down).

For example, an AC/20 normally has 5 shots per ton. Ammunition with a 0.5 multiplier would grant 2 shots per ton instead.

ARMOR-PIERCING (AP) AMMO

Game Rules: Apply a +1 modifier to the attack's Target Number.

So long as it deals damage, a hit with AP ammo has a chance at causing critical damage, even if the target's internal structure took no damage. After applying the attack damage, roll once on the Determining Critical Hits Table (see p. 46). This roll is modified based on the type of AC used, with the AC/2 applying a –4 modifier, the AC/5 a –3 modifier, the AC/10 a –2 modifier, and the AC/20 a –1 modifier.

If using a ProtoMech Autocannon (PAC), use the following modifiers instead: PAC 2 (–4), PAC 4 (–3), PAC 8 (–2).

If the attack scores a through-armor critical hit (see p. 45), make two checks for potential critical damage, applying the AP ammo roll modifier to only one of those checks. However, AP ammo does not trigger its special critical check if the attack damages internal structure directly (i.e. if there was no armor left in that location). Damage that transfers to a location with armor does not trigger a new critical damage check in that location.

This ammunition loses its critical-enhancing ability against hardened or ferro-lamellor armor. However, the ammunition treats both armor types in all other ways as standard armor (1 point of damage eliminates one armor circle).

Ammunition Multiplier: x0.5
Technology Base: Clan (PAC only), Inner Sphere
Usable By: AC, LAC, PAC

CASELESS AMMO

Game Rules: On an attack roll result of 2, the ammo feed breaks down: the weapon is useless for the rest of the scenario. In addition, the shot automatically misses and the controlling player must immediately roll once on the Determining Critical Hits Table (see p. 46), applying all critical hits to the affected autocannon (starting with the uppermost undamaged slot). This is not an ammunition explosion, and MechWarrior damage does not occur as a result.

A weapon using caseless ammo can use no other special munition types.

Ammunition Multiplier: x2
Technology Base: Inner Sphere
Usable By: AC, LAC

CLUSTER AMMO

Game Rules: Apply a –1 modifier to the attack's Target Number. An attack with cluster ammunition is treated as both a type C (Cluster) and type F (Flak) attack. The weapon size is always equal to the autocannon rating (e.g. an AC/5 has a weapon size of 5), and the grouping size is always C1.

Ammunition Multiplier: x1
Technology Base: Clan or Inner Sphere
Usable By: LB-X

FLECHETTE AMMO

Game Rules: Double damage against woods or jungle hexes. Half damage versus all other targets (round down).

Ammunition Multiplier: x1

Technology Base: Clan (PAC only), Inner Sphere

Usable By: AC, LAC, PAC

PRECISION AMMO

Game Rules: Reduce the target's movement modifier by 2, to a minimum of zero.

Ammunition Multiplier: x0.5

Technology Base: Inner Sphere

Usable By: AC, LAC

MISSILE MUNITIONS

ARTEMIS IV MISSILES

Game Rules: Artemis IV-equipped missiles used with an Artemis IV FCS (see p. 110) gain a +2 modifier when rolling on the Cluster Hits Table (to a maximum modified result of 12). If interfered with by ECM, they may still be fired as normal (unaugmented) missiles of their base launcher type.

Artemis IV-equipped missiles provide no benefits when fired from an Artemis V launcher (and vice versa).

Technology Base: Clan or Inner Sphere

Usable By: LRM, LRT, MML, NLRM, NLRT, SRM, SRT

ARTEMIS V MISSILES

Game Rules: Artemis V-equipped missiles used with an Artemis V FCS (see p. 110) apply a –1 modifier to their attack rolls, and add +3 to rolls on the Cluster Hits Table (to a maximum modified result of 12). If interfered with by ECM, they may still be fired as normal (unaugmented) missiles of their base launcher type.

Artemis V-equipped missiles provide no benefits when fired from an Artemis IV launcher (and vice versa).

Technology Base: Clan

Usable By: LRM, LRT, SRM, SRT

FRAGMENTATION MISSILES

Game Rules: Double damage against woods or jungle hexes. No damage versus all other targets.

Technology Base: Inner Sphere

Usable By: LRM, MML, NLRM, SRM

INFERNO MISSILES

Game Rules: Infernos do not have a Damage Value. Instead, each missile that strikes a 'Mech increases its heat level by 2 points during the Heat Phase of the turn in which the missiles hit. Any additional heat dealt by infernos lasts for the current turn.

When rolling to avoid an ammunition explosion due to its own heat level, a 'Mech carrying Inferno ammunition must roll an additional set of Heat Scale Avoid Rolls to determine whether or not the ammo explodes per the Inferno Ammo Explosion Table. The inferno Avoid Rolls at 19, 23, and 28 heat points must be made in addition to the normal Avoid Rolls required at these heat levels. If the inferno ammo explodes, it inflicts 2 heat per missile to the carrying 'Mech, to a maximum of 30 heat, along with the standard damage from an SRM explosion.

Building Hexes: Each inferno missile that strikes a building hex inflicts 2 points of damage to the hex.

INFERNO AMMO EXPLOSION TABLE

Heat Level	Avoid Number	Heat Level	Avoid Number
10	4+	23	10+
14	6+	28	12
19	8+		

Additionally, each missile that strikes a building hex may also affect 'Mechs inside the building on the level of the building that was hit by inferno missiles. For every 'Mech on the level of the building that was hit, roll 1D6. On a result of 1-4, the missile has no effect against that 'Mech. On a result of 5-6, the missile also strikes the 'Mech. It is possible for one missile to strike multiple 'Mechs on the same level of the building hex. Note that inferno missiles are an exception to the *Attacking 'Mechs Inside Buildings* rules (see p. 71).

Fire: If an attacker is intentionally trying to start a fire, one or more inferno missiles fired at any woods, jungle, or building hex will automatically set that hex on fire. See *Starting a Fire*, page 63.

Woods: Every inferno missile that strikes a woods hex inflicts 4 points of damage to the hex.

Technology Base: Clan or Inner Sphere

Usable By: MML, SRM

NARC-EQUIPPED MISSILES

Game Rules: Narc-equipped missiles can home in on Narc homing pods (see both *Homing Pods* entries, p. 108). If interfered with by ECM, they may still be fired as normal (unaugmented) missiles of their base launcher type.

Technology Base: Clan or Inner Sphere

Usable By: LRM, MML, NLRM, SRM

SEMI-GUIDED LRMS

Game Rules: Against any target successfully designated by friendly TAG (see p. 113), these missiles ignore the target movement modifier (if fired indirectly, also ignore all indirect fire, terrain, and spotter-related modifiers).

Technology Base: Inner Sphere

Usable By: LRM, MML, NLRM

NARC / INARC PODS

Game Rules: Fired from Narc missile beacons (see p. 104). Multiple pods of the same type do not have a cumulative effect; they only add redundancy.

A pod, once attached, remains in effect for the entire battle. However, if the location a pod is attached to is destroyed, the pod is also destroyed and its effects are lost at the end of the phase in which the location was destroyed.

Critical Hit: Unless stated otherwise, a critical hit to a slot that contains a Narc ammunition bin causes it to explode, inflicting 2 points of damage per pod (3 points per iNarc pod).

ECM PODS

Game Rules: On a successful attack, treat the target as if within an ECM field hostile to it (see *ECM Suites*, p. 112). The ECM pod has no effect radius—it only works on the target to which it is attached.

Technology Base: Inner Sphere
Usable By: iNarc

EXPLOSIVE PODS

Game Rules: On a successful attack, the target suffers 4 points of damage (6 for iNarcs).

Critical Hit: A critical hit to a slot that contains explosive pods causes the pods to explode, inflicting 4 points of damage (6 for iNarcs) per pod remaining.

Technology Base: Inner Sphere
Usable By: Narc, iNarc

HAYWIRE PODS

Game Rules: On a successful attack, the target suffers a +1 Target Number modifier to all weapon attacks (including TAG) and may not spot for indirect fire.

Technology Base: Inner Sphere
Usable By: iNarc

HOMING PODS (NARC)

Game Rules: On a successful attack, in subsequent turns all friendly Narc-equipped missile hits on the target gain a +2 modifier when rolling on the Cluster Hits Table, and Narc-equipped missiles capable of indirect fire can also be fired indirectly at the target even without a spotter; see *Narc Missile Beacon* on page 104 for details. Homing pods are non-functional as long as they are within the area of effect of an active enemy ECM suite (see *ECM Suites*, p. 112).

Technology Base: Clan or Inner Sphere
Usable By: Narc

HOMING PODS (iNARC)

Game Rules: On a successful attack, in subsequent turns all friendly Narc-equipped missile attack rolls against the target apply a –1 modifier to their Target Number, and a +2 modifier when rolling on the Cluster Hits Table, and Narc-equipped missiles capable of indirect fire can also be fired indirectly at the target even without a spotter; see *Narc Missile Beacon* on page 104 for details. Homing pods are non-functional as long as they are within the area of effect of an active enemy ECM suite (see *ECM Suites*, p. 112).

Technology Base: Inner Sphere
Usable By: iNarc

NEMESIS PODS

Game Rules: Attacks using Artemis IV-capable, Narc-equipped, or semi-guided missiles will strike a friendly 'Mech tagged by an enemy Nemesis pod instead of their intended enemy target, if:

- the Nemesis-tagged 'Mech is along the LOS between the attacker and the original target, and
- LOS exists between the attacker and the Nemesis-tagged 'Mech, and
- the Nemesis-tagged 'Mech is not underwater.

Apply a +1 Target Number modifier to such unintentional attacks, starting with the closest Nemesis-tagged friendly if more than one 'Mech meets the criteria (if there is more than one such friendly, determine the friendly randomly).

These Narc missile beacon munitions can also attract Arrow IV homing missiles (see p. 108). If a Nemesis-tagged 'Mech is in a hex adjacent to the intended target in the turn the homing missile arrives, roll 1D6. On a result of 1–3, the attack hits the Nemesis-tagged target. On a result of 4–6, the attack is resolved against the intended target. If there are multiple adjacent Nemesis-tagged targets, there is an equal chance the homing missile will strike any of them if it does not strike the intended target.

Technology Base: Inner Sphere
Usable By: iNarc

PHYSICAL WEAPONS

Physical weapons are a wide variety of cutting, bashing, or slashing weapons mounted onto BattleMechs. These range from 'Mech-sized claws to ornate blades to massive, pulverizing maces. The Clans initially demonstrated no interest in melee weapons, finding them a primitive and wasteful "barbarian" alternative to "proper" ranged combat. However, their recent employment of claws may indicate that this attitude is changing.

Game Rules: In a single turn, a 'Mech can either fire the weapons mounted on an arm or deliver a physical weapon attack with that arm, but not both. While a 'Mech can have multiple physical weapons, it can only ever attack with one of them each turn unless the weapon specifies otherwise. For more details on physical attacks, see page 34.

Hit Location Table: Some physical weapons use the standard Hit Location Table, and some the Punch Hit Location Table. Still others by default use the standard Hit Location Table, but have the option to use the Punch Hit Location Table if this is announced before the attack and a +4 modifier is applied to the attack's Target Number. For each weapon, the annotation of Standard, Punch, or Option will be used to indicate which of the above three hit location methods applies.

TSM: Some physical weapons benefit from triple-strength myomer (see p. 117). Those that do will be noted (see also chart on p. 38).

CLAWS

"RIP AND TEAR!"

Game Rules: A claw applies a +1 Target Number modifier, and deals 1 point of damage for every 7 tons of the attacking 'Mech's weight (rounded up). A 'Mech with two claws can make two claw attacks each Physical Attack Phase.

'Mechs with claws may lift an object as normal (see *Lifting Items (Simplified)* p. 21), but must make a successful Piloting Skill Roll (see p. 53) with a +2 modifier to avoid damaging the object. If the object is extremely fragile (such as a human being), an additional +1 modifier applies (at the gamemaster's discretion or mutual player agreement). A failed attempt to lift an object effectively destroys it.

Claws may be used to grab and wield a club (see p. 36), with an additional +2 modifier to the attack's Target Number.

Hit Location Table: Punch.
TSM: Yes.
Technology Base: Clan or Inner Sphere

FLAIL

"If you get over the whole 'crushing your own cockpit' thing, it's not a bad weapon."

Game Rules: A flail deals 9 damage when it hits. However, on a flail attack roll result of 2 the attacker strikes itself instead of the intended target, dealing 5 points of damage (resolve using the Front column of the standard Hit Location Table). The attacker must also make a Piloting Skill Roll (see p. 53) to avoid falling. Flail damage is never reduced by absent or damaged actuators.

Hit Location Table: Standard.
TSM: No.
Technology Base: Inner Sphere

HATCHET

"I could bury this, but you wouldn't like it."

Game Rules: Attacks with a hatchet apply a –1 Target Number modifier, and deal 1 damage per 5 tons of the attacking 'Mech's weight.

Hit Location Table: Option.
TSM: Yes.
Technology Base: Inner Sphere

LANCE

"At last I can joust with a fusion bomb between my legs."

Game Rules: Attacks with a lance apply a +1 Target Number modifier, and deal 1 damage per 5 tons of the attacking 'Mech's weight. After a lance hit, if armor remains in the location struck (or facing, for torso locations), the attacker rolls 2D6. On a result of 10+, the hit also deals 1 point of internal structure damage; the resulting Determining Critical Hits Table roll applies a –2 modifier. (A lance attack that strikes unarmored internal structure rolls for Determining Critical Hits normally.)

Hit Location Table: Standard.
TSM: Yes.
Technology Base: Inner Sphere

MACE

"Walk as softly as an 80-ton 'Mech can and carry this."

Game Rules: Attacks with a Mace apply a +1 Target Number modifier, and deal 1 damage per 4 tons of the attacking 'Mech's weight (rounded up).

If a mace attack misses its target, the attacker must make a Piloting Skill Roll (see p. 53) with a +2 modifier to avoid falling.

Hit Location Table: Option.
TSM: Yes.
Technology Base: Inner Sphere

RETRACTABLE BLADE

"The time-honored tradition of shanking, now for BattleMechs."

Game Rules: Attacks with a retractable blade apply a –2 Target Number modifier, and deal 1 damage per 10 tons of the attacking 'Mech's weight (rounded up).

Piercing (Optional): With this option, if a 'Mech hits with a punch attack, and that arm has a retractable blade which is currently retracted, its controller can immediately extend that blade. This inflicts a possible critical hit in the location struck by the punch attack (regardless of whether internal structure was damaged or not; if the attack damages the target's internal structure as well, make two checks). The attacker must then immediately roll 2D6: on a result of 10+, the blade is destroyed (mark off the topmost critical slot).

Hit Location Table: Option.
TSM: Yes.
Technology Base: Clan or Inner Sphere

SWORD

"The noble art of fencing reduced to a brutal form. Bashing someone's 'Mech apart has never been so elegant."

Game Rules: Attacks with a sword apply a –2 Target Number modifier, and deal 1 damage per 10 tons of the attacking 'Mech's weight (rounding up), plus 1 additional point of damage.

Hit Location Table: Option.
TSM: Yes.
Technology Base: Inner Sphere

TALONS

"Partial wings and now this. I bet the Diamond Sharks make a Jade Falcon beak-and-feathers kit next."

Game Rules: When performing a kick or Death From Above attack, multiply the damage by 1.5 (round to nearest whole number; .5 rounds up). On mixed-technology 'Mechs, talons can receive the TSM damage bonus for kick attacks, but not for DFA attacks. Critical hits to the talons or the foot actuator in the leg where they are mounted destroy the talons.

Hit Location Table: As per attack type.
TSM: With kicks.
Technology Base: Clan

VIBROBLADE

"Lets even light 'Mechs carve up assaults like a chainsaw on butter."

Game Rules: Attacks with a vibroblade apply a –2 Target Number modifier. When declaring a vibroblade attack, the attacker must announce if the weapon is activated or deactivated. Activated vibroblades generate heat and deliver a fixed damage amount (7 for a small vibroblade, 10 for a medium, and 14 for a large), regardless of actuator damage or equipment such as TSM.

When deactivated, a vibroblade creates no heat and deals damage equal to 1 plus the 'Mech's tonnage divided by 10 (rounding up to the nearest whole number), modified by equipment and critical damage as normal. The damage dealt by a deactivated vibroblade is capped at the damage an activated vibroblade of that type deals, unless TSM is used, in which case the TSM modifier is applied to the capped damage.

Hit Location Table: Option.
TSM: If deactivated.
Technology Base: Inner Sphere

ELECTRONICS

Game Rules: Any system in this section can be deactivated in the End Phase of any turn. It may then be reactivated in any following End Phase.

ACTIVE PROBES

'Mechs with multiple active probes (of any type) may use only one at a time. These systems can be activated, deactivated, or switched over during the End Phase of any turn.

Hidden 'Mechs (Expanded) (Optional): When using the *Hidden 'Mechs* rules, a player normally determines if any enemy 'Mechs lie within the detection radius of an active probe after the 'Mech has finished moving (see p. 82). Using this option, the detection radius is active throughout the 'Mech's entire movement. This allows a probe-equipped 'Mech to detect hidden 'Mechs during its movement, instead of potentially passing a hidden 'Mech without detecting it.

Targeting (Optional): If a target is within the probe's range and line of sight exists to the target, reduce the total Target Number modifier for firing through and into woods/jungles by 1 (just the total, and only by 1, no matter how many woods/jungle hexes are involved). This bonus applies only to the 'Mech with the probe, unless that 'Mech is part of a C^3 network, in which case the bonus is passed along to other 'Mechs in that network.

ACTIVE PROBE/BEAGLE ACTIVE PROBE/LIGHT ACTIVE PROBE

"I see you..."

Game Rules: An active probe only affects game play if players are using the *Hidden 'Mechs* rules (see p. 82). Under those rules, an active probe can detect any hidden 'Mech, if at the end of a Movement Phase the concealed 'Mech lies inside the probe's range) and line of sight would exist between the 'Mech carrying the probe and the hidden 'Mech (if that 'Mech was not concealed).

The ranges for each probe type are as follows: Light Active Probe (3), Beagle Active Probe (4), Active Probe (5). Otherwise, all three probes behave the same, with any reference to "active probes" applying equally to all three types listed here.

Water: If an active probe is in a location of a 'Mech that is above the surface of a water hex, it cannot detect a 'Mech hidden underwater. For an active probe to work underwater, it must have a clear LOS to a hidden 'Mech that passes only through underwater hexes.

Technology Base: Clan (Active Probe, Light Active Probe) or Inner Sphere (Beagle)

BLOODHOUND ACTIVE PROBE

"… Yes, even you."

Game Rules: A Bloodhound Active Probe follows the same rules as regular active probes (see above). However, in addition to its increased range (8 hexes), the Bloodhound is not defeated by regular ECM suites. Only Angel ECM can defeat a Bloodhound active probe.

Technology Base: Inner Sphere

ARTEMIS IV FIRE-CONTROL SYSTEM

"Give the gift of good missile grouping."

Game Rules: Successful attacks by launchers using this system and firing Artemis IV-equipped missiles (see p. 107) apply a +2 to the Cluster Hits Table roll (to a maximum modified result of 12). Indirect fire attacks do not receive this bonus.

An Artemis FCS cannot be used with Streak launchers or Narc-equipped missiles. It can be used with other specialty munitions, but those munitions will not gain the benefits of the Artemis system. Artemis may also be used with OS missile launchers.

Critical Hit: If the Artemis system assigned to a specific launcher is destroyed, the missile launcher can still be fired as a normal launcher.

Technology Base: Clan or Inner Sphere

ARTEMIS V FIRE-CONTROL SYSTEM

"Why don't Clan missile launchers just come standard with this?"

Game Rules: Attacks by launchers using this system and firing Artemis V-equipped missiles (see p. 107) apply a –1 Target Number modifier and +3 to the Cluster Hits Table roll (to a maximum modified result of 12). Indirect fire attacks do not receive these bonuses.

An Artemis FCS cannot be used with Streak launchers or Narc-equipped missiles. It can be used with other specialty munitions, but those munitions will not gain the benefits of the Artemis system. Artemis may also be used with OS missile launchers.

Critical Hit: As per Artemis IV (see above).

Technology Base: Clan

C^3 COMPUTER SYSTEM

"I thought this firmware upgrade was going to let me stay out of sight when shooting?"

Game Rules: A C^3 computer system can link up to twelve 'Mechs together in a communications network that shares targeting information. When using a C^3 computer network, range to a target is calculated from the networked 'Mech nearest the target. This means that a well-placed 'Mech may allow an attacker to use their weapon's short-range Target Number modifier, even though the attacker might be at long range.

A weapon attack using a C^3 network must conform to standard LOS restrictions and cannot fire beyond its maximum range. The C^3 network itself has no maximum range, but only 'Mechs actually on the playing area can benefit from the network or provide information to it, and the C^3 Master (or C^3 Masters, if using a company-sized network) must be on the playing area.

Indirect Fire: A C^3 network does not help when launching or spotting targets for indirect fire (see p. 30). Note that TAG (which conveys the benefits of spotting if it hits) does benefit, however.

LOS: While 'Mechs must have LOS to a target to make an attack using a C^3 system, the C^3 system itself need not have LOS.

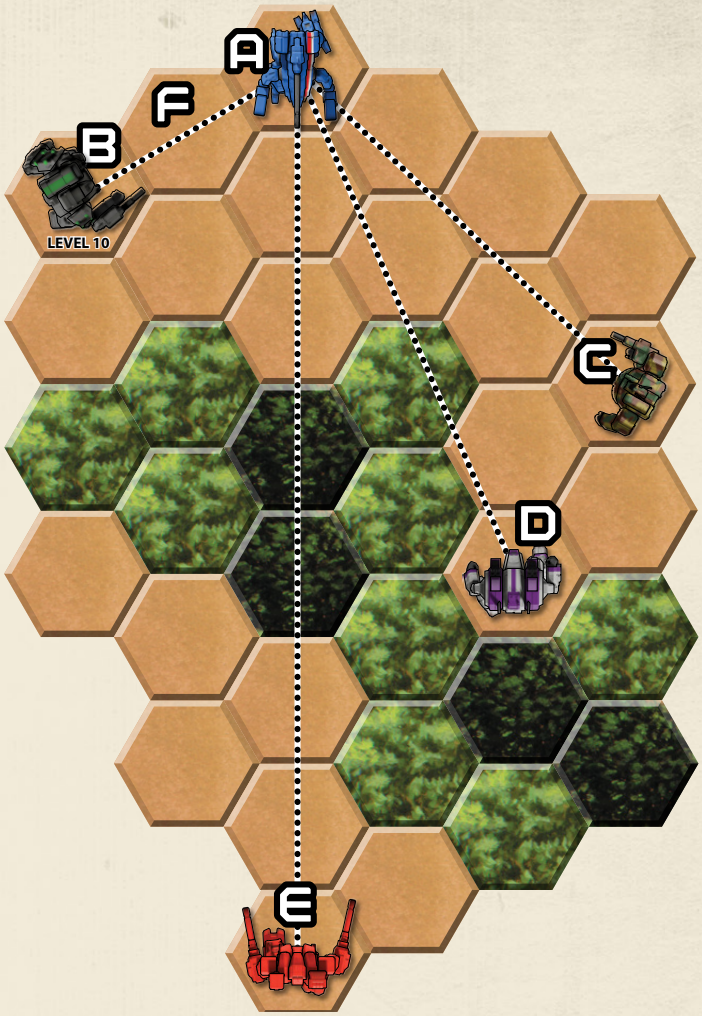
Minimum Range: This is calculated from the actual attacker.

Same Hex: If a networked 'Mech providing targeting data to other members of the network is in the same hex as a target (such as a building), attacks by those other members against that target are still legal.

Water: If the C^3 computer (any kind) is in a location above the surface of a water hex, it cannot be linked to a C^3 system underwater. Instead, C^3 underwater is only able to link to another C^3 that is also underwater.

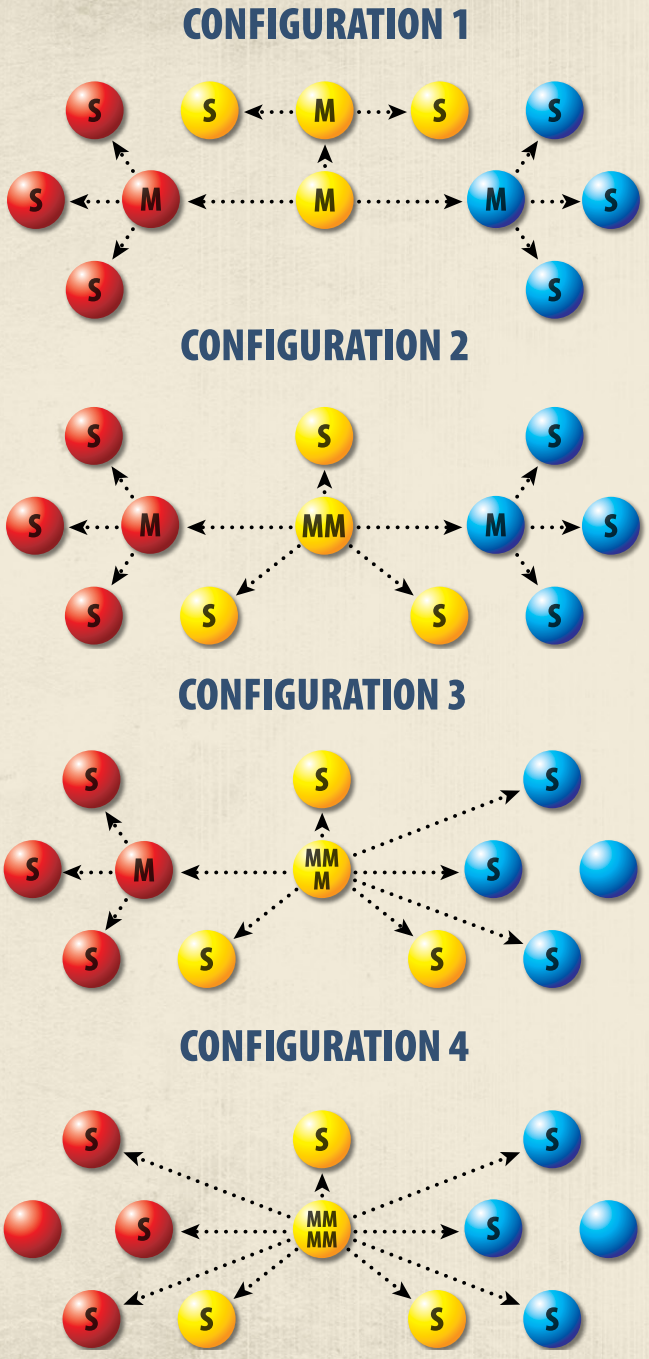
Weapons and Equipment: The following items and rules interact with C^3 systems in specific ways:

- **ECM:** ECM cuts off any C^3-equipped 'Mech from its network. Only those C^3 'Mechs able to draw a direct line to the master 'Mech that does not pass through an ECM's area of effect can access the network. See *ECM Suites*, page 112, for details.
- **Stealth Armor:** Stealth armor range penalties are calculated using the distance between the stealth-equipped 'Mech and the closest 'Mech to it in the network.
- **TAG:** A C^3 Master (but not C^3 Slaves) exactly duplicates the function of TAG (see *Target Acquisition Gear*, p. 113).
- **Variable Damage Weapons:** When determining the damage of a Type V (Variable Damage) weapon, always use the range from the attacking 'Mech to the target.



• C³ DIAGRAM •

• C³ CONFIGURATION DIAGRAM •



In the C³ Diagram, the enemy 'Mech in Hex A is facing 'Mechs in Hexes B, C, D, and E, which are connected to a C³ network. The 'Mech in Hex B is closest to the enemy, at a Range of 2; even though it is at Level 10, levels are never taken into consideration when determining range to target. The 'Mech in Hex C can attack as though it were at a Range of 2, provided the weapons it fires have a Maximum Range of 4 or more. The 'Mech in Hex D can also fire as though at a Range of 2, but must add the terrain modifier for firing through one hex of light woods. The 'Mech in Hex E cannot attack the enemy 'Mech in Hex A because no line of sight exists between the two.

In all cases, if the 'Mechs in Hex C, D, or E fired weapons with minimum ranges, those ranges must be taken into consideration, with the range determined from the attacker to the target, not the range from the 'Mech in Hex B to the target.

If there were a Level 2 hill in Hex F (blocking LOS between the 'Mech in Hex B and the target in Hex A), the 'Mechs in Hexes C and D would still be able to target the 'Mech in Hex A as though they were at Range 2.

DESIGNATING A C³ NETWORK

Prior to the start of play, the controller of a C^3 network must clearly designate on their 'Mech's record sheets which C^3 Masters controls which C^3 Masters and/or C^3 Slaves, and which C^3 Master connects three C^3 Masters together to form a company-level (12-'Mech) C^3 network; as noted above, the C^3 network can never be extended beyond 12 'Mechs.

To designate a C^3 network, use the following rules:

- A C^3 Master can only control one to three C^3 Slaves or one to three C^3 Masters.
- While a C^3 Master controls another C^3 Master, it cannot control a C^3 Slave.

The C^3 Configuration Diagram on page 111 illustrates the only four ways (see *All-C^3 Master Network* below for the exception) a C^3 network can be created. In each of the four diagrams, a circle represents a 'Mech; an "S" represents a C^3 Slave, while an "M" represents a C^3 Master. The arrows show the network links between C^3 Masters and C^3 Slaves, while the colors differentiate between each lance-level (4-'Mech) C^3 network.

All-C^3 Master Network: The only allowable variation on those four configurations is to replace C^3 Slaves with C^3 Masters.

For example, in Configuration 1, if the controlling player replaced any of the C^3 Slaves in the red lance with a C^3 Master, then all three C^3 Slaves in that lance would have to be replaced with C^3 Masters, as a C^3 Master cannot control another C^3 Master and control a C^3 Slave simultaneously. Such an exchanging of C^3 Slaves for C^3 Masters can happen at the lance level, without affecting the company level. In the cited example, even if the entire red lance was composed of C^3 Masters, the blue and yellow lances could still be composed of networks consisting of a C^3 Master and three C^3 Slaves.

Multiple Networks: If multiple C^3 networks are established at the start of the game and they are not designated as connected before play begins, they cannot share targeting information.

A 'Mech may only belong to a single C^3 network. Additionally, while a 'Mech may mount both a C^3 Master and a C^3 Slave, it may not use both at once. If a player fields a 'Mech equipped in this way, they must designate at the start of game which C^3 system is operational.

C³ BOOSTED SYSTEM (C³BS)

C^3BS is a variant of standard C^3 that functions identically to a standard C^3 network. However, the only ECM that can block it is the Angel ECM Suite.

Standard and boosted C^3 systems can be connected together into the same network. However, in such a network, a non-boosted member is still cut off as normal if in the effect radius of hostile ECM, and a non-boosted master cannot transmit data to boosted members in the effect radius of any kind of hostile ECM.

C³I COMPUTER SYSTEM

"The perfection of Sainted Blake's ideals. Unless you want to work as a team of seven."

Game Rules: The C^3i computer follows the standard C^3 computer rules above, with the following changes.

Every 'Mech in a C^3i network mounts the same C^3i computer: there are no master and slave variants. Up to six 'Mechs may be linked in a single C^3i network. Multiple networks cannot be linked together.

Because a C^3i network has no master computer, the network cannot be shut down by the loss of a single 'Mech. Only 'Mechs in the effect radius of enemy ECM or whose C^3i computer is destroyed are isolated from the network.

C^3i computers are not compatible with standard C^3 Masters/C^3 Slaves and do not have the TAG capability of the standard C^3 Master.

Technology Base: Inner Sphere

ECM SUITES

'Mechs with multiple ECM suites (of any type) may use only one at a time.

Water: If an ECM suite is in a location above the surface of a water hex, it cannot affect 'Mechs underwater. In other words, for an ECM suite to affect 'Mechs underwater, the suite must be in a location that is underwater.

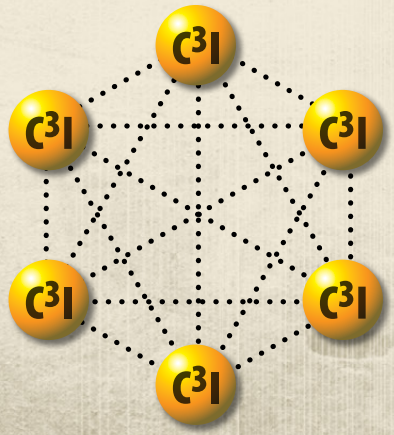
ECM SUITE/GUARDIAN ECM SUITE

"Only one man would dare *use Somerset jamming techniques on me. ADAM STEINER!" – Nicolai Malthus,* The Somerset Strikers *series, broadcast 08/12/3052*

Game Rules: ECM suites create a 6-hex bubble. While within the bubble or tracing line of sight through the bubble, many enemy systems become disrupted (friendly 'Mechs are unaffected). An ECM suite has the following effects on the following enemy systems:

- **Active Probe:** Active probes cannot penetrate an ECM's area of effect. The probing 'Mech would notice that it is being jammed, however. This does not apply to Bloodhound Active Probes.
- **Artemis IV/V FCS:** ECM blocks the effects of Artemis FCS. Artemis-equipped launchers may be fired through or into ECM, but they lose any Target Number and Cluster Hits Table bonuses.
- **C^3 and C^3i Computer:** ECM cuts off any C^3-equipped 'Mech from its network. Only those C^3 'Mechs able to draw an LOS to the master 'Mech that does not pass through the ECM radius can access the network. However, as long as a 'Mech is not cut off, enemy ECM does not affect its ability to share C^3 targeting data.

  If a C^3 Master is isolated from the network by ECM, the entire network below it is effectively shut off (all 'Mechs subordinate to it on the diagram on p. 111), though the Master's



• C³I DIAGRAM •

TAG capability still functions. For a Master 'Mech connecting the lances of a company, this means that the link between the lances is broken, but the network of each lance remains intact unless also subject to ECM.

- **Narc Missile Beacon:** Narc-equipped missiles lose the bonuses provided by their homing pod if the pod they are homing in on lies within an ECM bubble. A Narc launcher itself (standard and iNarc) is not affected by ECM.
- **Unaffected Systems:** Any system not listed above is unaffected by ECM. This includes Streak launchers, targeting computers, TAG, and MRM Apollo FCS.

Technology Base: Clan (ECM suite) or Inner Sphere (Guardian)

ANGEL ECM SUITE

"The good and bad angel, all in one lovely package."

Game Rules: The Angel ECM suite works like standard ECM (see above), but also blocks the Bloodhound Active Probe and even negates the locking systems of Streak missiles. Streak missiles fired into or through a hostile Angel ECM bubble will not fire if the attack roll fails (as normal), but on a successful Streak launcher attack, the attacker must roll on the Cluster Hits Table as though the launcher were a standard (non-Streak) model.

Technology Base: Clan or Inner Sphere

MRM "APOLLO" FIRE CONTROL SYSTEM

"Caution: not responsible for lost missiles."

Game Rules: An Apollo system negates the +1 Target Number modifier that applies to an equipped MRM launcher. However, on a successful hit with such a launcher, the attacker must subtract 1 from their roll on the Cluster Hits Table.

Critical Hit: An Apollo Fire Control System is an attachment to an MRM launcher, but if it suffers a critical hit the launcher continues to function.

Technology Base: Inner Sphere

TARGET ACQUISITION GEAR (TAG)

"You're it!"

Game Rules: Instead of making an attack roll for TAG during the Weapon Attack Phase, all 'Mechs making TAG attacks do so after the end of the Movement Phase, but before the beginning of the Weapon Attack Phase (initiative order is still followed, however).

To designate a target with TAG, calculate the Target Number as for a standard weapon attack (TAG does not benefit from a targeting computer). As using TAG occurs before the Weapon Attack Phase, it does not count towards the number of targets a 'Mech engages in the Weapon Attack Phase, or prevent physical attacks by the firing 'Mech.

If the attack roll is successful, the target is TAG-designated for that turn's Weapon Attack Phase (no hit location roll is made). Note that the designation causes no damage on its own; TAG only enables the use or enhanced functionality of certain ammunition types (semi-guided munitions, Arrow IV Homing missiles, and Copperhead artillery shells). Any number of attacks that turn from any number of 'Mechs (including the 'Mech with TAG itself) may utilize the TAG designation.

A target designated by TAG also counts as spotted for indirect fire (see *Indirect Fire*, p. 30).

Light TAG functions as TAG; only its range is altered, and any reference to TAG applies to Light TAG equally.

Technology Base: Clan (TAG, Light TAG) or Inner Sphere (TAG)

TARGETING COMPUTER

"All I'm saying is that it stops being just a computer when you've added four tons of servos and waldoes."

Game Rules: A targeting computer applies a –1 Target Number modifier to all weapon attacks of the DB, DE, or P types (but see *Weapon Clarifications & Exceptions*, below).

A targeting computer also allows eligible weapons to make aimed shots (see p. 30) against non-immobile 'Mechs. Follow the normal rules for aimed shots, except that the head cannot be targeted. The targeting computer's standard –1 Target Number modifier is ignored, and a +3 modifier is applied instead.

Below are the standard modifiers for a 'Mech with a targeting computer (TC) when making the various types of aimed shots:

- **Aimed Shot w/ TC, Mobile Target:** +3 Target Number modifier (head may not be targeted), no TC modifier applies
- **Aimed Shot w/ TC, Immobile Target, Head Shot:** +3 Target Number modifier, no TC modifier applies
- **Aimed Shot w/ TC, Immobile Target, any other location:** –5 Target Number modifier (–4 Immobile, –1 TC)

Weapon Clarifications & Exceptions:

Cluster and Flak Weapons / Attacks: Not eligible for TC benefits (but see *Hyper-Assault Gauss Rifles*, below).

Flamers and Machine Guns: Not eligible for TC benefits.

Hyper-Assault Gauss Rifles: Though a Cluster weapon, these apply the –1 Target Number TC modifier if not making a Flak attack.

Pulse Weapons: Cannot make an aimed shot against mobile targets. Never applies the –1 Target Number TC modifier for making aimed shots against immobile targets.

Rapid-Fire Weapons: Apply the –1 Target Number TC modifier even if rapid-firing. However, aimed shots while rapid-firing are impossible, per the standard aimed shot rules.

Technology Base: Clan or Inner Sphere

WATCHDOG COMPOSITE ELECTRONIC WARFARE SYSTEM (CEWS)

"By the time the probe picks up the enemy, I'm already jamming them."

Game Rules: The CEWS has the same radius, range, and function as the Clans' standard ECM and Light Active Probe (see pp. 112 and 110).

A Watchdog can be used as the ECM suite stealth armor requires.

Technology Base: Clan

STRUCTURAL COMPONENTS

Only equipment with special gameplay rules is listed here.

ARMOR

Critical Hit: Slots allocated to armor are inapplicable slots (see p. 46); critical hit rolls to them have no effect and are re-rolled.

PSRs: For armor types that can absorb two points of damage per armor circle, only full armor circles removed count for determining if the 'Mech has to make a PSR for taking 20+ points of damage.

BALLISTIC-REINFORCED ARMOR

"At last the autocannon's reign of terror is at an end."

Game Rules: Each point of ballistic-reinforced armor absorbs up to two points of damage from attacks of the DB or M types before being destroyed. Leftover single points of damage are discarded, but a damage source will always destroy at least one point of this armor.

Technology Base: Inner Sphere

FERRO-LAMELLOR ARMOR

"I said your 'Mech was cluster-munition proof, not LB-X-proof."

Game Rules: Ferro-lamellor armor reduces all damage by 1 point for every 5 points (or fraction thereof) delivered per hit. Excess damage affects internal structure per the normal rules for that structure type. In addition, ferro-lamellor armor negates the critical-enhancing effects of Armor-Piercing ammo (see p. 106).

Attacks reduced to zero damage (such as LB-X cluster munitions) do not inflict pilot injury in the event of a cockpit hit and cannot create through-armor critical hits.

Technology Base: Clan

HARDENED ARMOR

"I've fallen and I can't get up, but I'm INVINCIBLE!"

Game Rules: A point of hardened armor is only destroyed after taking two full points of damage (single points are recorded but do not destroy an armor circle by themselves). As long as there is armor in a facing/location, apply a –2 modifier when rolling on the Determining Critical Hits Table due to points of damage being inflicted there and the critical-enhancing effects of Armor-Piercing ammo (see p. 106) are negated.

Hardened armor impairs mobility: 'Mechs with it suffer a +1 Piloting Skill Roll modifier and –1 Running MP.

Technology Base: Clan or Inner Sphere

LASER-REFLECTIVE ARMOR

"Lasers bounce off like it's a mirror. If you fall it behaves the same."

Game Rules: Each point of laser-reflective armor absorbs up to two points of damage from flamers, lasers, PPCs, and plasma weapons before being destroyed. Leftover single points of damage are discarded, but a damage source will always destroy at least one point of this armor. Additionally, heat-causing effects from these weapons are halved (round down to a minimum of 1 heat point). However, damage to it from Area-Effect weapons, falls, physical attacks and moving into buildings is doubled, and Armor-Piercing ammo (see p. 106) applies a +2 modifier for critical hit checks when it strikes this armor.

Remaining single points of damage created as a result of the damage-doubling effect do not transfer to internal structure. For example, a 5-point hit would be doubled to 10 points of damage. A location with 9 points of armor would lose all 9 points to this hit, but the remaining single point would be lost, with no damage transfer.

Technology Base: Clan or Inner Sphere

STEALTH ARMOR

"WARNING: DOES NOT ACTUALLY TURN YOUR 'MECH INVISIBLE."

Game Rules: A 'Mech with the stealth armor system must also mount an ECM suite. When the stealth armor system is not engaged, the ECM suite functions normally. When the stealth armor system is engaged, the ECM does not function, but the 'Mech suffers effects as if in the radius of an enemy ECM suite (see p. 112) and cannot use any type of active probe (including the Bloodhound). If the ECM suite is destroyed, the stealth armor system cannot function.

A 'Mech may start the game with the system engaged; this must be indicated on the 'Mech's record sheet. Stealth armor may be turned on or off during the End Phase of any turn.

While the system is engaged, attacks against a 'Mech equipped with stealth armor receive additional Target Number modifiers of +1 at medium range and +2 at long range. In addition, a 'Mech with its stealth armor system engaged cannot be attacked as a secondary target (see *Multiple Targets Modifier*, p. 27).

Engaged stealth armor generates 10 heat points per turn.

Technology Base: Inner Sphere

COCKPIT SYSTEMS

FULL-HEAD EJECTION SYSTEM

"Something to remember your Hatchetman *by."*

Game Rules: This system may be deployed at any phase at the player's choosing, or the head may jettison automatically in the event of an internal explosion. Any critical hits to the 'Mech's Life Support slots disable this system, rendering it impossible to eject. A 'Mech with a submerged head that has suffered a hull breach also cannot use the system.

The MechWarrior automatically suffers 1 point of damage when the system is launched. If the 'Mech (or at least its head) is not submerged, the warrior may then choose any hex within 12 hexes of the 'Mech (if the 'Mech was upright at the time), or within 12 hexes in the 'Mech's forward arc (if it was prone) to make landfall. A Piloting Skill Roll (see p. 53) with a +3 modifier is required to land on target; failure means the head scatters 1D6 / 2 hexes (rounding down) in a random direction. Upon landing, a standard MechWarrior ejection roll (see p. 81) must be made with an additional +2 Target Number modifier to avoid taking a second point of damage from the ejection.

If the 'Mech is submerged (and the head location has suffered no armor breach), an ejected full-head pod automatically rockets to the water's surface and floats there as an immobile vessel, retaining the remaining armor and structure values of the 'Mech's head location.

Once the pod lands (or surfaces), the warrior may step out and make for safety, or—if floating on water or in an otherwise toxic environment—remain in the capsule to await rescue. This has no effect on standard gameplay, but may be important for certain scenarios or campaigns featuring role-playing elements.

Attacks against a jettisoned full-head pod can only be made after it has landed: for such attacks, apply the –4 modifier to the attack's Target Number for an immobile target. Attacks against a jettisoned head inflict damage to the head's remaining armor, internal structure, and critical hit locations per any other attack that strikes a 'Mech's head.

Once a full-head ejection system is jettisoned (whether successfully or not), the 'Mech is considered to have suffered a Head Blown Off critical hit and so is treated as destroyed.

Technology Base: Inner Sphere

SMALL COCKPIT

"Why yes, I'd love a Stinger*'s cockpit, only smaller."*

Game Rules: Apply a +1 modifier to all Piloting Skill Rolls.

Technology Base: Clan or Inner Sphere

TORSO-MOUNTED COCKPIT

"Knocking this 'Mech's block off doesn't really do much. Pity about the ejection system though."

Game Rules: Apply a +1 modifier to all Piloting Skill Rolls.

A MechWarrior using a torso-mounted cockpit receives no pilot damage from hits to the torso area where the cockpit is placed, though ammunition explosions, excess heat, falling damage, and cockpit destruction affect the warrior as normal. It is impossible to eject from a torso-mounted cockpit.

A 'Mech using this system may survive 2 sensor critical hits and still perform weapon attacks. Instead of the standard modifiers from sensor damage, the first sensor hit incurs a +2 Target Number modifier to all weapon attacks, while the second increases this to a +4 modifier. If both sensor hits are in the head, the +4 modifier for the second sensor hit also applies to all Physical Attacks and Piloting Skill Rolls, as the 'Mech is effectively blinded. Three sensor hits (or more) completely blind a 'Mech with a torso-mounted cockpit, rendering all weapon attacks impossible and applying a +4 modifier to all Piloting Skill Rolls.

Finally, warriors using a torso-mounted cockpit suffer more severe heat effects in the event of life support damage. If a 'Mech with a torso-mounted cockpit takes a life support critical hit, the MechWarrior suffers 1 point of damage for every turn the 'Mech overheats by 1 to 14 points, and 2 points of damage at the end of every Heat Phase of every turn the 'Mech overheats by 15 points or more. This replaces the standard MechWarrior heat effects.

If the head of a 'Mech with a torso-mounted cockpit is destroyed, excess damage does not transfer to other locations, and further strikes to the head have their location re-rolled.

Technology Base: Clan or Inner Sphere

ENGINES

XXL ENGINE

"The reactor shielding's basically aluminum foil, but it's nice to be able to heat your rations without leaving the cockpit."

Game Rules: XXL fusion engines run hot: standing still generates 2 heat points per turn, Walking movement 4 heat points, and Running movement 6 heat points. Heat generated per hex jumped is doubled, with a minimum of 6 points per jump (the heat modifiers for improved jump jet use and jumping with an XXL engine cancel each other out).

Technology Base: Clan or Inner Sphere

GYROS

HEAVY-DUTY GYRO

"Takes a licking, keeps on kicking."

Game Rules: It takes three critical hits to destroy a heavy-duty gyro. The first critical hit does not force a Piloting Skill Roll, but applies a +1 modifier to all future PSRs. On the second critical hit, replace the above with the effects of the first critical hit to a standard gyro. The third critical hit destroys a heavy-duty gyro with all the usual effects for gyro destruction.

Technology Base: Inner Sphere

MANEUVERABILITY AIDS

MYOMER ACCELERATION SIGNAL CIRCUITRY (MASC)

"It's the Six Million C-Bill 'Mech!"

Game Rules: A 'Mech with MASC can activate the system when it declares which movement mode it will use during the Movement Phase. If so, its controller immediately rolls 2D6 (before the 'Mech moves):

- On a result of 3 or higher, the 'Mech can run that turn at a speed equal to double its current Walking MP.
- On a result of 2, the player automatically assigns one critical hit to each leg that has not yet been destroyed; the player does not roll on the Determining Critical Hits Table. This damage (and any Piloting Skill Rolls that result) is applied before the 'Mech moves. Any speed reductions that occur apply immediately, with the controlling player recalculating the appropriate Walking MP and then doubling that new number to determine how many total MPs the 'Mech may expend that turn due to the MASC system activation.
- The player must roll 2D6 every turn the 'Mech is using MASC to determine whether or the system delivers a critical hit to each leg. On the second consecutive turn of MASC use, a result of 4 or less inflicts a critical hit to each leg. A result of 6 or less inflicts a critical hit to each leg on the third consecutive turn, 10 or less on the fourth, and the fifth turn of MASC use, a critical hit is automatically assigned to each leg.
- For each turn in which the system is not used, reduce the target number at which critical hits are assigned by one interval, but never below 3. For example, a player using MASC for three consecutive turns needs a result of 7 or higher on the third turn to avoid inflicting critical damage. After an intervening turn of not using the system, the player needs a 5 or higher to avoid inflicting critical hits on the 'Mech's legs. Two turns without using MASC reduces the threshold number to the original 3.
- If a 'Mech uses both MASC and a supercharger in the same round, its controller must make separate activation rolls for each. If successful, the 'Mech receives a Running MP equal to 2.5 times its Walking MP (rounded up).

Critical Hit: If critical damage occurs to MASC during a Movement Phase as it is being used (for example, due to a fall), the MASC immediately stops functioning and the player must recalculate the 'Mech's MP. If this recalculation results in the 'Mech having already expended more MP than it has available, its movement ends.

Technology Base: Clan or Inner Sphere

MASC TABLE

Turn	Avoid Number
1	3+
2	5+
3	7+
4	11+
5	Automatic

MECHANICAL JUMP BOOSTERS

"Go go Blakist Jump Booster!"

Game Rules: Mechanical jump boosters act as standard jump jets, with the following exceptions:

- Mechanical jump boosters generate no heat and may be used even underwater.
- A 'Mech using mechanical jump boosters cannot steer in mid-flight and so begins and ends its jumps with the same facing.
- Mechanical jump boosters may not be used to execute a Death From Above attack.
- A 'Mech may not use both jump jets and mechanical jump boosters in the same turn.

Technology Base: Inner Sphere

PARTIAL WING

"Not so much flying as falling with style."

Game Rules: A partial wing confers a bonus to a 'Mech's jump and heat sink capabilities.

The added Jumping MP bonus (+2 for light and medium 'Mechs, +1 for heavy and assault 'Mechs) does not apply if the 'Mech has no available Jumping MP, but does allow the 'Mech to jump beyond its normal maximum Jump MP limits.

A partial wing applies –3 heat points per turn, whether the 'Mech jumped with it or not. In addition, the bonus Jumping MP provided by a partial wing is not included when calculating heat caused by a jump: to calculate the heat generated, subtract the partial wing's Jumping MP bonus from the distance jumped. This cannot reduce the heat generated by a jump below the minimum heat points normally generated.

Each critical hit suffered by a partial wing reduces its Jump MP and heat dissipation bonuses by 1, to a minimum of 0.

Technology Base: Clan or Inner Sphere

SUPERCHARGER

"Well, sir, it all started when Technician Barton told Davis to hold his beer..."

Game Rules: A 'Mech with a supercharger can activate the system when it declares which movement mode it will use during the Movement Phase. A supercharger enables a 'Mech to move at twice its standard Walking MP as an enhanced run. It follows the same rules as MASC (see at left), with the following exceptions:

- On a failed supercharger activation roll, the controlling player rolls on the Determining Critical Hits Table (p. 46), ignoring any modifier from reinforced structure, to determine the number of critical hits that results. Apply any such hits to the upper-most undamaged Engine critical slot(s) in the center torso.
- Regardless of whether or not critical damage occurs, a failed supercharger activation roll destroys the supercharger system: mark off its critical slot.
- If a 'Mech uses both MASC and a supercharger in the same round, its controller must make separate activation rolls for each. If successful, the 'Mech receives a Running MP equal to 2.5 times its Walking MP (rounded up).
- If a 'Mech uses both triple-strength myomer and a supercharger in the same round, the TSM modifier is added before applying the supercharger modifier.

Technology Base: Clan or Inner Sphere

TRIPLE-STRENGTH MYOMER (TSM)

"Almost makes you want to laugh at inferno missiles. Almost."

Game Rules: Triple-strength myomer provides a benefit only when a 'Mech is running hot. If a 'Mech is equipped with TSM and ends a turn with a heat level of 9 or higher, the following effects take place the next turn; apply other modifiers that decrease movement, such as heat and damage, normally.

- Increase the 'Mech's Walking MP by 2 and recalculate its Running MP; multiply the new Walking MP by 1.5, rounding up. Jumping MP is unaffected. Note that any MP penalties due to heat are still applied.
- Double the 'Mech's lifting ability (see *Lifting Items*, p. 21).
- Double the damage for punch, kick, and club attacks (e.g. a punch from a 55-ton 'Mech with active TSM would do 12 damage), as well as some physical weapon attacks (see p. 38).
- If a 'Mech uses both triple-strength myomer and a supercharger in the same round, the TSM modifier is added before applying the supercharger modifier.

Critical Hit: Slots allocated to TSM are inapplicable slots (see p. 46); critical hits to them have no effect and must be re-rolled.

Technology Base: Inner Sphere

UNDERWATER MANEUVERING UNIT (UMU)

"Under the sea, where a watery doom is waiting for thee..."

Game Rules: To use UMUs, a 'Mech must use the jumping movement mode (and counts as jumping when calculating Target Number modifiers). Each UMU generates 1 UMU MP, and a 'Mech may move 1 hex underwater or 1 depth (up or down) for each UMU MP. No move with UMUs can take a 'Mech out of water or move it to a depth or 1 or less. The normal hull breach rules still apply (see *Hull Integrity And Breaches*, p. 65), and if the UMUs are disengaged, all standard rules for moving underwater apply.

UMUs generate only 1 heat point per turn total, regardless of the number of UMU MP used (2 total heat if used with an XXL engine).

A 'Mech using UMUs moves in an upright position and marks its submerged depth based on the location of its feet relative to the water surface. If this is Depth 2, the 'Mech is presumed to be just above the bottom of the hex.

Technology Base: Clan or Inner Sphere

STRUCTURE

Critical Hit: Slots allocated to internal structure are inapplicable slots (see p. 46). Critical hits to such slots have no effect and must be re-rolled.

COMPOSITE STRUCTURE

"Sure it's light, but it's got all the structural integrity of wet cardboard."

Game Rules: All damage to the internal structure of a 'Mech using composite structure is doubled.

Excess damage that transfers to a location still protected by armor does not apply this effect. Additionally, remaining points of damage created as a result of the damage-doubling effect do not transfer. For example, a 5-point hit to composite internal structure would be doubled to 10 points of damage. A location with 9 points of internal structure would lose all 9 points to this hit, but the remaining single point would be lost, with no damage transfer.

Technology Base: Inner Sphere

REINFORCED STRUCTURE

"A healthy diet of endo-composites will give your 'Mech strong bones."

Game Rules: A point of reinforced structure is only destroyed after taking two full points of damage (single points are recorded but do not destroy a structure circle by themselves). Only full structure circles removed count for determining if the 'Mech has to make a PSR for taking 20+ points of damage in a phase.

When rolling on the Determining Critical Hits Table against a 'Mech with reinforced structure, apply a –1 modifier to the roll result.

Technology Base: Clan or Inner Sphere

MISCELLANEOUS EQUIPMENT

ACTUATOR ENHANCEMENT SYSTEM (AES)

"New from Defiance Industries: the Axman*, now with Kung-Fu Grip!"*

Game Rules: An actuator enhancement system mounted in an arm provides a –1 Target Number modifier for all weapon attacks mounted in that arm (weapons split between an arm and torso location gain no bonus). Also apply a –1 Target Number modifier for all physical attacks with that arm (including physical weapons; for physical attacks that require both arms, the modifier only applies if both arms mount AES). When mounted in the legs, the AES provides a –2 modifier to Piloting Skill Rolls, while charges and kicks receive a –1 Target Number modifier.

'Mechs with AES in their legs lose all AES-provided modifiers if any of its leg-based AES slots suffer critical damage; arms only lose AES-provided modifiers if the AES in the affected arm takes a critical hit.

Technology Base: Clan or Inner Sphere

ANTI-MISSILE SYSTEM (AMS)

"The Yori Flyswatter is the last word in missile defense."

Game Rules: If a successful attack roll is made with a Missile weapon against a 'Mech carrying an AMS, and strikes in the attack direction covered by the firing arc of the location in which the AMS is mounted, the defending player can choose to engage the AMS. AMS only works against weapons that are Type M (Missile weapons: see p. 97). Weapons that lack this Type, regardless of their name or any other factor, are unaffected.

If the player chooses to engage, the result is as follows:

- The attacking player applies a –4 modifier to their roll on the Cluster Hits Table (an AMS never reduces a roll on the table below 2). This modifier is cumulative with other Cluster Table modifiers.
- If the missile weapon is a Streak launcher, treat the launcher as though the controlling player rolled a 7 on the appropriate column of the Cluster Hits Table to determine how many missiles struck the target.
- If the missile weapon attack consists of only a single missile (such as a Thunderbolt missile or Narc pod), roll 1D6: on a result of 1-3 the missile is destroyed, on a result of 4-6 the missile strikes the target.
- One shot of ammunition is marked off and 1 heat point generated each time the AMS engages a missile weapon.

Each AMS cannot engage more than one missile attack per turn. If, in a single Weapon Attack Phase, more than one missile weapon successfully strikes the target in the attack direction covered by the firing arc where an AMS is mounted, the defending player can choose which (if any) missile weapon to inflict the modifier against.

Only one AMS can engage each attack, regardless of how many AMS cover the attack direction. If a 'Mech mounts more than one AMS that covers the same attack direction, the defender chooses the order in which they activate and against which (if any) missile attack each inflicts their modifiers.

Firing an AMS is not an attack. As such, engaging arm-mounted AMS will not prevent a 'Mech from punching.

Enhanced Missile Defense (Optional): Using this optional rule, if an AMS engages a missile attack, and the modified roll result on the Cluster Hits Table is less than 2, all missiles in the attack are destroyed and no damage is applied.

Critical Hit: If a slot containing AMS ammo suffers a critical hit, it explodes for 2 points per shot.

Technology Base: Clan or Inner Sphere

LASER ANTI-MISSILE SYSTEM (L-AMS)

"L-AMS: fully compatible with LAMs!"

Game Rules: Standard anti-missile system rules apply when using the Laser AMS (see above), except the laser system requires no ammunition and generates 7 heat points (if an Inner Sphere model) or 5 heat points (if a Clan model). If a L-AMS reduces a missile flight's Cluster Hits roll to less than 2, it completely destroys the incoming flight.

A laser AMS used underwater affects torpedoes in the same manner as it does missiles.

Technology Base: Clan or Inner Sphere

ARMORED COMPONENT

"Armored item inside armored item. Is like matryoshka. Yuri approves."

Game Rules: Each critical slot of an armored component can suffer one critical hit with no effect. Armored components receive this benefit over all of the component's slots. Ammunition bins cannot be armored.

Armored shoulder and hip actuators that have not received a first critical hit also protect a 'Mech from a Limb Blown Off critical damage result. In such a case, the appropriate shoulder or hip slot merely loses its component armor as normal. Component armor cannot prevent a Blown Off result to a 'Mech's head, or critical damage caused by the use of MASC or a supercharger.

Technology Base: Clan or Inner Sphere

CELLULAR AMMUNITION STORAGE EQUIPMENT (CASE)

"Sure it's a flaming wreck. But it's a salvageable wreck."

Game Rules: If ammo (or any other explosive component, such as a Gauss rifle) in a CASE-equipped location explodes, resolve the resulting damage to the internal structure in that location as normal. However, any excess damage is ignored, rather than transferring to an additional internal structure location. Remember that the loss of all internal structure in a side torso location also blows off the corresponding arm, though the arm is not damaged by the explosion and can still be used as a club (see p. 36).

If an ammo explosion transfers into a location protected by CASE, the internal structure in that location takes damage as normal. All excess damage is ignored, as above. For example, if an Inner Sphere 'Mech suffered an arm ammo explosion and damage transferred to a side torso equipped with CASE, the internal structure of the side torso would be damaged as normal, but any excess damage beyond that would not transfer or count towards a PSR.

Critical Hit: Slots allocated to CASE are inapplicable slots (see p. 46): critical hits to them have no effect and must be re-rolled.

Technology Base: Clan or Inner Sphere

CASE II

"Poor Leftenant Weiss had all of his ammo set off. His 'Mech survived, but Weiss is a vegetable."

Game Rules: All rules references to CASE apply equally to CASE II, unless that reference or the rules below indicate otherwise.

When a location protected by CASE II takes damage from an ammunition explosion (including any ammunition-like explosions, such as critical hits to Gauss weapons), only 1 point of internal damage is inflicted to the location (with the normal chance of critical effects), while any remaining damage is applied to the location's rear armor (if any). If the location is a limb, remove all remaining armor in the location, or half the location's total original armor, whichever is less. Any remaining damage from the explosion does not transfer after that.

Furthermore, for any critical hits resulting from an ammunition explosion vented by CASE II, the controlling player rolls 2D6 again for each one, and disregards the critical effects on a result of 8+. Otherwise, the critical hit applies as normal. However, CASE II does not stop MechWarrior damage from feedback (see *Ammunition*, p. 47).

If a 'Mech carries ammunition in its head protected by CASE II, an ammunition explosion causes one point of internal structure damage. Then, half of the starting head armor (rounding up) is destroyed (e.g. if a 'Mech was designed with 9 points of head armor then it will always lose 5 points of head armor, even if the head only had 6 points remaining at the time of the explosion). No damage is transferred.

Critical Hit: Slots allocated to CASE II are inapplicable slots (see p. 46): critical hits to them have no effect and must be re-rolled.

Technology Base: Clan or Inner Sphere

COOLANT POD

"Ha! It looked *like I was overheating, but this means my 5 ER PPCs are ready to go!"*

Game Rules: Though a 'Mech may mount multiple coolant pods, only one may be triggered per turn, at the start of the 'Mech's Heat Phase. A coolant pod increases the cooling capacity of all of the 'Mech's active heat sinks by 1 point each (regardless of heat sink type). Each coolant pod can only be used once per game.

Critical Hit: A critical hit on an unused coolant pod results in a 10-point ammunition explosion.

Technology Base: Clan or Inner Sphere

WEAPONS AND EQUIPMENT TABLES

The *Weapons and Equipment* chapter of the *BattleMech Manual* covers the most common weapons and equipment as of the in-universe year of 3145.

Primitives and Prototypes: There are a host of primitive and prototype weapons in *BattleTech*, spanning many eras. As this book is focused on the most common weapons and equipment, they are not included in the *BattleMech Manual*.

Note: While it is beyond the scope of the *Manual*, every weapon and piece of equipment within this book has a very specific Introduction date (when it first reached the battlefield). Additionally, some have an Extinct date (when the depredations of war robbed the Great Houses of their abilities to manufacture the most high-tech of weaponry) and then a Recovered date (when a technological renaissance brought such technologies back into the modern eras). These tables, however, do not employ such precise dates.

Once more, all such decisions were made to make these tables easy to use. Players wishing more exacting details for era play can find them within the various *BattleTech* books noted in the Introduction.

WEAPON TYPES

The following is a complete list of weapon Types (see pp. 96-97 for the full descriptions of each Type's effects):

- **AE: Area-Effect**
- **AI: Anti-Infantry**
- **C: Cluster**
- **DB: Direct-Fire Ballistic**
- **DE: Direct-Fire Energy**
- **E: Electronics**
- **F: Flak**
- **H: Heat-Causing**
- **M: Missile**
- **OS: One-Shot**
- **P: Pulse**
- **PB: Point-Blank**
- **R: Rapid-Fire**
- **S: Switchable Ammo**
- **V: Variable Damage**
- **X: Explosive**

INNER SPHERE WEAPONS

Item	Type*	Heat	Damage Value	Minimum Range	Short Range	Medium Range	Long Range	Tons	Critical Slots	Shots Per Ton	Page Ref
Direct-Fire Ballistic Weapons											
Autocannon/2	DB, S	1	2	4	1–8	9–16	17–24	6	1	45	98
Autocannon/5	DB, S	1	5	3	1–6	7–12	13–18	8	4	20	98
Autocannon/10	DB, S	3	10	0	1–5	6–10	11–15	12	7	10	98
Autocannon/20	DB, S	7	20	0	1–3	4–6	7–9	14	10	5	98
Light Gauss Rifle	DB, X	1	8	3	1–8	9–17	18–25	12	5	16	100
Gauss Rifle	DB, X	1	15	2	1–7	8–15	16–22	15	7	8	99
Heavy Gauss Rifle	DB, V, X	2	25/20/10	4	1–6	7–13	14–20	18	11	4	100
Improved Heavy Gauss	DB, X	2	22	3	1–6	7–12	13–19	20	11	4	100
Magshot Gauss Rifle	DB, X	1	2	0	1–3	4–6	7–9	0.5	2	50	100
Silver Bullet Gauss	C, DB, F, X	1	C1/15	2	1–7	8–15	16–22	15	7	8	100
LB 2-X AC	DB, S	1	2	4	1–9	10–18	19–27	6	4	45	98
LB 5-X AC	DB, S	1	5	3	1–7	8–14	15–21	8	5	20	98
LB 10-X AC	DB, S	2	10	0	1–6	7–12	13–18	11	6	10	98
LB 20-X AC	DB, S	6	20	0	1–4	5–8	9–12	14	11	5	98
Light AC/2	DB, S	1	2	0	1–6	7–12	13–18	4	1	45	98
Light AC/5	DB, S	1	5	0	1–5	6–10	11–15	5	2	20	98
Light Machine Gun	AI, DB	0	1	0	1–2	3–4	5–6	0.5	1	200	101
Machine Gun	AI, DB	0	2	0	1	2	3	0.5	1	200	101
Heavy Machine Gun	AI, DB	0	3	0	1	2	—	1	1	100	101
Light Rifle (Cannon)	DB	1	3**	0	1–4	5–8	9–12	3	1	18	106
Medium Rifle (Cannon)	DB	2	6**	1	1–5	6–10	11–15	5	2	9	106
Heavy Rifle (Cannon)	DB	4	9**	2	1–6	7–12	13–18	8	3	6	106
Rotary AC/2	DB, R	1/Sht	2/Sht, R6	0	1–6	7–12	13–18	8	3	45	98
Rotary AC/5	DB, R	1/Sht	5/Sht, R6	0	1–5	6–10	11–15	10	6	20	98
Ultra AC/2	DB, R	1/Sht	2/Sht, R2	3	1–8	9–17	18–25	7	3	45	99
Ultra AC/5	DB, R	1/Sht	5/Sht, R2	2	1–6	7–13	14–20	9	5	20	99
Ultra AC/10	DB, R	4/Sht	10/Sht, R2	0	1–6	7–12	13–18	13	7	10	99
Ultra AC/20	DB, R	8/Sht	20/Sht, R2	0	1–3	4–7	8–10	15	10	5	99
Direct-Fire Energy Weapons											
Binary (Blazer) Cannon	DE	16	12	0	1–5	6–10	11–15	9	4	—	100
Flamer	AI, DE, H	3	2**	0	1	2	3	1	1	—	99
ER Flamer	AI, DE, H	4	2**	0	1–3	4–5	6–7	1	1	—	99
Heavy Flamer	AI, DE, H	5	4**	0	1–2	3	4	1.5	1	10	99
Small Laser	DE	1	3	0	1	2	3	0.5	1	—	100
Medium Laser	DE	3	5	0	1–3	4–6	7–9	1	1	—	100
Large Laser	DE	8	8	0	1–5	6–10	11–15	5	2	—	100
ER Small Laser	DE	2	3	0	1–2	3–4	5	0.5	1	—	101
ER Medium Laser	DE	5	5	0	1–4	5–8	9–12	1	1	—	101
ER Large Laser	DE	12	8	0	1–7	8–14	15–19	5	2	—	101
Plasma Rifle	DE, H	10	10**	0	1–5	6–10	11–15	6	2	10	106
Light PPC	DE	5	5	3	1–6	7–12	13–18	3	2	—	105
PPC	DE	10	10	3	1–6	7–12	13–18	7	3	—	105
Heavy PPC	DE	15	15	3	1–6	7–12	13–18	10	4	—	105
ER PPC	DE	15	10	0	1–7	8–14	15–23	7	3	—	105
Snub-Nose PPC	DE, V	10	10/8/5	0	1–9	10–13	14–15	6	2	—	105
Pulse Weapons											
Small Pulse Laser	AI, P	2	3	0	1	2	3	1	1	—	101
Medium Pulse Laser	P	4	6	0	1–2	3–4	5–6	2	1	—	101

INNER SPHERE WEAPONS (CONTINUED)

Item	Type*	Heat	Damage Value	Minimum Range	Short Range	Medium Range	Long Range	Tons	Critical Slots	Shots Per Ton	Page Ref
Pulse Weapons (continued)											
Large Pulse Laser	P	10	9	0	1–3	4–7	8–10	7	2	—	101
Small X-Pulse Laser	AI, P	3	3	0	1–2	3–4	5	1	1	—	101
Medium X-Pulse Laser	P	6	6	0	1–3	4–6	7–9	2	1	—	101
Large X-Pulse Laser	P	14	9	0	1–5	6–10	11–15	7	2	—	101
Small Re-engineered Laser	P	4	4	0	1	2	3	1.5	1	—	101
Medium Re-engineered Laser	P	6	6	0	1–3	4–6	7–9	2.5	2	—	101
Large Re-engineered Laser	P	9	9	0	1–5	6–10	11–15	8	5	—	101
Small VSP Laser	AI, P, V	3	5/4/3	0	1–2	3–4	5–6	2	1	—	101
Medium VSP Laser	AI, P, V	7	9/7/5	0	1–2	3–5	6–9	4	2	—	101
Large VSP Laser	AI, P, V	10	11/9/7	0	1–4	5–8	9–15	9	4	—	101
Missile Weapons[M]											
LRM 5	C, M, S	2	1/Msl, C5/5	6	1–7	8–14	15–21	2	1	24	103
LRM 10	C, M, S	4	1/Msl, C5/10	6	1–7	8–14	15–21	5	2	12	103
LRM 15	C, M, S	5	1/Msl, C5/15	6	1–7	8–14	15–21	7	3	8	103
LRM 20	C, M, S	6	1/Msl, C5/20	6	1–7	8–14	15–21	10	5	6	103
Enhanced LRM 5	C, M, S	2	1/Msl, C5/5	3	1–7	8–14	15–21	3	2	24	102
Enhanced LRM 10	C, M, S	4	1/Msl, C5/10	3	1–7	8–14	15–21	6	4	12	102
Enhanced LRM 15	C, M, S	5	1/Msl, C5/15	3	1–7	8–14	15–21	9	6	8	102
Enhanced LRM 20	C, M, S	6	1/Msl, C5/20	3	1–7	8–14	15–21	12	9	6	102
Extended LRM 5	C, M	3	1/Msl, C5/5	10	1–12	13–22	23–38	6	1	18	103
Extended LRM 10	C, M	6	1/Msl, C5/10	10	1–12	13–22	23–38	8	4	9	103
Extended LRM 15	C, M	8	1/Msl, C5/15	10	1–12	13–22	23–38	12	6	6	103
Extended LRM 20	C, M	10	1/Msl, C5/20	10	1–12	13–22	23–38	18	8	4	103
MML 3	C, M, S							1.5	2		103
LRM ammo	—	2	1/Msl, C3/3	6	1–7	8–14	15–21			40	
SRM ammo	—	2	2/Msl, C2/3	0	1–3	4–6	7–9			33	
MML 5	C, M, S							3	3		103
LRM ammo	—	3	1/Msl, C5/5	6	1–7	8–14	15–21			24	
SRM ammo	—	3	2/Msl, C2/5	0	1–3	4–6	7–9			20	
MML 7	C, M, S							4.5	4		103
LRM ammo	—	4	1/Msl, C5/7	6	1–7	8–14	15–21			17	
SRM ammo	—	4	2/Msl, C2/7	0	1–3	4–6	7–9			14	
MML 9	C, M, S							6	5		103
LRM ammo	—	5	1/Msl, C5/9	6	1–7	8–14	15–21			13	
SRM ammo	—	5	2/Msl, C2/9	0	1–3	4–6	7–9			11	
MRM 10	C, M	4	1/Msl, C5/10	0	1–3	4–8	9–15	3	2	24	103
MRM 20	C, M	6	1/Msl, C5/20	0	1–3	4–8	9–15	7	3	12	103
MRM 30	C, M	10	1/Msl, C5/30	0	1–3	4–8	9–15	10	5	8	103
MRM 40	C, M	12	1/Msl, C5/40	0	1–3	4–8	9–15	12	7	6	103
Narc Missile Beacon	M, S	0	**	0	1–3	4–6	7–9	3	2	6	104
Improved Narc Launcher	M, S	0	**	0	1–4	5–9	10–15	5	3	4	104
Rocket Launcher 10	C, M, OS	3	1/Msl, C5/10	0	1–5	6–11	12–18	0.5	1	OS	103
Rocket Launcher 15	C, M, OS	4	1/Msl, C5/15	0	1–4	5–9	10–15	1	2	OS	103
Rocket Launcher 20	C, M, OS	5	1/Msl, C5/20	0	1–3	4–7	8–12	1.5	3	OS	103

INNER SPHERE WEAPONS (CONTINUED)

Item	Type*	Heat	Damage Value	Minimum Range	Short Range	Medium Range	Long Range	Tons	Critical Slots	Shots Per Ton	Page Ref
SRM 2	C, M, S	2	2/Msl, C2/2	0	1–3	4–6	7–9	1	1	50	103
SRM 4	C, M, S	3	2/Msl, C2/4	0	1–3	4–6	7–9	2	1	25	103
SRM 6	C, M, S	4	2/Msl, C2/6	0	1–3	4–6	7–9	3	2	15	103
Streak SRM 2	C, M	2	2/Msl, C2/2	0	1–3	4–6	7–9	1.5	1	50	104
Streak SRM 4	C, M	3	2/Msl, C2/4	0	1–3	4–6	7–9	3	1	25	104
Streak SRM 6	C, M	4	2/Msl, C2/6	0	1–3	4–6	7–9	4.5	2	15	104
Thunderbolt 5	M	3	5	5	1–6	7–12	13–18	3	1	12	104
Thunderbolt 10	M	5	10	5	1–6	7–12	13–18	7	2	6	104
Thunderbolt 15	M	7	15	5	1–6	7–12	13–18	11	3	4	104
Thunderbolt 20	M	8	20	5	1–6	7–12	13–18	15	5	3	104
Artillery Weapons											
Arrow IV	**	10	20**	0	**	**	**	15	15	5	97
Physical Attack Weapons											
Vibroblade, Small	—	3	7**	—	—	—	—	3	1	—	109
Vibroblade, Medium	—	5	10**	—	—	—	—	5	2	—	109
Vibroblade, Large	—	7	14**	—	—	—	—	7	4	—	109
Special											
Active Probe, Beagle	E	0	—	—	—	—	4	1.5	1	—	110
Active Probe, Bloodhound	E	0	—	—	—	—	8	2	3	—	110
Anti-Missile System	PB	1	—	—	—	—	—	0.5	1	12	118
Anti-Missile System, Laser	PB	7	—	—	—	—	—	1.5	2	—	118
ECM Suite, Angel	E	0	—	—	—	—	6	2	2	—	112
ECM Suite, Guardian	E	0	—	—	—	—	6	1.5	2	—	112
M-Pod	C, OS, V, X	0	15/10/5; C1	0	1	2	3	1	1	OS	101
Target Acquisition Gear (TAG)	E	0	—	0	1–5	6–9	10–15	1	1	—	113

* See *Weapon Types*, pages 96-97, for an explanation of each Type.

** Follow the Page Ref entry for this item to see the full details.

M Missile Weapons display their Damage Value uniquely; e.g. an LRM 20 is "1/Msl, C5/20", meaning each missile does 1 point of damage (1/Msl), the largest Damage Value grouping is 5 (C5), and that 20 missiles are fired with each shot (/20).

CLAN WEAPONS

Item	Type*	Heat	Damage Value	Minimum Range	Short Range	Medium Range	Long Range	Tons	Critical Slots	Shots Per Ton	Page Ref
Direct-Fire Ballistic Weapons											
LB 2-X AC	DB, S	1	2	4	1–10	11–20	21–30	5	3	45	98
LB 5-X AC	DB, S	1	5	3	1–8	9–15	16–24	7	4	20	98
LB 10-X AC	DB, S	2	10	0	1–6	7–12	13–18	10	5	10	98
LB 20-X AC	DB, S	6	20	0	1–4	5–8	9–12	12	9	5	98
AP Gauss Rifle	AI, DB, X	1	3	0	1–3	4–6	7–9	0.5	1	40	99
Gauss Rifle	DB, X	1	15	2	1–7	8–15	16–22	12	6	8	99
HAG 20	C, DB, F, X	4	C5/20	2	1–8	9–16	17–24	10	6	6	100
HAG 30	C, DB, F, X	6	C5/30	2	1–8	9–16	17–24	13	8	4	100
HAG 40	C, DB, F, X	8	C5/40	2	1–8	9–16	17–24	16	10	3	100
Light Machine Gun	AI, DB	0	1	0	1–2	3–4	5–6	0.25	1	200	102
Machine Gun	AI, DB	0	2	0	1	2	3	0.25	1	200	101
Heavy Machine Gun	AI, DB	0	3	0	1	2	—	0.5	1	100	101
ProtoMech AC/2	DB, S	1	2	0	1–7	8–14	15–20	3.5	2	40	98
ProtoMech AC/4	DB, S	1	4	0	1–5	6–10	11–15	4.5	3	20	98
ProtoMech AC/8	DB, S	2	8	0	1–3	4–7	8–10	5.5	4	10	98
Rotary AC/2	DB, R	1/Sht	2/Sht, R6	0	1–8	9–17	18–25	8	4	45	98
Rotary AC/5	DB, R	1/Sht	5/Sht, R6	0	1–7	8–14	15–21	10	8	20	98
Ultra AC/2	DB, R	1/Sht	2/Sht, R2	2	1–9	10–18	19–27	5	2	45	99
Ultra AC/5	DB, R	1/Sht	5/Sht, R2	0	1–7	8–14	15–21	7	3	20	99
Ultra AC/10	DB, R	3/Sht	10/Sht, R2	0	1–6	7–12	13–18	10	4	10	99
Ultra AC/20	DB, R	7/Sht	20/Sht, R2	0	1–4	5–8	9–12	12	8	5	99
Direct-Fire Energy Weapons											
Chemical Laser, Small	DE	1	3	0	1	2	3	0.5	1	60	100
Chemical Laser, Medium	DE	2	5	0	1–3	4–6	7–9	1	1	30	100
Chemical Laser, Large	DE	6	8	0	1–5	6–10	11–15	5	2	10	100
Flamer	AI, DE, H	3	2**	0	1	2	3	0.5	1	—	99
ER Flamer	AI, DE, H	4	2**	0	1–3	4–5	6–7	1	1	—	99
Heavy Flamer	AI, DE, H	5	4**	0	1–2	3	4	1.5	1	10	99
ER Micro Laser	DE	1	2	0	1	2	3–4	0.25	1	—	101
ER Small Laser	DE	2	5	0	1–2	3–4	5–6	0.5	1	—	101
ER Medium Laser	DE	5	7	0	1–5	6–10	11–15	1	1	—	101
ER Large Laser	DE	12	10	0	1–8	9–15	16–25	4	1	—	101
Heavy Small Laser	DE	3	6	0	1	2	3	0.5	1	—	101
Heavy Medium Laser	DE	7	10	0	1–3	4–6	7–9	1	2	—	101
Heavy Large Laser	DE	18	16	0	1–5	6–10	11–15	4	3	—	101
Improved Heavy Small Laser	DE, X	3	6	0	1	2	3	0.5	1	—	101
Improved Heavy Medium Laser	DE, X	7	10	0	1–3	4–6	7–9	1	2	—	101
Improved Heavy Large Laser	DE, X	18	16	0	1–5	6–10	11–15	4	3	—	101
Plasma Cannon	DE, H	7	0**	0	1–6	7–12	13–18	3	1	10	106
ER PPC	DE	15	15	0	1–7	8–14	15–23	6	2	—	105

CLAN WEAPONS (CONTINUED)

Item	Type*	Heat	Damage Value	Minimum Range	Short Range	Medium Range	Long Range	Tons	Critical Slots	Shots Per Ton	Page Ref
Pulse Weapons											
Micro Pulse Laser	AI, P	1	3	0	1	2	3	0.5	1	—	101
Small Pulse Laser	AI, P	2	3	0	1–2	3–4	5–6	1	1	—	101
Medium Pulse Laser	P	4	7	0	1–4	5–8	9–12	2	1	—	101
Large Pulse Laser	P	10	10	0	1–6	7–14	15–20	6	2	—	101
ER Pulse Small Laser	AI, P	3	5	0	1–2	3–4	5–6	1.5	1	—	101
ER Pulse Medium Laser	P	6	7	0	1–5	6–9	10–14	2	2	—	101
ER Pulse Large Laser	P	13	10	0	1–7	8–15	16–23	6	3	—	101
Missile Weapons[M]											
ATM 3	C, M, S	2	2/Msl, C5/3	4	1–5	6–10	11–15	1.5	2	20	102
ATM 6	C, M, S	4	2/Msl, C5/6	4	1–5	6–10	11–15	3.5	3	10	102
ATM 9	C, M, S	6	2/Msl, C5/9	4	1–5	6–10	11–15	5	4	7	102
ATM 12	C, M, S	8	2/Msl, C5/12	4	1–5	6–10	11–15	7	5	5	102
ATM ER Ammo	—	**	1/Msl, C5/**	4	1–9	10–18	19–27	**	**	**	
ATM HE Ammo	—	**	3/Msl, C5/**	0	1–3	4–6	7–9	**	**	**	
Narc Missile Beacon	M, S	0	**	0	1–4	5–8	9–12	2	1	6	104
LRM 5	C, M, S	2	1/Msl, C5/5	0	1–7	8–14	15–21	1	1	24	103
LRM 10	C, M, S	4	1/Msl, C5/10	0	1–7	8–14	15–21	2.5	1	12	103
LRM 15	C, M, S	5	1/Msl, C5/15	0	1–7	8–14	15–21	3.5	2	8	103
LRM 20	C, M, S	6	1/Msl, C5/20	0	1–7	8–14	15–21	5	4	6	103
SRM 2	C, M, S	2	2/Msl, C2/2	0	1–3	4–6	7–9	0.5	1	50	103
SRM 4	C, M, S	3	2/Msl, C2/4	0	1–3	4–6	7–9	1	1	25	103
SRM 6	C, M, S	4	2/Msl, C2/6	0	1–3	4–6	7–9	1.5	1	15	103
Streak SRM 2	C, M	2	2/Msl, C2/2	0	1–4	5–8	9–12	1	1	50	104
Streak SRM 4	C, M	3	2/Msl, C2/4	0	1–4	5–8	9–12	2	1	25	104
Streak SRM 6	C, M	4	2/Msl, C2/6	0	1–4	5–8	9–12	3	2	15	104
Streak LRM 5	C, M	2	1/Msl, C5/5	0	1–7	8–14	15–21	2	1	24	104
Streak LRM 10	C, M	4	1/Msl, C5/10	0	1–7	8–14	15–21	5	2	12	104
Streak LRM 15	C, M	5	1/Msl, C5/15	0	1–7	8–14	15–21	7	3	8	104
Streak LRM 20	C, M	6	1/Msl, C5/20	0	1–7	8–14	15–21	10	5	6	104
Artillery Weapons											
Arrow IV	**	10	20**	0	**	**	**	12	12	5	97
Special											
Active Probe	E	0	—	—	—	—	5	1	1	—	109
Active Probe, Light	E	0	—	—	—	—	3	0.5	1	—	110
Anti-Missile System	PB	1	—	—	—	—	—	0.5	1	24	118
Anti-Missile System, Laser	PB	5	—	—	—	—	—	1	1	—	118
ECM Suite	E	0	—	—	—	—	6	1	1	—	112
ECM Suite, Angel	E	0	—	—	—	—	6	2	2	—	113
Target Acquisition Gear (TAG)	E	0	—	0	1–5	6–9	10–15	1	1	—	113
TAG, Light	E	0	—	0	1–3	4–6	7–9	0.5	1	—	114
Watchdog CEWS	E	0	—	—	—	—	3	1.5	2	—	114

* See *Weapon Types*, pages 96-97, for an explanation of each Type.

** Follow the Page Ref entry for this item to see the full details.

M Missile Weapons display their Damage Value uniquely; e.g. an LRM 20 is "1/Msl, C5/20", meaning each missile does 1 point of damage (1/Msl), the largest Damage Value grouping is 5 (C5), and that 20 missiles are fired with each shot (/20).

INNER SPHERE EQUIPMENT

Equipment	Tons	Critical Slots	Page Ref
Armor			
Standard	—	0	N/A
Primitive	—	0	N/A
Ballistic-Reinforced	—	10	114
Ferro-Fibrous, Light	—	7	114
Ferro-Fibrous	—	14	114
Ferro-Fibrous, Heavy	—	21	114
Hardened	—	0	114
Laser-Reflective	—	10	114
Stealth	—	12*	115
Cockpits			
Standard	3	—	47
Primitive	3	—	47
Small	2	*	47, 115
Torso-Mounted	4	*	47, 115
Electronics			
Artemis IV FCS	1	1*	110
C^3 Computer, Master	5	5	110
C^3 Computer, Slave	1	1	110
C^3 Boosted System, Master	6	6	112
C^3 Boosted System, Slave	3	2	112
C^3I Computer	2.5	2	112
MRM "Apollo" FCS	1	1*	113
Targeting Computer	*	*	114
Engines			
Standard	—	6*	48
Compact	—	3*	48
Light	—	10*	48
XL	—	12*	48
XXL	—	18*	115
Gyros			
Compact	—	2*	48
Standard	—	4*	48
Heavy-Duty	—	4*	48, 115
XL	—	6*	48
Maneuverability Aids			
Jump Jet, Standard	*	1*	14, 18
Jump Jet, Improved	*	2*	14, 18
MASC	*	*	116
Mechanical Jump Boosters	*	*	116
Partial Wing	*	8*	116
Supercharger	*	1*	116
Triple-Strength Myomer	0	6	117
UMU	*	1*	19, 117

Equipment	Tons	Critical Slots	Page Ref
Munition Bins			
AC – Armor-Piercing	1	1	106
AC – Caseless	1	1	106
AC – Cluster	1	1	106
AC – Flechette	1	1	107
AC – Precision	1	1	107
Missile – Artemis IV	1	1	107
Missile – Fragmentation	1	1	107
Missile – Inferno	1	1	107
Missile – Narc-Equipped	1	1	107
Missile – Semi-Guided	1	1	107
Narc – ECM	1	1	108
Narc – Explosive	1	1	108
Narc – Haywire	1	1	108
Narc – Homing (Narc)	1	1	108
Narc – Homing (iNarc)	1	1	108
Narc – Nemesis	1	1	108
Physical Weapons			
Claws	*	*	108
Flail	5	4	109
Hatchet	*	*	109
Lance	*	*	109
Mace	*	*	109
Retractable Blade	*	*	109
Sword	*	*	109
Vibroblade	*	*	109
Structure			
Standard	—	0	N/A
Composite	—	0	117
Endo-Composite	—	7	117
Endo-Steel	—	14	117
Reinforced	—	0	117
Miscellaneous Equipment			
AES	*	*	118
Armored Component	*	0	118
CASE	0.5	1	118
CASE II	1	1	119
Coolant Pod	1	1	119
Full-Head Ejection System	0	0	115
Heat Sink	1	1	51
Heat Sink, Double	1	3	51
Machine Gun Array	0.5	1*	102
OS Missile Launcher	—	—	103
OS Missile Launcher, Improved	—	—	103
PPC Capacitor	1	1*	105

*This item's tonnage and/or critical slot placement requires construction rules not covered in this book.

CLAN EQUIPMENT

Equipment	Tons	Critical Slots	Page Ref
Armor			
Standard	—	0	N/A
Ferro-Fibrous	—	7	114
Ferro-Lamellor	—	12	114
Hardened	—	0	114
Laser-Reflective	—	5	114
Cockpits			
Standard	3	—	47
Small	2	*	47, 115
Torso-Mounted	4	*	47, 115
Electronics			
Artemis IV FCS	1	1*	110
Artemis V FCS	1.5	2*	110
Targeting Computer	*	*	114
Engines			
Standard	—	6*	48
XL	—	10*	48
XXL	—	14*	115
Gyros			
Standard	—	4*	48
Maneuverability Aids			
Jump Jet, Standard	*	1*	14, 18
Jump Jet, Improved	*	2*	14, 18
MASC	*	*	116
Partial Wing	*	6*	116
Supercharger	*	1*	116
UMU	*	1*	19, 117

Equipment	Tons	Critical Slots	Page Ref
Munition Bins			
AC – Cluster	1	1	106
Missile – Artemis IV	1	1	107
Missile – Artemis V	1	1	107
Missile – Inferno	1	1	107
Missile – Narc-Equipped	1	1	107
Narc – Homing (Narc)	1	1	108
PAC – Armor-Piercing	1	1	106
PAC – Flechette	1	1	107
Physical Weapons			
Claws	*	*	108
Talons	*	*	109
Structure			
Standard	—	0	N/A
Composite	—	0	117
Endo-Composite	—	4	117
Endo-Steel	—	7	117
Reinforced	—	0	117
Miscellaneous Equipment			
AES	*	*	118
Armored Component	*	0	118
CASE	0*	0*	118
CASE II	0.5	1	119
Coolant Pod	1	1	119
Full-Head Ejection System	0	0	115
Heat Sink	1	1	51
Heat Sink, Double	1	2	51
Machine Gun Array	0.25	1*	102
OS Missile Launcher	—	—	103
OS Missile Launcher, Improved	—	—	103
PPC Capacitor	1	1*	105

* This item's tonnage and/or critical slot placement requires construction rules not covered in this book.

COMMON MISCONCEPTIONS

BattleTech has been played for more than thirty years, and one of its great strengths is that the rules, especially related to BattleMechs, haven't really changed all that much—greatly expanded, yes, but not radically changed. Nonetheless, over time a number of rules misconceptions have arisen, sometimes due to readings of more vaguely worded rules text in much earlier rulebooks, and sometimes simply due to the rules drift that can occur across the history of any popular game.

The following lists some of the most common misconceptions we've encountered, broken down in chapter order. The initial statement that is bolded and in italics is the misconception.

PLAYING THE GAME

1. ***"Immobile 'Mechs can't be given a move order."***

Unless a 'Mech is destroyed, it can be always be the subject of a movement selection. Thus, immobile 'Mechs (see p. 18) can indeed be given a movement selection, even though they can't actually move. This means they can be used as "initiative sinks" to allow moveable 'Mechs to act later in Initiative order.

2. ***"Rolling a 12 is always a hit, even if the Target Number for an attack, Piloting Skill Roll, etc. is higher than 12. Also, rolling a 2 is always a failure, even if the Target Number is 2 or lower."***

If the Target Number is higher than a 12, it is impossible to succeed. Likewise, if the Target Number is lower than a 2, it is impossible to fail the roll.

However, even with auto-success or auto-failure, you still need to roll if using a rapid-fire weapon or other items that have effects dependent on your roll result.

MOVEMENT

1. ***"A 'Mech that is reduced to 0 MP for any reason is considered immobile."***

There are several conditions under which a 'Mech has 0 MP, but is not considered immobile (as each hex covers thirty meters, there's lots of room for a 'Mech to thrash around in even if it can't actually leave that space). Immobile is a very specific rules status, as defined on page 18; if the rules don't specifically state that a 'Mech is immobile, then it is not.

2. ***"A 'Mech can make level changes while moving backwards."***

'Mechs may not move up or down hills (i.e. make level changes) while moving backwards (unless the optional *Backward Level Changes* rule is in effect; see p. 15).

3. ***"A 'Mech can run into water."***

'Mechs are not allowed to run into a water hex of Depth 1 or greater. 'Mechs may only walk or jump into such hexes. Note that they can use running movement to change facings in water, and/or to move directly from a water hex to a land hex.

4. ***"Entering Depth 1 water from a level 0 hex costs 2 MP."***

Presuming standard ground movement (i.e. not jumping), entering Depth 1 water from a Level 0 hex costs 3 MP.

It always costs 1 MP to enter any hex. In addition, it costs 1 MP for changing from Level 0 to Depth 1, and 1 MP to enter a Depth 1 water hex. Thus, it costs 3 MP to enter Depth 1 water, and they must be Walking MP, not Running MP, as explained above.

If a 'Mech attempts to enter a Depth 2 water hex from Level 0, it would cost 1 MP to enter any hex, 2 MP for changing two levels from 0 to Depth 2 and 3 MP for entering a Depth 2 water hex. Total MP spent = 6 MP.

5. ***"When jumping, you have to pay 1 MP extra for each level you move over. So jumping over a Level 6 hex would cost 7 MP."***

Jumping costs 1 MP per hex regardless of the terrain in that hex. So long as a 'Mech has sufficient MP to legally move over the level of the hex (see *Jumping Movement*, p. 18), moving through/over it costs 1 MP, whether it be a clear hex or a Level 10 hex.

6. ***"A 'Mech can crawl from one hex to another hex."***

If a 'Mech is prone for any reason, it may change facing in its hex or it may attempt to stand, but it may not crawl to an adjacent hex.

COMBAT

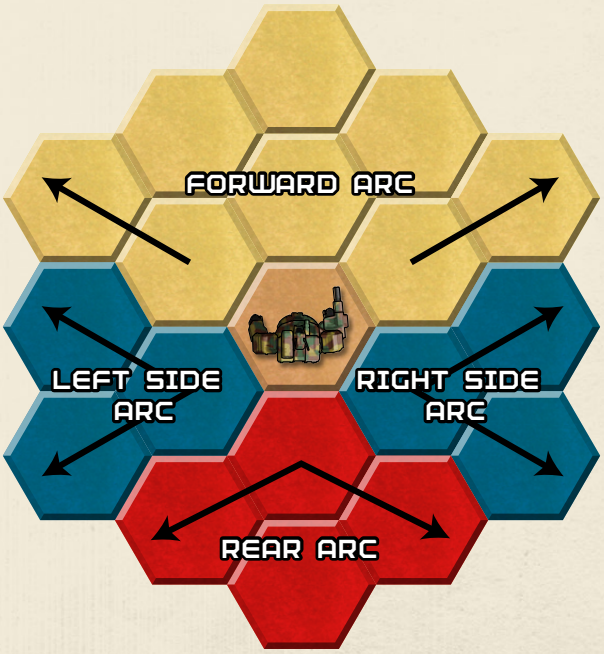
1. *"A 'Mech standing in light woods can't be seen by an enemy if there's two hexes of light woods between the two 'Mechs, but it can still shoot out at the enemy."*

LOS is only affected by *intervening* terrain (see p. 22), and any terrain a 'Mech is in (whether as attacker or target) isn't intervening. Therefore, standing in one hex of light woods and having two hexes of light woods between a 'Mech and its attacker does not block LOS in either direction.

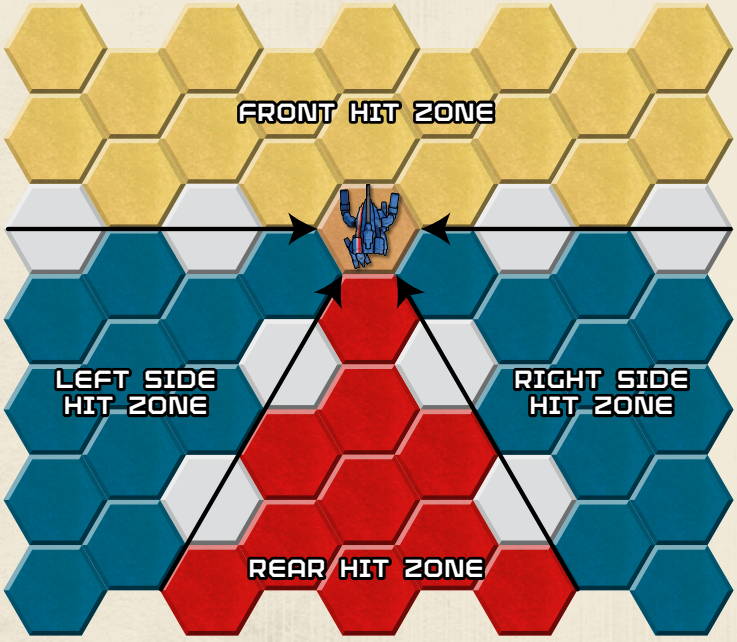
Remember that LOS is always mutual. If you can see a target, it can see you.

2. *"Firing arcs are the same as damage arcs."*

The two arcs are not the same. However, we recognize that while every rules publication on the matter has provided two separate diagrams outlining the differences, the diagram to illustrate a 'Mech's damage arc left a lot to be desired. We've placed the Firing Arcs Diagram and the Attack Direction Diagram side by side below, so that you can see how they differ.



• FIRING ARCS DIAGRAM •



• ATTACK DIRECTION DIAGRAM (WHITE HEXES = TARGET'S CHOICE) •

Note that there are no blind spots in firing arcs: every hex belongs to one of the four firing arcs.

3. *"A 'Mech can torso twist to change where an attacker's damage will strike it. If 'Mech A moves into the rear damage arc of 'Mech B, 'Mech B can torso twist and any damage from 'Mech A will instead strike 'Mech B's left or right side (depending on which way it twisted)."*

Torso twisting only affects a 'Mech's upper-body firing arcs, never the damage arcs against it. Thus, a 'Mech cannot torso twist to avoid getting shot in the back.

4. *"The only 'Mechs that can flip their arms are those specifically mentioned as such in their TRO text/other source (like the* Rifleman*)."*

Any 'Mech designed without *both* hand and lower arm actuators (like the *UrbanMech*)—as well as any OmniMech configuration lacking these actuators—can flip *both* its arms (not just one arm). But it must be capable of flipping both arms: a 'Mech with just one arm lacking these actuators cannot flip even that one arm—it's all or nothing.

5. *"Minimum range modifiers are not applied against a target that is exactly at minimum range. For example, if firing a PPC (minimum range of 3), if the target is 3 hexes away then no minimum range penalty is applied."*

Minimum range always includes the full number of hexes. If a weapon has a minimum range of 6 hexes, then the minimum range Target Number modifier begins being calculated there: +1 at 6 hexes, +2 at 5, and so forth.

6. *"If I spot a target for indirect fire, that lets me use TAG-guided munitions against that target."*

Hitting a target with TAG (see p. 58) also spots that target for indirect fire (see p. 30). However, the reverse is not true: spotting a target for indirect fire in no way helps TAG-guided munitions.

7. *"You declare and immediately resolve attacks on a 'Mech by 'Mech basis."*

Declaration and resolution are two separate stages. Only after all attacks by all players are declared do you begin resolving them.

8. *"If a 'Mech has partial cover, then attackers roll on the Punch Hit Location Table when assigning hits to it."*

This was true until 2006, when *Total Warfare* was released. Now 'Mechs with partial cover gain a +1 Target Number modifier, the standard Hit Location Table is used, and any hit striking the legs is ignored (and not re-rolled).

9. *"A prone 'Mech that has lost an arm can still fire any torso weapons it has during the Weapon Attack Phase."*

A prone 'Mech requires both arms to fire its torso weapons, although the weapons in one arm can fire as well in this case (unless using the *One-Armed Prone Fire* option; see p. 30).

10. *"Piloting Skill Rolls and rolling to hit for physical attacks are the same, so, for example, the +1 PSR modifier from having a small cockpit makes a 'Mech's physical attacks more difficult."*

Physical attacks are not Piloting Skill Rolls: modifiers that only affect Piloting Skill Rolls do not affect physical attacks, and vice versa.

11. *"Charge attacks must be made in a straight line to the target, and must be made using Running movement."*

A charge need neither be made in a straight line or by the shortest route. So long as the charging 'Mech has sufficient MP, it is allowed to go around terrain or enemy 'Mechs that are blocking its path to the target.

A charge can be made using either Walking or Running movement; only Jumping is forbidden (unless the charge is being made underwater; in this case, it can be made if the charging 'Mech uses UMUs; see p. 35).

12. *"When calculating charge attack damage, you include the hex the target occupies."*

While you must be able to pay the MP required to enter the target's hex in order to declare a charge attack, when determining damage, the hex the target occupies is not counted.

13. *"A 'Mech can attack with two physical weapons in the same phase."*

A 'Mech may only ever make one physical weapon attack in the Physical Attack Phase, unless using claws (see p. 108). A 'Mech may make two punch attacks in the same phase, but a punch is not a physical weapon attack (it's a punch attack).

If using the optional *Piercing* rules (see p. 109), a 'Mech can immediately make a follow-on retractable blade attack after landing a punch with the same arm.

14. *"A successful hatchet attack always rolls on the Punch Hit Location Table to determine damage location."*

Hatchet attacks (as well as mace, sword, or retractable blade attacks) by default roll on the regular Hit Location Table. However, if attacking with one of these four weapons, the attacker can choose to roll on the Punch or Kick Hit Location Table instead, but *only* if this intention was declared, and a +4 Target Number modifier applied, *prior* to the attack roll.

Only claws always roll on the Punch Hit Location Table. Flails (and clubs, which are not physical weapons but occupy their own unique category) always roll on the regular Hit Location Table.

15. *"Push attacks can be made against any enemy 'Mech that is within the three hexes of the pusher's front arc. A 'Mech is allowed to torso twist to push an enemy located in its side arc."*

Push attacks may only be made against an enemy 'Mech that is directly in front of the pushing 'Mech, and torso twists have no effect on this. Remember that at the end of a successful push attack, the pusher moves into the enemy's hex.

DAMAGE

1. *"The player making the attack is the one that rolls the locations of any critical hits that result."*

Since the very earliest editions of the game, while the attacker rolls on the Determining Critical Hits Table, the controller of the 'Mech that is taking the damage is the one that rolls to see where any critical hits to that 'Mech go.

However, as far as we can tell, nobody ever plays that way. As such, we've changed the rule to match common usage: the attacker now does indeed roll the location of any critical hits.

2. *"Critical hits on locations that don't have anything in them are wasted."*

If a 'Mech suffers a critical hit in a location that, at the start of the phase, had nothing that could be damaged (i.e. all slots in the location were inapplicable at the start of the phase; see p. 46), then the critical hit is applied to an adjacent location, as dictated by the Damage Transfer Diagram (see p. 43).

3. *"Since you resolve damage as it happens, if my 'Mech is destroyed in an attack phase before I've resolved my attacks, I don't get to attack back."*

There is a difference between damage resolution and applying the gameplay effects of damage. Always remember that all declared attacks must be resolved; they are only resolved one at a time as a way of figuring out exactly what happens. In the end, every player always gets the chance to make the attacks they declared in a phase, even if an attacking 'Mech loses the weapons it is attacking with in that phase, or is destroyed in that phase.

4. *"If you destroy the gyro, you destroy the 'Mech."*

While a destroyed gyro is very detrimental, a 'Mech without a gyro is not even considered immobile (see p. 18), let alone destroyed under standard *BattleTech* rules. Not even if using the optional *Forced Withdrawal* rules (see p. 81) does destroying a gyro also destroy that 'Mech.

5. *"When rolling for critical damage against a 'Mech's torso or arm, if the 'Mech has items in both blocks of slots (i.e. the first block of six slots, labeled "1–3", and the second block of six, labeled "4–6"), check to see if the second block only has one item in it. If it does, and the first D6 roll for critical damage indicates that equipment in the lower block was damaged, then the lone item is automatically damaged."*

The second D6 roll is always made. If the result indicates a hit on a "roll again" result, the entire process for determining which block and which item is hit starts over.

6. *"All combat is simultaneous, so if 'Mech A hits 'Mech B with two weapons (say, two PPCs) it doesn't matter which order the damage is resolved."*

"All combat is simultaneous" is a slightly misleading way of stating the rule that all declared attacks must be resolved. However, the order of resolution matters.

If a PPC hits 'Mech B's left rear torso and does exactly enough damage to eliminate all the remaining internal structure there (which also destroys the attached left arm) and then a second PPC hits 'Mech B's left arm, the damage will be transferred all the way to 'Mech B's center torso rear. If the rear armor and internal structure of the center torso could absorb less than 10 points of damage, 'Mech B would be destroyed.

If, however, the order of resolution were reversed (the first PPC hits the left arm and then the second PPC hits the left torso), so long as the left arm had enough armor and/or internal structure remaining to absorb 10 points of damage, no damage from those attacks would be done to 'Mech B's center torso and 'Mech B would not be destroyed.

7. *"Destroyed 'Mechs remain in the hex they were destroyed in and block other 'Mechs from entering the hex."*

Destroyed 'Mechs are removed from the map. Their carcasses have no effect on the game (unless using the optional *Wreckage* rules; see p. 49).

An Archer *ARC-2R's left torso contains 7 equipment items. The upper block of equipment contains an LRM 20 launcher (5 slots) and one of its tons of LRM 20 ammo (1 slot). The lower block of equipment contains a lone ton of LRM 20 ammo (1 slot). Should the* Archer *suffer a critical hit in the left torso, its attacker would roll a D6 first to determine which block—upper or lower—the critical damage is to be applied, and roll another D6 to determine which slot in that block to apply the critical damage to. A final result of "roll again" (or an inapplicable slot such as endo-steel, TSM, CASE, etc.; see* Inapplicable Slots, *p. 46) means that the entire process starts over with the* Archer's *attacker rolling to see which block—upper or lower—to apply the critical damage.*

Left Torso

1-3	1.	LRM 20
	2.	LRM 20
	3.	LRM 20
	4.	LRM 20
	5.	LRM 20
	6.	Ammo (LRM) 6
4-6	1.	Ammo (LRM) 6
	2.	Roll Again
	3.	Roll Again
	4.	Roll Again
	5.	Roll Again
	6.	Roll Again

HEAT

1. *"A player must make all rolls for shutdown avoidance and ammo explosion avoidance that are below their 'Mech's heat level."*

A player must only pass the shutdown avoidance roll and the ammo explosion avoidance rolls with the highest applicable Target Number. For example, a player whose 'Mech is at 22 heat would have to make two rolls, one needing an 8+ to avoid shutting down, and the other at 4+ to avoid an ammo explosion.

A MechWarrior does have to make all Consciousness Rolls required (see p. 58), however, which is perhaps where the confusion stems from.

OTHER ACTIONS

1. ***"Piloting Skill Rolls don't stack. If a 'Mech suffers a gyro hit and a foot actuator hit in the same turn that it takes 20+ points of damage, it only makes a single PSR with a +5 modifier."***

PSRs do indeed stack. A 'Mech that suffers a gyro hit, foot actuator hit, and 20 or more points of damage in the same phase would have to make three PSRs, and each one would have a +5 modifier (+3 for the gyro, +1 for the foot actuator, and +1 for the 20+ points of damage).

The only case where PSRs do not stack is when a 'Mech is missing a leg and makes an attempt to stand. In this case, only one PSR is required by the 'Mech, no matter how many might be triggered by the attempt. However, all applicable modifiers are still applied to that one roll.

2. ***"A 'Mech doesn't need to make a Piloting Skill Roll for jumping into water hexes."***

A PSR is required every time a 'Mech enters a water hex.

3. ***"A 'Mech needs to make a Piloting Skill Roll for jumping into woods hexes."***

No PSR is required for a 'Mech to jump into a woods hex.

4. ***"Once a 'Mech is in water, it doesn't need to make a Piloting Skill Roll for entering adjacent water hexes."***

A PSR is required every time a 'Mech enters a water hex.

5. ***"The modifier to Piloting Skill Rolls for taking 20+ points of damage in a phase is cumulative. So, it's +1 if I take 20 or more points of damage, but +2 if I take 40, and so on."***

In the standard rules, the modifier is only ever +1, whether the damage taken in a phase is 20 points or 200. Only if using the *Expanded Damage Modifiers* optional rule (see p. 53) does it work as presumed above.

6. ***"When a 'Mech falls, it falls into an adjacent hex as indicated by the Facing After Fall Table."***

A 'Mech falls into its own hex. The Facing After Fall Table is only used to determine which side the 'Mech fell onto, so that the damage from the fall can be applied to the correct locations.

7. ***"Piloting Skill Rolls made to see if the MechWarrior is injured after a fall aren't modified."***

All modifiers applied to the PSR that caused the fall are also applied to the PSR to see if the MechWarrior is damaged by that fall, with an additional +1 modifier applied for every level above 1 fallen.

SPECIAL-CASE RULES

1. ***"Hidden 'Mechs can't spot for indirect fire."***

Hidden 'Mechs can indeed spot for indirect fire without being revealed.

WEAPONS & EQUIPMENT

1. ***"ECM affects Streak missiles."***

Standard ECM has no effect whatsoever on Streak launchers; only Angel ECM (see p. 113) affects them. For the list of the equipment affected by standard ECM, see page 112.

INDEX

• A •
Active probes, 109-110
  Beagle, 110
  Bloodhound, 110
  ECM suites and, 112
  hidden 'Mechs and, 82
  light, 110
Actuator
  enhancement system, 118
  exposed, 87
  hyper-extended, 84
  protected, 85
Aerospace support
  defensive light air cover, 76
  defensive heavy air cover, 76-77
  offensive bombing, 76
  offensive strafing, 76
  offensive strikes, 76
Aimed shot, 30
  heat and, 52
  indirect fire and, 30
Ammo feed, static, 89
Ammunition, 96. *See also* Munitions
  critical hits and, 47
  dumping, 58
  expenditure, shots, 31
  explosion, 44
  explosion and MechWarrior damage, 58
  feed problem, 86
  heat and, 52
  switchable, 97
  use timing for, 31
Anti-aircraft targeting, 82
Anti-Infantry weapons (AI), 96
Anti-missile system, 118
Area-Effect weapons (AE), 96
  battlefield support damage, 75
  clearing woods, jungle, 61
  damage, 72
  engine explosion, 47
  hidden 'Mechs and, 82
Arm
  blown off hit, 47
  critical hits, 46
  low-mounted, 87
  no/minimal, 87
  overhead, 85
Armor
  ballistic-reinforced, 114
  diagram, 8, 44
  ferro-lamellor, 114
  hardened, 114
  laser-reflective, 114
  stealth, 115
  value, 6
Armored component, 118
  critical hit, 45
Armor-piercing ammo, critical hit, 45
Artemis FCS, indirect fire and, 30
Artemis IV/V fire-control systems, 110
  ECM suites and, 112
  indirect fire and, 30
Artillery, 96
  Arrow IV homing missiles, 97
Artillery support, 77-79
  Copperhead, 78
  pre-designated hexes, 77
  radius diagram, 77
  scatter diagram, 77
  targeting, 77
Attack. *See also specific attack types*
  arm flipping expanded, 25
  making, 25
  PSR timing, 55
  resolution order, 25
  reversing (flipping) arms, 24
  rolls, 31
Attack declaration, 24
  battlefield support, 75
Attack direction
  battlefield support, 75
  diagram, 32, 129
Attack modifiers table, 28-29, 134-135
Attacker movement modifier, 6
Autocannon, 98
  LB-X, 98
  light, 98
  ProtoMech, 98
  rotary, 98-99
  ultra, 99
Autocannon munitions
  armor-piercing, 106
  caseless, 106
  cluster, 106
  flechette, 107
  precision, 107

• B •
Backward
  level changes, 15
  movement, 15
Barrel fist, 82
Basement
  collapse, 73
  table, 140
Battle computer, 82
Battlefield support
  illumination/smoke artillery, 79
  one-time use, 75
  playing area, 75
  points, 75, 78
  table, 78, 142
  types of, 76-79
Battlefield tracking
  GM record sheet, 146
  record sheet, 145
Battlefists, 83
BattleMech. *See* 'Mech
Bridge
  intervening, 23
  movement, 20
Building
  attacking, 32, 35, 71-72
  basements, 66
  basements table, 66
  combat diagram, 72
  combat effects on, 71
  construction factor and, 66
  damage and, 66, 73. *See also* Building collapse
  displacement and, 55
  falling, and, 67
  fire in, 64
  hex, Inferno missiles and, 107
  intervening, 23
  involuntary entry and, 67
  jumping and, 67
  levels, 66
  line of sight and, 22
  modifiers table, 67, 140
  movement and, 13, 67
  movement modifiers table, 67, 140
  multiple hex, 66
  simplified, 66
  skidding into, 70-71
  stacking and, 67
  targets in, 35
  types, 66
Building collapse, 67, 73
  damage and, 73
  diagram, 74
  displacement and, 73, 74
  falling and, 74
  hex, 73-74
  movement and, 73

• C •
C^3 computer system, 110
  boosted, 112
  configuration diagram, 111
  designating, 112
  diagram, 111
  ECM and, 110, 112-113
  master network, 112
  multiple networks, 112
  stealth armor and, 110
  TAG and, 110
C^3I computer system, 112
  diagram, 112
  ECM and, 112-113
Cannons, 106
Careful stand, 19
Cargo, dropping, 21
Cellular ammunition storage equipment (CASE), 43, 47, 118
Cellular ammunition storage equipment II (CASE II), 43, 47, 119
Charge attacks, 35-36
  falls and, 36
  location after, 36
  multiple targets, 35
  multiple, 35
  UMUs and, 35
Clan
  equipment, 127
  weapons, 124-125
Claws, 36, 38, 108
Clear terrain, 7, 60
Club attack, 36
Clubs, 25
Cluster attack, 31, 32
Cluster hits table, 33, 136
Cluster weapons (C), 96
  clearing woods, jungle, 61
Cockpit, cramped, 86
Cockpit systems
  full-head ejection system, 115
  small, 115
  torso-mounted, 115
Collision
  dodging, 71
  with 'Mechs, 71
  skidding and, 70
  with stationary objects, 70-71
Combat
  building diagram, 72
  computer, 83
  lifting, 21
  line of sight, 22
  line of sight diagram, 22
  misconceptions, 129-130
  target, 22
  within buildings, 72
Command 'Mech, 83
Communications, improved, 84
Compact 'Mech, 83
Components, 7-8
Consciousness
  recovering, 58
  rolls, 55, 58
Construction factor, buildings, 66
Coolant pod, 51, 119
Cooling jacket
  improved, 84
  no, 87
  poor, 88
Cooling system flaws, 86
Counters, 7
Cowl, 83
Critical damage
  jumping, 14
  kick attacks, 38
  physical attacks, 34
  running, 14
Critical hit, 43, 45
  ammunition and, 47
  applying, 46
  arm blown off, 47
  cockpit, 47
  destroyed locations, 46
  determining, 32
  determining table, 140
  effects, 96
  engine, 47
  engine explosions, 47
  floating, 45
  foot actuator, 47
  gyro, 48
  hand actuator, 48
  head blown off, 48
  heat sinks, 48
  hip, 48
  inapplicable slots, 46
  jump jet, 48
  leg blown off, 48
  life support, 48
  lower arm actuator, 48
  lower leg actuator, 49
  multiple, 46
  multi-slot components, 46
  sensors, 49
  shoulder, 49
  table, 8, 46
  through-armor, 45
  transferring, 46
  upper arm actuator, 49
  upper leg actuator, 49
  weapons, equipment, 49

• D •
Damage
  attacks, 41
  crippling, 81
  explosive components and, 41
  'Mech, 41
  MechWarrior, 58
  misconceptions, 130-131
  modifiers, 27
  skidding, 68
  step-by-step, 41-43
  timing, 43
  transfer, 44
  transfer diagram, 42
  value, 6
Damage resolution, 32, 41
  ammunition, internal components explosion, 41
  armor and, 41
  CASE, CASE II and, 43
  critical hit components and, 41
  flow chart, 42, 138
  internal structure and, 41
  physical attacks, 40
Damage value groupings
  battlefield support, 75
  cluster weapons, 96
  Copperhead artillery, 78
Death from above attacks, 26, 36
  damage to attacker, 37
  diagram, 37
  jumping, 19
  location after, 37
  multiple, 36
  prohibited terrain, 37
  shutdown targets, 37
  stacking and, 37
  UMUs and, 37
  during weapon attack, 37
Depths, 6
  line of sight and, 22
Destroyed 'Mech, 49
  displacement, 55
Dice, 7
  movement, 13
Direct-Fire Ballistic weapons (DB), 97
Direct-Fire Energy weapons (DE), 97
Directional torso mount, 83
Displacement, 55
  diagram, 56
  physical attacks, 34
  prohibited, 55
  PSR timing, 55
  timing, 55
Distracting quirk, 83
Diversionary raid support, 75
Domino effect, 55
  dodging, 55-56
  multiple, 56
Dropping 'Mech, 79-80

• E •
ECM suites, 112-113
  Angel, 113
  Guardian, 112-113
Ejection, 81
  auto-, 81
  difficult, 86
  full-head, 81
  modifiers table, 81, 142
  no system, 87
  torso-mounted cockpit, 81
  water and, 81
Electromagnetic interference, 87
Electronics, 109-114
End phase, 12
Engine
  critical hits heat, 50
  explosions, 47
  XXL, 115
Equipment
  Clan, 127
  consciousness rolls and, 58
  critical hit, 49
  heat, 50
  Inner Sphere, 126
  misconceptions, 132
Explosive
  components, 49
  weapons (X), 97
Extraction raid support, 75
Extreme temperatures, 62

• F •
Facing, 14
  change diagram, 14
  after fall, 56
  after fall table, 57, 140
Falling, 56
  accidental, 70
  accidental, from above, 57
  building collapse and, 73
  charge attacks and, 36
  collisions and, 56
  diagram, 56
  facing after, 56
  facing after table, 57

INDEX

location after, 56
'Mech damage from, 57-58
'Mech hits target, 57
'Mech misses target, 57
Fast reload, 83
Fine manipulators, 84
Fire
accidental, 63
determining spread, 64
effects of, 63-64
extinguishing, 64
illumination and, 64
Inferno missiles, 107
intentional, 63
segmented, 11
spreading, 64
starting, 63
table, 64, 141
wind direction and, 63
Firing arc, 24
diagram, 24, 129
forward, 24
left side, 24
multiple, 27
rear, 24
right side, 24
torso twists and, 24
Firing weapons, 25
Flail, 38, 109
Flak (F), 97
dropping 'Mech and, 80
Flamers, 99
Flechette ammunition, clearing woods, jungle, 61
Floating criticals, 45
Fog, 62
Forced off-board, displacement and, 55
Forced withdrawal, 81
Four-legged 'Mech, 5
firing, 30
hip critical hit, 48
kick attacks, 38
leg destruction, 45
partial cover modifier, 26
piloting skill rolls, 53
standing up, 19

• G •

Game playing
deploying 'Mechs, 10
misconceptions, 128
sequence of play, 10-12
GATOR
attack movement modifier, 25
base target number, 25
damage modifiers, 27
heat modifier, 26
immobile target modifiers, 27
multiple targets modifier, 27
partial cover modifier, 26
range modifier, 27
target movement modifier, 25
target number modifier, 25
terrain modifiers, 26
weapon, equipment modifiers, 26
Gauss rifles, 99
anti-personnel, 99
heavy, 100
hyper assault, 100
improved heavy, 100
light, 100
Magshot, 100
silver bullet, 100
Girder clubs, 36
Good reputation, 84
Ground movement, 15
backward, 15
direction diagram, 15
Gyro, 48
heavy-duty, 115

• H •

Hand
actuator damage, 48
vestigial, 86
Hardened armor, critical hit, 45
Hatchet, 38, 109
Head
armor, weak, 89
critical hits, 46
destruction, 44
hits, 'Mech damage, 58
shots, 30
Heat
ammunition and, 52
building up, 50
data, 8
data table, 51
dissipating, 51
effects, 51, 96
'Mech damage, 58
misconceptions, 131
movement and, 50-52
outside sources of, 50-51
overflow, 51
phase, 12
point, 50
point table, 50, 140
recording, 51
scale, 8
shutdown and, 52
standing up, 19
timing, 51
weapon attacks, 52
Heat sink
critical hit, 48
shutting off, 50
types, 51
Heat-causing weapon (H), 97
Hexes
occupied, 16
partially occupied, 16
Hidden 'Mech, 21, 82
combat, 22
forbidden hexes, 82
jumping, 82
misconceptions, 132
point-blank shots, 82
Hills, intervening, 23
Hip critical hit, 48
Hit. *See also* Critical hit
attack direction and, 32
location, 32
rolling for, 31
Hull integrity
breaches, 43, 65

• I •

Ice, 61
Immobile 'Mech, 18
piloting skill rolls, 18, 53
target modifiers, 27
Inapplicable slots, 46
iNarc pod, 107
explosive, 108
homing, 108
Indirect fire, 30-31
C^3 computer systems, 110
line of sight, 22
Inferno missiles, heat and, 52
Initiative, 10
Inner Sphere
equipment, 126
weapons, 121-123
Internal structure diagram, 44

• J •

Jettison-capable weapon, 84-85
Jump jet hit, 48
Jumper, nimble, 85
Jumping, 14, 18-19
diagram, 18
heat and, 50-51
Jungle
clearing, 30, 61
heavy, 7, 60
intervening, 23
light, 7, 60
line of sight and, 22
partial cover modifier, 26
terrain modifiers, 26

• K •

Kick attacks, 38

• L •

Lance, 38, 109
Landing failures, 80
Laser, 100
anti-missile system, 118
binary (Blazer) cannon, 100
chemical, 100
extended-range, 100
extended-range pulse, 101
heavy, 101
improved heavy, 101
pulse, 101
re-engineered, 101
variable-speed pulse, 101
X-pulse, 101
Lateral shift, 15
Leg
critical hits, 46
reinforced, 85
weak, 89
Leg destruction, 44-45
minimum movement and, 45
standing up, 19
Leg-mounted weapons, partial cover and, 24
Level, 6, 8-9
collisions and, 71
different, physical attacks and, 34
different, physical attacks table, 34
range modifier, 27
table, 137
Level change, 15
backward movement, 15
prohibited, 15
Lifting, 21
Light, 62
Limb clubs, 36, 43
Line of sight, 22
adjacent 'Mech, 22
bordering two hexes, 22
C^3 computer systems, 110
clearing woods, jungle, 61
diagram, 23
indirect fire, 22
inside building, 72
intervening terrain, 22-23
levels and height, 22
other 'Mechs, 23
underwater table, 65
Location, 6
destruction, 43

• M •

Mace, 38, 109
Machine guns, 101
array, 102
heavy, 101
light, 102
Maintain
easy to, 83
difficult to, 87
Mapsheet, 6, 7
Margin of failure, 6
Margin of success, 6
MASC, 116
Master unit list, 89
'Mech, 5
abandoned, 49
data, 8
forced withdrawal, 49
heights, LOS and, 22
illegal design, 87
movement of hidden, 13
record sheet, 7, 8
skills of, 9
unequal numbers, 12
'Mech damage, 52, 58
ammunition explosions, 52
destroying, 49
falling damage, 57
gunnery skill rating, 9
hit location table, 33, 136
kick location table, 136
piloting skill rolls, 9, 53
PSR timing, 55
punch location table, 137
wreckage, 49
'Mech design quirks
base design vs. variants, 89
listing, 89
negative, 86-89
negative table, 144
positive, 82-86
positive table, 143
table, 90-95
Mechanical jump boosters, 116
Minefield support, 79
Minefield
clearing, 79
detecting, 79
dropping 'Mech and, 79, 80
hidden 'Mech, 82
Mines, 96
Missile weapons (M), 97
Missile launcher
advanced fire control, 102
advanced tactical, 102
ammo explosions, 102
enhanced LRM, 102
extended LRM, 103
improved one-shot, 103
indirect fire, 102
long range, 103
medium range, 103
multi-, 103
one-shot, 103
rocket, 103
short range, 103
special munitions, 102
streak LRM, 104
streak SRM, 104
thunderbolt, 104
torpedo, 104
Missile munitions
Artemis IV/V, 107
fragmentation, 107
inferno, 107
inferno explosion table, 107
Narc-equipped, 107
semi-guided LR, 107
Modified target number, 6
Morale, broken, 81
Movement, 13-14
basics, 14-15
building and, 13, 67-71
costs table, 17, 133
dice, 13
hidden 'Mechs and, 13
minimum, 13, 16, 19
misconceptions, 128
modes, 13-14
multiple 'Mechs and, 10
pavement, 20
phase, 10
phase, PSR timing, 55
points, 6
reckless, 20-21
under-bridge, 20
M-Pods, 101
MRM "Apollo" fire control system, 113
Mud, 61
Multiple locations, critical hit, 45
Multiple target modifiers, 27
Multi-trac 'Mech, 85
Myomer acceleration signal circuitry (MASC), 116
running and, 14
table, 116

• N •

Narc missile beacons, 104
ECM suites and, 113
indirect fire and, 30
iNarc and, 104-105
Narc pods, 107
ECM, 108
explosive, 108
homing, 108
Nemesis, 108
Night, 62
Non-functional item, 87
Non-standard parts, 88

• O •

Objective raid support, 75
Obsolete 'Mech, 88
One-armed prone fire, 30
One-legged 'Mech, walking, 13
One-shot weapons (OS), 97
heat and, 52
Oversized 'Mech, 88

• P •

Partial cover
aimed shot, 30
critical hit, 45
hit location and, 32
indirect fire, 30
LOS and, 23
modifier, 26
terrain modifiers, 26
water and physical attacks, 35
Partial hexes, ground movement, 15
Partial wing, 116
Particle projector cannons, 105
capacitor, 105-106
ER, 105
heavy, 105
light, 105
snub-nose, 105
Pavement, 7, 60
movement on, 20
running and, 14

INDEX

Physical attack, 12, 34
comparative modifier, 34
damage resolution, 40
on different levels, 34, 35
on different levels table, 34
inside building, 72
making, 34
modifiers, 28, 29
modifiers table, 34
multiple, 34
multiple targets, 27
prone 'Mechs, 35
water and, 35
Physical weapon attack, 38
table, 38, 137
Pilot
easy to, 83
hard to, 87
Piloting skill roll (PSR), 13, 53
consciousness and, 58
damage timing and, 43
expanded damage modifiers, 53
immobile, 18
making, 53
modifiers, 47, 48, 49
multiple rolls and, 53
table, 54, 139
timing, 55
Planetary assault support, 75
Planetary conditions, 59
dropping 'Mechs, 80
Plasma weapons, 106
clearing woods or jungle and, 61
Playing area, 6
Point-blank
shot, 82
weapons (PB), 97
Poor
performance, 88
workmanship, 88
Primitive 'Mech, critical hit, 45
Profile, narrow/low, 85
Prone 'Mech, 19
attack direction, 32
attacking, 30
combat in water, 27
damage arc for falling, 56
dropping to ground, 19
firing, 30
minimum movement, 16
movement and, 13, 19
one-armed fire, 30
partial cover modifier, 26
physical attacks, 35
piloting skill rolls, 53
reversing arms prohibition, 24
standing up, 19
target movement modifier, 25
ProtoMech autocannon, 98
Pulse weapons (P), 97
Punch attack, 38, 40
location table, 39
missing/destroyed actuators, 40
multiple targets, 40
Push attack, 40

• Q •

Quad 'Mech, see *Four-legged 'Mech*

• R •

Rain, 62
Ramshackle 'Mech, 88
Range, 27
Rapid-fire
attack, 31
hit location and, 32
weapons (R), 97
Rear
attacks vs. secondary targets, 27
damage
accidental falls from above and, 57
dumping ammunition and, 58
transfer, 44
vs. prone 'Mechs, DFAs and, 37
firing arc, 24
hit zone, 32
kicks, four-legged 'Mechs and, 38
Record sheet
battlefield tracking, 145
GM battlefield tracking, 146
Reinforced structure, critical hit, 45
Reputation
bad, 86
good, 84
Retractable blade, 38, 109
Rifles, 106
Road movement, 20
Rough terrain, 7, 60
Rubble, 7, 60
Rugged 'Mech, 85
Rules
optional, 5
strategic, 5
Running, 13-14
heat, 50

• S •

Sand, 7, 60
Scale, 8
realism and, 8
Scenario, 6
Sealing, poor, 88
Searchlight, 86
Segmented fire, 11
Semi-guided missiles, 107
Sensor ghosts, 88
Sensors
critical hit, 49
improved, 84
Shutdown, 52
Skidding, 20, 68
collisions diagram, 69
combat effects of, 68
diagram, 68, 70
initial movement diagram, 68
interrupting, 70
modifiers table, 68, 140
Slot, 6
inapplicable, 46
Slow traverse, 89
Smoke
fire spread and, 64
heavy, 61, 64
light, 61, 64
wind direction, 64
Snow
deep, 61
heat dissipation, 51
heavy, 63
moderate, 62
Specialized attack, 30
Sprinting, 14
Stable 'Mech, 86
Stacked terrain, 15
Stacking, 18
death from above and, 37
dropping 'Mechs, 80
landing failures, 80
Standing still, 13
Standing up, heat, 50
Stealth armor, 27
heat, 50
Structural components, 114-117
Structure, 117
Sublevels, 6, 9
LOS and, 22
Submerged 'Mechs, 23
Supercharger, 116
Swamp, 61
Switchable ammo weapons (S), 97
Sword, 38, 109

• T •

Talons, 38, 109
Target, 6, 22
hex as, 22
unusual, 35
Target acquisition gear (TAG), 58, 113-114
designation, 21
indirect fire and, 31
Target movement modifier (TMM), 6, 25
Target number, 6
Target number modifiers, 48, 49
battlefield support, 75
Targeting
computer, 114
computer, aimed shot, 30
improved, 84
poor, 88
variable range, 86
Terrain, 15
difficult, 15
height, depth, 22
hex icons, 7
intervening, 22-23
modifications, 61
modifiers, 26
skidding, 70
types, 59-60
water, 15
Terrain factor and conversion table, 61, 141
Through-armor critical (TAC), 45
Torpedoes, indirect fire, 31
Torso
critical hit, 46
destruction, 45
Torso twist, 11, 14, 25
extended, 83
firing arc and, 24
Torso-mounted cockpit, 44, 47, 48, 49
critical hit, 52
Total Warfare, 5
Tree clubs, 36
Triple-strength myomer (TSM), 117

• U •

Ubiquitous 'Mech, 86
Unbalanced 'Mech, 89
Unconscious warrior, 58
Underwater
ammunition dumping, 58
combat, attacking, 64
hull integrity and breaches, 65
line of sight table, 65, 141
physical attacks, 65
range, 64
range table, 65, 141
Underwater maneuvering unit (UMU), 117

• V •

Variable damage weapons (V), 97
Vestigial hands, 86
Vibroblade, 38, 109
Victory conditions, 12

• W •

Walking, 13
heat, 50
Warrior data, 8
Watchdog composite electronic warfare system, 114
Water, 7
active probes, 110
battlefield support, 75
C^3 computer systems, 110
combat in, 27
depth, 60
depth 2+, 35
engine explosion, 47
falling damage and, 57
heat dissipation and, 51
hexes, 23
intervening, 23
jumping into/out of, 19
'Mech falling and, 57
movement in, 15, 19-20
partial cover modifier, 26
physical attacks and, 35
running and, 14
skidding and, 70
terrain modifiers, 26
Weapon, 96
accurate, 82
Clan, 124-125
critical hit, 49
declaring fire, 10-11
exposed linkage, 87
heat, 50
inaccurate, 87
Inner Sphere, 121-123
jettison-capable, 84-85
misconceptions, 132
modifiers, 26
modular, 85
physical, 108-109
rear firing arc, 24
resolving fire, 11
stabilized, 86
types, 96-97, 120
Weapon attack, 10-11
death from above, 37
heat and, 52
modifiers, 28-29
Weather conditions
extreme temperatures, 62
fire and smoke, 63-64
fog, 62
light, 62
rain, 62
snowfall, 62-63
Wind
direction, 63
fire spread and, 64
moderate gale, 63
storm, 63
strong gale, 63
Woods
clearing, 30, 61
heavy, 7, 60
inferno missiles, 107
intervening, 23
light, 7, 60
line of sight and, 22
partial cover modifier, 26
terrain modifiers, 26
Wreckage, 49

• X •

XXL engines
heat, 50
movement, 13

MOVEMENT COSTS TABLE

Movement Action/Terrain Type	MP Cost Per Hex/Terrain Cost	Piloting Skill Roll
Cost to Enter Any Hex	1	—
Terrain Cost When Entering Any New Hex		
Clear	+0	—
Paved/Bridge	+0	—
Road	+0*	—
Rough	+1	—
Sand	+0	+1
Light woods	+1	—
Heavy woods	+2	—
Light jungle	+2	+1
Heavy jungle	+3	+2
Water		
Depth 1	+1† (Level change MP cost not included)	–1
Depth 2+	+3† (Level change MP cost not included)	+0 (+1 if Depth 3+)
Level change (up or down)		
1 level	+1	—
2 levels	+2	—
Rubble	+1	+0
Light building	+1	‡
Medium building	+2	‡
Heavy building	+3	‡
Hardened building	+4	‡
Additional Movement Actions		
Facing change	1/hexside	—
Dropping to the ground	1	—
Standing up	2/attempt	+0
Lateral Shift (quad 'Mechs only)	+1	—
Terrain Modifications		
Ice	+1	+3§
Mud	+1	+1§
Deep Snow	+1	+1§
Swamp	+1	+1§
Weather Conditions		
Light Fog	+1	—
Heavy Fog	+2	—
Full Moon Night	+1**	—
Moonless Night	+2**	—

* If traveling along road; otherwise cost of underlying terrain.
† MP cost to move along the bottom of the water hex.
‡ Piloting Skill Roll required to prevent damage. The roll is modified for the 'Mech's movement per the Building Movement Modifiers Table (see p. 67).
§ This penalty applies to all PSRs made in this hex (including the PSR made when entering it).
**A 'Mech with an activated searchlight ignores this penalty.

ATTACK MODIFIERS TABLE

All Attacks: Weapons and Physical	Modifier
Attacker	
Movement (modifiers are cumulative)	
Stationary	None
Walked	+1
Ran	+2
Jumped	+3
Prone	+2 (does not apply to four-legged 'Mechs)
Skidding	+1
Terrain *(modifiers are cumulative)*	
Light Woods/Jungle	+1 per intervening hex; +1 if target in light woods/jungle
Heavy Woods/Jungle	+2 per intervening hex; +2 if target in heavy woods/jungle
Water*	
Depth 1	+1; see *Partial Cover Modifier*, p. 26
Depth 2	Underwater 'Mechs cannot target units that are not underwater (see *Terrain Modifiers*, p. 26).
Partial Cover	+1; see *Partial Cover Modifier*, p. 26
Target *(modifiers are cumulative)*	
Prone	–2 from adjacent hex; +1 from all others
Immobile	–4
Skidding	+2
Jumped	+1
Hexes Moved *(highest applicable modifier only)*	
Moved 0–2 hexes	0
Moved 3–4 hexes	+1
Moved 5–6 hexes	+2
Moved 7–9 hexes	+3
Moved 10–17 hexes	+4
Moved 18–24 hexes	+5
Moved 25+ hexes	+6
Stealth Armor	+1 at medium range, +2 at long range

Weapon Attacks Only	Modifier
Attacker	
'Mech Damage	
Sensor hit	+2
Shoulder hit	+4 for weapons in arm, disregard other damaged actuators in arm
Upper or lower arm actuator (each)	+1 for weapons in arm
Heat	
0–7	None
8–12	+1
13–16	+2
17–23	+3
24+	+4
Making Indirect Fire attack	+1 (+2 if the spotter also attacked this turn)
Spotting for Indirect Fire	+1

ATTACK MODIFIERS TABLE (CONTINUED)

Weapon Attacks Only (Continued)	Modifier
Range and Terrain	
Range	
Short	None
Medium	+2
Long	+4
Minimum range	[Minimum] – [Target Range] +1 (see *Minimum Range Modifier*, p. 27)
Each Intervening Hex/Level between Attacker and Target (as well as target's hex) in same multi-hex building	+1 per hex/level (maximum +3; see *Combat Within Buildings*, p. 72)
Target	
Secondary target in forward arc	+1
Secondary target in side or rear arc	+2
Weather Conditions	
Heavy Fog	+1 to all attacks of the pulse (P) and direct-fire energy (DE) types
Light	Modifiers reduced if the target has high heat (see p. 62)
Dusk/Dawn	+1
Full Moon Night	+2; ignore if the target is illuminated
Moonless Night	+3; ignore if the target is illuminated
Moderate/Heavy Rain	+1
Torrential Downpour	+2
Moderate/Heavy Snowfall	+1
Wind	
Moderate Gale	+1 to all attacks of the missile (M) type (see p. 97)
Strong Gale	+2 to all attacks of the missile (M) type, and +1 to all attacks of the direct-fire ballistic (DB) type (see pp. 96-97)
Storm	+3 to all attacks of the missile (M) type, and +2 to all attacks of the direct-fire ballistic (DB) type (see pp. 96-97)

Physical Attacks Only	
Attacker	
'Mech Damage	
Shoulder hit	No punching or physical weapon attack with arm; no clubbing attacks; +2 to pushing attack (each)
Upper or lower arm actuator hit (each)	+2 to punching and physical weapon attack with arm; half damage for punching attack with arm; +2 to clubbing attacks
Hand actuator hit	+1 to punching attack with arm; no clubbing attacks; no physical weapon attack with arm
Hip actuator hit	No kicking attacks
Upper or lower leg actuator hit (each)	+2 and half damage to kicking attack with that leg
Foot actuator hit	+1 to kicking attack with that leg
Other Modifiers	
Charging attack	Modify for relative Piloting Skills (see *Comparative Modifier*, p. 34)
Death from above attack	Modify for relative Piloting Skills (see *Comparative Modifier*, p. 34)
Rear kick (quad 'Mechs only)	+1
Unintentional charge	+3
Weather Conditions	
Moonless Night	+1; ignore if the target is illuminated

*See *Terrain Modifiers*, page 26, for exceptions.

CLUSTER HITS TABLE

Roll (2D6)	Weapon Size 2	3	4	5	6	7	9	10	12	15	20	30	40
2	1	1	1	1	2	2	3	3	4	5	6	10	12
3	1	1	2	2	2	2	3	3	4	5	6	10	12
4	1	1	2	2	3	3	4	4	5	6	9	12	18
5	1	2	2	3	3	4	5	6	8	9	12	18	24
6	1	2	2	3	4	4	5	6	8	9	12	18	24
7	1	2	3	3	4	4	5	6	8	9	12	18	24
8	2	2	3	3	4	4	5	6	8	9	12	18	24
9	2	2	3	4	5	6	7	8	10	12	16	24	32
10	2	3	3	4	5	6	7	8	10	12	16	24	32
11	2	3	4	5	6	7	9	10	12	15	20	30	40
12	2	3	4	5	6	7	9	10	12	15	20	30	40

'MECH HIT LOCATION TABLE

Roll (2D6)	Left Side	Biped (Four-legged) Front/Rear	Right Side
2*	Left Torso [critical]	Center Torso [critical]	Right Torso [critical]
3	Left Leg (Left Rear Leg)	Right Arm (Right Front Leg)	Right Leg (Right Rear Leg)
4	Left Arm (Left Front Leg)	Right Arm (Right Front Leg)	Right Arm (Right Front Leg)
5	Left Arm (Left Front Leg)	Right Leg (Right Rear Leg)	Right Arm (Right Front Leg)
6	Left Leg (Left Rear Leg)	Right Torso	Right Leg (Right Rear Leg)
7	Left Torso	Center Torso	Right Torso
8	Center Torso	Left Torso	Center Torso
9	Right Torso	Left Leg (Left Rear Leg)	Left Torso
10	Right Arm (Right Front Leg)	Left Arm (Left Front Leg)	Left Arm (Left Front Leg)
11	Right Leg (Right Rear Leg)	Left Arm (Left Front Leg)	Left Leg (Left Rear Leg)
12	Head	Head	Head

*A result of 2 may inflict a critical hit. Apply damage to the armor in that section in the normal manner, but the attacking player also rolls once on the Determining Critical Hits Table, p. 46.

'MECH KICK LOCATION TABLE

Roll (1D6)	Left Side	Biped Front/Rear	Right Side
1–3	Left Leg	Right Leg	Right Leg
4–6	Left Leg	Left Leg	Right Leg

Roll (1D6)	Left Side	Four-legged Front/Rear	Right Side
1-3	Left Front Leg	Right Front Leg/Right Rear Leg	Right Front Leg
4-6	Left Rear Leg	Left Front Leg/Left Rear Leg	Right Rear Leg

'MECH PUNCH LOCATION TABLE

BIPED			
Roll (1D6)	Left Side	Front/Rear	Right Side
1	Left Torso	Left Arm	Right Torso
2	Left Torso	Left Torso	Right Torso
3	Center Torso	Center Torso	Center Torso
4	Left Arm	Right Torso	Right Arm
5	Left Arm	Right Arm	Right Arm
6	Head	Head	Head

FOUR-LEGGED			
Roll (1D6)	Left Side	Front/Rear	Right Side
1	Left Torso	Left Front Leg/Left Rear Leg	Right Torso
2	Left Torso	Left Torso	Right Torso
3	Center Torso	Center Torso	Center Torso
4	Left Front Leg	Right Torso	Right Front Leg
5	Left Rear Leg	Right Front Leg/ Right Rear Leg	Right Rear Leg
6	Head	Head	Head

DIFFERENT LEVELS TABLE

Target is:	Allowed Physical Attack
Standing 'Mech 1 level higher	Charge, Club (Kick Table), DFA, Physical Weapon (Kick Table), Punch (Kick Table)
Standing 'Mech 1 level lower	Charge, Club (Punch Table), DFA, Kick (Punch Table), Physical Weapon (Punch Table)
Prone 'Mech 1 level higher	Club, DFA, Physical Weapon, Punch
Prone 'Mech 1 level lower	DFA

Note: A 'Mech can always make a death from above attack if it has the necessary Jumping MP, provided the target is valid.

PHYSICAL WEAPON ATTACKS TABLE

Weapon Type	To-Hit Modifier	Damage Value**	Hit Location Table	Affected by TSM?
Claws	+1	1 per 7 tons	Punch	Yes
Flail	0	9†	Standard	No
Hatchet	–1	1 per 5 tons	Option‡	Yes
Lance	+1	1 per 5 tons	Standard	Yes
Mace	+1	1 per 4 tons	Option‡	Yes
Retractable Blade	–2*	1 per 10 tons	Option‡	Yes
Sword	–2	(1 per 10 tons) +1	Option‡	Yes
Talons	0	×1.5	As per attack	Kicks only
Vibroblade				
Activated	–2	See p. 109	Option‡	No
Deactivated	–2	(1 per 10 tons) +1††	Option‡	Yes

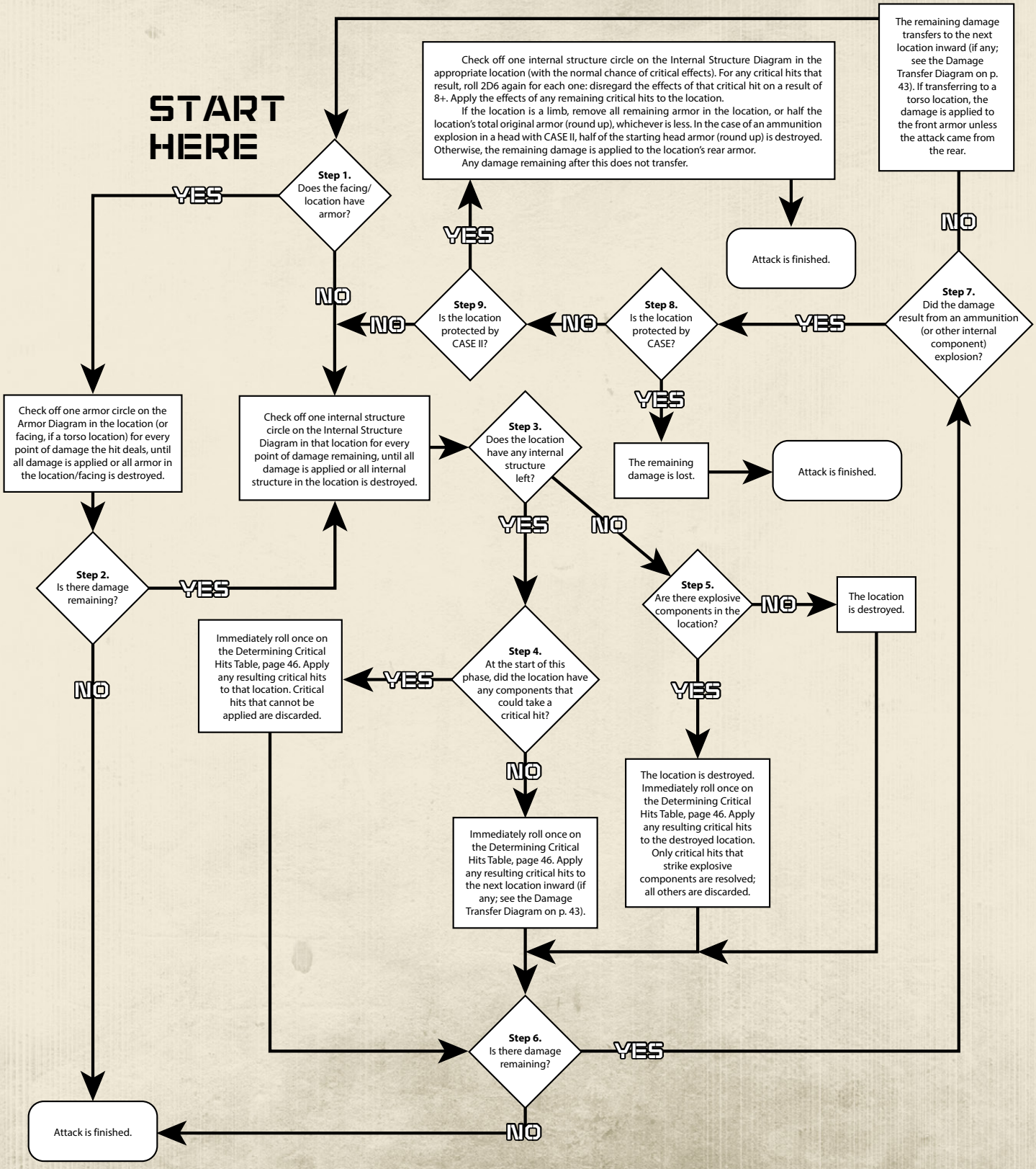
*Not affected by an absent or damaged hand actuator.
**Assign all damage as a single damage grouping. "1 per X tons" mean that the weapon does 1 point of damage for every X tons the 'Mech carrying it weighs (round fractional damage up); swords and deactivated vibroblades then add 1 point to this total.
†Not reduced by absent or missing actuators.
††Cannot be higher than the damage an activated vibroblade of the same type deals.
‡By default the weapon rolls on the regular Hit Location Table. However, when the player declares the physical weapon attack, they may also declare that it will use the Punch or Kick Hit Location Table instead. If so, apply a +4 Target Number modifier in addition to the weapon's standard modifier (the +4 modifier does not apply when forced to roll on a Punch/Kick Location Table due to attacks from different levels; see p. 34).

PHYSICAL ATTACK MODIFIERS TABLE

Attack Type	Modifier
Charging	+0
Clubbing	–1
Death From Above (DFA)	+0*
Kicking	–2
Physical Weapon	See p. 38
Punching	+0
Pushing	–1

*All the normal attack modifiers apply, including the attacker's jumping movement, but the roll is not modified for terrain.

• DAMAGE RESOLUTION FLOW CHART •



PILOTING SKILL ROLL TABLE

Cause of PSR	Modifier
Damage to 'Mech	
'Mech takes 20+ damage points in one phase	+1[1]
Involuntary fusion reactor shutdown	+3[2]
Leg/foot actuator destroyed	+1
Hip actuator destroyed	+2[3]
Gyro hit	+3
Gyro destroyed	Automatic fall[4]
Leg destroyed	Automatic fall[5]
Physical Attacks Against 'Mech	
'Mech was kicked	0
'Mech was pushed	0
'Mech was successfully charged/hit by death from above	+2
'Mech's Actions	
Missed kick	0
Made a successful charging attack	+2
Made a successful death from above attack	+4[6]
Entered Depth 1 water hex	–1
Entered Depth 2 water hex	0
Entered Depth 3+ water hex	+1
Attempted to stand	0
Entered sand hex	+1
Entered light jungle hex	+1
Entered heavy jungle hex	+2
Entered rubble hex	0
Running after facing change while on pavement	See *Skidding*, p. 68.
Jumped with damaged gyro or leg/foot/hip actuators	per Preexisting Damage, at right
Jumped with destroyed leg	per Preexisting Damage, at right
Ran with damaged hip or gyro	per Preexisting Damage, at right
Terrain Modifications	
Ice	+3[7]
Mud	+1[7]
Deep Snow	+1[7]
Swamp	+1[7]

[1]This is damage actually taken, after any damage modifiers are applied.

[2]Only during the phase that the reactor shuts down. If the MechWarrior must make a Piloting Skill Roll for a 'Mech with a shutdown reactor, the 'Mech automatically falls; in either case, if the 'Mech falls, the warrior automatically takes 1 point of damage (see *Falling Damage to a MechWarrior*, p. 57).

[3]This modifier can replace other modifiers. See *Hip (Leg)* on page 48 for details.

[4]The modifier for a destroyed gyro is +6 when making a Piloting Skill Roll to avoid damaging the MechWarrior during an automatic fall.

[5]The modifier for a destroyed leg is +5 when making a Piloting Skill Roll to avoid damaging the MechWarrior during an automatic fall.

Cause of PSR	Modifier
Skidding Movement	
Hexes moved in turn	
0–2	–1
3–4	0
5–7	+1
8–10	+2
11–17	+4
18–24	+5
25+	+6
Building Movement[8]	
'Mech entering light building hex	0
'Mech entering medium building hex	+1
'Mech entering heavy building hex	+2
'Mech entering hardened building hex	+5
Hexes moved in turn	
1–2	0
3–4	+1
5–6	+2
7–9	+3
10–17	+4
18–24	+5
25+	+6

Special Cases[9]	Modifier
MechWarrior trying to avoid damage when their 'Mech is falling	+1/every level above 1
Four-legged 'Mech with all legs	–2
Unintentional charge	+3
'Mech mounts AES in its legs	–2
'Mech mounts Hardened Armor	+1
'Mech mounts small or torso-mounted cockpit	+1

Preexisting Damage	
Per leg/foot actuator previously destroyed	+1
Per hip actuator previously destroyed	+2[10]
Gyro previously hit	+3
Leg previously destroyed	+5[11]

[6]Automatic fall if death from above attack is unsuccessful.

[7]This penalty applies to all PSRs made in this hex (including the PSR made when entering it).

[8]To avoid damage only; does not result in a fall if Piloting Skill Roll fails (see *Moving Into Buildings*, p. 67). Add an additional +1 modifier if 'Mech is charging or being charged (in addition to the +2 modifier normally required in that situation). A 'Mech displaced into a building automatically fails this PSR.

[9]These cases do not cause PSRs, but their modifiers apply to all PSRs made.

[10]Do not add modifiers for other damaged actuators in the leg.

[11]Ignore all other critical hit modifiers to that leg from previous phases (see p. 48).

FACING AFTER FALL TABLE

Roll (1D6)	New Facing	Hit Location
1	Same Direction	Front
2	1 Hexside Right	Right Side
3	2 Hexsides Right	Right Side
4	Opposite Direction	Rear
5	2 Hexsides Left	Left Side
6	1 Hexside Left	Left Side

BASEMENTS TABLE

Roll (2D6)	Effect
2	Double basement. The 'Mech falls 2 levels.
3-4	Basement. The 'Mech falls 1 level.
5-8	No basement.
9	Small basement. No effect on 'Mechs.
10-11	Basement. The 'Mech falls 1 level.
12	Double basement. The 'Mech falls 2 levels.

HEAT POINT TABLE

Activity	Heat Points
Walking	+1 per turn
Running	+2 per turn
Jumping	+1 per hex (min. of 3 per turn)
Attempting to stand	+1 per attempt
Weapons fire	Per Weapons Tables, pp. 120-125
Equipment	Per entry in Weapons and Equipment, pp. 120-127
Heat sink	–1 per operational heat sink –2 per operational double heat sink –1 additional per operational single heat sink under water (maximum 6 points) –2 additional per operational double heat sink underwater (maximum 6 points)
First engine hit	+5 per turn
Second engine hit	+10 (total) per turn

SKID MODIFIERS TABLE

Hexes Moved	Piloting Skill Modifier
0-2	–1
3-4	0
5-7	+1
8-10	+2
11-17	+4
18-24	+5
25+	+6

DETERMINING CRITICAL HITS TABLE

Roll (2D6)	Effect
2–7	No Critical Hit
8–9	Roll 1 Critical Hit Location
10–11	Roll 2 Critical Hit Locations
12	Head or Limb Blown Off / Roll 3 Critical Hit Locations*

*Roll 3 critical hit locations if the hit struck a torso location.

BUILDING MODIFIERS TABLE

Building Type	Original CF	MP Cost Per Hex	Piloting Skill Modifier
Light	1–15	+1	0
Medium	16–40	+2	+1
Heavy	41–90	+3	+2
Hardened	91–150	+4	+5
Cost to Enter Any Hex		*1*	

BUILDING MOVEMENT MODIFIERS TABLE

Hexes Moved In Turn	Piloting Skill Modifier
1–2	0
3–4	+1
5–6	+2
7–9	+3
10–17	+4
18–24	+5
25+	+6

POSITIVE QUIRK TABLE

Type	Cost	Brief Description
Accurate Weapon	Variable	All Target Numbers for that weapon receive a –1 modifier.
Anti-Aircraft Targeting	Variable	'Mech is more effective against airborne targets.
Barrel Fist	1	Arm with this Quirk does not apply the +1 TN modifier for punching without a hand actuator.
Battle Computer	5	'Mech provides a +2 Initiative bonus.
Battlefists	2	'Mech receives –1 TN modifier for punching.
Combat Computer	3	'Mech generates four points of heat less than normal.
Command 'Mech	2	'Mech provides a +1 Initiative bonus.
Compact 'Mech	1	Allows 2 'Mechs with this same Quirk to occupy one transport slot.
Cowl	4	+3 points of head armor except against attacks that originate along the row of hexes directly in front of the cowled 'Mech.
Directional Torso Mount	2 or 3	Weapons in mount may shoot in either the forward torso arc or the rear arc, depending on the mount's current facing. The mount's facing can be changed at the same time torso twists are made.
Distracting	1	+1 Special Source modifier to any Morale Checks when using the *Morale* rules (see pp. 179-180, *TO:AR*). If using the Demoralizer SPA (see p. 74, *CO*), reduce its Target Number modifier to +3.
Easy to Maintain	1	All repair or replacement rolls made for a 'Mech with this quirk receive a –1 Target Number modifier.
Easy to Pilot	2	A MechWarrior with a Piloting Skill higher than 3 will receive a –1 Target Number modifier for Piloting Skill Rolls required as a result of damage or terrain. More skilled MechWarriors receive no benefit.
Extended Torso Twist	3	When torso twisting, the 'Mech can change its facing by one or two hexsides.
Fast Reload	1	Affected weapons can reload in half the normal time.
Fine Manipulators	0	May mount up to one Manipulator (see p. 245, *TM*) in each side torso and/or arm location.
Good Reputation	0	For each level of this quirk a 'Mech has, it costs 10% more C-bills than normal.
Hyper-Extending Actuators	1	Can still flip arms to fire all arm-mounted weapons into its rear arc.
Improved Communications	1	The 'Mech is more resistant to electronic warfare.
Improved Cooling Jacket	1	When fired, this weapon generates 1 point less heat (never less than 1 point).
Improved Life Support	1	Treat the 'Mech's heat level as being 5 points lower when determining MechWarrior damage as a result of heat following a life support critical hit.
Improved Sensors	3	A 'Mech with this quirk is treated as if it has an active probe. If it is equipped with an actual active probe, instead add 2 to that active probe's range.
Improved Targeting, Short	3	All Short ranged attack Target Numbers receive a –1 modifier.
Improved Targeting, Medium	4	All Medium ranged attack Target Numbers receive a –1 modifier.
Improved Targeting, Long	5	All Long ranged attack Target Numbers receive a –1 modifier.
Jettison-Capable Weapon	1	Weapon can be jettisoned in combat and—if recovered—remounted on the same 'Mech with ease.
Modular Weapons	1	Equipment can be replaced in half the normal time.
Multi-Trac	2	Attack any number of targets in its front and arm firing arcs in the same turn without adding the secondary-target modifier. Secondary targets in the rear arc are treated as normal.
Narrow/Low Profile	3	Some attacks deliver only glancing blows to this 'Mech.
Nimble Jumper	3	When jumping, the 'Mech can be harder to hit and has more flexibility when choosing a jump path.
Overhead Arms	2	If it already has LOS to the target, the 'Mech treats its arm-mounted weapons as one level higher when determining terrain effects.
Protected Actuators	1	+1 TN against Leg and Swarm attacks by infantry.
Reinforced Legs	1	'Mech suffers half the normal damage to the legs when performing Death From Above successfully.
Rugged	1 or 2	1 point: Can go for two times as long between maintenance periods; 2 points: Three times as long.
Searchlight	1	'Mech gains a searchlight.
Stabilized Weapon	Variable	If the 'Mech runs or sprints all Target Numbers for that weapon receive a –1 modifier.
Stable	2	–1 Target Number modifier when forced to make a Piloting Skill Roll as a result of a physical attack.
Ubiquitous	1	When attempting to locate replacement parts for this 'Mech, add +2 to the Target Number.
Variable Range Targeting	Variable	All weapon attacks at the designated range receive a –1 Target Number modifier, but all weapon attacks at the other range receive a +1 Target Number modifier (medium range remains unmodified).
Vestigial Hands	0	'Mech with vestigial hands may lift and drop items if a scenario objective requires it.

NEGATIVE QUIRK TABLE

Type	Cost	Brief Description
Ammunition Feed Problem	1	After making attack roll with weapon, roll 2D6. On a 10+, the weapon jams and cannot be fired again in this battle. On a 12, the new round of ammunition will explode in the weapon for normal damage.
Bad Reputation	1	Worth only half the normal resale value. A Clan 'Mech begins the battle with 1 *dezgra* point.
Cooling System Flaws	3	Whenever the 'Mech executes or receives a physical attack, falls, or is forced to make a PSR because it received 20+ points damage, roll 2D6. On a 10+ the 'Mech will generate 5 points more heat each turn.
Cramped Cockpit	2	'Mech receives a +1 Piloting Skill Roll modifier.
Difficult Ejection	1	If the MechWarrior fails a PSR when ejecting, they suffer an additional point of damage.
Difficult to Maintain	1	All repair or replacement rolls made for this 'Mech receive a +1 Target Number modifier.
EM Interference	1 or 2	The turn after the weapon has been fired, Electronics aboard the 'Mech will not function.
Exposed Actuators	1	'Mech is more susceptible to anti-'Mech infantry attacks
Exposed Weapon Linkage	2	When the weapon's location is hit, roll 2D6: on a 10+, the weapon can't fire for the rest of the game.
Hard to Pilot	2	'Mech receives a +1 Target Number modifier to all Piloting Skill Rolls.
Illegal Design	0	'Mech may spontaneously break down at any point; much harder to repair (see p. 87).
Inaccurate Weapon	Variable	Weapon receives a +1 Target Number modifier.
Low-Mounted Arms	2	LOS from arms are as if they are from/to the 'Mech's legs; arms still rotate with the torso position.
No/Minimal Arms	2	+2 TN when making a PSR to stand up. 'Mech cannot make physical attacks with its arms.
No Cooling Jacket	2	Weapon generates 2 extra points of heat when fired.
No Ejection System	2	'Mech's pilot may not use Ejection rules
Non-Functional Item	5	Item simply does not work and cannot be repaired or replaced.
Non-Standard Parts	1	When attempting to locate replacement parts, add +2 to the Target Number.
Obsolete	4	'Mech suffers penalties to rolls to repair or locate replacement parts.
Oversized	2	PSRs to avoid damage when moving into buildings have a +1 Target Number modifier; the 'Mech never receives the +1 Target Number modifier for partial cover.
Poor Cooling Jacket	1	Weapon generates 1 extra point of heat when fired.
Poor Life Support	1	Treat the 'Mech's heat level as being 5 points higher when determining MechWarrior damage as a result of heat following a life support critical hit.
Poor Performance	3	'Mech must spend one turn expending MP equal to its Walking speed before it can use MP up to its Running speed the following turn.
Poor Sealing	2	+2 modifier to the roll when making a Hull Breach check. Must also make a Hull Breach check when first entering a hostile environment.
Poor Targeting (Short)	2	All Short ranged attack Target Numbers receive a +1 modifier.
Poor Targeting (Medium)	2	All Medium ranged attack Target Numbers receive a +1 modifier.
Poor Targeting (Long)	2	All Long ranged attack Target Numbers receive a +1 modifier.
Poor Workmanship	1	Rolls for critical hits are modified by +1.
Prototype	2	Rolls for critical hits are modified by +2.
Ramshackle	3	The 'Mech likely receives a random negative quirk at the start of each battle.
Sensor Ghosts	3	All ranged attack Target Numbers receive a +1 modifier.
Slow Traverse	1	Any turret on the 'Mech may only rotate one hexside a turn.
Static Ammo Feed	1	Weapon is incapable of switching between multiple ammunition bins.
Unbalanced	1	'Mech receives a +1 Target Number modifier to PSRs when entering a hex that requires PSRs.
Weak Head Armor	Variable	A number of head armor points missing equal to the value of the Quirk.
Weak Legs	1	When the 'Mech is kicked or executes a Death From Above attack, the 'Mech may suffer critical hits to one or more legs.

BATTLEFIELD SUPPORT TABLE

Support Type	Target Number*	Damage Value Groupings**	Weapon Type	Battlefield Support Point (BSP) Cost
Offensive Aerospace Support				
Light Strike	5	2	N/A	2
Light Bombing†	5	3	AE	3
Heavy Strike	6	4	N/A	3
Heavy Bombing†	7	6	AE	4
Strafing	7	3	DE	5
Defensive Aerospace Support				
Light Air Cover		N/A	N/A	1
Light Strike	3			
Light Bombing	4			
Heavy Strike	9			
Strafing/Heavy Bombing	11			
Heavy Air Cover		N/A	N/A	2
Light Strike/Bombing	9			
Heavy Strike	5			
Strafing/Heavy Bombing	6			
Artillery Support†				
Thumper	8	3/1††	AE	3
Sniper	8	4/2††	AE	4
Long Tom	8	5/3/1††	AE	6
(Copperhead)	‡	‡	N/A	‡
(Illumination/Smoke)	8	‡‡	N/A	‡‡
Minefield Support				
Light Density	9	2	AE	0.5
Medium Density	8	4	AE	2
Heavy Density	7	6	AE	4

*No modifiers are applied to this Target Number, except for Artillery Support (see p. 77).
†Scatters if misses; see page 76.
‡See rules, page 78.
**All Damage Value groupings are five points of damage.
††Affects multiple hexes; see Artillery Support, page 77.
‡‡See rules, page 79.

EJECTION MODIFIERS TABLE

Landing Terrain	Modifier
Clear	–2
Water	–1
Deep Snow	–2
Rough	0
Rubble	0
Light Jungle/Woods	+2
Heavy Jungle/Woods	+3
Per Level of Building	+1

Planetary Conditions	Modifier
Heavy Snowfall/Strong Gale/ Torrential Downpour	+2
Storm	+3
Situation	**Modifier**
BattleMech Prone	+5
Pilot Unconscious	+3
Per Point of Head Internal Structure Damage	+1
Automatic Ejection	+1

BATTLETECH™

BATTLEFIELD TRACKING (FOR ALL EYES)

TURN NUMBER

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

OPTIONAL RULES CHECKLIST

- ❑ Alternate Initiative (p. 10)
  _____
- ❑ Playing Card Initiative (p. 10)
- ❑ Lance Movement (p. 10)
- ❑ Segmented Fire (p. 11)
- ❑ Movement Dice (p. 13)
  - Walk Color _____
  - Run Color _____
  - Jump Color _____
  - Sprint Color _____
- ❑ Sprinting (p. 14)
- ❑ Backward Level Changes (p. 15)
- ❑ Partially-Occupied Hexes (p. 16)
- ❑ Careful Stand (p. 19)
- ❑ Extended Arm Flipping (p. 25)
- ❑ One-Armed Prone Fire (p. 30)
- ❑ Floating Criticals (p. 45)
- ❑ Engine Explosions (p. 47)
- ❑ Wreckage (p. 49)
- ❑ Expanded Damage Mods. (p. 53)
- ❑ Terrain Modifications (p. 61)
- ❑ Weather Conditions (p. 62)
- ❑ Fire and Smoke (p. 63)
- ❑ Underwater Combat (p. 64)
- ❑ Simplified Buildings (p. 66)
- ❑ Battlefield Support (p. 75)
- ❑ Dropping 'Mechs (p. 79)
- ❑ Ejection (p. 81)
- ❑ Forced Withdrawal (p. 81)
- ❑ Hidden 'Mechs (p. 82)
- ❑ Hidden 'Mechs (Expanded, p. 110)
- ❑ Design Quirks (p. 82)
- ❑ Enhanced Flamers (p. 99)
- ❑ Rapid-Fire Mode MGs (p. 101)
- ❑ Piercing (Retractable Blade, p. 109)
- ❑ Targeting (Active Probe, p. 110)
- ❑ Enhanced Missile Defense (AMS, p. 118)

G.A.T.O.R.

G **Gunnery Skill Rating** of the attacker (the base Target Number for the attack). Then add:
A **Attacker** movement modifier
T **Target** movement modifier
O **Other** modifiers (potentially a big category, but typically only includes woods, partial cover, and heat)
R **Range** modifiers

FORCED WITHDRAWAL (p. 81)

Any 'Mech that suffers crippling damage must withdraw from the map. Crippling damage is defined as one or more of the following:

- **Four or more points of damage to the MechWarrior**
- **The destruction of all sensor critical slots**
- **One gyro and one engine critical hit**
- **Two engine critical hits**
- **A side torso location is destroyed**
- **Internal structure damage in either three or more limbs or two or more torso locations (torso internal structure damage does not count towards crippling damage if that location still has front armor)**
- **The loss of all weapons. This is triggered if a 'Mech loses all weapons (either through damage or ammunition depletion) with a range greater than five hexes and if it can no longer cause more than 5 points of combined weapons damage. This does not apply if the 'Mech did not start with the ability to do 5 or more damage, or the ability to do damage at a range greater than five hexes; in this case the 'Mech can never trigger this condition.**

A 'Mech making a forced withdrawal must move at least one hex toward its home map edge. However, a 'Mech need not spend Running MP; it can also move backward if its controller wishes. Also, if equipped with MASC, a supercharger, or some other system that increases speed at the risk of damage, a 'Mech need not engage that system when forced to withdraw. Withdrawing 'Mechs may still attack an enemy 'Mech that closes within range of a weapon or physical attack.

WEATHER CONDITIONS

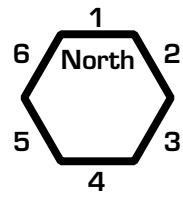
LIGHT (p. 62)
- ❑ Daylight (No additional Rules)
- ❑ Dusk/Dawn
- ❑ Full Moon Night
- ❑ Moonless Night

❑ **EXTREME TEMPERATURE** (p. 62)
C°_____Heat/Turn _____

FOG (p. 62)
❑ Light Fog ❑ Heavy Fog

RAIN (p. 62)
❑ Moderate ❑ Heavy ❑ Torrential

Wind Direction
Indicate which direction is North (1) on map (p. 63)



WIND (p. 63)
❑ Moderate Gale ❑ Strong Gale
❑ Storm

SNOW (p. 63)
❑ Moderate ❑ Heavy

TERRAIN MODIFICATIONS (p. 61)

❑ Ice Hexes _____
❑ Mud Hexes _____
❑ Snow, Deep Hexes _____
❑ Swamp Hexes _____

FIRE AND SMOKE (pp. 63-64)

Fire Hexes _____
Smoke Hexes _____

BATTLEFIELD SUPPORT (p. 75)

BATTLEFIELD SUPPORT POINTS*:
- ❑ Extraction Raid Support: 5 BSPs
- ❑ Diversionary Raid Support: 20 BSPs
- ❑ Planetary Assault Support: 35 BSPs
- ❑ Objective Raid Support: 12 BSPs
- ❑ Other _____
- Number of Maps* _____

Support Types:
- ❑ Offensive Aerospace Support
- ❑ Artillery Support
- ❑ Defensive Aerospace Support
- ❑ Minefield Support

*For every two maps beyond the first four, add 50% to the BSPs provided.

NOTES

© 2023 The Topps Company, Inc. Classic BattleTech, BattleTech, 'Mech and BattleMech are trademarks of The Topps Company, Inc. All rights reserved.Catalyst Game Labs and the Catalyst Game Labs logo are trademarks of InMediaRes Production, LLC. Permission to photocopy for personal use.

BATTLETECH™

BATTLEFIELD TRACKING (GM/TEAM EYES ONLY)

OBJECTIVE

BATTLEFIELD SUPPORT (p. 75)

TOTAL BATTLEFIELD SUPPORT POINTS: _____

OFFENSIVE AEROSPACE SUPPORT

Type	TN*	Dmg**	Cost/Each	Uses Remaining
❑ Light Strike	5	2	2	○○○○○○○○○○
❑ Light Bombing†	5	3 (AE)	3	○○○○○○○○○○
❑ Heavy Strike	6	4	3	○○○○○○○○○○
❑ Heavy Bombing†	7	6 (AE)	4	○○○○○○○○○○
❑ Strafing	7	3 (DE)	5	○○○○○○○○○○

DEFENSIVE AEROSPACE SUPPORT

Type	TN*	Cost/Each	Uses Remaining
❑ Light Air Cover	see below	1	○○○○○○○○○○
Light Strike	3		○○○○○○○○○○
Light Bombing	4		
Heavy Strike	9		
Strafing/Heavy Bombing	11		
❑ Heavy Air Cover	see below	2	○○○○○○○○○○
Light Strike/Bombing	9		○○○○○○○○○○
Heavy Strike	5		
Strafing/Heavy Bombing	6		

ARTILLERY SUPPORT†

Type	TN	Dmg**	Cost/Each	Uses Remaining
❑ Thumper	8	3/1†† (AE)	3	○○○○○○○○○○
Target Hex/Turn:				
Type	TN	Dmg**	Cost/Each	Uses Remaining
❑ Sniper	8	4/2†† (AE)	4	○○○○○○○○○○
Target Hex/Turn:				
Type	TN	Dmg**	Cost/Each	Uses Remaining
❑ Long Tom	8	5/3/1†† (AE)	6	○○○○○○○○○○
Target Hex/Turn:				
Type	TN	Dmg**	Cost/Each	Uses Remaining
❑ (Copperhead)	‡	‡	‡	○○○○○○○○○○
Target Hex/Turn:				
Type	TN	Dmg**	Cost/Each	Uses Remaining
❑ (Illumination/Smoke)	8	‡‡	‡‡	○○○○○○○○○○
Target Hex/Turn:				

MINEFIELD SUPPORT

Type	TN*	Dmg**	Cost/Each	Total Minefields
❑ Light Density	9	2 (AE)	0.5	
Mined Hexes:				
Type	TN*	Dmg**	Cost/Each	Total Minefields
❑ Medium Density	8	4 (AE)	2	
Mined Hexes:				
Type	TN*	Dmg**	Cost/Each	Total Minefields
❑ Heavy Density	7	6 (AE)	4	
Mined Hexes:				

*No modifiers are applied to this Target Number.
**All Damage represents groupings of five points of damage.
††Affects multiple hexes; see page 77
†Scatters if misses; see page 77.
‡See rules, page 78.
‡‡See rules, page 79.

OBJECTIVES/OTHER LOCATIONS

Objective	Hex	Notes

HIDDEN 'MECHS (P. 82)

'Mech	Hex	Facing	Miscellaneous

NOTES

© 2023 The Topps Company, Inc. Classic BattleTech, BattleTech, 'Mech and BattleMech are trademarks of The Topps Company, Inc.
All rights reserved.Catalyst Game Labs and the Catalyst Game Labs logo are trademarks of InMediaRes Production, LLC. Permission to photocopy for personal use.